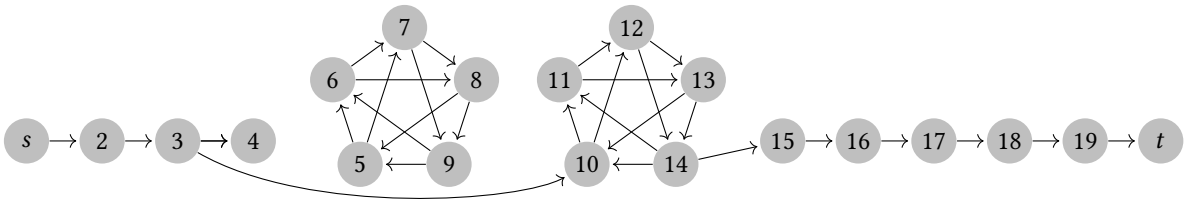
Climate Intelligence for Extreme Environments

A macroeconomic feasibility report on the *Climate Intelligence & Decision Twin for Dubai* and the *Africa Blue Intelligence Grid*

Stakeholders

Dr Uchi Dominic Terhile	PhD Graduate, Federal University Dutse, Nigeria
Dr Chakradhar Iyyunni	BITS Pilani Dubai Campus, United Arab Emirates
Abdullah Al Salmani	SpacePoint, United Arab Emirates
Arti Ahluwalia	Country Chair, Ecocivilisation.earth, India
Mahesh Shastry	Kalman Systems Pty Ltd, Sydney, Australia



Prepared from the stakeholders' own working documents, the rescoped Letter of Intent for the Dubai Future Foundation RDI Grant under the Cognitive Cities priority and the Africa Blue Intelligence Grid strategic concept paper of 20 September 2026, together with the sixty-seven published sources registered in Appendix A.

Private and confidential
For stakeholder circulation

Executive Summary

Keywords: urban heat; decision twin; outdoor tourism capacity; minimum cooling network; blue economy; Earth observation; life-cycle cost; stochastic dynamic programming; spiral development; synthetic data; risk modelling; policy

Two proposals are assessed here: the *Climate Intelligence & Decision Twin for Dubai*, artefact is designed to be presented to the Dubai Future Foundation Research, Development and Innovation grant ([1,400,000.00:388,888.89](AED:USD) over 36 months from 15 March 2027) [Iyyunni, 2026], and the *Africa Blue Intelligence Grid*, a continental ocean, environment, security and space-data programme convened and financed through the UAE [Terhile et al., 2026]. They are assessed as one portfolio because they share one architecture (physical system, data, semantic layer, knowledge graph, intelligence, decision twin, action, learning), the Dubai storyboard supplying its upper steps and the ABIG paper its lower ones.

Of sixty seven archived documents only two are measured with controls, both from the Phoenix cool pavement trials: coated residential streets ran about 12 °F cooler at noon and 10.5 °F cooler in the afternoon than adjacent aged asphalt [ASU and City of Phoenix, 2021], while mean radiant temperature at the road centre rose 4.5-5.8 °F at midday [ASU and City of Phoenix, 2024], a load, its authors note, unrepresentative of the adjoining foot path. Shade therefore ranks ahead of coatings. Abu Dhabi's 26-station network is the region's only measured urban heat study [Manandhar et al., 2019]; Paris OASIS [Urban Innovative Actions, 2023] published a controlled before and after design but no effect sizes. The gap is defensible: no archived study ties activity hours lost to heat, at corridor scale, to a built intervention tested against a decision twin's prediction.

Costs are reported in one unit (AED per restored activity hour for Dubai, cost per decision improved for ABIG) over a nine category breakdown from pre-award to renewal [ISO, 2017a, GAO, 2020]. Money contingency is unavailable: every plausible reference class uplift breaches the grant ceiling [Flyvbjerg, 2014]; contingency must be scope, phase and partner capital, and the month twelve gate is the programme's principal cost instrument.

Risk concentrates before submission: the written confirmation of usefulness from RTA or DEWA, the one gate condition the team cannot satisfy alone; the missing black globe instrumentation; the [500,000.00:138,888.89](AED:USD) annual budget cap against a derived annual average of [466,667.00:129,629.72](AED:USD); and an undocumented third funding cycle, since the archived programme rules *[Dubai RDI Programme, 2025]*[a] and the 2025 call *[Dubai Future Foundation, 2025]*[b] describe only that round. ABIG's exposure is structural, treated only by naming operational users. Policy follows measurement: the Special Report on Cities is approved before the grant begins, so AR7 Working Group II is the publication window *[IPCC news, 2024]*[c].

Calendar Boundaries We seek the consideration of the board members to approve the 180-day sequence of Chapter 5, adopt or grant the guidance on the metric definitions and reporting unit of Chapters 7 and 10, and confirm the funding calendar before committing to submission.

[a] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

[b] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[c] Intergovernmental Panel on Climate Change, *IPCC approves outlines of the first two reports in the seventh assessment cycle*, IPCC news, 2 August 2024. https://www.ipcc.ch/2024/08/02/ipcc-approves-outlines-of-the-first-two-reports-in-the-seventh-assessment-cycle/

Your children are not your children.
They are the sons and daughters of Life's longing for itself.
They come through you but not from you,
And though they are with you yet they belong not to you.

You may give them your love but not your thoughts,
For they have their own thoughts.

You may house their bodies but not their souls,
For their souls dwell in the house of tomorrow.

Kahlil Gibran, *The Prophet*, 1923

أولادكم ليسوا لكم
أولادكم أبناء الحياة المشتاقة إلى نفسها
بكم يأتون إلى العالم، ولكن ليس منكم
ومع أنهم يعيشون معكم، فهم ليسوا ملكاً لكم

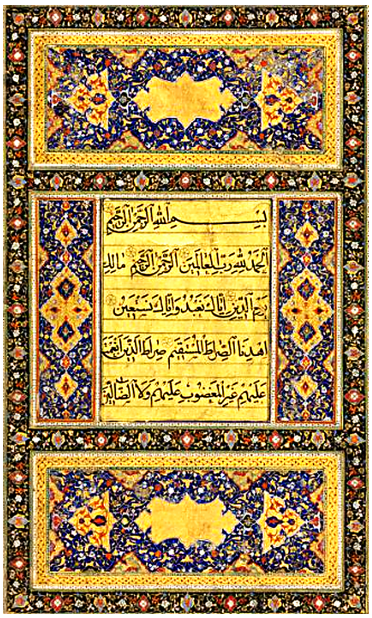
أنتم تستطيعون أن تمنحوهم محبتكم
ولكنكم لا تقدرون أن تغرسوا فيهم بذور أفكاركم
لأن لهم أفكاراً خاصة بهم

وفي طاقتكم أن تصنعوا المساكن لأجسادكم
ولكن نفوسهم لا تقطن في مساكنكم
فهي تقطن في مسكن الغد

جبران خليل جبران

Qur'an Manuscript Folio, 1500s

Afghanistan, Herat, Safavid period (1501–1722)



Calligraphy, the art of beautiful writing, was elevated above all other art forms in the Islamic world because Allah (God) revealed the divine word of Islam to the Prophet Muhammad (570–632) in the Arabic language. This beautiful double page forms the opening pages, or *unwan*, of a Qur'an, the sacred book of Islam. Read from right to left, the verses are written in Arabic in elegant *naskhi* script on a gold ground with florets identifying the ends of the verses. Calligraphers who specialized in beautiful writing often dedicated their lives to copying the Qur'an to grow closer to Allah and receive his blessings. The pages are enhanced with splendid illumination: ornamentation in colors and gold with scrolling vines, blossoms, lozenges, and cartouches within bordered rectangles. Because the book arts were held in high esteem in the Islamic world, decorative motifs created by illuminators were often adopted in other art forms such as metalwork, textiles, and carpets.

THE CLEVELAND MUSEUM OF ART
11150 East Boulevard, Cleveland, OH 44106 · 216-421-7350
https://www.clevelandart.org/art/1924.746

كنت في السابعة، ويدي في يد أبي، والسوق يضيق بنا. رأيتُ أولاً الدخان: عموداً رقيقاً يصعد من مقلاة نحاسية واسعة، وفيه رائحة لم أكن أعرف لها اسماً بعد — كمون وثوم وشيء أخضر يشبه رائحة المطر على التراب.

كان الرجل واقفاً خلف الطاولة، وفي يده ملعقة معدنية تغرف من طين أخضر وتقذفه في الزيت. تنفجر الحبّات بصوت كصوت المطر الأول. تدور، تعلو، ثم تصعد ذهبيةً كأنها تذكّرت شيئاً.

وقفتُ لا أتكلّم. كان أمامنا رجلان، وكنت أعدّ ثوانيهما وأنا أرتفع على أطراف قدميّ لأرى. لم أكن أريد شيئاً في الدنيا كما أردتُ أن أرى ما سيصير إليه ذلك الطين الأخضر.

ثم شاهدته يلفّها. فتح الرغيف كما يُفتح كتاب. مرّ خطّ الطحينة أبيض هادئاً، ورصّ الحبّات فوقه وكسرها كسراً رحيماً. جاء اللفت الأحمر — لون لم أرَ مثله في طعام قبلها — ثم البندورة، ثم البقدونس أخضر كأنه أُقطف في تلك اللحظة.

ولفّها. بحركة واحدة، بيدين تعرفان ما تفعلان منذ ثلاثين سنة، لفّها وشدّها بالورق ومدّها إليّ.

كانت ساخنة. كانت أثقل مما توقّعت، كأنّ فيها شيئاً غير الطعام.

قال أبي: «امسكها بيديك الاثنتين».

ولم أفهم لماذا حتى عضضتُ العضّة الأولى، فسال الطراطور على معصمي حتى المرفق، وضحك الرجل، وضحك أبي، ولم أضحك أنا؛ كنت مشغولاً. وحين رفعتُ رأسي عرفتُ أنني صرتُ من أهل هذا السوق.

Authors

Mahesh Shastry

Kalman Systems Pty Ltd, Sydney, New South Wales, Australia

Interdisciplinary knowledge across science, engineering, industry and business, and wide-ranging experience that equips him to read the landscapes already in place. He brings to the work clarity, considered thinking and simplicity.

He is an advocate for the NSPE Code of Ethics for Engineers and works to the standards of ISO, IEC, IEEE-SA and BSI. He is a member of Engineers Australia, the Institute of Electrical and Electronics Engineers, the Institution of Engineering and Technology, the Association for Computing Machinery, ASM International, the American Mathematical Society and SPIE.

Kalman Systems Pty Ltd, ACN 660 552 670.

kalmansystems.com.au
github.com/mshastryieeeorg
linkedin.com/in/mshastryacmorg

Dr Chakradhar Iyyunni

Professor of Practice, BITS Pilani Dubai Campus, United Arab Emirates

Thirty years and more across academia and industry, working at the intersection of engineering, sustainability and systems innovation. His work at BITS Pilani Dubai Campus covers digital twins and automation, industry collaboration for energy transformation, and sustainable mobility, infrastructure and regenerative cities.

He is a Fellow at the European School of Governance and an Advisory Council Member at Ecocivilisation, India. Earlier roles with L&T Technology Services, the University of Houston and SmithKline Beecham span engineering consulting, leadership in research and development, and academic programme design, and carry a commitment to systems thinking and human-centred development.

linkedin.com/in/ciyyunni

'If you want to build a ship, don't drum up people to collect wood and don't assign them tasks and work, but rather teach them to long for the endless immensity of the sea.'

Antoine de Saint-Exupéry

Attributed to Antoine de Saint-Exupéry. The English wording circulates as a paraphrase and is not a sentence of *Citadelle* (translated by Stuart Gilbert as *The Wisdom of the Sands*, Harcourt, Brace and Company, 1950).

Acknowledgements

Five stakeholders agreed to place two proposals, written for two continents, under one architecture and one standard of evidence. They are Dr Uchi Dominic Terhile (PhD Graduate, Federal University Dutse, Nigeria); Dr Chakradhar Iyyunni (BITS Pilani Dubai Campus, United Arab Emirates (UAE)); Abdullah Al Salmani (SpacePoint, UAE); Arti Ahluwalia (Country Chair, Ecocivilisation.earth, India); and Mahesh Shastry (Kalman Systems Pty Ltd, Sydney, New South Wales, Australia). The storyboard for the Climate Intelligence & Decision Twin for Dubai [Iyyunni, 2026] and the strategic concept paper for the Africa Blue Intelligence Grid, with its addendum [Terhile et al., 2026], are their work, and the discipline of grading evidence is applied to their documents as strictly as to anyone else's.

Our current analysis draws on the public plans, data descriptions, funding rules and announcements of agencies to which we have not yet presented it for endorsement. In Dubai we have the Roads and Transport Authority, the Dubai Electricity and Water Authority, Dubai Municipality, Dubai Culture, the National Center of Meteorology and the Dubai Future Foundation, whose Research, Development and Innovation grant programme frames the Dubai proposal. The written confirmations that the storyboard anticipates from several of these agencies are still to be sought, and nothing here should be read as their view.

For the African and ocean material the sources are the published strategies, declarations and programme statements of the African Union; GOOS-Africa, the African regional alliance of the Global Ocean Observing System; the Ocean Prediction Enhancement in Regions of Africa programme, OPERA; GMES & Africa; the International Union for Conservation of Nature; and Global Fishing Watch. The urban heat and labour evidence rests on the field reports of Arizona State University and the City of Phoenix, the schoolyard programme of the Ville de Paris as documented by the Urban Innovative Actions initiative, the Cooling Singapore project, the International Labour Organization, the Lancet Countdown on Health and Climate Change, the United Nations Environment Programme and the Intergovernmental Panel on Climate Change. Every source was secured and dated before it was read, and each is graded (measured-with-controls, measured, modelled, projection, policy-fact, case-report or method-reference) on what its text supports. The grades are a judgement about evidence and not a comment on the care of the original authors; where a source is incomplete, that is stated rather than guessed at.

Responsibility for what remains (every derived figure, every indicative assumption and every error) rests with the stakeholders alone.

22 September 2026

Foreword

Two proposals sit at the centre of this report, and at first sight they have little in common. One asks for [1,400,000.00:388,888.89](AED:USD) over 36 months to instrument roughly one kilometre of Dubai Creek with twenty microclimate nodes and a reference station, to model heat at street scale, and to build one shade structure whose effect is predicted before construction and measured after it [Iyyunni, 2026]. The other proposes a continental grid that connects the ocean, freshwater, coastal and security information Africa already produces into decisions taken by ministries, coast guards and fisheries authorities, with the United Arab Emirates as a convening and financing hub [Terhile et al., 2026]. A pedestrian corridor and an ocean; a heritage quarter and, by the concept paper's count, thirty-eight coastal member states of the African Union. Why treat them in one book?

The answer is that they are the same machine at two scales. In both, a physical system (a street in the sun, a fishing ground under a satellite pass) is observed; observations become data; data are given a shared meaning in a semantic layer; meanings are connected in a knowledge graph; the graph feeds models that reason and predict; the predictions are assembled into a decision twin that compares options before capital is committed; a decision is acted on; and the outcome is measured and fed back so that the whole system learns. Physical system, data, semantic layer, knowledge graph, intelligence, decision twin, action, learning. The Dubai storyboard names the last four steps explicitly: a Decision Twin that is 'not a 3D rendering', a stop-or-go gate at month twelve, one built intervention, and a before-and-after summer compared with the twin's prediction [Iyyunni, 2026]. The ABIG concept paper names the first four: assemble existing data, produce useful knowledge products, test them with operational users, and only then invest in sensors and payloads where the evidence shows they are needed [Terhile et al., 2026]. Each proposal is strongest where the other is thinnest, which is the practical reason for reading them together. Dubai is the first application of the architecture, not its purpose.

What should a reader take from the chapters that follow? Three things. First, that the question worth funding is not 'how hot is the city' but 'which activities, routes and places are lost to heat, for how many hours, and what restores the most of them per dirham, litre and kilowatt-hour' [Iyyunni, 2026], and, for Africa, 'which decisions improve, and at what cost, when observation is connected to the people who must act'. Second, that no published study yet reports the measurement the Dubai proposal promises (activity-hours lost to heat at corridor scale, tied to a built intervention and validated against a twin's prediction) and that this gap, not the technology, is the claim. Third, that both programmes stand or fall on written commitments from operational users: the letter from the Roads and Transport Authority or the Dubai Electricity and Water Authority that gate condition 6.2.4 demands, and a named operational user for every ABIG pilot. The chapters on where to start, on circles of influence and on risk return to this point.

Feasibility reports are prone to borrowing numbers from press releases, so this one imposes two disciplines on itself. The first is evidence grading. Every figure carries one of seven evidence grades: measured-with-controls, measured, modelled, projection, policy-fact, case-report or method-reference. The grades did real work. When the fifteen source analyses were checked against the agreed baseline, a Phoenix surface-temperature result turned out to belong to a different year's report from the one cited, a Makkah road-cooling figure proved to rest on an agency

statement with no published protocol, and the Paris schoolyard programme that the baseline had graded ‘measured-with-controls’ had published its design but not its effect sizes (Chapter 4). Those corrections are applied throughout; where a document could not be archived, the text says so and cites it by title only. The second discipline is vocabulary. Each key concept is explained at three tiers (plain words, professional vocabulary, technical vocabulary) and, where it helps, as a mathematical model, so that a minister, a transport planner and a modeller can read the same page and disagree about the same thing.

An invitation, then. Every source used here was secured and dated, and is registered in Appendix A with its extent, its evidence grade and the chapters that rely on it; the sources that could not be obtained are registered there as well, so that a reader can tell which claims rest on a document and which rest on a title. The retrieval date, 22 September 2026, is this report’s evidence horizon. Readers are asked to challenge every number against the sources themselves, and to prefer the source to this report wherever the two disagree. Derived figures are labelled derived or indicative, with the arithmetic shown, so that anyone can recompute them. A feasibility report earns its place by making it easy to prove wrong.

The five of us who commissioned this report came to it from a university in Dutse, a campus in Dubai, SpacePoint in the Emirates, a civil-society network in India and a systems practice in Sydney. My part was to convene the group, to insist on one architecture and one grading scheme, and to turn a conversation into a structure others could test. The substance (the corridor, the grid, the metrics and the roadmap) belongs to my colleagues. If the book persuades a funder, an agency or a sceptical reader to ask for the archive before accepting a claim, it will have done its work.

Mahesh Shastry
Kalman Systems Pty Ltd, Sydney
September 2026

Preface

This book is a macroeconomic feasibility report on two proposals that share one architecture: the *Climate Intelligence & Decision Twin for Dubai*, the storyboard for a rescoped Letter of Intent to the Dubai Future Foundation RDI Grant under the Cognitive Cities priority, budgeted at [1,400,000.00:388,888.89](AED:USD) over 36 months [Iyyunni, 2026], and the *Africa Blue Intelligence Grid* (ABIG), a strategic concept paper of 20 September 2026 [Terhile et al., 2026]. Both were written by the stakeholders named on the title page. This report, dated 22 September 2026, tests them against the public evidence before either is submitted for funding.

The genre and what it must contain

A feasibility report answers whether a proposal can be done, whether it is worth doing and what could stop it. The adjective *macroeconomic* widens the second question from the proposal's own budget to the economy around it: the outdoor activity that heat removes from a city and the value of Africa's blue economy. Six assessments are required. *Need* (Chapters 1 and 2): what the proposals are for and which gap in published practice they fill. *Technical* (Chapters 3, 4 and 8): comparable efforts, their evidence and the data architecture required. *Economic and financial* (Chapters 9 and 10): life-cycle costs in the sense of ISO 15686-5 [ISO, 2017a], reduced to one reporting unit for each programme: AED per restored activity-hour for Dubai, and cost per decision improved for ABIG. *Institutional* (Chapters 5, 6 and 7): who must agree, in writing, before each phase begins. *Risk* (Chapter 11): risk modelling, SWOT analysis, loss remediation and recovery, and force majeure. *Policy* (Chapter 12): how work led from the Gulf can inform planetary policy.

Understanding: Feasibility

TIER 1: IN PLAIN WORDS

Can it be done, is it worth doing, and what could stop it?

TIER 2: IN PROFESSIONAL TERMS

A proposal is tested against need, technical capability, whole-life economic and financial return, institutional readiness, risk and policy fit.

TIER 3: IN TECHNICAL TERMS

The expected net present value of life-cycle benefits and costs is estimated under uncertainty, subject to technical, institutional and legal constraints.

AS A MATHEMATICAL MODEL

Proceed $\iff \mathbb{E}\left[\sum_t \gamma^t (B_t - C_t)\right] > 0$ subject to $g_k \leq 0,\ k = 1, \ldots, K$

Audience

The report is written first for the five stakeholders; second for the funders they will approach (the Dubai Future Foundation RDI Grant, the New Development Bank and the UAE Space Agency); and third for the partner agencies whose written confirmations the proposals depend on: the Roads and Transport Authority (RTA), the Dubai Electricity and Water Authority (DEWA), Dubai Municipality and Dubai Culture; GOOS-Africa, the Ocean Prediction Enhancement in Regions of Africa programme (OPERA) and GMES & Africa. A funder can read the chapter abstracts and the Conclusion alone; a reviewer can follow every number to its source through Appendix A.

From the stakeholders' questions to the chapters

The stakeholders put fourteen questions; Table 1 maps them to the chapters. Questions 4 to 6 are answered together in Chapter 4. Questions 9 and 10 are combined in Chapter 7 because spiral development and stochastic dynamic programming draw the same boundary from two sides. Chapter 11 answers no numbered question: the list contained none on risk, and a feasibility report without a risk assessment is incomplete. Question 14 is answered by Chapter 12.

Table 1: The stakeholders' fourteen questions and the chapters that answer them.

Question	Subject	Chapter
1	What the source documents contain	1
2	The purpose of the two papers	2
3	Similar efforts around the world	3
4	Which feasibility studies bear on the proposals	4
5	What those studies found	4
6	Where the sources are, with URLs	4 and Appendix A
7	Where to start	5
8	Circles of influence	6
9	System boundaries under spiral development	7
10	System boundaries under stochastic dynamic programming	7
11	Synthetic-data identifiers critical to life-cycle measurement	8 and Appendix B
12	Ways to forecast life-cycle costs	9
13	Life-cycle cost breakdown as a strategic roadmap	10
n/a	Risk modelling, SWOT, loss remediation and recovery, force majeure (added)	11
14	How GCC-led work can lead planetary policy	12

How the evidence is graded

Every number, date and claim is traced to a named source and carries one of seven evidence grades. *Measured-with-controls*: an effect measured in the field against a simultaneous or paired control, such as a treated street beside an untreated one on the same day. *Measured*: quantities observed with instruments, without an intervention or a control. *Modelled*: model output driven by observations for a present or past period. *Projection*: model output for a future date or under an assumed adoption rate. *Policy-fact*: a rule, date, budget, target or institutional fact stated by the

responsible authority. *Case-report*: an account of a single episode or programme by its own actors or by the press, without controls or a published protocol. *Method-reference*: a source used for its method, framework or standard, not for a result. A figure takes the grade of its weakest link; the report's own arithmetic is shown and labelled *derived* or *indicative*. Appendix A registers the grade of every source and records which sources are held in the archive and which are cited by title only.

Three tiers of explanation

Every key concept is explained at three tiers of vocabulary, after the three tiers in which Beck, McKeown and Kucan classify words: everyday words, the general vocabulary of mature readers across domains, and the vocabulary of one discipline [Beck et al., 2002]. Tier 1 uses plain words, Tier 2 general professional vocabulary and Tier 3 technical vocabulary; where a model applies, a fourth line states the concept mathematically. Each chapter contains two to four boxes headed *Understanding*, like the one above, and the prose outside them keeps the same order.

Conventions

The text is in British English: 'Dr' takes no full stop, quotations take single quotation marks, and proportions are written 'per cent'. Dates are written day, month, year (22 September 2026) and ranges take an en-dash (2027–2030); 'Year 1' of the Dubai programme is the first grant year, 15 March 2027 to 14 March 2028 [Iyyunni, 2026]. Dubai costs are in dirhams (AED), international figures in United States dollars (USD), and every money figure states its year; nothing is converted between currencies or adjusted for inflation unless the conversion is shown. Where a present value is computed, the base year, discount rate and period are stated beside it as the report's own assumptions. Temperatures are in Celsius except where a United States source is quoted as printed. Chapter headings show the Eastern Arabic numeral followed by the Western numeral in braces, for example ٣ {3}, without the word 'Chapter'; cross-references use the Western numeral. Acronyms are written in plain text and expanded at first use in each chapter; OTC means Outdoor Tourism Capacity here but Outdoor Thermal Comfort in the Singapore sources. Named metrics are set in small capitals (HLAH, Heat-Lost Activity Hours) and data fields in monospace (`place_id`).

Scope and exclusions

The report assesses the proposals as they stood on 22 September 2026: the storyboard for the Dubai Letter of Intent [Iyyunni, 2026] and the ABIG concept paper with its addendum [Terhile et al., 2026]. It performs no new field measurement and no modelling beyond the arithmetic it shows. It does not audit the budgets line by line, give legal advice (the force-majeure clauses in Chapter 11 are drafting guidance subject to counsel's review) or offer investment advice. Partner agencies are described from their public documents only. The indoor and space extensions raised in the stakeholders' discussions are noted but not assessed.

Where the evidence comes from

Every source in the archive was secured and dated on 22 September 2026, and that date is the evidence horizon of this study: nothing after it is claimed. Sources fall into two kinds, and they are cited differently. A source with a normative identity (a journal article or book, cited by its digital object identifier; a numbered standard, whose designation is itself the citation; a report formally issued by a named institution) is a bibliography entry and is listed at the back, with no web address, because it can be requested from a library or a publisher by the details printed there. A source without that identity (a press release, a news report, a programme page) appears instead as a

bracketed label in the text with a footnote giving its full reference and a live link. Those footnoted sources carry announcements and context rather than measured results: every statement graded *measured* or *measured-with-controls* rests on a bibliography entry. Appendix A registers all sixty-seven secured sources with the grade each carries, names the thirteen that could not be obtained, and records the defects that make particular numbers unsafe to quote.

Standing and revision

This is Version 1.0, dated 22 September 2026, and it should be cited by version and date. It is a statement of what the evidence supported on that day. Three things would justify a revision before anything else: confirmation of the grant cycle on which the Dubai calendar depends, which no available document describes; the first measured results from either programme; and the arrival of the alignment exercise that would turn the African concept paper into a costed plan. Each would change conclusions in Chapters 5, 10 and 11 rather than merely add to them.

Using the Glossary and the Index

The Glossary is grouped by taxonomy rather than as one alphabet, in nine sections: G.1 Climate and thermal comfort; G.2 Urban systems, mobility and tourism; G.3 Ocean, Earth observation and the blue economy; G.4 Data, semantics and synthetic data; G.5 Decision science and modelling; G.6 Economics, costs and finance; G.7 Programme management and risk; G.8 Policy and governance; G.9 Institutions and programmes. Within each section the terms are alphabetical, defined in two to four sentences with a cross-reference to the chapter that uses them; acronyms appear as aliases pointing to the full term. The Index lists concepts, metrics, methods, places, institutions, programmes and standards under the same spelling as the Glossary, with sub-entries under the main term (for example *Decision Twin*, gate conditions); a bold page number marks the Glossary definition. Authors are not indexed; they are found through the Bibliography. The acronym list and the List of Symbols (introduced in Chapter 7) complete the apparatus.

Governance and Expectations

Every agreement that matters is written twice: once in the document, and once in the understanding between the people who sign it. Where the two differ it is the understanding that decides what actually happens, and the document merely records what can later be argued about. What follows is the understanding, set down in advance: who we take each other to be, what we will say without being asked, what we will not do, and how a partner may object, refuse or be heard. It stands before the evidence because it governs how the evidence should be read, and before the money because it governs how the money should be discussed.

On who we are to each other

- **We are people before we are parties.** The document that eventually gets signed is a description of a relationship, not a substitute for one. If the relationship is honest the document can be simple; if it is not, no amount of drafting will save it.
- **We intend friendship, and we understand what friendship costs.** Friendship is not the absence of hard conversations. It is the condition that makes hard conversations survivable. A partner who cannot tell us our number is wrong is not being kind to us.
- **Our common ground is human, and it is not in dispute.** The subject is heat that shortens working lives and ends some of them early [UN Secretary-General, 2024] (grade: modelled). No tradition present in this work, and no absence of tradition, holds a different view about whether that is worth preventing. Everything else is method.

On money and its purposes

- **An investment model is not one model.** It varies with the investor's expectations (return type, horizon, tolerance for loss) and with the purpose of the investment (commercial return, sovereign strategic interest, development mandate, endowment, philanthropy). The same programme, costed identically, is a different proposition to each.
- **So we will ask what kind of money this is, early, and we will answer the same question about what kind of money we need.** That conversation belongs at the beginning and on the record, not as an assumption carried silently into a term sheet.

On the frameworks in the room

- **More than one framework is present, and each binds only those it binds.** Partners in this work are Muslim, Christian, Hindu, Jewish, of other faiths and of none, across the Gulf, Africa, Asia and Europe, alongside institutions that are wholly secular by charter. No framework is imposed on a party it does not govern, and none is treated as the default against which others are exceptions.

- **Where Islamic law governs a financing relationship, we accept it as a condition of the dealing**, not an obstacle to it, and not a flourish. For a Shariah-compliant structure raised from or based in the United Arab Emirates, that is the applicable law of the transaction.
- **We distinguish first-order from second-order requirements, and we say which is which.** First order: the settled prohibitions and requirements, which do not vary and which we can design against today. Second order: the derived contractual structures, the standards that codify them, and the rulings of the board with authority over the specific transaction, which do vary and which are not ours to decide.
- **Naming a framework is not advocating it.** We will explain what governs a structure and why, in the vocabulary that structure actually uses. We will not use the moral language of any tradition to press a partner who does not hold it, and we do not invite anyone into anyone's faith. That restraint extends past religion: we will not proselytise for our methodology, our politics or our theory of development either.

On how we will behave

- **Transparency.** We will state our purpose, our need, and the limits of what we can do. We ask our partners for the same, and we will ask plainly rather than hint.
- **Non-obfuscation.** We will not hide behind volume, jargon or complexity. A long document can be a form of concealment. If what we send you is too long to act on, ask us for the two pages, and the failure is ours.
- **Objections upfront.** Anyone's objection belongs at the start of the work, not at the end of it. That includes our objections to our own work, which are published in the Limitations rather than discovered by a reviewer.
- **Self-reporting.** We will report our own problems before we are asked: the missed milestone, the assumption that failed, the overrun, the result that did not replicate. A partner should never learn of a difficulty from a third party.
- **No duress, including the silent kind.** No party should be placed under pressure in a commercial decision, whether beneficiary or benefactor. Duress includes intrinsic motivations left unspoken out of politeness or diplomatic sentiment, which commit a party to terms they have not actually examined.
- **The right of response, and to be heard.** Any person or institution named, assessed or affected may respond, and the response will be carried rather than filed. Disagreement that survives our reply stays visible in the document.
- **Respect for the granting authorities.** Funders and agencies have their own mandates, their own constraints and their own accountability to the public. We will follow their process rather than route around it, accept a refusal without treating it as an injury, and remember that public money is entrusted, never owed.
- **Humour.** We would like to be able to laugh with the people we work with, and we take that more seriously than it sounds. A partnership in which nobody has laughed for six months has usually developed a problem that the minutes do not record. The one boundary: humour is for easing the room, not for deflecting a question that deserves an answer.

An investor is a shape, not an amount

The ask ladder of Table 13.3 gives a range at each tier. That is a start and it is not enough, because a tier is a number and an investor is a shape. The same [700,000,000.00:194,444,444.44](AED:USD), the floor of the tier-4 ask in 2026 prices (derived), is four different propositions, as Table 2 sets out.

Table 2: The same sum read by four kinds of investor. The classification is this report's own. Cost per restored activity-hour is the one measure that serves all four rows, which is the argument for making it the spine of the investment case rather than a section inside it.

Investor purpose	What they need to see	What kills it
Commercial return	Revenue line, exit, internal rate of return, service contracts	No cash-generating asset
Sovereign strategic	Capability retained in country, jobs, ownership of standards	Foreign dependency
Development mandate	Beneficiary count, additionality, measured outcome	No counterfactual
Endowment or charitable	Perpetuity, preservation of the corpus, public benefit	An asset that decays unfunded

There is also a plainer reason to ask the question early. An investor who says 'return' and means 'positioning' will apply tests nobody was expecting and withdraw for reasons nobody can address. That is not bad faith. It is what happens when a real motive goes unstated and a proxy motive gets written down instead. Both sides then argue about the proxy.

First order and second order, and why the distinction is a courtesy

People conflate these constantly, and the conflation is what makes Islamic finance feel opaque to those outside it and feel misrepresented to those inside it.

First order is settled and does not vary between jurisdictions or boards. *Ribā* (الربا): no interest, because return comes from risk genuinely borne and not from time applied to a loan. *Gharar* (الغرر): no excessive uncertainty or ambiguity in the essential terms. *Maysir* (الميسر): no wagering. Asset-backing and risk-sharing: the transaction attaches to real assets or activity, and the financier carries genuine ownership risk rather than passing all of it across. And certain sectors are excluded outright.

Behind these sits a maxim that says it more economically than any paraphrase: *al-ghunm bi-l-ghurm* (بالغرم الغنم), gain goes with liability. Whoever takes the profit takes the exposure.

Second order is the applied layer and it genuinely varies. It holds the named contracts, *murābaḥa* (المرابحة), *ijāra* (الإجارة), *istiṣnāʿ* (الاستصناع), *muḍāraba* (المضاربة), *mushāraka* (المشاركة), *wakāla* (الوكالة) and *ṣukūk* (الصكوك); the codifying standards of the Accounting and Auditing Organization for Islamic Financial Institutions and of the Islamic Financial Services Board, and in the United Arab Emirates the Higher Shariah Authority at the central bank; and the ruling of the particular board supervising the particular transaction. Two boards of equal standing can differ on the same question and both be reasoning properly.

So: first order tells you what you cannot do, and you can plan against it before you meet anybody. Second order tells you which instrument you may actually use, and nobody knows until a board rules. Saying so is not hedging. It is the difference between describing a law and pretending to be its judge, and claiming to be its judge would repeat, in a new place, exactly the overclaiming this report has spent two chapters correcting.

Three structures fit this programme unusually well, and they are offered as questions for a board rather than as available facts.

- *Istiṣnāʿ* (الاستصناع): commissioned manufacture or construction, designed for a thing that does not yet exist. That is the shade canopy of Chapter 10.
- *Ijāra* (الإجارة): lease, where the lessor retains ownership and with it the obligations of ownership, including major maintenance. The hardest question in this whole study is who carries the cost of running a public cooling asset from 2030 to 2045, and no chapter answers it. This instrument answers it structurally: the form of the contract attaches the obligation to a named party and will not let the field be left blank.
- *Waqf* (الوقف): perpetual endowment. A corpus is dedicated, never spent, and its yield maintains a public benefit indefinitely. For an asset whose operating cost exceeds its construction cost several times over, whose benefit is public and unpriced, and whose funding must outlast governments, this is not an exotic option.

Endowment is not one tradition's idea

The *waqf* point is often heard as a sectarian one. It should not be, and the honest way to make it is comparative.

Dedicating property in perpetuity so that its yield serves strangers is close to a human universal. Islamic cities built *sabīl* (سبيل) fountains and shaded resting places by *waqf*. Jewish law has *hekdesh* (הקדש), property consecrated to a sacred or charitable purpose. Christian Europe built and still runs charitable foundations on the same instinct. Hindu and Buddhist temple endowments and the *dharmaśālā* (धर्मशाला) housed travellers arriving out of the heat, and the *piyāū* (प्याऊ) gave water to anyone who passed. Roman and Byzantine cities maintained public fountains, and English parishes maintained the pump. It has even been argued, in a contested line of legal scholarship this report has not reviewed, that the English charitable trust itself owes something to *waqf*, carried back through the Crusader period. We would not lean on it.

What is not contested is the pattern. Providing water and shade to strangers, funded in a way designed to outlive the donor, is one of the oldest things human beings organise money to do. A cooling programme is a modern instance of an old institution, and every partner in this work can find it inside their own tradition. Naming *waqf* first is not a claim of priority. It is a matter of where the money is most likely to come from, and every cognate deserves the same standing in the document.

What fiduciary duty becomes, and why it helps everyone

Under Shariah-compliant structures the duty owed is wider than maximising return within the law.

- Compliance is part of the duty, not a limit on it. A profitable but non-compliant return is a failure, and may require *taṭhīr* (التطهير), purification of the tainted portion by giving it away.
- The Shariah supervisory board (هيئة الرقابة الشرعية) is a governance organ, not an advisory one. Its rulings bind, and it sits alongside the audit function rather than beneath management.
- *Amāna* (الأمانة), the trust relationship, imposes duties of care and candour on whoever holds another's capital that exceed an arm's length commercial standard.
- *Zakāt* (الزكاة) and purification are real line items with real accounting consequence.

- In a *muḍāraba* (المضاربة), loss allocation is fixed by the form: financial loss falls on the provider of capital, and the manager loses their labour. Who bears which loss is settled before signature, not discovered during a bad year.

Two of these are worth adopting as behaviour regardless of which law governs, and we intend to. A duty of candour heavier than the commercial minimum is simply a good way to work. And settling who bears which loss before signing, rather than negotiating it while it is happening, is the difference between a partnership and a dispute.

Non-proselytisation, in both directions

- Nobody is being invited into anyone's faith, and no framework carries moral seniority in this work.
- We will describe *istiṣnāʿ*, *ijāra* and *waqf* in their own vocabulary because that is what they are called, in the same way a French instrument keeps its French name. Explaining a term is not preaching it.
- Where a framework's higher objectives genuinely bear on the substance, we will say so and stop there. The preservation of life sits among the *maqāṣid al-sharīʿa* (مقاصد الشريعة), and a programme about lethal heat sits inside that. Stating this to an investor for whom it is decisive is information. Deploying it to make a partner who does not hold it feel unreasonable would be coercion wearing a nicer coat, and we will not do it.
- The same restraint applies to our secular commitments. We have views about measurement, about controls and about what counts as evidence. Those go in the document as arguments to be contested, not as conditions of fellowship.

Parallel-compliant by design

The continental programme spans thirty-eight coastal states, Muslim-majority and Christian-majority and plural, with partners in countries where Islamic finance has no standing at all.

The practical form of respect here is a financing architecture built from the outset so that a Shariah-compliant tranche and a conventional tranche can fund the same programme side by side without either contaminating the other. This is ordinary practice in large project finance. What is not ordinary, and what we are committing to, is declaring it as design intent at the beginning rather than improvising it at the end. Retrofitting compliance is where partners get told, late, that the structure they agreed to cannot include them.

On duress, and the courtesy that causes it

This is the heart of it, and it has a precise name inside the framework the Gulf applies.

Islamic contract law requires free consent, *riḍā* (الرضا). Duress, *ikrāh* (الإكراه), vitiates it. And *gharar*, the prohibition on excessive uncertainty or ambiguity in a contract's essential terms, reaches further than people expect: if a party is committing for a reason they have not stated, the written terms do not describe the transaction actually occurring.

So an unspoken motive is not merely discourteous. Under the applicable law it is a defect. That converts a question of manners into a question of validity, which is why it belongs in a document rather than in an understanding.

Understanding: Gharar

TIER 1: IN PLAIN WORDS

Both sides have to know what they are agreeing to. A bargain with a hidden term in it is not really a bargain.

TIER 2: IN PROFESSIONAL TERMS

Excessive uncertainty or ambiguity in the essential terms of a contract, prohibited because it hands one party a risk the other cannot see and did not price; the remedy is disclosure, not compensation.

TIER 3: IN TECHNICAL TERMS

A defect arising where the material terms are not common knowledge between the parties. The extension from undisclosed *terms* to undisclosed *motives* is this report's reading and not a ruling: an agreement whose written terms omit the reason a party is entering it is an incomplete description of the transaction actually occurring, and is defective in the same way and for the same reason.

AS A MATHEMATICAL MODEL

Let $\mathcal{T}$ be the terms that in fact govern and $\mathcal{T}_A, \mathcal{T}_B$ those known to each party. The contract is sound where $\mathcal{T}_A = \mathcal{T}_B = \mathcal{T}$; *gharar* is present wherever that equality fails, and the party holding the larger set holds the defect.

What has to be said out loud, symmetrically:

- **The benefactor's actual purpose.** Strategic positioning, portfolio compliance, national capability, reputational return and financial return are different motives with different tests.
- **The beneficiary's actual need.** The recurring failure in this material is asking for capital while needing operating cover. Asking for the wrong thing politely is still asking for the wrong thing.
- **Obligation of kinship, country, faith or long acquaintance.** A benefactor who feels bound to support a fellow national or co-religionist, and a recipient who feels unable to refuse a senior figure, are both under pressure. Naming it dissolves it. Leaving it unnamed does not make it absent; it only makes it unexaminable.
- **The right to decline.** A frank discussion is frank only if 'no' is genuinely available. *No* is a complete sentence and needs no reason attached. We will establish that before terms are discussed, not after.

And the connection back to everything else in this work: this is the same failure as the [1,400,000.00:388,888.89](AED:USD) of Chapter 2. A number chosen because it seemed acceptable rather than because it was true is an unspoken motive operating on a commercial decision. The diplomacy that produced that figure and the diplomacy that leaves an investor's real purpose unstated are one habit in two coats. The first has now been corrected. This is the correction of the second.

Self-reporting, response, and the authorities

Self-reporting is a discipline, not a virtue, and it is cheap compared to what it prevents. We will publish our own overruns, failed assumptions and unreplicated results before anyone asks. A partner learning of a difficulty from a third party has learned two things, and the second one is worse.

The right of response means any institution named or assessed here may reply, and the reply travels with the document. Where disagreement survives our answer, it stays visible. A report that only carries the objections it has defeated is an advertisement.

Respect for the granting authorities is partly procedural and partly a matter of memory. The Dubai Future Foundation, Dubai Municipality, the Roads and Transport Authority, the Dubai Electricity and Water Authority, the African Union institutions and the development banks each carry a mandate and an accountability to a public that did not choose us. We will use their process rather than route around it through personal access, accept refusal without treating it as injury, and keep in view that public money is entrusted and never owed. Where we think a rule produces a bad outcome we will say so through the channel that exists for saying so, which is also the only way to be heard the next time.

Why humour belongs in a document about money

Because it is diagnostic. People who can joke with each other are people who feel safe enough to disagree, and a partnership where the laughter has stopped has usually developed a problem some months before anyone writes it down. We would rather notice early.

It also has a limit worth stating, since charm is the most respectable way to avoid a question. Humour is for easing a difficult conversation, never for ending one. If a joke is doing the work a straight answer should be doing, the answer is still owed.

What healthy looks like

If this goes well, the observable signs will be ordinary ones.

- Bad news travels fast and upward, and arrives from the person responsible for it.
- Objections appear in month one rather than month eleven.
- Both sides can say what they actually want without rehearsing it first.
- Refusals are given and received without damage.
- Documents get shorter as the relationship matures, because less has to be written down defensively.
- The people involved are still speaking to one another after the programme ends, including in the version where it did not work.

That last one is the real test. Programmes fail for reasons nobody controls. Relationships fail for reasons everyone controls, and almost always for the same reason: something true that would have been awkward to say in the first month, and became impossible to say in the sixth.

Nothing in this chapter is a ruling, and none of it is offered as one. The settled prohibitions are reported because they are settled. Every question of structure, and every instrument named above, is referred to a qualified Shariah supervisory board, whose determination governs. The undertakings on disclosure, duress, response and conduct require no board at all. They are ours to keep from the first conversation.

Limitations

The authority of a feasibility report rests on what it declines to claim. This one was written without a field campaign, without a vendor quotation, without a signed partner agreement and without a funded budget, and every conclusion in it inherits those four absences. The chapters that follow the front matter are careful about the evidence they cite and, in one respect, were not careful enough about the evidence they assumed. This chapter states the shortcomings plainly, because a reader who is asked to commit money is entitled to know the weakest joint before being shown the strongest argument.

No measurement was made

Nothing in this report was observed. No instrument was placed on the Al Fahidi corridor or beside the creek, no pedestrian was counted, no surface temperature was read, no canopy was priced by a supplier and no salary was offered to a candidate. Every thermal, behavioural and monetary quantity here is inherited from a published source, derived by arithmetic from such a source, or assumed and labelled as an assumption.

The reason is structural rather than editorial. A field study in a heritage quarter and on a municipal right-of-way requires written site access from the heritage and municipal authorities, and a study that draws on transport and utility records requires data agreements with the authorities that hold them. Those instruments were not in hand: they sit in the proposals' own calendars as milestones for December 2026 and February 2027 [Iyyunni, 2026]. Nor had any solicitation been made to a funder, so no budget existed from which instruments, technicians, calibration or insurance could have been paid. Without the endorsements there was no lawful access, and without the solicitation there was no money. A study conducted in that condition can survey, grade and reason. It cannot measure.

The consequence should be held in view when reading Chapters 3 and 4. Where those chapters report that a coating in Phoenix lost reflectivity within ten months, or that a Paris schoolyard programme expects a service life beyond fifty years, they report what other investigators measured in other climates, under other procurement regimes and on other materials. Not one archived programme measured a Gulf summer on a Gulf street with a paired control. The nearest analogues are structural, not thermal, and none of them transfers a number.

The analytical instrument and the corpus

The synthesis was carried out with the assistance of a large language model, Claude, made by Anthropic, applied to a corpus of publicly available material: peer-reviewed literature, economic and statistical series, engineering and metrological standards, institutional reports, programme pages and press statements. The model was used to locate candidate sources, to extract quantities and their context, to cross-check a claim in one document against its statement in another, to apply the grading scheme of the Preface consistently across several hundred assertions, and to compose the prose. Every quantity that survived into the text was traced to a document that could be opened and read.

Three limits of that instrument bear directly on how the report should be used, and the third is the one that did damage.

First, a model applied to public sources cannot exceed its sources. Where the literature does not publish a unit cost, no amount of synthesis produces one. The absence of a published cost per microclimate node, per square metre of canopy or per restored activity-hour is a property of the archive, and it survived into this report as an absence.

Second, thirteen sources that would have borne on the argument could not be obtained, and one of the most important, the international life-cycle costing standard, was held only as a preview of its first four clauses [ISO, 2017a]. Its definitions are used; its discounting procedure and worked annexes are known by heading alone and are cited as requirements nowhere.

Third, and most seriously, fluency is not verification. A model composes coherent argument around a weak input as readily as around a strong one, and the grading scheme guards the sources without guarding the premises. That is precisely what occurred with the budget. The figures of [1,400,000.00:388,888.89](AED:USD) and [250,000.00:69,444.44](AED:USD) entered the report as though they were costed estimates. They are neither. They appear in the source as drafting directives to the author of a Letter of Intent, instructing that equipment be ‘reduced to’ about [250,000.00:69,444.44](AED:USD), that computing be ‘raised to a realistic figure’ and that the total be ‘held near’ [1,400,000.00:388,888.89](AED:USD) [Iyyunni, 2026]. An instruction to raise a figure to a realistic one is a statement that the figure is not yet realistic. Every other source in this study was graded with care; the one input on which two chapters were built was not graded at all, and the machinery of the report then made it look solid.

The financial limitation

Chapters 9 and 10 supply a complete life-cycle costing method and an incomplete life-cycle cost. The vocabulary is the standard’s, the procedural discipline is the cost-estimating guide’s, the ten forecasting methods are set out with their inputs and outputs, the nine-category cost breakdown structure is defined so that each element is decidable from a single invoice, and the method-selection matrix assigns a primary method and a cross-check to every cost element in every phase [ISO, 2017a, GAO, 2020]. What is missing is the numbers to put in it.

Between them those two chapters contain four programme specific monetary figures. Two are the storyboard’s drafting targets and two are the funder’s published envelope. Every other figure in either chapter is one of those four divided or multiplied by something. There is no vendor quotation for a sensor node, no salary table for a staff of eleven, no service life from a supplier warranty, no measured unit cost for any intervention, and no cost of any kind attached to the physical asset that Chapter 10 correctly identifies as the largest item in the whole life cycle. The cumulative distribution drawn in Chapter 9 carries a spread chosen to display the method rather than estimated from the programme, and although it is labelled illustrative, conclusions are drawn from where its percentiles fall.

Two further shortcomings belong here. A capital-to-operating ratio was available and was not computed. The archived Paris programme publishes an annual programme cost and a capital figure for the same intervention type: about [9,000,000.00:37,800,000.00:10,500,000.00](EUR:AED)(AED:USD) a year for around 25 yards, against an ERDF contribution of [4,996,000.00:20,983,200.00:5,828,666.67](EUR:AED)(AED:USD) for 10 pilot yards (2018–2022) [Urban Innovative Actions, 2023]*[EEA Climate-ADAPT, 2023]*[1]. That is [360,000.00:1,512,000.00:420,000.00](EUR:AED)(AED:USD) a year against [499,600.00:2,098,320.00:582,866.67](EUR:AED)(AED:USD)

[1] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

of capital per yard, a ratio of roughly seven-tenths (derived, indicative, 2018–2022 prices). Over a fifteen-year period of analysis that is a present value of between six and eleven times the cost of construction across the nought to eight per cent real discount band of the assumption register. And the report states in several places that no comparable reference class exists. For the continental programme that is not so. GMES & Africa, the African Union–European Union Earth-observation service programme, appears in the concept paper's own stakeholder list [Terhile et al., 2026] (grade: *policy-fact*), and its southern marine consortium is operating *[MarCOSIO, 2025]*[2]. Its country coverage and phase budgets are not asserted here: the programme pages could not be captured *[GMES and Africa, 2025]*[3], and no cost, uptake or benefit figure for any of its marine services exists in the archive. A metropolitan urban-cooling programme in the region entered procurement during 2026 (verified 23 September 2026, after the evidence horizon and therefore outside the source register of Appendix A). Both are reference-class members, and their absence from the analysis is an oversight rather than a gap in the literature.

Understanding: Conditional return

TIER 1: IN PLAIN WORDS

You cannot say what an investment earns until you know the terms on which it is made.

TIER 2: IN PROFESSIONAL TERMS

A return is not a property of a project alone but of a project and a term sheet together: the same cost and benefit streams yield different returns under different tenors, tranche sizes, costs of capital and allocations of who pays to run the asset afterwards.

TIER 3: IN TECHNICAL TERMS

The net present value of a programme is a function of an agreed term structure as well as of its cash flows; with the period of analysis, the discount rate and the operating liability unassigned, the life-cycle model returns a family of outcomes rather than a figure, and remains sound in structure while indeterminate in value.

AS A MATHEMATICAL MODEL

$\mathrm{NPV}(\theta) = \sum_t \frac{B_t - C_t(\theta)}{\left(1 + r(\theta)\right)^t}$, where θ is the term sheet: tenor, tranche, cost of capital, and the party carrying operation after handover.

The right conclusion is narrower than a dismissal. The life-cycle strategy set out in those chapters is a valid instrument for calculating a return, and it is not yet a calculation. It becomes one at the moment investment terms are fixed, because the quantities it still lacks are exactly the quantities a term sheet supplies: the period of analysis, the discount rate, the tranche structure and, above all, the identity of the party that carries operating cost after the research ends. Until those are settled the chapters should be read as the apparatus into which a funded programme pours its own outturns, and not as a forecast of what either programme will cost.

The evidence horizon has already moved

Every source was secured and dated on 22 September 2026, and nothing after that date is claimed. One day beyond it the funding position had already changed in ways that bear on Chapters 10 and 11, which carry the third grant cycle as an undocumented dependency. Verification

[2] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[3] GMES & Africa, *About Us – Global Monitoring for Environment and Security & Africa*, programme site, 2025. https://gmes.rmc.africa/about-us

on 23 September 2026, after the horizon and therefore outside the source register of Appendix A, found that cycle open and its terms revised: the share of grant funding to be spent within Dubai raised from one half to three-fifths; the two-phase structure and the non-automatic progression between phases imposed as a condition of the award rather than adopted by the applicant; eligibility restricted to a Principal Investigator not previously funded as such under the initiative; environmental science added as a third priority; and the Letter of Intent deadline set at 6 September 2026, with the full proposal due on 28 October 2026.

Three consequences follow. The fifty per cent rule stated in several chapters is superseded. The month-twelve gate, which Chapter 9 treats as the programme's own design and its most valuable instrument of cost control, is a condition the funder imposes, and the analysis of it survives while the claim of authorship does not. And the funding-timing risk in Chapter 11 is no longer a risk of an undocumented cycle but a question of whether a submission was lodged before a date that has passed. These are not corrections that can be made silently inside a version; they are the substance of the revision the Preface anticipates.

Limits of scope

Two programmes are examined in one frame. They share an architecture, and treating them together shows that, but they are two applications to two sponsors under two instruments, with two cost structures, two periods of analysis and two entirely different orders of magnitude. The shared treatment has cost the continental programme most: it carries no monetary figure in any archived document, its own concept paper states that an investment-grade version remains to be prepared, and this report reproduces that silence and describes it as a gap in the evidence rather than as an uncosted programme it was in a position to cost.

The report also assesses the proposals as they stood on one day, performs no modelling beyond the arithmetic it shows, audits no budget line by line, gives no legal advice and offers no investment advice. Partner agencies are described from their public documents only, and none was approached. The indoor and orbital extensions raised in the stakeholders' discussions are noted and not assessed.

A register of the limitations

Table 3 sets each limitation against what it prevents the reader from concluding and the specific act that would remove it. The third column is the useful one: none of these limitations is permanent, and most are removed by a signature, a quotation or a summer of measurement rather than by further analysis.

What survives

The limitations are serious and they are specific, which means they are bounded. What survives them is the part of the work that does not depend on measurement: the survey of what has been attempted elsewhere and what it demonstrated, the grading of every source so that a reader can see which claims rest on controlled measurement and which on a press release, the institutional map of who must agree in writing and when, the data architecture and its identifiers, the risk analysis, and the costing machinery itself. Those are the durable parts. The numbers are not, and they should be replaced before this report is used to ask anyone for money.

Table 3: Register of limitations, what each prevents the reader from concluding, and the act that removes it. The third column names an obtainable document, quotation or measurement in every case; no limitation here is removed by further desk analysis.

Limitation	What it prevents	What removes it
No field measurement	Any claim about thermal conditions, behaviour or comfort on the corridor itself	Site access letters, then one instrumented summer with a paired control
No vendor quotation	Any statement of what the sensing network costs, beyond a built-up estimate	Three quotations against a written node specification, including the radiation and globe instruments
Personnel never priced	Any statement that the ask is sufficient, since people are the largest element	A costed staffing table: eleven named roles, monthly rate, share of time, months
Budget figures ungraded	Treating [1,400,000.00:388,888.89] (AED:USD) or [250,000.00: 69,444.44](AED:USD) as estimates rather than drafting targets	Regrading both as design-document proposals, not findings (Appendix A), and replacing them with a bottom-up budget
No service life from any supplier	Any replacement schedule, and therefore any whole-life total	Supplier warranties for coating, node and canopy before a candidate set is ranked
Cost distribution spread assumed	Reading any percentile of the cumulative curve as a forecast	Three-point ranges per cost element with a documented correlation assumption
Operating cost unassigned	Any return, because return depends on who pays after handover	A written sustainment commitment from the owning authority
Standard held as a preview	Citing any discounting procedure or service life to the standard	The full text of the standard
Thirteen sources unobtained	Closing several arguments that are left open in Chapters 3 and 4	Institutional access or purchase; each is listed in Appendix A
Evidence horizon of one day	Relying on the grant calendar, envelope or eligibility rules as stated	Re-verification at each submission, and a dated revision
Reference classes treated as absent	The conclusion that no comparable programme exists	Admitting GMES & Africa and the regional urban-cooling programme to the class
Two programmes in one frame	Reading either budget as independently complete	Separate cost structures, separate ask ladders, separate sponsors

Assumptions

An assumption is a claim carried without evidence because the work cannot proceed without it. A study that admitted none would be either trivial or dishonest, and the object of this chapter is not to remove them but to make every one visible, to say where it does its work and to state what fails if it is wrong. Assumptions are not distributed evenly. A handful carry most of the structure, and the rest are load-shedding detail. The ranking matters more than the count.

The numbered chapters flag their own assumptions in a consistent vocabulary. Where the text says *this report's own*, *indicative*, *illustrative* or *presupposes*, an assumption is being declared, and the registers below collect them. Where a claim instead carries one of the seven evidence grades of the Preface, it is not an assumption and does not appear here. The distinction is the whole of the reader's protection: a graded claim can be checked against a document, and an assumption can only be tested by doing something.

Understanding: Load-bearing assumption

TIER 1: IN PLAIN WORDS

Some guesses hold the roof up. If those turn out wrong, the building has to be redesigned, not patched.

TIER 2: IN PROFESSIONAL TERMS

An assumption on which a conclusion depends so directly that its falsification changes the recommendation rather than merely adjusting a figure; it is distinguished from an incidental assumption, whose failure moves a number without moving a decision.

TIER 3: IN TECHNICAL TERMS

An input to which the study's output is strongly sensitive: following the cost-estimating guide's own test, a parameter is sensitive where a change of ten to fifty per cent alters the preferred alternative and very sensitive where a change below ten per cent does [GAO, 2020]; the load-bearing assumptions are those whose plausible range spans a change of decision.

AS A MATHEMATICAL MODEL

Assumption a is load-bearing if $\exists\, a' \in A$ with $\Pr(a') > 0$ such that $\arg\max_x V(x \mid a') \neq \arg\max_x V(x \mid a)$, where A is the assumption's plausible range and V the value of the decision.

Financial and funding assumptions

These are the assumptions that carry the most and are supported by the least. Table 4 lists them.

Currency and conversion. Every money figure in this report is shown as a pair in the form [1,400,000.00:388,888.89](AED:USD), giving the dirham amount and its dollar equivalent together, so that a reader working in either currency can check a figure without doing arithmetic. Where a figure originates in a third currency it is carried through both steps, as [1,000,000.00:4,200,000.00:1,166,666.67](EUR:AED)(AED:USD), so that the chain from the source currency to the dirham and on to the dollar stays visible rather than being collapsed into a single conversion.

Two rates are applied throughout: USD 1 = AED 3.6 and EUR 1 = AED 4.2. Both are rounded working conventions, close to market levels over 2025 and 2026 but not averages of them, adopted so that a reader can check a conversion without a calculator. The dirham's peg to the dollar is deliberately not used, and neither rate should be read as a peg, a central-bank reference rate or a rate obtainable on any particular day. Foreign exchange rates vary, a rate quoted on any given day will differ from the ones used here, and the difference grows with the length of the period being considered. These rates are adopted for ease of comparison across the two programmes and are not a treasury assumption, a forecast or a hedging position. No conclusion in this report turns on either of them, and a reader who prefers different rates can restate every converted figure by applying them to the dirham figure printed alongside.

Currency variation and contracting. The variation in these rates is not an immediate concern at the stage this report addresses. Nothing here commits money, and no figure in it is an offer, a quotation or a price. Any agreed transaction, and any exchange of monies arising from one, will be governed by the foreign exchange terms agreed between the parties at the time, including the rate basis, the date on which it is struck and which party carries the movement between agreement and settlement. Those terms belong to contractual development and not to a feasibility assessment, and they should be settled there rather than assumed here.

Two cautions attach to the pair. The first is false precision: both sides are shown to two decimal places for consistency of notation, but a dirham figure that was rounded to the nearest hundred thousand in its source does not become exact by being divided, and the cents on the dollar side of such a figure carry no information. The second is that the dirham is the currency in which the Dubai programme would actually be funded, so the dirham figure is the primary one throughout and the dollar figure is a convenience for comparison.

Physical, service-life and instrument assumptions

Nothing in the archive states how long a microclimate node, a reflective coating or a canopy fabric lasts in a hot, dusty, low-rainfall climate. The Phoenix trial defers its own life-cycle analysis precisely because the service life is unknown, while measuring reflectivity falling from thirty-three to thirty-eight per cent at installation to nineteen to thirty per cent within ten months [ASU and City of Phoenix, 2021]. The service lives used in this report are therefore its own, adopted so that the replacement rows of the later phases exist at all, and they must not be attributed to any source.

Institutional and consent assumptions

The programme is gated by letters rather than by science, and every letter is assumed. Table 6 lists them in the order the calendar needs them.

Technical and modelling assumptions

These are the assumptions the modelling carries, and most can be tested inside the grant. Table 7 lists them.

Strategic and scale assumptions

These reach beyond the grant, and none can be retired by the pilot alone. Table 8 lists them.

Table 4: Financial and funding assumptions. The first five are load-bearing: each, if false, changes a recommendation rather than a figure. Budget figures are the storyboard's drafting targets [Iyyunni, 2026]; envelope and cap are the funder's published rules *[Dubai RDI Programme, 2025]*.

Assumption	Operates in	If it is wrong
[1,400,000.00:388,888.89](AED:USD) is the cost of the programme rather than a figure the proposal was written towards	2, 9, 10, 11	The financial chapters describe a budget that does not exist; every derived figure in them inherits the error
[250,000.00:69,444.44](AED:USD) buys twenty nodes, a reference station and a field workstation	9, 10	The network is unaffordable as specified, or the specification must lose instruments it needs
Personnel fit inside whatever the equipment line leaves	everywhere implicitly; priced nowhere	The largest cost element exceeds the grant on its own, and the ask is not a budget but a fraction of one
The built intervention will be carried by a partner who has not yet been asked	10, 11	The programme reaches Year 3 with a design and no construction budget
An agency will carry operation, cleaning, water, energy and repair after March 2030	10	The asset is built and then degrades unfunded; the whole-life claim collapses
The annual cap admits a workable profile across project years	9, 10	Year 1, which must buy the network and stand up the platform before the gate, cannot be funded
Real discount rates of nought to eight per cent bracket the rate that will be applied	9	Present values, and therefore the ranking of a coating against a canopy, change
The spread of the cost distribution is narrow enough that the funding ceiling lies near its median	9	The percentile arguments, including the claim that money contingency is unavailable, do not follow
Megaproject overrun rates are context and never an uplift at this scale	9, 10	The illustrative uplifts of Chapter 9 must be believed, and every one of them breaches the funding ceiling
No figure requires adjustment for price level or inflation once converted at the two stated rates	Preface conventions	Comparisons between a Gulf cost and a European or American one are not like for like

Table 5: Physical, service-life and instrument assumptions. Every service life in the first three rows is this report's own placeholder, stated so that replacement costs have somewhere to sit, and each must be replaced by a supplier warranty before any candidate intervention is ranked.

Assumption	Operates in	If it is wrong
A field sensor node lasts about five years	9, 10	The replacement schedule and its present value are wrong in either direction
A reflective coating is renewed at about five-year intervals	10	On the measured reflectivity decline the true interval may be far shorter, so the surface-temperature benefit decays within months while the cost recurs, making the coating a subscription rather than a purchase
A canopy fabric lasts about fifteen years	10	The period of analysis, and therefore the ranking of canopy against coating, changes
The node specification of air temperature, humidity, wind, radiation and surface temperature is sufficient	8, 10	Wet-bulb globe temperature cannot be computed at a node, and the denominator loses comparability with the international labour-hours series [ILO, 2019]
Ninety per cent sensor uptime is achievable across a Gulf summer	7, 11	The gate's first condition fails on instrumentation rather than on science
Dust, heat and vandalism losses stay inside the spares and calibration provision	11	Field attrition consumes the equipment line before the second summer
Interventions behave in Dubai as they behaved where they were measured	3, 4	The evidence base transfers neither the effect size nor its sign

The three that carry the most

Of the assumptions above, three hold up most of the structure, and each fails in a different way.

The first is that [1,400,000.00:388,888.89](AED:USD) is a cost. It is a target, written into the source as an instruction, and two chapters of life-cycle analysis rest on it. Its failure is not a shortfall to be covered by contingency; it is the discovery that the financial chapters have been describing a budget nobody prepared.

The second is that someone else will pay for the asset. The grant explicitly does not carry the built intervention, and the roadmap assigns it to a municipal or heritage authority that has been asked for permission to build and not for a construction budget. If that assumption fails, the programme arrives at Year 3 with a validated design, a ranked candidate set and no means of building anything, which converts a demonstration into a report.

The third is that someone will keep paying after the grant ends. The research closes in March 2030; a structure raised in 2029 stands for as long as its fabric lasts. On the one comparator that publishes an annual programme cost at all (grade: case-report with policy-fact content), running such an asset costs a large sum every year, and Chapter 10 explains why that annual figure cannot be set against the pilot's capital figure *[EEA Climate-ADAPT, 2023]*[1]. No party has been asked to carry it. This is the assumption with the longest tail and the least documentation, and it is the one a funder is most likely to ask about first.

Retiring an assumption

An assumption is retired by an act, not by an argument, and the acts are cheap relative to what they de-risk. A service life is retired by a supplier warranty. A unit cost is retired by three quotations

Table 6: Institutional and consent assumptions. Most are instruments that must be signed by a named party on a named date; the first seven attach to the Dubai programme [Iyyunni, 2026], and the last two to the continental programme: its governance structure [Terhile et al., 2026] and the common reporting this report proposes for it. None had been obtained when this assessment was made, and the first two gate everything after them.

Assumption	Operates in	If it is wrong
The Principal Investigator is eligible under the rules of the cycle actually open	11	The application is inadmissible irrespective of its quality
A funding cycle exists on the assumed calendar	10, 11	The start date, and therefore both summers, move
Site access is granted by the heritage and municipal authorities by December 2026	5, 11	No instrument may lawfully be placed; the first summer is lost and a lost summer is a lost year
Transport and utility data agreements conclude by February 2027	5, 11	The model loses its exogenous inputs and the optimiser its tariffs
At least one authority will confirm in writing at month twelve that the output is useful	7, 10	The gate fails on a condition the programme does not control
The collaborating entity's agreement is concluded and its budget share is real	5	The downscaling and optimisation capability is unfunded
The municipal digital twin platform will grant access to geometry and layers	9	A platform integration cost reverts to a platform build cost
Seven continental governance layers can be convened and will meet	10	The continental programme's standing overhead is unfunded and unstaffed
Member states will report against one common chart of accounts	10	No continental total is computable and no reference class is created

Table 7: Technical and modelling assumptions. The error threshold in the second row is this report's own indicative suggestion and is also one of the four conditions on which the release of the second phase depends, which is why it belongs in a register rather than in a footnote.

Assumption	Operates in	If it is wrong
A district model at twenty-five to fifty metres and a street model at one to five metres resolve the corridor adequately	4, 8	The comfort map is not decision-grade at the scale a pedestrian experiences
Agreement within roughly one degree on bias-corrected hourly air temperature is the right gate threshold	7	The gate is set where the programme can pass rather than where the decision needs it
Machine-learning downscaling transfers between zones of different form	8	Each zone requires its own training data and the method does not generalise
Two summers are sufficient to validate	7, 11	Interannual variability is mistaken for model error, or the reverse
Restored activity-hours are measurable against a paired untreated control	10	The reporting unit has no denominator and the benefit is modelled rather than observed
Pedestrian counts and dwell time capture the behaviour that matters	8	Displacement is recorded as improvement
The comfort index chosen is the one that predicts behaviour	4	Comfort improves on the index while the street stays empty
Compute demand for two summers of street-scale simulation is bounded by the budget	9	Runs are truncated, scenarios are dropped, or the optimiser searches a smaller candidate set
The terminal value at the end of the decision horizon is correctly specified	7	The policy optimises towards the wrong end state
Likelihood and impact scores are approximately right	11	The risk ranking, and therefore where mitigation money goes, is misordered

Table 8: Strategic and scale assumptions. These are the least examined and the furthest reaching, because each concerns not whether the work can be done but whether doing it leads anywhere.

Assumption	Operates in	If it is wrong
A result on one corridor transfers to a district and then to the city	2, 10	The citywide product and the scaling case do not follow from the pilot
A second corridor and agency procurement follow the grant	10	The programme ends at March 2030 with a dataset and no route onward
No comparable reference class exists for either programme	9, 10	Comparators that do exist are ignored, and the cost case is built on their absence
Named development and sovereign financiers will engage at the scale proposed	10	Institutions whose minimum commitment is orders of magnitude larger are addressed with an ask they cannot act on
The two programmes are similar enough to share one analysis	Preface, 1	The continental programme is under-served, and its budget is never built
The continental programme's sequencing will hold under pressure to buy hardware	10	Capital is committed before demand is demonstrated, which is the failure the concept paper names [Terhile et al., 2026]
Published continental valuations rest on sound method	4	Benefit projections built on assumed adoption rates are treated as measured value

against a written specification. A consent is retired by a signature on a letter that already has a date beside it in the programme calendar. An eligibility rule is retired by reading the rules of the cycle that is actually open. A cost distribution is retired by eliciting three-point ranges per element from the people who will do the work. A transfer assumption is retired by instrumenting a paired control and measuring.

Each numbered chapter should be read with its register open beside it, and the registers should shrink between versions. A feasibility report improves not by adding argument but by converting assumptions into graded evidence, and the measure of the next version is how many rows of these five tables it can delete.

Contents

Authors iv

Acknowledgements v

Foreword vi

Preface viii

Governance and Expectations xii

Limitations xix

Assumptions xxiv

List of Figures xxxviii

List of Tables xl

Acronyms xlii

List of Symbols xliv

١ {1} The Source Documents 1

1.1 The working documents and their standing . . . 1
1.2 The Dubai storyboard . . . 3
1.3 The discussion notes behind the Dubai proposal . . . 4
1.4 The African concept paper . . . 5
1.5 The supporting civil-society documents . . . 6
1.6 Assessing the documents against one standard . . . 7
1.7 Implications for the programmes . . . 8

٢ {2} Purpose of the Papers 10

2.1 The Dubai storyboard . . . 10
2.1.1 The problem the project exists to answer . . . 10
2.1.2 The idea and its three constructs . . . 11
2.1.3 The testbed and the partners . . . 13
2.1.4 Method by year and the month-12 gate . . . 13
2.1.5 Team, budget, impact and the ask . . . 14

2.2 The ABIG concept paper . . . 15
2.2.1 Core proposition and the five domains . . . 15
2.2.2 Capability-first sequencing and the roadmap . . . 15
2.2.3 Governance, financing and the UAE hub . . . 16
2.2.4 The addendum: an integration layer, not a parallel programme . . . 18
2.3 The unifying thesis . . . 18
2.3.1 One layered architecture and six distinctions . . . 18
2.3.2 From 30 m to 3 m . . . 19
2.3.3 Climate-resilient urban cells and low-exposure mobility networks . . . 20
2.3.4 The indoor extension . . . 20
2.3.5 From the arid world to space . . . 20
2.4 What the papers do not yet say . . . 21
2.5 Implications for the programmes . . . 21

٣ {3} Similar Efforts Around the World 23

3.1 How the line items are graded . . . 23
3.2 Urban heat and decision twins . . . 25
3.2.1 Cooling Singapore and the Digital Urban Climate Twin . . . 25
3.2.2 Destination Earth . . . 26
3.2.3 Dubai Digital Twin Platform . . . 26
3.2.4 Dubai Walk Master Plan, the Future Loop and The Loop . . . 27
3.2.5 Green Riyadh . . . 27
3.2.6 Makkah and Arafat cooled roads and Hajj cooling measures . . . 28
3.2.7 The Abu Dhabi local climate zone study and the Dubai paper . . . 29
3.2.8 Phoenix cool pavement, Year 1 and Phase II . . . 29
3.2.9 Paris OASIS . . . 30
3.3 Heat economics and policy . . . 31
3.3.1 UNEP Beating the Heat . . . 32
3.3.2 Singapore Heat Stress Advisory . . . 32
3.3.3 Arsht-Rock, Hot Cities, Chilled Economies . . . 32
3.4 Tourism-climate indices . . . 33
3.4.1 Holiday Climate Index and Tourism Climate Index . . . 33
3.5 Ocean and blue intelligence . . . 34
3.5.1 GMES & Africa, MarCOSIO and MarCNoWA . . . 34
3.5.2 Digital Earth Africa . . . 35
3.5.3 Global Fishing Watch in Africa . . . 36
3.5.4 Great Blue Wall . . . 36
3.5.5 European Digital Twin Ocean . . . 37
3.6 Finance . . . 37
3.6.1 Seychelles blue bond . . . 37
3.6.2 AU Blue Economy Strategy and the Luanda Declaration . . . 38
3.7 Global policy instruments . . . 39
3.7.1 Global Cooling Pledge . . . 39
3.7.2 UN Secretary-General's Call to Action on Extreme Heat . . . 39
3.7.3 IPCC Special Report on Climate Change and Cities . . . 40
3.8 Comparison matrix and landscape . . . 40
3.9 Implications for the programmes . . . 46

٤ {4} Feasibility Studies: Evidence, Results and Sources **48**
4.1 What counts as a feasibility study here . . . 48
4.2 The evidence-grading scheme . . . 49
4.3 Urban-heat studies with field measurement . . . 50
4.3.1 Phoenix cool pavement, Year 1 . . . 51
4.3.2 Phoenix cool pavement, Phase II . . . 52
4.3.3 Paris OASIS . . . 53
4.3.4 The Abu Dhabi local climate zone study . . . 54
4.3.5 The Makkah and Arafat cooled roads . . . 55
4.3.6 Cooling Singapore and the Digital Urban Climate Twin . . . 56
4.3.7 Green Riyadh . . . 57
4.4 Heat-economics and heat-policy studies . . . 57
4.4.1 Arsht-Rock, Hot Cities, Chilled Economies . . . 57
4.4.2 ILO, Working on a Warmer Planet . . . 58
4.4.3 Lancet Countdown 2024 . . . 58
4.4.4 The incumbent policy and the incumbent index . . . 59
4.5 Blue-economy and Earth-observation studies . . . 59
4.6 Method references used as evidence about evidence . . . 61
4.7 Master evidence table . . . 61
4.8 Three conclusions . . . 64
4.8.1 The defensible gap . . . 64
4.8.2 Cool pavement lowers the surface and raises the radiant load . . . 64
4.8.3 Projections, measurements, and what informs Year 3 . . . 65
4.9 Running the Year-3 before/after study . . . 65
4.10 Sources that could not be archived, and why . . . 66
4.11 Implications for the programmes . . . 69

٥ {5} Where to Start **70**
5.1 The ordering rule . . . 70
5.2 Dubai: eight steps before the grant starts . . . 71
5.2.1 Step 1: write the semantic layer . . . 71
5.2.2 Step 2: define the metrics . . . 72
5.2.3 Step 3: fix the comfort index and close the instrument gap . . . 73
5.2.4 Step 4: anchor the twin on the Dubai Digital Twin Platform . . . 73
5.2.5 Step 5: instrument a paired-control design . . . 73
5.2.6 Step 6: turn the storyboard into the written proposal . . . 75
5.2.7 Step 7: the letters of December 2026 and February 2027 . . . 75
5.2.8 Step 8: carry the third funding cycle as a dependency . . . 75
5.3 ABIG: four first steps . . . 76
5.3.1 Step 1: the alignment and mapping exercise . . . 76
5.3.2 Step 2: build the capability map . . . 77
5.3.3 Step 3: two or three use cases with named operational users . . . 78
5.3.4 Step 4: put the ninety-day list in dependency order . . . 78

5.4 The 180-day action table . . . 79
5.5 Dependencies . . . 82
5.6 Putting the documents in order . . . 84
5.7 Implications for the programmes. . . . 84

٦ {6} Circles of Influence 86
6.1 Three rings, and who decides . . . 86
6.2 Dubai: actors and levers by ring . . . 87
6.3 ABIG: actors and levers by ring . . . 90
6.4 The escalator. . . . 91
6.5 Why gate condition 6.2.4 matters . . . 93
6.6 The five stakeholders and their reach . . . 94
6.7 The influence plan . . . 94
6.8 Implications for the programmes. . . . 99

٧ {7} System Boundaries: Spiral Development and Stochastic Dynamic Programming 100
7.1 Part A: boundaries under the spiral model. . . . 101
7.1.1 The spiral as a boundary-drawing instrument . . . 101
7.1.2 The Dubai timeline as four loops. . . . 102
7.1.3 How much is enough: the risk boundary . . . 103
7.1.4 Win conditions: the commitment boundary. . . . 105
7.1.5 ABIG as five loops . . . 105
7.1.6 The scale spiral . . . 107
7.2 Part B: boundaries under stochastic dynamic programming . . . 107
7.2.1 The five elements . . . 107
7.2.2 The four metrics as functions of the state . . . 108
7.2.3 Decisions at two timescales . . . 110
7.2.4 Exogenous information and the transition . . . 110
7.2.5 The objective and the Bellman recursion . . . 111
7.2.6 The information boundary and the value of Year 1 . . . 111
7.2.7 Policy classes and the curse of dimensionality . . . 112
7.2.8 Resource coupling: water, energy and the annual cap . . . 113
7.2.9 ABIG as a sequential decision problem . . . 114
7.3 Part C: a unified boundary taxonomy . . . 114
7.4 Implications for the programmes. . . . 116

٨ {8} A Taxonomy of Synthetic Data Across the Systems Architecture 118
8.1 Why identifiers decide what can be measured . . . 118
8.2 The master tree of the architecture . . . 120
8.3 Sub-trees: the synthetic-data classes of each layer . . . 120
8.4 The identifier schema . . . 123
8.5 Synthetic-record flags and the `is_synthetic` firewall. . . . 126
8.6 Data lineage . . . 128
8.7 Life-cycle indicators measured through the identifiers . . . 129
8.8 The same recursion for the ocean twin . . . 130

8.9 Implications for the programmes. . . . 132

٩ {9} Forecasting Life-Cycle Costs 133

9.1 What is being forecast . . . 134
9.2 Ten ways to forecast . . . 135
9.2.1 Deterministic life-cycle costing with net present value . . . 135
9.2.2 Parametric estimating . . . 136
9.2.3 Analogy and engineering build-up . . . 136
9.2.4 Reference class forecasting . . . 136
9.2.5 Monte Carlo simulation, percentiles and S-curves . . . 137
9.2.6 Service-life and degradation curves . . . 138
9.2.7 The month-12 gate as a real option. . . . 139
9.2.8 Total cost of ownership for the digital platform . . . 140
9.2.9 Learning curves . . . 141
9.2.10 The value function as expected life-cycle cost . . . 141
9.3 Choosing a method. . . . 141
9.4 Worked indicative examples. . . . 143
9.5 Discount rates and the period of analysis . . . 144
9.6 Reporting uncertainty . . . 145
9.7 Implications for the programmes. . . . 146

١٠ {10} Life-Cycle Cost Breakdown and Strategic Roadmap 148

10.1 The cost breakdown structure . . . 149
10.2 Phases and stage gates . . . 149
10.3 The breakdown crossed with the phases. . . . 151
10.4 What the grant covers and what it does not . . . 154
10.5 One reporting unit . . . 155
10.6 The roadmap. . . . 156
10.7 The same structure for ABIG . . . 157
10.8 Implications for the programmes. . . . 159

١١ {11} Risk Modelling, SWOT Analysis, Loss Remediation and Recovery, and Force Majeure 161

11.1 A process, and what counts as a risk . . . 162
11.2 A taxonomy for two programmes . . . 163
11.3 The Dubai risk register . . . 165
11.4 The ABIG risk register . . . 167
11.5 The likelihood and impact heat map . . . 169
11.6 Quantitative risk modelling . . . 169
11.6.1 Monte Carlo on schedule and cost . . . 169
11.6.2 A Bayesian model of the month-12 gate . . . 170
11.6.3 Sensitivity and the tornado . . . 171
11.7 Key risk indicators . . . 171
11.8 What the models do not capture . . . 173
11.9 SWOT analysis. . . . 174

11.10 Loss remediation and recovery . . . 174
11.10.1 Six playbooks . . . 178
11.10.2 Recovery objectives for the data platform . . . 178
11.10.3 Insurance, warranty and other transfers . . . 180
11.10.4 The gate-failure fallback . . . 180
11.11 Force majeure . . . 181
11.11.1 What force majeure is . . . 181
11.11.2 The elements a clause must contain . . . 182
11.11.3 Which events belong in the list . . . 183
11.11.4 Drafting for the three instruments, and the question for counsel . . . 183
11.12 Implications for the programmes. . . . 184

١٢ {12} Policy **186**
12.1 The laboratory argument . . . 186
12.2 Policy follows measurement . . . 188
12.3 The instruments the Gulf already holds . . . 189
12.4 The science–policy window . . . 191
12.5 South–South transfer, and whose knowledge counts . . . 194
12.6 A note on planetary boundaries . . . 195
12.7 Implications for the programmes. . . . 197

١٣ {13} Conclusion and Recommendations **199**
13.1 Synthesis of findings by chapter . . . 199
13.2 Recommendations . . . 201
13.2.1 Dubai: Climate Intelligence & Decision Twin . . . 201
13.2.2 ABIG: Africa Blue Intelligence Grid . . . 202
13.2.3 Joint recommendations . . . 202
13.3 Decisions required from each stakeholder. . . . 203
13.4 The next ninety days . . . 204
13.5 Open questions and further work . . . 204
13.6 Implications for the programmes. . . . 206

Next Steps **207**

A {A} Source Archive and Evidence Register **213**
A.1 What the register contains and why . . . 213
A.2 How to read the registers . . . 214
A.3 Register I: primary feasibility studies and evidence . . . 216
A.4 Register II: announcements and reporting. . . . 216
A.5 Register III: methods, standards and policy instruments . . . 218
A.6 Sources that could not be archived . . . 218
A.6.1 Bibliography entries held as metadata only . . . 219
A.6.2 The stakeholders' own working papers . . . 221

A.7 The evidence horizon . . . 221
A.8 Known defects in the sources . . . 222
A.9 Implications for the programmes. . . . 222

B {B} Data Dictionary and Identifier Schema **224**
B.1 How to read this dictionary . . . 224
B.1.1 Cardinality . . . 225
B.1.2 Stewardship . . . 226
B.1.3 Retention . . . 226
B.2 The field dictionary . . . 227
B.3 Controlled vocabularies . . . 231
B.4 The join path from a sensor record to a decision . . . 232
B.5 The metric dictionary . . . 232
B.6 What is design and what is evidence . . . 234
B.7 Implications for the programmes. . . . 235

Glossary **237**
G.1 Climate and thermal comfort . . . 237
G.2 Urban systems, mobility and tourism . . . 239
G.3 Ocean, Earth observation and the blue economy . . . 241
G.4 Data, semantics and synthetic data . . . 242
G.5 Decision science and modelling . . . 244
G.6 Economics, costs and finance . . . 246
G.7 Programme management and risk . . . 247
G.8 Policy and governance . . . 249
G.9 Institutions and programmes . . . 251

Bibliography **254**

Index **258**

List of Figures

1.1 How the source documents relate 8

2.1 The layered architecture shared by the Dubai Decision Twin and ABIG, drawn from the discussion notes 19

3.1 Landscape of similar efforts by domain (rows) and spatial scale of the product (columns) 45
3.2 Resolution ladder from planetary twins to the Dubai street model 46

4.1 The grading decision path applied to every quantity cited in this chapter. 51
4.2 Phoenix Year-1 mean differences between cool pavement and adjacent untreated asphalt 53
4.3 The Year-3 before-after-control-impact design 66

5.1 Dependency structure of the first steps 83

6.1 Circles of influence for the Dubai Decision Twin 88
6.2 Circles of influence for the Africa Blue Intelligence Grid 90

7.1 The Dubai programme as a risk-driven spiral 104
7.2 The scale spiral: the boundary of the Dubai method enlarged one turn at a time . . 107
7.3 One epoch of the sequential decision model 108

8.1 Master tree of the systems architecture 121
8.2 Sub-trees of the physical-environment and sensing layers 123
8.3 Sub-trees of the data, semantic, knowledge-graph and model layers 124
8.4 Sub-tree of the upper layers 124
8.5 The identifier schema: twelve canonical keys in four families and the seven flags every synthetic record carries 125
8.6 Data lineage for one indicator 129
8.7 The recursion applied to the ocean twin 131

9.1 Cumulative cost S-curve with clearly labelled illustrative data 139
9.2 The month-12 gate read as a commitment structure 140

10.1 Strategic roadmap for the two programmes on one calendar 156
10.2 The Africa Blue Intelligence Grid financing ladder read as a gated commitment structure 157

11.1 The eight-branch risk taxonomy used to build both registers 164
11.2 Likelihood-by-impact heat map for both registers 169
11.3 Tornado diagram of the Dubai cost estimate's sensitivity to six factors 173
11.4 The two continuity objectives on one timeline 178

12.1 The policy pathway proposed in Proposition 12.2.1, with the two programmes' mileston 189
12.2 The science–policy calendar 193

B.1 The join path from an instrument reading to a decision, with the keys each step adds 232

List of Tables

1 The stakeholders' fourteen questions and the chapters that answer them. ix
2 The same sum read by four kinds of investor xiv
3 Register of limitations, their consequences and what removes each xxiii
4 Financial and funding assumptions, where each operates and what fails if it is wrong xxvi
5 Physical, service-life and instrument assumptions xxvii
6 Institutional and consent assumptions xxviii
7 Technical and modelling assumptions xxix
8 Strategic and scale assumptions xxx

1.1 The working documents of the two programmes on 22 September 2026 1
1.2 Assessment of the stakeholders' own documents 7

2.1 Milestone summary of the Dubai storyboard, as stated in the rescoped Letter of Intent 14
2.2 ABIG roadmap horizons as stated in the concept paper 17

3.1 Comparison matrix of similar efforts 41

4.1 The seven evidence grades and the test applied to each quantity. 50
4.2 Master evidence table: archived feasibility documents, what each tested, the headline result, the evidence grade and the citation. 62
4.3 Specification of the Year-3 before/after campaign, with the archived design each element is drawn from. 67
4.4 Sources that could not be archived, the reason, and how they are cited in this report. 68

5.1 The 180-day action table 80

6.1 The five stakeholders and their documented institutional reach 95
6.2 Influence plan for the two programmes 96

7.1 The Dubai storyboard as four risk-driven loops, with anchor points, boundaries and exit criteria 103
7.2 The ABIG roadmap as five risk-driven loops 106
7.3 The four policy classes mapped onto the decisions of the Decision Twin 113
7.4 A unified taxonomy of eight boundaries, drawn from the spiral model and the sequential decision model 115

8.1 Layers of the architecture, the identifiers born at each, and the observed and synthetic records eac 122
8.2 The identifier schema in detail 127
8.3 Life-cycle indicators as joins over the identifier schema 129

9.1 Ten ways to forecast a life-cycle cost: input required, output produced, and the support available i . . . 135
9.2 Method-selection matrix: primary method with cross-check in brackets, by cost element and programme phase . . . 142
9.3 Present value of an indicative [250,000.00:69,444.44](AED:USD) node replacement falling due in year five, at four illust . . . 144

10.1 The nine-category cost breakdown structure for the Dubai Decision Twin, with the owner accountable f . . . 150
10.2 Phases, anchor points and gate conditions for the Dubai Decision Twin . . . 152
10.3 Cost breakdown structure against phase for the Dubai Decision Twin . . . 153
10.4 The Africa Blue Intelligence Grid financing ladder against the cost breakdown structure, with the ow . . . 158

11.1 Dubai Decision Twin: gate-critical risks . . . 165
11.2 Dubai Decision Twin: programme, resource, institutional and external risks . . . 166
11.3 Africa Blue Intelligence Grid risk register . . . 168
11.4 Key risk indicators for the highest exposures in both registers . . . 172
11.5 SWOT analysis: Climate Intelligence and Decision Twin for Dubai . . . 175
11.6 SWOT analysis: Africa Blue Intelligence Grid . . . 176
11.7 SWOT analysis: the combined portfolio . . . 176
11.8 TOWS strategy matrix for the combined portfolio . . . 177
11.9 Recovery playbooks for the six most likely losses . . . 179
11.10 Proposed recovery time and recovery point objectives for the Dubai data platform 180
11.11 Force-majeure clause elements and where each attaches . . . 182

12.1 Policy instruments within reach of the two programmes: who owns each, where a small programme can en . . . 192
12.2 Policy asks by stakeholder: what each is asked to carry, where, on what evidence, and from when . . . 196

13.1 Proposed ninety-day action plan from 22 September 2026 (day 0), with the fixed storyboard milestones beyond day 90. . . . 205
13.2 A three-year budget that fits the annual cap, with the descopes that make it fit . . . 208
13.3 The ask ladder: amount, sponsor and owner at each tier . . . 211

A.1 Distribution of the sixty-seven secured sources by evidence grade . . . 215
A.2 Register I: primary feasibility studies and evidence . . . 216
A.3 Register II: landing pages, press releases and news reports. . . . 217
A.4 Register III: methods, standards and policy instruments. . . . 218
A.5 Sources that could not be archived, with the reason and the chapters that refer to them by title only. . . . 219
A.6 Sources cited from their publication details alone: standards, classic texts and reference works. . . . 220
A.7 The stakeholders' own documents, cited as working papers. . . . 221

B.1 Field-level data dictionary for the canonical keys and synthetic-record flags . . . 228
B.2 Controlled vocabularies . . . 231
B.3 Metric dictionary. Formulas are given in compact form . . . 232

Acronyms

ABIG Africa Blue Intelligence Grid.

ADP approximate dynamic programming.

AIS Automatic Identification System (vessels).

AU African Union.

BRICS Brazil, Russia, India, China, South Africa and the 2024–25 accession members (BRICS-11).

CBS cost breakdown structure.

COP Conference of the Parties (UN Framework Convention on Climate Change).

DEWA Dubai Electricity and Water Authority.

DFF Dubai Future Foundation.

DUCT Digital Urban Climate Twin (Cooling Singapore).

EO Earth observation.

GCC Gulf Cooperation Council.

GFW Global Fishing Watch.

GMES & Africa Global Monitoring for Environment and Security & Africa.

GOOS-Africa Global Ocean Observing System, Africa regional alliance.

HCI Holiday Climate Index.

HLAH Heat-Lost Activity Hours.

IOC Initial Operational Capability (spiral anchor point).

IoT Internet of Things.

IPCC Intergovernmental Panel on Climate Change.

IUU illegal, unreported and unregulated (fishing).

LCA Life-Cycle Architecture (spiral anchor point).

LCC life-cycle cost.

LCO Life-Cycle Objectives (spiral anchor point).

LCZ local climate zone.

LST land-surface temperature.

MCN Minimum Cooling Network.

MRT mean radiant temperature.

NCM National Center of Meteorology (UAE).

NDB New Development Bank.

NPV net present value.

OPERA Ocean Prediction Enhancement in Regions of Africa.

OTC Outdoor Tourism Capacity.

POMDP partially observable Markov decision process.

RAH Restored Activity Hours.

RCF reference class forecasting.

RDI Research, Development and Innovation (Dubai programme).

REC Regional Economic Community (Africa).

RTA Roads and Transport Authority (Dubai).

SAR synthetic-aperture radar.

SDP stochastic dynamic programming.

SWOT strengths, weaknesses, opportunities, threats.

TCO total cost of ownership.

UHI urban heat island.

UNEP United Nations Environment Programme.

UTCI Universal Thermal Climate Index.

WBGT wet-bulb globe temperature.

WIO Western Indian Ocean.

List of Symbols

Costs and Economics

$\bar{B}_y, \bar{W}_y, \bar{E}_y$ Budget, water and energy limits binding in capital epoch y

γ Discount factor, $0 < \gamma \leq 1$

EVPI, EVSI Expected value of perfect information; expected value of sample information

RE Risk exposure: probability of loss times size of loss

$c^{\text{cap}}, c_h^{\text{om}}$ Capital cost of an intervention; operating and maintenance cost in hour h

$C_t(S_t, x_t)$ Contribution (reward) function at epoch t

v_c Value of one restored activity-hour for cohort c (AED per person-hour)

$V_t(S_t)$ Value function: optimal expected contribution from epoch t onward

Decisions and Policies

$\mathcal{X}_t$ Set of feasible decisions at epoch t

$\phi_c(\cdot)$ Acceptability function of cohort c, mapping the comfort index onto $[0, 1]$

π, Π A policy; the set of admissible policies

θ Tunable parameters of a policy function or cost function approximation

x_y^{cap} Capital decision in epoch y: which intervention, and where

x_t^{op} Operating decision (hourly): misting, route advisory, cooling-load schedule

$X^\pi(S_t)$ Decision rule of policy π evaluated at state S_t

x_t Decision at epoch t, $x_t = (x_t^{\text{op}}, x_{y(t)}^{\text{cap}})$

Other Symbols

τ Risk-exposure tolerance written down by the stakeholders

$\varepsilon_{\text{gate}}$ Stated model-error threshold of gate condition 6.2.2

$S^M(\cdot)$ Transition function (the system model)

W_{t+1} Exogenous information arriving after x_t: weather, dust, demand, prices, agency decisions

$y(t)$ Capital epoch (programme year) containing hour t

Z An observation programme; in Dubai, the twenty nodes and one reference station

State and Sets

$\mathbb{E}[\cdot]$ Expectation operator

κ_p Physical capacity of place p (persons)

$\mathscr{P}$ Set of places (H3 cells or route segments) inside the corridor boundary

$\mathscr{R}_k, \mathscr{M}_k$ Major risks identified in spiral cycle k; those covered by the risk-management plan

$\mathbb{N}$ Set of natural numbers

$\mathbb{R}$ Set of real numbers

$A_{p,h,c}$ Counterfactual activity of cohort c at place p in hour h (persons)

B_t Belief state: distribution of model error and of behavioural-response parameters

H, H_y Set of evaluated hours; the hours falling in capital epoch y

I_t Information state: comfort field per place-hour, footfall and dwell, forecasts in hand

R_t Physical state: interventions installed and their condition; remaining capital, water and energy budgets

S_t^x Post-decision state, after decision x_t and before W_{t+1} arrives

S_t State variable at epoch t, $S_t = (R_t, I_t, B_t)$

U^* Comfort threshold defining an acceptable place-hour on a priority route

$U_{p,h}$ Comfort index at place p in hour h (UTCI for visitors and residents, WBGT for workers)

١ {1}

The Source Documents

Abstract

A feasibility study is only as good as the documents beneath it, so this chapter examines the material on which the two programmes currently rest, as it stood on 22 September 2026. Between them the stakeholders hold one finished paper, the Africa Blue Intelligence Grid strategic concept paper and its addendum of 20 September 2026; one complete but unformalised storyboard for the rescoped Dubai Letter of Intent; a raw discussion log of unusual intellectual richness that also carries a personal telephone number and a private letter; and four civil-society documents that supply values and policy context rather than technical input. Each is set against six criteria (complete, attributed, sourced, free of sensitive content, readable and gradable) and assigned the evidence grade it can carry. The pattern is asymmetric: Dubai has a plan but not a document, Africa has a document but not a costed plan, and neither yet carries the sourcing that a funder or a reviewer will expect. The closing section turns those findings into the housekeeping agenda that must precede any submission.

A feasibility study inherits the evidence of the documents it is built on, and it should say plainly what that evidence is worth before it says anything else. The two programmes examined here have been developed largely in conversation and in draft, and the drafts are uneven in a way that matters for what can be claimed on their behalf. This chapter therefore records what exists, what each document establishes, and what each cannot be asked to establish.

1.1 The working documents and their standing

Table 1.1 sets out the material. It is a small body of work: two working texts for Dubai, one finished paper with an addendum for Africa, and four supporting documents. Extents are stated as the documents themselves present them.

Table 1.1: The working documents of the two programmes on 22 September 2026, with the standing each can claim.

Document	What it contains	Extent	Standing
Storyboard for the rescoped Dubai Letter of Intent [Iyyunni, 2026]	Ten numbered sections set against a timeline: problem, idea, testbed, partners, method by year, gate, team, budget, impact and ask, with a milestone summary	About 1,400 words	Complete as a plan; not yet a document that could be submitted; undated and unsourced

Table 1.1 (continued)

Document	What it contains	Extent	Standing
Discussion notes behind the Dubai proposal	Advisory exchange on knowledge architecture, spatial resolution, an indoor extension, indigenous knowledge and commercial arrangements	About 3,700 words	Raw; intellectually the richest source; carries a telephone number, a private letter and third-party names
Africa Blue Intelligence Grid strategic concept paper, with addendum, 20 September 2026 [Terhile et al., 2026]	Nineteen numbered sections: core proposition, five information domains, capability-first sequence, ten applications, five-horizon roadmap, financing ladder, governance, first-90-day list and nine risks; addendum on alignment with existing African programmes	About 23 pages, the addendum occupying the last eight	The only finished paper; dated, attributed and internally sourced; carries no budget
Unity: A Pathway to Achieving a Thriving Planet, post-HLPF 2024 edition [Commons Cluster, 2024]	Unity-building initiatives at global, national and grassroots level; the Supplementary Income for All proposal	25 pages	Values and policy context
Building Unity as a Prerequisite to Achieving the SDGs, 2023 [Commons Cluster, 2023]	Proposal for the 2023 Sustainable Development Goals Summit, organised around nature, society and economy	43 pages	Values and policy context
Integrating the Laws and Qualities of Nature to Implement the 2030 Agenda, 2023	Civil-society document prepared for the High-Level Political Forum	60 pages	Context only; the available copy carries no machine-readable text, so its content cannot be verified or cited
BE WELL: Overview for YOU, 2026	Behaviour and well-being framework for adolescent students, India-focused by its author's description	35 pages	Context only; an overview rather than a full framework

Two observations follow at once, and they shape everything after. The body of work is small, which is not a criticism of a programme at this stage but does mean that a single unsourced number travels a long way. And the balance is uneven: the Dubai proposal exists as a plan but not as a document, while the African proposal exists as a document but not yet as a plan with a budget. The two programmes are at opposite stages of the same journey, and each can borrow from the other's strength.

Understanding: Evidence and its standing

TIER 1: IN PLAIN WORDS

Not every document proves the same kind of thing. A plan proves what somebody intends. A conversation proves what somebody said. Only a measurement proves what actually happened, and even then only where it was measured.

TIER 2: IN PROFESSIONAL TERMS

Each source carries a grade describing what its text will support: a target in a plan is a fact

about the plan, never evidence of an effect; a record of a discussion is evidence of an idea, not of a result. Applying that discipline to one's own documents as strictly as to anyone else's is what makes the rest of the assessment credible.

TIER 3: IN TECHNICAL TERMS

Every claim is traceable to a source, and every source is graded on the seven-step scale set out in the Preface. Provenance is recorded for each in the manner of the standard provenance vocabulary [W3C, 2013]: what the source is, who issued it, when, and what derivation was applied to reach the claim.

AS A MATHEMATICAL MODEL

A claim c is admissible only where some source s supports it; its grade is then $g(c) = \max_{s \vdash c} g(s)$ under the ordering measured-with-controls $\succ$ measured $\succ$ modelled $\succ$ projection $\succ$ policy-fact $\succ$ case-report $\succ$ method-reference. A target drawn from a plan is never graded above policy-fact.

1.2 The Dubai storyboard

The formal Dubai document is the storyboard for the rescoped Letter of Intent to the Dubai Future Foundation Research, Development and Innovation Grant under the Cognitive Cities priority [Iyyunni, 2026]: ten numbered sections, each item set against a timeline. It is the proposal in outline, and Chapter 2 examines its substance. Recorded here are only the facts that fix its identity and its standing.

The idea rests on three constructs: Outdoor Tourism Capacity (OTC), the Minimum Cooling Network (MCN) and the Decision Twin, defined as a simulation environment that compares interventions before they are built, 'not a 3D rendering'. The testbed is the Al Fahidi Historical Neighbourhood on Dubai Creek, a heritage core used only as a calibration baseline, together with a corridor of about 800 metres from Al Ghubaiba metro station used as the intervention site. Four agencies (the Roads and Transport Authority; the Dubai Electricity and Water Authority; Dubai Culture; Dubai Municipality) are each asked for one dataset or one permission. Twenty microclimate nodes and one reference station record five variables every ten minutes; models run at 25 to 50 m and at 1 to 5 m; one intervention, most likely a shade canopy over the most exposed 200 metres, is built in Year 3 and measured before and after. The gate at month twelve carries four conditions (sensor uptime of at least 90 per cent across the summer, model error within a stated threshold, an hour-by-hour comfort map of both zones, and written confirmation from the Roads and Transport Authority or the Dubai Electricity and Water Authority that the output is useful) and, if it fails, the work closes with the dataset and the platform as deliverables. The ask is [1,400,000.00:388,888.89](AED:USD) over 36 months in two phases, equipment held at about [250,000.00:69,444.44](AED:USD), for a grant period running from 15 March 2027 to 14 March 2030 [Iyyunni, 2026].

Three things about its standing deserve record. It is complete as a plan. It is not submittable: it carries no bibliography, no figure and no site map, although its own third section argues that one site with a map and marked sensor positions beats a citywide promise. And its calendar rests on an assumption that no available document supports: the published material describes only the 2025 funding cycle, with a Letter of Intent deadline of 13 June 2025, awards on 10 September 2025, and grants of [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD) for up to

three years *[Dubai Future Foundation, 2025]*[1] *[Dubai RDI Programme, 2025]*[2]. Nothing describes a cycle receiving a Letter of Intent in September 2026. Nor is there a record of whether that Letter was submitted, or a copy of the earlier proposal that the word 'rescoped' presupposes. This funding-timing dependency is carried into Chapters 5 and 11; the missing earlier submission is carried into the housekeeping list below.

Understanding: The storyboard

TIER 1: IN PLAIN WORDS

The storyboard is the plan written out as a list: what will be done, in which order, by whom, for how much, and what has to be true after twelve months for the money for the next two years to be released.

TIER 2: IN PROFESSIONAL TERMS

It is a phased project plan with a stage gate. Phase 1, of twelve months, delivers observation and a validated model; Phase 2, of twenty-four months, delivers prediction, optimisation and one built intervention. Four pass conditions and a defined fallback bound the funder's exposure at the end of Phase 1 if the science does not perform [Iyyunni, 2026].

TIER 3: IN TECHNICAL TERMS

In the vocabulary of Chapter 7 it is the pre-award loop of a risk-driven spiral. It contains an operational concept, a life-cycle plan and a budget, but not yet the requirements description, architecture description or feasibility rationale that an anchor-point review expects [Boehm and Hansen, 2000]; this study supplies the last of these.

AS A MATHEMATICAL MODEL

With $c_1, \ldots, c_4$ the gate conditions, release of Phase 2 is $G_{12} = \bigwedge_{i=1}^{4} c_i$; the deliverable set is $\{\text{twin},\ \text{MCN},\ \text{intervention}\}$ if G_{12} holds and $\{\text{dataset},\ \text{platform}\}$ otherwise.

1.3 The discussion notes behind the Dubai proposal

Behind the storyboard lies an advisory exchange recorded as it happened. It is raw: contributions appear in sequence, and the register moves between advisory prose, pasted analysis, personal remarks and a letter. It carries no date in a form that can be relied on, being ambiguous between early September and early August 2026. Its intellectual content is nevertheless the richest of any document considered here, and much of it has been carried into this study.

Five threads matter. The first is the knowledge-architecture argument: that the project is not an artificial-intelligence model placed on top of a climate dataset but a chain running from sensors and models through a data layer, a semantic layer, a knowledge graph, reasoning and decision intelligence to a digital twin, intervention and learning, six distinct things that must not be conflated. This is the unifying thesis of Chapter 2 and the reason Chapter 5 puts the semantic layer first. The second is the resolution argument: a 25 to 30 m citywide layer is more defensible than either a 100 m layer or a citywide claim at 1 to 3 m, because the multispectral satellite data commonly used are at 30 m while the thermal band is coarser, and because the Abu Dhabi local climate zone study coupled classification at that scale with street modelling at 3 m [Manandhar et al., 2019]. The third is an extension the storyboard does not contain: a building climate and infection layer running from outdoor conditions through the building envelope and ventilation to airborne exposure, and

[1] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

a vision of the city as climate-resilient urban cells linked by low-exposure mobility networks. The fourth concerns legitimacy: Bedouin and indigenous knowledge of living with heat as something to learn from rather than to romanticise, and *nemawashi*, the practice of preparing the ground with each party before any formal proposal. The fifth is commercial: a private letter to an investor, a non-disclosure agreement and candidate collaborators.

Its standing is the problem. The record contains a personal mobile telephone number, the full text of a private letter to a named third party, the names of private individuals who are not stakeholders of this study, and informal language that should not reach agencies or funders. None of it is reproduced here. The remedy is a scrubbed derivative that keeps the five threads, removes every personal identifier and the letter, carries a date in unambiguous form and can be cited, so that the ideas travel while the record does not.

Circulation of the raw discussion record

The ideas in the discussion record have already been extracted into this study, but the record itself remains as first written. If it is shared with an agency for due diligence, or included in any wider release, the telephone number and the private letter go with it. The mitigation is to produce the scrubbed derivative before any widening of access, and to treat the original as confidential to the stakeholders. The exposure is registered in Chapter 11.

1.4 The African concept paper

The African material is the mirror image of the Dubai material: a finished paper where Dubai has a plan. The paper is the Africa Blue Intelligence Grid (ABIG) strategic concept paper, dated 20 September 2026 and authored by Dr Uchi Dominic Terhile, Dr Chakradhar Iyyunni, Abdullah Al Salmani and Arti Ahluwalia, with an addendum prepared from Dr Terhile's perspective [Terhile et al., 2026].

The main body runs to nineteen numbered sections. Its core proposition is that Africa should treat its ocean, freshwater, land, atmosphere and coastal systems as a single information-and-capability system, and its stated purpose is not initially to build a satellite constellation but to connect what Africa already has (space agencies, GMES & Africa, universities, oceanographic institutions, coast guards, fisheries authorities, ports and communities) into a repeatable continental capability. It defines five information domains and a capability-first sequence in which decisions are defined, information requirements derived and existing data assembled before any sensor, payload or factory is funded. It sets out ten priority applications, a roadmap in five horizons from three months to three years, a six-stage financing ladder, a seven-layer governance model, a twelve-item list for the first ninety days and nine principal risks with design responses [Terhile et al., 2026]. It anchors itself on the African Union's Luanda Declaration, adopted on 24 July 2026 *[African Union, 2026]*[3]. The figures it quotes from that Declaration (38 of 55 member states coastal, seven Large Marine Ecosystems, inland waters of about 305,000 km^2 and transboundary basins covering almost 64 per cent of the land area) are quotations rather than the paper's own measurements, and since the Declaration's own text was not available for review this study attributes them to the paper.

The addendum repositions ABIG as an integration and application layer, complementary to what exists rather than parallel to it: observation through the Global Ocean Observing System's African alliance; forecasting through the OceanPrediction Decade Collaborative Centre's African Seas Regional Team and the Ocean Prediction Enhancement in Regions of Africa programme;

[3] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

Earth-observation services through GMES & Africa. It proposes eight regional hubs following African marine systems, a federated data hub in which African institutions retain ownership of their data, a co-design chain running from user decision to service improvement and, for the first twelve months, an alignment and mapping exercise whose output is a continental capability map [Terhile et al., 2026]. Chapter 5 adopts that exercise as ABIG's first deliverable.

What the paper lacks matters as much as what it holds. It states no budget, cost, staffing number or benefit estimate, offering only second-hand scale indicators that it explicitly flags as indicative, and it closes by noting that an investment-grade version would need a country-by-country capability database of sources, sensors, institutions, personnel, facilities and budgets. In the vocabulary used throughout this study the paper is a method-reference: it defines an architecture that Chapters 2, 7 and 10 evaluate, and supplies no measured or modelled result of its own. It has also not yet been prepared as a formal proposal, with the bibliography and the costed plan that a funder would expect.

Understanding: The concept paper

TIER 1: IN PLAIN WORDS

The paper says what the African programme is for, what it would connect, in what order things should be done and who would govern it. It does not yet say what it would cost or what it would measure.

TIER 2: IN PROFESSIONAL TERMS

It is a strategic design document with a phased roadmap, a governance model, a financing ladder and a risk register, positioned by its addendum as an integration layer above three existing African programmes rather than as a new observing or forecasting system [Terhile et al., 2026].

TIER 3: IN TECHNICAL TERMS

It specifies a socio-technical architecture: a knowledge-production cycle running observe, integrate, model, predict, understand, decide, act, measure and learn; a federated data hub with provenance and controlled access; and a co-design chain in which each observation and modelling requirement derives from a named user decision, the decision-to-sensor traceability that Chapter 8 formalises through identifiers.

AS A MATHEMATICAL MODEL

The programme is the map $\Phi : (O_{\mathrm{GOOS}}, F_{\mathrm{OPERA}}, E_{\mathrm{GMES}}) \mapsto \mathcal{K} \to \mathcal{X}$ from observation, forecast and Earth-observation services to knowledge products $\mathcal{K}$ and decisions $\mathcal{X}$; the co-design chain runs the other way, $x \to i \to o \to m \to k \to t \to f$ (decision, information requirement, observation requirement, modelling requirement, product, test, feedback).

1.5 The supporting civil-society documents

Four further documents accompany the concept paper. Three come from the Commons Cluster of the NGO Major Group, a civil-society grouping that convenes side events at the United Nations High-Level Political Forum; the fourth is a framework overview by one of the stakeholders. The connection runs through Arti Ahluwalia, listed among the Cluster's contributors as Country Chair for Ecocivilisation and named as the developer of the BE WELL framework.

Unity: A Pathway to Achieving a Thriving Planet is the post-HLPF 2024 edition of a summary generated at an official side event on 10 July 2024; it collects unity-building initiatives at global, national and grassroots level and closes with the Supplementary Income for All proposal, a self-financing scheme for a sustainable global marketplace [Commons Cluster, 2024]. *Building Unity as a Prerequisite to Achieving the SDGs* is the Cluster's proposal for the Sustainable Development Goals

Summit of September 2023, organised around nature, society and economy [Commons Cluster, 2023]. *Integrating the Laws and Qualities of Nature to Implement the 2030 Agenda* is the Cluster's 2023 document; the available copy carries no machine-readable text, so its contents could not be verified and it is cited nowhere in this study. *BE WELL: Overview for YOU* presents a behaviour and well-being framework for adolescent students, India-focused by its author's own description.

These are values and policy context rather than technical inputs. They contain no ocean, climate, sensor, cost or economic data that could be cited as evidence, and they are not so used. What they supply is a reading of the concept paper's language (human rights, lawful decision support, community observation, African ownership) as continuous with a civil-society tradition in which one stakeholder has long taken part. That continuity is worth stating, because it explains a cast of language in the African proposal that a technical reviewer might otherwise find puzzling.

1.6 Assessing the documents against one standard

Every external source in this study is graded on a seven-step scale (measured-with-controls; measured; modelled; projection; policy-fact; case-report; method-reference), set out in the Preface. Applying the same discipline to the stakeholders' own documents is a matter of consistency: a plan's targets are facts about the plan and never evidence of an effect, and a record of a discussion is evidence of an idea. Table 1.2 sets each document against six criteria.

Table 1.2: Assessment of the stakeholders' own documents. 'Attributed' means author and date stated in the document; 'Sourced' means the document references the evidence behind its own claims; 'Sensitive' flags content to be removed before wider circulation.

Document	Complete	Attributed	Sourced	Sensitive	Readable	Evidence grade
Dubai storyboard	Yes, as a plan	Author known; no date stated	No	No	Yes	Policy-fact: plan targets and budget, not evidence of effect
Discussion record	Raw, unstructured	Ambiguous date; no author line	No	Yes: telephone number, private letter, third-party names	Yes	Case-report: a record of a conversation, evidence of ideas only
ABIG concept paper and addendum	Yes; no budget	Yes: 20 September 2026, four authors	Partly: fourteen references, none verifiable from the paper	Partly: personal e-mail addresses in the title block	Yes	Method-reference: a design document with second-hand scale figures
Unity (2024) and *Building Unity* (2023)	Yes	Yes: contributors listed	Partly	No	Yes	Policy-fact: civil-society positions
Laws and Qualities of Nature (2023)	Yes	Yes on the cover	Not checkable	No	No machine-readable text	Not assessable
BE WELL overview (2026)	Overview only	Yes: author named	Partly	No	Yes	Case-report: a framework overview

Three judgements follow. The concept paper is the strongest document on every criterion but one: complete, dated, attributed and internally sourced, and honest about the limits of its own numbers; its weakness is the absence of any cost or benefit figure. The storyboard is complete and precise but undated and unsourced; it should carry a date and a version, and its numbers should be quoted only as plan targets. The discussion record is the richest source of ideas and the least fit for circulation; its ideas are already extracted here, and the record itself should not travel.

Figure 1.1 shows how the documents relate, and which derivations are still missing.

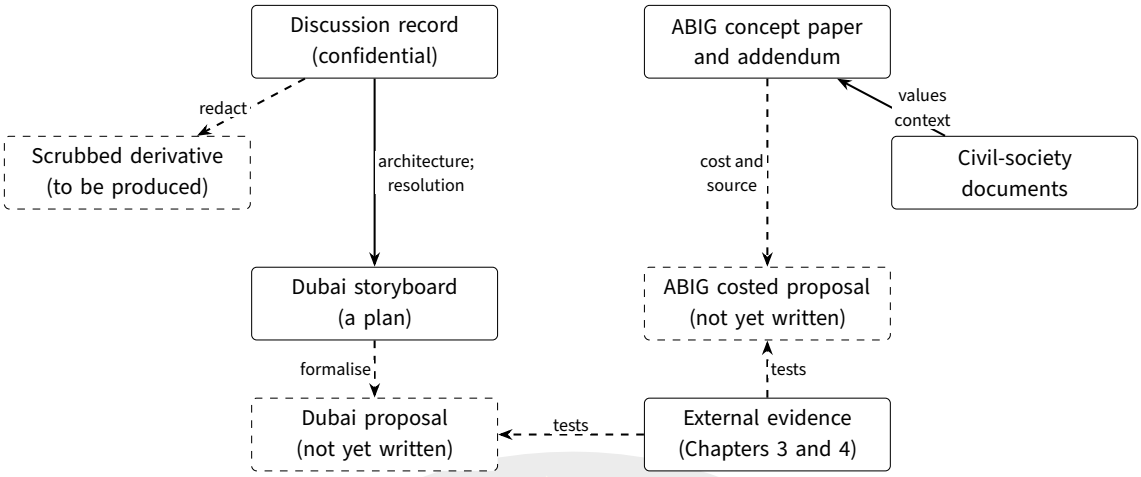


Figure 1.1: How the source documents relate. Solid arrows mark relationships that exist; dashed boxes and arrows mark the derivations this chapter recommends, none of which existed on 22 September 2026.

1.7 Implications for the programmes

The findings of this chapter are housekeeping findings, and housekeeping is where a feasibility exercise should begin. For the **Climate Intelligence & Decision Twin for Dubai**, four actions follow. *Redact*: produce a scrubbed derivative of the discussion record, dated unambiguously, free of the telephone number, the private letter and third-party names, and citable, and treat the original as confidential to the stakeholders. *Formalise*: turn the storyboard into the Letter of Intent and then the full proposal, adding the site map with marked sensor positions that the storyboard itself calls decisive, the references behind its claims, and the requirements, architecture and feasibility rationale that Chapters 5 to 10 provide. *Date and version*: record whether and when the September 2026 Letter of Intent was submitted, and keep the earlier submission, so that the difference between the original and the rescoped proposal is documented. *Register the funding-timing dependency*: the published material describes only the 2025 cycle *[Dubai Future Foundation, 2025]*[1] *[Dubai RDI Programme, 2025]*[2], and the March 2027 start presupposes a cycle whose calendar no document records.

For the **Africa Blue Intelligence Grid**, four actions follow. *Formalise*: the concept paper is finished but is not yet a proposal; preparing it as one makes it reviewable on the same terms as the Dubai submission, and requires the one thing the paper never states, the quantity each priority application would measure and the grade that measurement would carry. *Source*: the paper's fourteen references should become verifiable citations, beginning with the Luanda Declaration itself, which the paper quotes for its headline figures but whose own text was not available for review; only the African Union's press release of 24 July 2026 could be obtained *[African Union, 2026]*[3]. *Redact lightly*: remove the authors' personal e-mail addresses from any version that circulates beyond the stakeholders. *Cost*: the paper carries no budget, and the country-by-country capability database it calls for is the precondition of one; Chapters 9 and 10 therefore build the ABIG cost structure from analogy and parametric estimate, and Chapter 5 makes the addendum's alignment and mapping exercise the first deliverable, because it is what turns a concept paper into a costed plan.

For both programmes, the civil-society documents should be cited, if at all, as the values context they are. And for both, the discipline applied to external evidence in Chapters 3 and 4 (every source secured, dated, graded, and its limits stated) is the discipline the programmes' own documents should adopt. A programme whose first artefact is an evidence register is a programme that can

be audited, and that, more than any single number in the chapters that follow, is what a funder is buying.

٢ {2}

Purpose of the Papers

Abstract

What does each of the two proposals exist to do, and what purpose do they share? It reads the Dubai storyboard and the Africa Blue Intelligence Grid (ABIG) concept paper as design documents (evidence grade: method-reference) and states their purpose in their own terms: for Dubai, the problem, the idea, the three constructs, the testbed, the partners, the method by year, the month-12 gate, the team, the budget, the impact and the ask; for ABIG, the core proposition, the five domains, capability-first sequencing, the roadmap, governance and financing, and the addendum that places ABIG above GOOS-Africa, OPERA and GMES & Africa rather than beside them. It then extracts the unifying thesis from the discussion notes: one layered architecture in which data model, semantic model, knowledge graph, AI model, digital twin and decision system are six different things; a 30 m to 3 m resolution architecture; climate-resilient urban cells; an indoor extension; and an arid-world and space ambition. Readers should treat the chapter as the statement of intent against which every later chapter tests feasibility.

The purpose of a paper is what it commits its authors to do, for whom, by when and at what cost. Read that way, the documents of Chapter 1 are two papers that state a purpose. The first is the storyboard for the rescoped Letter of Intent, *Climate Intelligence & Decision Twin for Dubai*, addressed to the Dubai Future Foundation RDI Grant under its Cognitive Cities priority and asking for [1,400,000.00:388,888.89](AED:USD) over 36 months, 15 March 2027 to 14 March 2030 [Iyyunni, 2026]. The second is the ABIG strategic concept paper of 20 September 2026 by Dr Uchi Dominic Terhile, Dr Chakradhar Iyyunni, Abdullah Al Salmani and Arti Ahluwalia, with an addendum by Dr Terhile [Terhile et al., 2026]. A third document, the group discussion notes of 8 and 9 September 2026, states no purpose of its own but records the reasoning that links the two; it is cited through the working paper that carries it [Iyyunni, 2026]. Both papers are design documents (method-reference in the grading scheme of the Preface) and every number in them is a plan, a target, a rule or a date, not a measurement. Where a plan is compared with an archived study, the study carries its own grade; testing the plans is the work of Chapters 3 and 4.

2.1 The Dubai storyboard

2.1.1 The problem the project exists to answer

The storyboard opens with an economic claim, not a climatic one: Dubai's economy depends on people being able to walk, wait, visit and work outdoors for more of the year than heat currently allows [Iyyunni, 2026]. Climate data already exist at the National Center of Meteorology (NCM), the Dubai Electricity and Water Authority (DEWA) and Dubai Municipality, but in separate systems

that describe heat rather than what heat prevents. The unanswered question has three parts: which activities, routes and places are lost to heat, and for how many hours a year; what economic and social value attaches to restoring them; and which single intervention restores the most activity per dirham, litre of water and kilowatt-hour.

The framing matches metrics governments already accept. The International Labour Organization projects that the United Arab Emirates will lose more than 2 per cent of gross domestic product to heat stress at work by 2030 (grade: projection) [ILO, 2019], and the UAE midday break bans work under direct sunlight and in open spaces from 12:30 to 15:00 between 15 June and 15 September (grade: policy-fact) *[The National, 2026]*[1]. Both encode hours lost to heat at national scale; the storyboard proposes the same quantity for a place and an hour. Chapter 4 finds no archived study that reports it at that scale, and that gap is the project's claim.

2.1.2 The idea and its three constructs

The idea is to shift urban climate work from mapping exposure to optimising the activity that exposure constrains [Iyyunni, 2026]. Three constructs carry the shift, each stated here at three vocabulary tiers and as a model. The models are this report's formalisation of prose that gives no equations; the symbols are provisional until Chapter 7 fixes them.

Understanding: Outdoor Tourism Capacity (OTC)

TIER 1: IN PLAIN WORDS

How many visitors can comfortably use a place, for how long, at which hours, on a given kind of day.

TIER 2: IN PROFESSIONAL TERMS

A destination capacity metric combining physical carrying capacity with the share of each hour in which the outdoor thermal environment is acceptable to its users; hourly and spatially explicit, unlike the incumbent daily, destination-scale Holiday Climate Index [Scott et al., 2016].

TIER 3: IN TECHNICAL TERMS

A place-hour indicator on a physiological comfort index (this report recommends the Universal Thermal Climate Index for pedestrians, with wet-bulb globe temperature reported alongside for comparability [Jendritzky et al., 2012, ISO, 2017b]), with an explicit heat-stress override and cohort-specific thresholds. Note the collision: in the Singapore sources OTC means outdoor thermal comfort *[Centre for Liveable Cities, 2024]*[a].

AS A MATHEMATICAL MODEL

For place p, hour h and cohort c, with κ_p the physical capacity (persons), $U_{p,h}$ the comfort index and $\phi_c : U \mapsto [0,1]$ an acceptability function equal to 0 above the cohort's heat-stress limit, $\mathrm{OTC}_{p,c} = \sum_{h \in H} \kappa_p \, \phi_c(U_{p,h})$ visitor-hours; the loss metric is $\mathrm{HLAH}_{p,c} = \sum_{h \in H} \big(1 - \phi_c(U_{p,h})\big) A_{p,h,c}$, the Heat-Lost Activity Hours, with $A_{p,h,c}$ the activity that would otherwise take place.

[a] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

[1] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

Understanding: Minimum Cooling Network (MCN)

TIER 1: IN PLAIN WORDS

The smallest set of shade, trees, materials and rest points that makes the routes people most need comfortable enough to use.

TIER 2: IN PROFESSIONAL TERMS

An optimised portfolio of cooling interventions along priority routes, chosen so that a comfort target is met at least cost across capital, energy, water and maintenance, not a list of everything that could be built.

TIER 3: IN TECHNICAL TERMS

The output of a multi-objective optimiser whose decision variables are candidate interventions at candidate locations (shade canopy, tree planting, cool pavement, misting rest point, revised route) and whose objectives are comfort hours gained, capital cost, energy, water and maintenance [Iyyunni, 2026]. The archived evidence ranks shade first and makes cool pavement conditional, because a reflective surface raises the pedestrian's mean radiant temperature at midday [ASU and City of Phoenix, 2021].

AS A MATHEMATICAL MODEL

With binary decisions $x \in \{0,1\}^n$ over n (intervention, location) pairs, cost vector $C(x)$ over capital, energy, water and maintenance, weights w, and $\Delta\mathrm{RAH}_p(x)$ the Restored Activity Hours predicted at priority place p: $x^\star = \arg\min_x w^\top C(x)$ subject to $\Delta\mathrm{RAH}_p(x) \geq R_p^\star$ for all $p \in \mathcal{P}_{\text{priority}}$; the MCN is the support of $x^\star$.

1

Understanding: Decision Twin

TIER 1: IN PLAIN WORDS

A working model of the corridor on which the city can try a canopy, a row of trees or a new route and see the comfort gained and the water and power used, before anything is built.

TIER 2: IN PROFESSIONAL TERMS

A simulation environment that compares interventions before they are built and reports the comparison as a short brief; the storyboard insists it is 'not a 3D rendering' [Iyyunni, 2026]. It is distinct from the Dubai Digital Twin Platform launched on 2 July 2026, with three-dimensional models of more than 195,000 buildings and 280,000 infrastructure assets in more than 1,500 geospatial layers and rainfall as its named simulation (grade: policy-fact) *[Dubai Media Office, 2026]*[a]: the geometry on which the Decision Twin should sit, not a competitor.

TIER 3: IN TECHNICAL TERMS

A coupled district model at 25–50 m and street model at 1–5 m, driven by twenty microclimate nodes and one reference station recording air temperature, humidity, wind, radiation and surface temperature every ten minutes, with machine-learning downscaling and bias correction, validated against the sensors to a stated error and then used to predict the change a candidate intervention would make; the Year-3 test compares that prediction with the measured before/after change, with Al Fahidi as the untreated control.

AS A MATHEMATICAL MODEL

Let F_θ map corridor state S and intervention x to a predicted comfort field, and $\hat{\Delta}(x) = F_\theta(S,x) - F_\theta(S,0)$. Gate condition 6.2.2 requires $\mathrm{RMSE}\big(F_\theta(S,0), \text{sensors}\big) \leq \varepsilon_{\text{gate}}$ for air and surface temperature; the Year-3 proof requires $\|\hat{\Delta}(x^\star) - \Delta_{\text{measured}}\| \leq \varepsilon_{\text{build}}$ under a paired control.

2

[a]Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

The storyboard adds an institutional reason: Dubai is the right place because tourism, mobility, energy, water and heat adaptation are decided by a small number of agencies that can act on results quickly [Iyyunni, 2026]. Chapter 6 takes that sentence as its seed.

2.1.3 The testbed and the partners

The testbed is Al Fahidi Historical Neighbourhood and Al Seef on Dubai Creek, roughly one kilometre of walkable waterfront and lanes, in two zones [Iyyunni, 2026]. Zone 1, the heritage core of narrow lanes, coral-stone walls and wind towers, is already passively cooled and serves only as the calibration baseline. Zone 2, about 800 m of open road and pavement from Al Ghubaiba metro station to the heritage core, is the intervention site. One site with a map and marked sensor positions, the storyboard argues, beats a citywide promise; a citywide 100 m product is a Year-3 stretch goal. The arrangement is in effect the paired-control design that Paris OASIS instrumented for its schoolyards (Chapter 4).

Each partner is asked for one thing. The Roads and Transport Authority (RTA): Al Ghubaiba station exit counts plus bus and abra ridership. DEWA: the Bur Dubai district cooling load, hourly, for one summer. Dubai Culture and Dubai Municipality: written permission to install sensors and later to build one shade structure. Public baselines need no agreement: NCM station data, Copernicus land-surface and vegetation products, municipal GIS geometry. Curtin University Dubai is a collaborating entity with a signed budget share for Green AI, downscaling and optimisation. Site-access letters are due by December 2026 and data letters by February 2027.

2.1.4 Method by year and the month-12 gate

The method runs in three yearly loops. Year 1, *Observe and Model*: install the nodes and reference station; build the data platform (ingestion, quality checks, one spatial-temporal store, one API); capture a full summer of observations, June to September 2027; run the district and street models with named, established tools; validate against the sensors; produce the first hour-by-hour pedestrian comfort map of both zones and present it to RTA and DEWA. Year 2, *Predict and Optimise*: combine the comfort index with RTA flow data to compute dwell time, route abandonment and OTC; define the candidate interventions; build the optimiser that outputs the MCN; deliver Decision Twin version 1; add a second summer of data; select the single intervention to build, agreed with Dubai Culture and Dubai Municipality. Year 3, *Build and Prove*: construct it with a partner, most likely a shade canopy over the most exposed 200 m; measure before and after across one summer; compare the measured change with the Twin's prediction; repeat pedestrian counts and dwell-time measurement; deliver decision briefs, a transferable method document, publications and a follow-on proposal [Iyyunni, 2026].

Between the first and second loops stands the gate. At month twelve (March 2028) Phase 2 funds are released only if all four conditions hold: sensor network live with at least 90 per cent uptime across the summer; model within the stated error of the sensors; hour-by-hour comfort map of both zones delivered; and at least one of RTA or DEWA confirming in writing that the output is useful for a decision it faces. If the gate fails, the project closes with the dataset and platform as deliverables [Iyyunni, 2026]. Table 2.1 gives the milestone summary. Chapter 7 maps the gate onto the spiral model's Life-Cycle Architecture anchor point [Boehm and Hansen, 2000]; Chapter 6 argues that the fourth condition is the most important line in the proposal.

Table 2.1: Milestone summary of the Dubai storyboard, as stated in the rescoped Letter of Intent [Iyyunni, 2026]. Phase labels follow the two-phase award (Phase 1 = 12 months, Phase 2 = 24 months).

Date	Phase	Milestone
September 2026	Pre-award	Letter of Intent submitted
December 2026	Pre-award	Site-access letters from Dubai Culture and Dubai Municipality
February 2027	Pre-award	RTA and DEWA data letters; Curtin Dubai agreement; budget final
March 2027	Phase 1	Grant start (15 March 2027)
May 2027	Phase 1	Sensor network live (twenty nodes, one reference station)
September 2027	Phase 1	First summer dataset complete (June–September)
January 2028	Phase 1	Model validated against sensors
March 2028	Gate	Phase 1 gate: four stop-or-go conditions
November 2028	Phase 2	Minimum Cooling Network for the corridor
February 2029	Phase 2	Decision Twin version 1
May 2029	Phase 2	Intervention built (most likely a canopy over the most exposed 200 m)
September 2029	Phase 2	Before/after measurement complete; comparison with the Twin's prediction
February 2030	Phase 2	Decision briefs and transferable method document
March 2030	Phase 2	Grant end (14 March 2030)

2.1.5 Team, budget, impact and the ask

BITS Pilani Dubai leads, with a principal investigator for thermal and flow modelling and four co-investigators for mobility, built environment, edge sensing and systems integration; Curtin University Dubai holds its own budget line; a postdoctoral researcher and a doctoral student are funded for 36 months; a platform partner, DataAlt Dynamics, is either contracted or replaced by a dashboard built in-house; advisers are named, unpaid or on small honoraria [Iyyunni, 2026]. The budget is held near [1,400,000.00:388,888.89](AED:USD) (2026 prices): equipment about [250,000.00:69,444.44](AED:USD) for twenty nodes, a reference station and a field workstation; personnel; computing for two summers of street-scale simulation; travel; the Curtin share; ethics, data-governance and publication lines. The archived rules of the second RDI round allow [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD) per project, cap annual spend at [500,000.00:138,888.89](AED:USD), permit one to three years and require at least 50 per cent of the grant to be spent within Dubai *[Dubai RDI Programme, 2025]*[2]. The storyboard fits ([1,400,000.00:388,888.89](AED:USD) over three years averages [466,667.00:129,629.72](AED:USD) a year (derived)) but a Year-1 equipment purchase of [250,000.00:69,444.44](AED:USD) leaves at most [250,000.00:69,444.44](AED:USD) for all other Year-1 costs if the cap applies per project year (Chapter 10). The call names 'Climate-neutral and resilient urban systems' and 'Urban health intelligence for proactive well-being' as topics and 'Internet of Things (IoT), edge computing, and digital twins' as cross-cutting technologies *[Dubai RDI Programme, 2025]*[2]; the first cycle funded 24 of 374 proposals *[Dubai Future Foundation, 2025]*[3], about one in sixteen (derived).

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

[3] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

The impact section names four beneficiaries and a sequel: government compares shade, cooling and public-realm options digitally before capital is committed; tourism gains measured comfortable hours at heritage and waterfront destinations; mobility gains the repair of the worst first- and last-mile links; resources are protected by preferring interventions that add comfort without adding energy and water load; and after the grant come a second corridor, procurement by a Dubai agency and a method transferable to other hot cities [Iyyunni, 2026]. The ask is [1,400,000.00: 388,888.89](AED:USD) in two phases with a hard gate at month twelve, and approval to proceed to a full proposal on the rescoped basis of one site, named partners, named tools, funded people and one built intervention. One dependency lies outside the storyboard's control: the archived sources document only the 2025 second cycle (Letter of Intent deadline 13 June 2025; awards 10 September 2025) *[Dubai Future Foundation, 2025]*[3] *[Dubai RDI Programme, 2025]*[2], so a September 2026 submission presupposes a third cycle that appears in no archived document, a funding-timing risk for Chapter 11 and a dependency for Chapter 5.

2.2 The ABIG concept paper

2.2.1 Core proposition and the five domains

The core proposition is that Africa should treat its ocean, freshwater, land, atmosphere and coastal systems as a single information-and-capability system, using space, marine sensors, local science, artificial intelligence and institutional networks to convert observation into stewardship, maritime security, livelihoods and economic value [Terhile et al., 2026]. The purpose 'is not initially to build a new African satellite constellation' but to connect what Africa already has (the African Space Agency and national space institutions, GMES & Africa, universities, oceanographic institutions, coast guards and navies, fisheries authorities, ports, environmental agencies, companies and communities) into a repeatable continental capability.

The political anchor is the Luanda Declaration, adopted on 24 July 2026 at the Third Africa Blue Economy Week, which commits AU Member States to accelerate the Blue Economy Strategy, strengthen governance, mobilise blue finance, protect ecosystems, deepen regional cooperation and expand value chains, and highlights science, innovation and digital transformation *[African Union, 2026]*[4]. Quoting the Declaration, the paper records that 38 of the 55 Member States are coastal, that seven Large Marine Ecosystems border the continent, that inland waters cover about 305,000 km^2 and that transboundary basins cover almost 64 per cent of the land area [Terhile et al., 2026]; the 38 coastal states are independently given by the AU Blue Economy Strategy, which estimates the collective ocean territory at 13 million km^2 [AU-IBAR, 2019]. The Declaration itself is not archived, so the other three figures are cited as quoted.

Five domains are to be integrated: ocean and freshwater systems; environmental management and climate resilience; maritime and human-security intelligence, including lawful detection and referral of suspected human-trafficking patterns; blue-economy economic intelligence; and space, remote sensing and human-capability development. One principle governs the third: satellite analytics support lawful, accountable decisions and never replace them; the analytical flow ends in human review and lawful referral, not automated accusation [Terhile et al., 2026].

2.2.2 Capability-first sequencing and the roadmap

The sequencing is capability-first: define priority decisions; identify the information required; assemble existing data; produce useful knowledge products; test them with operational users; create

[4] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

common standards and training; only then invest in sensors, ground infrastructure, payloads and manufacturing where the evidence shows the need [Terhile et al., 2026]. The end state is judged not by satellites launched but by whether African institutions can repeatedly turn observations into better decisions. This report reads the rule as a value-of-information policy.

Understanding: Capability-first sequencing

TIER 1: IN PLAIN WORDS

Decide what you need to know before you decide what to buy; build the sensor only when the decision it serves has been shown to need it.

TIER 2: IN PROFESSIONAL TERMS

A demand-led investment rule: information requirements derive from named decisions and users, existing data are exhausted first, and hardware is commissioned only where a gap in a decision, not in the inventory, justifies it. Global Fishing Watch's work in Gabon, where an intelligence report from existing satellite data led to the seizure of three unlicensed trawlers on 18 July 2026 (grade: case-report), shows the pattern working without new hardware *[Global Fishing Watch, 2026]*[a].

TIER 3: IN TECHNICAL TERMS

A sequential decision problem whose state is a capability and coverage map, whose decisions are which pilot, fellowship or sensor to fund, and in which each observation investment is valued by the decisions it improves: an active-learning step in Powell's vocabulary [Powell, 2019], priced by the expected value of information of standard decision analysis [Howard, 1966].

AS A MATHEMATICAL MODEL

For decision x, uncertain outcome ω, utility $u(x, \omega)$ and candidate observation y (a sensor, a survey, a pilot), $\mathrm{EVI}(y) = \mathbb{E}_y[\max_x \mathbb{E}[u(x, \omega) \mid y]] - \max_x \mathbb{E}[u(x, \omega)]$; commission y only if $\mathrm{EVI}(y)$ exceeds its life-cycle cost. Hardware follows evidence because EVI is zero for an observation that would change no decision.

[a] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

The roadmap runs over five horizons (Table 2.2). The first twelve months are a discovery-to-demonstration phase across eight workstreams, targeting five to seven measurable use cases, three to five rapid studies, two to three prototypes from existing data, an Africa Blue Intelligence Baseline and an African Blue Intelligence Academy blueprint; a twelve-item ninety-day list opens with a small founding team and closes with a workshop with actual users [Terhile et al., 2026]. Ten priority applications are tabulated with their observations, product and user, from sustainable fisheries and illegal, unreported and unregulated fishing to tourism and human security.

2.2.3 Governance, financing and the UAE hub

Governance is light: seven layers, a political and strategic council, a scientific and technical council, an operational user council, a data and ethics board, a programme office, regional hubs, and an industry and investment forum [Terhile et al., 2026]. Financing is a six-rung ladder: policy mandate, project-preparation grant, pilot, independent validation, blended finance, commercial service; sources include the New Development Bank, UAE sovereign and institutional capital, African budgets, climate and biodiversity finance, private capital, philanthropy and BRICS co-financing. The paper lists eleven BRICS members and two existing mechanisms (the Virtual Constellation of Remote Sensing Satellites and the Working Group on Polar and Ocean Science and

Table 2.2: ABIG roadmap horizons as stated in the concept paper [Terhile et al., 2026]. Targets are the paper's own; no costs are attached to any horizon.

Horizon	Goals and outcomes	Key processes
0–3 months	Vision agreed; core coalition formed; baseline completed; priority use cases selected; governance and ethics principles adopted	Stakeholder interviews; capability and data inventory; use-case scoring; legal and privacy review; pilot-site selection; BRICS and UAE engagement brief
3–6 months	First prototypes live; human-capability network launched; initial knowledge products published; data-sharing arrangements under negotiation	Earth-observation and data pipelines; model development; analyst and user co-design; fellowships; hackathons; common metadata and provenance standards
6–12 months	Two to three operational pilots demonstrate measurable value; continental capability map and training catalogue established; investment-ready programme defined	Independent validation; user acceptance tests; benefits measurement; curriculum roll-out; project preparation; financing round tables
Year 2	Regional hubs operational; multiple countries use common products; joint research and satellite or ground-sensor missions defined; first scaled infrastructure projects financed	Regional centres of excellence; shared cloud and data infrastructure; joint missions; procurement standards; blended finance; operational service agreements
Year 3	Continental network with repeatable services; African researchers and engineers lead significant parts of the stack; measurable environmental, economic and security outcomes	Service-level agreements; recurring budgets; regional manufacturing and integration capability; advanced AI and physics models; cross-border operating protocols; independent impact evaluation

Technology) as technical entry points; the membership count rests on one web source and should be verified before it is repeated.

The UAE's role is not to own Africa's data but to provide a trusted platform, capital, partnerships and market access while African institutions keep ownership of priorities and datasets. Eight levers are named, from an Africa–UAE analytics gateway on the UAE Space Agency's Space Data Center to an annual Africa–BRICS–UAE Blue Intelligence Forum and the co-sponsorship of payloads only after application requirements exist [Terhile et al., 2026]. One lever is the bridge to the Dubai paper: 'Use UAE ports, logistics, energy, tourism and climate applications as demanding test environments for products intended for African markets.' Nine risks carry design responses; 'becoming another coordination forum' is answered by operational pilots with measurable user outcomes, and 'pilot fatigue' by the rule that every pilot needs a pathway to an operational budget, procurement or investment.

2.2.4 The addendum: an integration layer, not a parallel programme

The addendum repositions ABIG as complementary to GOOS-Africa, the OceanPrediction Decade Collaborative Centre's African Seas Regional Team, the Ocean Prediction Enhancement in Regions of Africa programme (OPERA) and GMES & Africa, not parallel to them [Terhile et al., 2026]. GOOS-Africa provides the observing community; OceanPrediction and OPERA strengthen forecasting and operational ocean services; GMES & Africa provides Earth-observation services and institutional capacity; ABIG connects these with space systems, data infrastructure, AI, environmental and economic intelligence and decision-making; African institutions remain owners and primary users. The five GOOS-Africa working groups (in-situ observation, remote sensing, modelling, fundraising, end-user relations) become explicit interfaces, and the cycle Observe, Integrate, Model, Predict, Understand, Decide, Act, Measure, Learn replaces an observation-centric chain.

Three commitments follow. Regionalisation follows African marine systems, eight hubs from the Gulf of Guinea to the Red Sea and the island systems. The data hub is federated: institutions keep ownership and control while enabling discovery, interoperability, controlled access, provenance and reproducibility. Co-design is formal and ordered (user decision, information requirement, observation requirement, modelling requirement, knowledge product, operational test, feedback, service improvement), so that a fisheries authority first defines its management decision and the observing requirements follow. The addendum's concrete change to the first twelve months is a GOOS-Africa, OceanPrediction and OPERA alignment and mapping exercise whose output, an Africa Ocean Observation, Prediction and Intelligence Capability Map, becomes a foundational dataset; Chapter 5 adopts that exercise as ABIG's first deliverable.

2.3 The unifying thesis

2.3.1 One layered architecture and six distinctions

The discussion notes state what the two papers share. The advice recorded there is to explain the Dubai system not as an AI model placed on a climate dataset but as a chain of layers: sensors, satellites, climate models, mobility and infrastructure feeding a data layer; a semantic layer that gives every concept (heat exposure, pedestrian, tourism destination, shade, road, building, energy demand, vulnerability, intervention) a consistent meaning and defines how the concepts relate; a knowledge graph or city knowledge model; AI reasoning and prediction; decision intelligence; a digital twin; and intervention, decision and learning [Iyyunni, 2026]. The chain of meaning the system must understand runs from heat through pedestrian exposure, tourism route, dwell time, attraction capacity, cooling intervention and energy and water requirement to investment and outcome. The ABIG paper draws the same chain for the ocean, from observation sources through

a knowledge graph or semantic layer and a digital-twin layer to sector applications and learning [Terhile et al., 2026]. Figure 2.1 draws the common architecture.

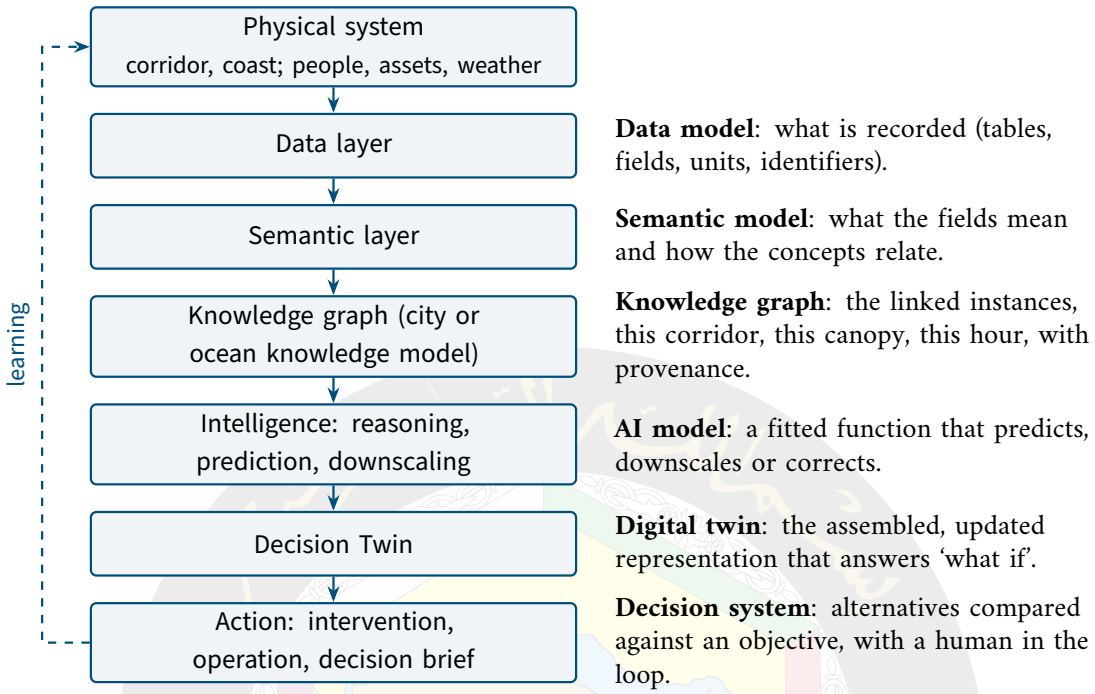


Figure 2.1: The layered architecture shared by the Dubai Decision Twin and ABIG, drawn from the discussion notes [Iyyunni, 2026] and the concept paper's process chains [Terhile et al., 2026]. The right-hand notes are the six distinctions the notes ask the proposals to keep. The dashed arrow is the learning loop: every decision produces evidence that improves the next model.

The six distinctions matter because each is a different deliverable with a different owner and failure mode. A data model can be complete while the semantic model is absent, which is the storyboard's own diagnosis of the present state; a knowledge graph can be populated while no AI model is validated; a model validated while no twin can be queried; a twin queried while no decision is taken. The month-12 gate tests one boundary, AI model against data, and its fourth condition tests the last: a decision maker states in writing that the output is useful. Chapter 8 gives each layer its identifiers; Chapter 5 makes the semantic layer the first thing to write.

2.3.2 From 30 m to 3 m

The notes' second argument concerns resolution. Landsat-8 provides multispectral data at 30 m while its thermal sensor is coarser, at 100 m [Manandhar et al., 2019]; a '30 m temperature map' is not what it seems, and the credible product is a fused, physics-constrained, AI-downscaled field at about 25–30 m [Iyyunni, 2026]. The precedent is the Abu Dhabi local climate zone study, which classified Abu Dhabi island from Landsat-8's eight 30 m bands, checked the zones against a 26-station urban network and a suburban reference station over four seasonal windows in 2017–18, and coupled them to an ENVI-met street model at 3 m per pixel on a 100 by 100 grid; a model calibrated on one district predicted another of the same class with a root-mean-square error of 0.24–0.68°C (grade: measured) [Manandhar et al., 2019]. A local climate zone definition for Dubai exists as a conference paper [Bande et al., 2020]; it is not archived and was not reviewed. The notes turn the precedent into a research question (how to infer a physically meaningful, temporally continuous 25–30 m climate field from observations whose native resolutions differ) and a two-level architecture: 30 m is city intelligence (satellites, stations, morphology, land cover, mobility, energy, water); 1–3 m is

human experience (UTCI, WBGT, mean radiant temperature, shade, wind, solar exposure, dwell time, routes).

The storyboard's figures are 25–50 m for the district model and 1–5 m for the street model; the notes recommend about 25–30 m Dubai-wide with hourly prediction and nowcasting, corridors refined to 5–10 m and 1–3 m in targeted spaces, and ask two questions: whether hourly output is enough, and whether 15-minute updates are possible. The ten-minute sensor cadence answers the second; the first is a scope choice the proposal must make explicit. Against continental twins the position is clear: Destination Earth's climate twin produces hourly output at 5–10 km and its extremes twin reaches 4.4 km globally and 500–750 m regionally (grade: method-reference) *[ECMWF, 2024]*[5]; the corridor twin begins two to three orders of magnitude below where they stop.

2.3.3 Climate-resilient urban cells and low-exposure mobility networks

The notes generalise the corridor into a spatial thesis: instead of one continuous fabric in which everyone is exposed to the outdoor climate, design interconnected climate-resilient urban cells joined by low-exposure mobility corridors, so that essential activities are reachable within a low-exposure network. The research question is whether urban areas can be organised into such cells, linked by optimised networks, to reduce exposure to extreme heat, humidity and air pollution while maintaining accessibility, tourism activity and productivity [Iyyunni, 2026]. The Al Ghubaiba corridor is one link; the MCN is the optimiser that would design the links; the Dubai Walk master plan's 3,300 km of new and 2,300 km of renovated walkways by 2040 (grade: policy-fact) is the demand at scale *[Dezeen, 2024]*[6]. Efficiency, the notes add, 'should be visible in the work, not stated'; the optimiser is where it becomes visible.

2.3.4 The indoor extension

A third extension moves from outdoor to whole-environment resilience: outside, building, occupant, health. The notes propose a layer called Building Climate and Infection Intelligence, in which outdoor temperature, humidity, wind, dust and radiation drive the building envelope, which drives the indoor environment (carbon dioxide, temperature, humidity, particulates, ventilation, filtration, operating mode), which combines with occupancy and movement to determine airborne exposure and healthcare demand. The question is framed not as detecting infection in buildings, which the notes call technically and ethically difficult, but as predicting the building conditions that raise the likelihood of airborne transmission and finding interventions that reduce exposure without unnecessarily increasing energy use; the objective is named Minimum-Energy Infection-Resilient Building Operation, a deliberate sibling of the MCN [Iyyunni, 2026]. The extension is unfunded and outside both the storyboard and the archived evidence; it is recorded because the semantic layer must carry indoor concepts from the start.

2.3.5 From the arid world to space

The widest statement of purpose is an investor letter reproduced in the notes under the theme 'Building human resilience: from extreme Earth environments to space' [Iyyunni, 2026]. Space is the observation backbone and all-weather tourism one spectacular first application, but the larger idea is a country's capability to thrive in extreme climates and to export it; the letter asks whether technologies developed for resilience in the UAE's environments can become part of the knowledge architecture for resilience beyond Earth, and proposes challenges and hackathons

[5] European Centre for Medium-Range Weather Forecasts, *Fact sheet: ECMWF and the EU's Destination Earth initiative*, ECMWF media centre, 2024. https://www.ecmwf.int/en/about/media-centre/focus/2024/fact-sheet-ecmwf-and-eus-destination-earth-initiative

[6] Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

linked to space-technology incubators. The same conversation records the convening stakeholder's insistence that the solution must help people in any arid environment, that Dubai's ability to fund the exercise is 'the least of my concerns', and that measurement alone will not suffice. Two threads belong to purpose rather than method. The first is heritage: the Gulf cities are sites of cultural practice whose Bedouin repertoire for heat, scarcity, mobility, seasonality and community predates air-conditioning and satellites, which raises the question of what technology can learn from cultures that evolved in extreme environments, with a caution not to romanticise. The second is process: Chapter 12 adopts *nemawashi*, the preparation of the ground before a formal proposal. The report records these ambitions as ambitions; none is funded, costed or evidenced in the archive, and Dubai is, in the notes' own words, the first application, not the goal.

2.4 What the papers do not yet say

Stating a purpose precisely shows where it is not yet precise. Six omissions recur in later chapters.

Gaps in the stated purpose

1. **No comfort index is named.** The storyboard promises a 'comfort map' but not the index; the ILO and Lancet labour series use WBGT [ILO, 2019, Romanello et al., 2024], Singapore's advisory uses WBGT bands [NEA Singapore, 2023], the pedestrian literature uses UTCI [Jendritzky et al., 2012]. The node specification lacks the black-globe and natural wet-bulb instruments that Singapore's published WBGT formula requires *[MSE Singapore, 2023]*[a]: an item for the [250,000.00:69,444.44](AED:USD) line.
2. **The gate's error threshold is unstated.** 'Within the stated error' is a placeholder; the Abu Dhabi precedent reached 0.27–1.62°C after bias correction [Manandhar et al., 2019]; Chapter 4 reads that as the order of magnitude and Chapter 5 proposes an indicative threshold on that basis.
3. **OTC has no formula and no valuation.** The models in this chapter are the report's; no source values a visitor-hour.
4. **ABIG has no budget, no cost per decision and no cost breakdown per hub**; its outcome indicators are qualitative except 'hundreds of trained practitioners' [Terhile et al., 2026]. Chapters 9 and 10 must build the cost structure by analogy.
5. **The Dubai timeline rests on an undocumented third RDI cycle** *[Dubai Future Foundation, 2025]*[3] *[Dubai RDI Programme, 2025]*[2].
6. **Sustainment after March 2030 is unassigned.** Procurement by a Dubai agency is the stated sequel, but no agency is committed; in the WinWin case study only the application with a sustaining budget survived [Boehm et al., 1998], so Chapter 7 makes a written sustainment commitment a condition of the Year-2 intervention choice.

[a] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

2.5 Implications for the programmes

Dubai. The purpose is narrow on paper and wide in intent, and should stay that way. Narrow: one corridor, twenty nodes, one summer, one comfort map, one written confirmation, one

built intervention, one before/after comparison, [1,400,000.00:388,888.89](AED:USD). Wide: OTC generalises to any destination, MCN to any route network and the Decision Twin to any cell of the climate-resilient city, and the layered architecture lets indoor, arid-world and off-Earth extensions attach to one semantic layer. Four consequences: fix the comfort index and the gate threshold before the December 2026 site letters (Chapter 5); put black-globe instruments in the equipment list; phase Year-1 spend against the [500,000.00:138,888.89](AED:USD) annual cap; and treat the fourth gate condition as the purpose test: the project succeeds when an agency uses the output, not merely when the model matches the sensors.

ABIG. The purpose is correctly bounded by the addendum, an integration layer above services that other programmes own, and the danger is drift back towards a parallel programme or a coordination forum. Three consequences: the alignment and mapping exercise, not a prototype, is the first deliverable, because it fixes the boundary in writing; every early pilot needs a named operational user and a written sustainment pathway, the paper's own answer to pilot fatigue; and capability-first should be stated as the value-of-information rule above, so that every hardware proposal in Years 2 and 3 carries the decision it serves and the evidence that the decision needs it.

Jointly. The concept paper's sentence (use UAE climate applications as test environments for products intended for African markets) makes the Dubai corridor the first proof of the architecture ABIG would run at continental scale. Chapter 3 asks whether anyone has built either; Chapter 4 whether the evidence supports the constructs; the remaining chapters whether the purpose can be started, bounded, identified, costed, insured and carried into policy.

۳ {3}

Similar Efforts Around the World

Abstract

Comparable work is under way in several countries, but what do those efforts actually show? It inventories twenty-three programmes, studies and instruments in six groups (urban heat and decision twins; heat economics and policy; tourism-climate indices; ocean and blue intelligence; finance; and global policy instruments) and for each records who ran it, when, who paid, at what scale, by what method, with what measured or projected result, at what evidence grade, whether it is still running, and what it means for the Climate Intelligence & Decision Twin for Dubai or for the Africa Blue Intelligence Grid (ABIG). Every figure is taken from an archived source or labelled as this report's own derivation; documents that could not be archived are named by title and marked as not reviewed. The chapter closes with a comparison matrix, a landscape figure and resolution ladder, and implications for the two programmes. Its central finding is that the landscape is crowded at planetary and city scale and empty at corridor scale: no effort yet ties activity-hours lost to heat, measured hour by hour along a walking route, to a built intervention validated against a twin's prediction.

The question 'has anyone done this before?' is the first that a funder asks and the one that a proposal most often answers loosely. A Letter of Intent to the Dubai Future Foundation must show that the Decision Twin is neither a duplicate of the Dubai Digital Twin Platform launched on 2 July 2026 nor a smaller copy of Singapore's island-wide climate twin; the ABIG concept paper must show that it adds an integration layer above GMES & Africa, GOOS-Africa and OPERA rather than another coordination forum [Terhile et al., 2026]. Both cases rest on a precise reading of what the existing efforts measured, modelled or merely announced. This chapter therefore treats each effort as a line item with the same eight fields (who, when, funder, scale, method, results and their evidence grade, status, and relevance), so that the two programmes can be positioned against the landscape rather than beside it. The material comes from the archived sources listed in Chapter 1 and Appendix A; the grading scheme is the one set out in the Preface.

3.1 How the line items are graded

The efforts described here range from a peer-reviewed field study with paired reference measurements to a press release announcing a pledge. Placing them in one table without saying which is which would mislead, so each item carries one of seven evidence grades: *measured-with-controls*, *measured*, *modelled*, *projection*, *policy-fact*, *case-report* and *method-reference*. A programme may carry more than one grade where its documents differ in kind; the Phoenix cool-pavement reports, for instance, are measured-with-controls for their temperature traverses and modelled for their extrapolated energy savings.

Understanding: Evidence grade

TIER 1: IN PLAIN WORDS

A grade says how much a number can be trusted and why. A measured number was read off an instrument; a modelled number came out of a computer; a projection is a forecast built on assumptions; a policy-fact is a date, rule or target an organisation has announced; a case-report is one story told by the people involved; a method-reference explains how something is done without giving results.

TIER 2: IN PROFESSIONAL TERMS

The grade records the design behind a figure: whether a comparison group or reference station existed (measured-with-controls), whether the quantity was observed at all (measured), simulated under stated inputs (modelled), extrapolated to a future year (projection), asserted as an institutional fact (policy-fact), narrated without a counterfactual (case-report), or specified as a procedure (method-reference).

TIER 3: IN TECHNICAL TERMS

Grades are assigned per document and per quantity, not per programme. A secondary citation inherits the grade of its primary source; a self-reported programme output without baseline or attribution method is a case-report however large the number; a scenario result is modelled even when quoted as an observation; a target is a policy-fact even when expressed in degrees.

AS A MATHEMATICAL MODEL

Ordering by inferential strength, measured-with-controls $\succ$ measured $\succ$ modelled $\succ$ projection; policy-fact, case-report and method-reference are categories rather than ranks, because they answer different questions.

Two conventions apply throughout. A figure that appears in the agreed baseline analysis but in no archived text has been dropped or is labelled as unverified press reporting. A document that could not be archived is cited by title only and flagged as not reviewed: the 2026 journal paper on Singapore's coupled climate and anthropogenic-heat models in a digital-twin framework [Aydt et al., 2026], the 2023 *Climate Services* article on Destination Earth [Hoffmann et al., 2023], the Dubai local-climate-zone conference paper [Bande et al., 2020], the GMES & Africa programme page *[GMES and Africa, 2025]*[1] and MarCNoWA news item *[GMES and Africa, 2023]*[2], the Great Blue Wall Impact Report itself [Great Blue Wall Initiative, 2026], the *Marine Policy* critique of the Seychelles blue bond [Kilic, 2024], and the 2018 World Bank press release on that bond *[World Bank, 2018]*[3], whose capture returned no content.

Understanding: Digital twin and decision twin

TIER 1: IN PLAIN WORDS

A digital twin is a detailed, continually updated computer copy of a place. A decision twin uses such a copy to try out choices before they are made (a canopy here, trees there, a different route) and to say which is better and by how much.

[1] GMES & Africa, *About Us – Global Monitoring for Environment and Security & Africa*, programme site, 2025. https://gmes.rmc.africa/about-us

[2] GMES & Africa, *GMES Phase 2 launched MarCNoWA project covering 18 countries in Africa*, news, 2023. https://gmes.rmc.africa/news/gmes-phase-2-launched-marcnowa-project-covering-18-countries-in-africa

[3] World Bank, *Seychelles launches World's First Sovereign Blue Bond*, press release, 29 October 2018. https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond

TIER 2: IN PROFESSIONAL TERMS

A digital twin is a data-integration platform that mirrors assets and their state (geometry, layers, live feeds). A decision twin adds validated process models, an explicit objective (comfort hours gained per dirham, litre and kilowatt-hour) and a comparison of intervention scenarios against a measured baseline with a control site.

TIER 3: IN TECHNICAL TERMS

The distinction is between representation and optimisation: the platform supplies state variables and boundary conditions; the decision twin supplies a transition model validated against sensors to a stated error, a policy that maps state to intervention, and a before/after evaluation with a paired control that tests the prediction. The storyboard's phrase 'not a 3D rendering' marks exactly this boundary [Iyyunni, 2026].

AS A MATHEMATICAL MODEL

$x^* = \arg\max_{x \in \mathcal{X}} \mathbb{E}\big[V(x, W) \big]$ subject to $|\hat{T}_{c,h} - T^{\text{obs}}_{c,h}| \le \varepsilon$ on the validation set, where x is an intervention set, W the weather and demand uncertainty, V the value of restored activity-hours net of life-cycle cost, and ε the gate error.

3.2 Urban heat and decision twins

3.2.1 Cooling Singapore and the Digital Urban Climate Twin

Who, when, funder, scale. Cooling Singapore was launched in 2017 with support from the National Research Foundation under the Prime Minister's Office; the Centre for Liveable Cities names the participants as 'including' the Singapore-ETH Centre, the Singapore-MIT Alliance for Research and Technology, TUMCREATE, Singapore Management University, the National University of Singapore and Cambridge CARES *[Centre for Liveable Cities, 2024]*[4]. Cooling Singapore 2.0 is led by the Singapore-ETH Centre with SMU, NUS and SMART, funded through NRF CREATE, and its flagship is the Digital Urban Climate Twin (DUCT), built by more than 20 researchers over four years to June 2024 and run on the National Supercomputing Centre's ASPIRE 2A, which completes up to 80 hours of climate-model simulation within 24 hours [NSCC Singapore, 2024, Cooling Singapore 2.0, 2024].

Method. The DUCT couples island-wide meso-scale and neighbourhood micro-scale climate models driven by land use, anthropogenic heat profiles, meteorological data and urban geometry, and exposes what-if scenarios (vegetation, wind potential, heat-wave mitigation, electric-vehicle share, building efficiency, district cooling, parks) through the browser-based DUCT Explorer, 'being developed in close collaboration with Singapore government agencies' [Cooling Singapore 2.0, 2024]*[Centre for Liveable Cities, 2024]*[4].

Results and grade. The urban heat island is 'up to 7°C'; the poster attributes the map to a 2021 project presentation and labels it 2016, the Centre for Liveable Cities says 'observed', and neither gives a method or period [Cooling Singapore 2.0, 2024]*[Centre for Liveable Cities, 2024]*[4]. The Centre reports projections of 1.4–4.6°C additional warming by 2100 without a stated scenario *[Centre for Liveable Cities, 2024]*[4]. The poster's own result is a modelled whole-island cooling of up to 0.13°C in average 2 m air temperature on one heat-wave day under an approximate Green Plan 2030 scenario, peaking at 06:00–08:00, with the caution that 'effectiveness of different strategies vary by district' [Cooling Singapore 2.0, 2024]. Grades: *modelled* for the scenario; *case-report* for the programme

[4]Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

descriptions. No validation error, sensor specification or cost figure exists in the archived texts. In these sources OTC means Outdoor Thermal Comfort; in this report it means Outdoor Tourism Capacity.

Status and relevance. The Explorer was still in development in the archived texts, and the 2026 paper was not reviewed [Aydt et al., 2026]. The lesson for Dubai is scope: a city-scale twin took a multi-institute team four years and a national supercomputer, which argues for a 36-month, one-corridor, one-intervention proposal; the district-sensitivity finding justifies a corridor with a paired control rather than a city-mean target, and warns that the citywide 100 m product, the stretch goal, must be reported per district or it will reproduce the 0.13°C problem, an island average that hides the local effect.

3.2.2 Destination Earth

Destination Earth (DestinE) is the European Union's programme, under DG CNECT, to build digital twins of the Earth system by 2030; it started in December 2021, entered its second phase in June 2024 (which 'ramps up operations'), and works through three entrusted entities (ECMWF for the twins and the Digital Twin Engine, ESA for the platform, EUMETSAT for the data lake) with 90 partners *[ECMWF, 2024]*[5]. The Weather-Induced Extremes twin runs globally at 4.4 km for four-day simulations and, in a Météo-France-led regional component over Europe, at 500–750 m for two days ahead; the Climate Change Adaptation twin produces multi-decadal projections at 5–10 km with hourly output from ICON, IFS with NEMO and IFS with FESOM in a twelve-institution partnership led by CSC Finland, on the EuroHPC machines LUMI, Leonardo, MareNostrum5 and MeluXina *[ECMWF, 2024]*[5]. Grade: *method-reference*; no skill scores, users, budget or access terms for non-EU users are stated. Its relevance is the resolution ladder of Figure 3.2 (DestinE stops at 500 m regionally and hourly at 5–10 km for adaptation, where the Dubai twin proposes 25–50 m for the district, 1–5 m for the street and ten-minute cadence [Iyyunni, 2026]) and, for ABIG, the entrusted-entity governance that separates policy owner, modelling body, platform and data federation.

3.2.3 Dubai Digital Twin Platform

Dubai Municipality launched the Dubai Digital Twin Platform on 2 July 2026 at the 'Partners in Building the Digital Twin' workshop attended by Sheikh Hamdan bin Mohammed. Its third phase delivered 3D models of more than 195,000 buildings, more than 280,000 infrastructure assets and more than 330,000 public facilities and assets; the platform carries more than 1,500 geospatial layers and more than 100 applications, its named uses are urban planning, infrastructure and asset management, rainfall simulation and decision support, and memoranda of understanding with Al-Futtaim Group and Huawei cover the next phase *[Dubai Media Office, 2026]*[6]. Grade: *policy-fact*. Nothing is said about accuracy, update cadence, licensing or third-party access, and no microclimate, shade or comfort layer is named. This is the decisive positioning fact: the municipal platform is a geometry-and-data twin, and the Decision Twin's contribution is the hour-by-hour comfort layer at 1–5 m for one corridor that the platform lacks. Re-anchoring the twin on the platform converts a platform-build risk into a data request; because access terms are undocumented,

[5] European Centre for Medium-Range Weather Forecasts, *Fact sheet: ECMWF and the EU's Destination Earth initiative*, ECMWF media centre, 2024. https://www.ecmwf.int/en/about/media-centre/focus/2024/fact-sheet-ecmwf-and-eus-destination-earth-initiative

[6] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

the December 2026 site letters should include a written data-access confirmation from Dubai Municipality.

3.2.4 Dubai Walk Master Plan, the Future Loop and The Loop

The Dubai Walk Master Plan was launched on 7 December 2024 by Sheikh Mohammed bin Rashid Al Maktoum *[Khaleej Times, 2025]*[7] *[Dezeen, 2024]*[8]. It comprises 3,300 km of new and 2,300 km of rehabilitated walkways by 2040, more than 900 km beyond 2040, and 110 bridges and underpasses *[Dezeen, 2024]*[8]; the 6,500 km headline is thus a total including the post-2040 tranche (derived: 3,300 + 2,300 = 5,600 km by 2040; 5,600 + 900 = 6,500 km). Khaleej Times adds 160 areas and a target to raise pedestrian and soft mobility from 13 to 25 per cent by 2040, under the Quality of Life Strategy 2033 and the 20-minute city goal of 80 per cent of residents reaching essential services within a 20-minute commute *[Khaleej Times, 2025]*[7] *[Dezeen, 2024]*[8]. Its flagship, the Future Loop, is a 2 km elevated, climate-controlled walkway at the Museum of the Future linking Dubai World Trade Centre, Emirates Towers, DIFC and nearby metro stations, with a 30,000 m^2 air-conditioned level and 30,000 m^2 of shaded open space, ten key locations, a width of 6–15 m, commercial space and a public–private delivery model; it was among the projects reviewed by Sheikh Hamdan in November 2025 *[Khaleej Times, 2025]*[7] *[Dezeen, 2024]*[8]. No archived source names the Roads and Transport Authority as owner or gives a cost, energy or comfort figure. Grade: *policy-fact*. A different project, The Loop, is URB's private concept unveiled in early 2023 for a 93 km climate-controlled corridor connecting more than 3 million residents, with URB's own aim that more than 80 per cent of residents cycle or walk daily by 2040 within a 1,000 km Dubai Cycle City 2040 vision; it is unfunded and unapproved *[URB, 2025]*[9]. Grade: *projection*. URB's 80 per cent modal-share aim is not the government's 80 per cent access goal.

The three form a cost-coverage spectrum: 93 km fully conditioned (concept), 2 km conditioned and shaded (approved), and the Minimum Cooling Network (MCN), the smallest set of shade, tree, material and rest-point interventions that makes a priority route comfortable [Iyyunni, 2026]. Dubai Walk commits to 5,600 km by 2040 without a published method for deciding which kilometres need shade, where and at what hour; the MCN is that method, and because conditioned outdoor space carries an energy cost the storyboard's objective already prices, the twin can report comfort hours gained per kilowatt-hour for shade-only versus conditioned options. The Al Fahidi–Al Ghubaiba testbed does not overlap with the Future Loop, which avoids confounding and adds a heritage, low-rise counterpart to a high-rise loop.

3.2.5 Green Riyadh

Green Riyadh was launched on 19 March 2019 by King Salman as one of four major Riyadh projects; the archived description is the Saudipedia article, which names the Royal Commission for Riyadh City and the Saudi Press Agency as its sources *[Saudipedia, 2025]*[10], the Commission's own page being cited only for the programme's existence *[RCRC, 2025]*[11]. Targets: more than 7.5 million trees by 2030; per-capita green space from 1.7 to 28 (unit printed 'm', evidently square metres),

[7] Khaleej Times, *Walk in the sky? How Dubai's Future Loop will change commuting*, 2025. https://www.khaleejtimes.com/uae/transport/dubai-future-loop-project-connect-10-landmarks-ease-traffic

[8] Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

[9] URB, *The Loop Dubai: 93 km Climate-Controlled Urban Corridor*, project page, 2025. https://urb.ae/projects/theloop/

[10] Saudipedia, *Green Riyadh Project*, encyclopaedia article (sources: Royal Commission for Riyadh City; Saudi Press Agency), 2025. https://saudipedia.com/en/green-riyadh-project

[11] Royal Commission for Riyadh City, *Green Riyadh Program*, project page, 2025. https://www.rcrc.gov.sa/en/projects/green-riyadh-project/

'sixteen times'; green share of the city from 1.5 to 9 per cent; 16,400 km of streets planted; 43 parks; 3,330 neighbourhood gardens; 1,100 km of green belts; 1,350 km of irrigation networks fed by a 100 per cent increase in treated-wastewater reuse; 72 species; shaded sidewalks *[Saudipedia, 2025]*[10]. Claimed benefits: 1.5–2°C cooler in the city and 8–15°C 'in intensively afforested locations' (air or surface unspecified); 3–6 per cent less carbon dioxide; and 650 GWh of power saved, which the source attributes to green-building techniques, not trees *[Saudipedia, 2025]*[10]. Grade: *policy-fact* for the targets, *projection* for every benefit; no cost, monitoring protocol or progress figure is archived. It is the GCC's largest live example of the tree-planting intervention class, and its temperature claims are precisely the hypotheses a corridor twin with a paired control can test; its 1,350 km irrigation network is the clearest evidence that water is the binding resource for greening in the Gulf, which is why litres per restored activity-hour belong in the optimiser's objective.

3.2.6 Makkah and Arafat cooled roads and Hajj cooling measures

Who, when, scale. The Saudi Roads General Authority (RGA) first trialled a locally developed road-cooling surface in the Jamarat corridors at Hajj 2023, extended it for Hajj 2024 to Mina, Arafat and Muzdalifah including about 25,000 m^2 near Namira Mosque, and paved more than 84,000 m^2 across the Arafat plains for Hajj 2025, an 82 per cent year-on-year expansion; a separate rubberised 'Flexible Rubber Roads' initiative began in 2024 and grew by 33 per cent in 2025 *[Arab News, 2024]*[12] *[Gulf News, 2025]*[13]. The material was developed with the Ministry of Hajj and Umrah, the Ministry of Municipal and Rural Affairs and Housing, the Pilgrim Experience Program and the Quality of Life Program and is 'made in the Kingdom' under the RGA research centre *[Arab News, 2024]*[12]. The Saudi Gazette adds, for Hajj 2024, misting columns and spray nozzles on 'mist-covered walkways', air-conditioned tents, 'over 600,000' Neem trees on the Arafat plain 'according to some sources', about 1.8 million pilgrims, temperatures up to 48°C, and annual measurement of temperature, wind and humidity inside and outside the tents by the National Center for Meteorology *[Saudi Gazette, 2024]*[14].

Results, grade, status. The agency claims a surface-temperature reduction of 12–15°C (2024) or 'around 12°C' (2025), 30–40 per cent more reflected sunlight 'in the mornings', and untreated surfaces retaining up to 70°C after sunset *[Arab News, 2024]*[12] *[Gulf News, 2025]*[13]; the Saudi Gazette's 'about 20°C' for the white coating does not match the agency's range and is treated as unverified press reporting, as is its relay of a hospital analysis claiming a 74.6 per cent fall in heat-stroke cases over forty years *[Saudi Gazette, 2024]*[14]. Grade: *case-report*: no instrument, site, period or control is published, and no air-temperature, mean radiant temperature, WBGT, UTCI, comfort, cost or durability figure exists. The programme is operational and expanding; Gulf News notes that Hajj 2025 was the last in peak summer for about sixteen years as the lunar calendar moves the pilgrimage into spring and winter *[Gulf News, 2025]*[13].

Relevance. Makkah is the nearest-climate precedent for the cool pavement candidate and supplies a GCC prior for the street model's surface layer, but because nothing pedestrian-level was published it cannot show that reflective pavement improves comfort; read with Phoenix it supports ranking shade first. It reports outputs (square metres, per cent expansion) rather than outcomes and

[12] Arab News, *Road-cooling tech will provide Hajj pilgrims with respite from heat*, Saudi Arabia section, 2024. https://www.arabnews.com/node/2530731/saudi-arabia

[13] Gulf News, *Hajj 2025: Saudi Arabia expands cooled, rubberised roads*. https://gulfnews.com/world/gulf/saudi/hajj-2025-saudi-arabia-expands-cooled-rubberised-roads-1.500139040

[14] Saudi Gazette, *Hajj 2024: Saudi Arabia implements advanced cooling techniques in holy sites amid scorching heat*. https://saudigazette.com.sa/article/643620/SAUDI-ARABIA/Hajj-2024-Saudi-Arabia-implements-advanced-cooling-techniques-in-holy-sites-amid-scorching-heat

expanded after an internal judgement that the 2023 trial 'was a success' with no published criterion: the pattern the Dubai gate avoids. That more than half of the 2025 pilgrims were expected to be elderly *[Gulf News, 2025]*[13] is a reminder that Outdoor Tourism Capacity must be cohort-specific, and as the largest recurring heat-management operation leaves the summer window, Dubai's fixed tourism season becomes the natural place to validate a year-round summer-comfort metric.

3.2.7 The Abu Dhabi local climate zone study and the Dubai paper

Who, when, funder, method. Manandhar, Bande, Tsoupos, Marpu and Armstrong of UAE University's National Space Science and Technology Center and Khalifa University published *A Study of Local Climate Zones in Abu Dhabi with Urban Weather Stations and Numerical Simulations* on 24 December 2019 in *Sustainability*, with no external funding and with Abu Dhabi Municipality approval for lamp-post-mounted stations [Manandhar et al., 2019]. Twenty-six stations recorded air temperature at 3–8.5 m, humidity, wind, surface and façade temperature and irradiance at one-minute resolution against a suburban reference at the Masdar Institute Field Station over four two-week windows from June 2017 to March 2018 (one week to calibrate, one to validate). Landsat-8 imagery at 30 m (thermal 100 m) was classified into nine Abu Dhabi-specific local climate zone classes (five built, four land cover), whose numbering does not follow the seventeen-class standard of Stewart and Oke [Stewart and Oke, 2012]; ENVI-met 4.0 street models at 3 m per pixel were bias-corrected with a quadratic fit and validated on the held-out week [Manandhar et al., 2019].

Results and grade. Measured daily-mean urban-minus-rural air temperature ranged from −0.23 to +3.21°C across five districts and four seasons: September −0.23 to +0.29, June +0.06 to +0.68, December +2.44 to +3.21, March +1.25 to +1.85; the December five-district mean is $(2.80+2.57+3.21+2.44+2.56)/5 = 2.72$°C (derived). The largest daily-mean island is thus in the cool season and near zero in summer: an artefact of daily averaging that hides the daytime inversion the paper reports, not evidence of no summer problem. Uncorrected ENVI-met error was 1.33–2.71°C RMS, 0.27–1.62°C after correction, and a model calibrated on one district predicted another of the same class within 0.24–0.68°C [Manandhar et al., 2019]. Grade: *measured*, a reference station and a hold-out week, but no intervention tested and no uncertainty reported. Only air temperature was analysed. The Dubai counterpart, *Local Climate Zones Definition in Relation to ENVI-met in the City of Dubai, UAE*, was not archived and was not reviewed [Bande et al., 2020].

Relevance. This is the UAE precedent for the nested-resolution design (Landsat at 30 m for zoning, a 3 m street model for the canyon) and a sensor-network template: municipal permission, a reference outside the urban fabric, siting by prior simulation. It gives the month-12 gate its only UAE benchmark, roughly 1°C error for a bias-corrected street model judged on corrected output, and its class-transfer result is the mechanism by which a corridor twin could later extend citywide without instrumenting every district.

3.2.8 Phoenix cool pavement, Year 1 and Phase II

Who, when, funder, method. The City of Phoenix coated 36 miles of residential streets in eight neighbourhoods with the reflective product CoolSeal in 2020 and commissioned Arizona State University to evaluate it. The Year-1 report (15 July 2020 to 14 July 2021), funded by the City and ASU's Healthy Urban Environments programme, ran three one-day campaigns in August–September 2020 with a six-directional radiometer cart on street and sidewalk and a vehicle with 2 m thermocouples and an infrared radiometer, looping through each treated neighbourhood and an adjacent untreated one from pre-sunrise to after sunset, with monthly reflectivity readings against a control road and paired subsurface sensors [ASU and City of Phoenix, 2021]. The

Phase II report (2022 data, October 2024) added vertical air-temperature arrays, a same-day radiant comparison with a stationary reference cart (adopted after an intended before/after design was abandoned because the two days' weather differed), a test bed against a dark sealant, building-energy modelling and laboratory tests [ASU and City of Phoenix, 2024].

Results and grade. Year 1: surface temperature 12.0°F lower at noon and 10.5°F lower in the afternoon than adjacent aged asphalt (range 9–16°F; derived, about 6.7 and 5.8°C), 2.4°F lower at sunrise; 2 m air temperature about 0.5°F lower in the evening and 0.3°F by day; road-centre mean radiant temperature about 5.5°F higher at noon and afternoon (range 2.6–9.2°F; derived, about 3.1°C), while sidewalk differences of −5.2 to +1.9°F show no consistent penalty beside the coated road; subsurface 4.8°F cooler; solar reflectivity 33–38 per cent at installation falling to 19–30 per cent within ten months, against 12 per cent for asphalt [ASU and City of Phoenix, 2021]. Phase II: road-centre mean radiant temperature +5.8°F at noon and +4.5°F in the afternoon, 'not representative of a pedestrian on an adjacent sidewalk', and a standing person needing 12–13 W m^{-2} more heat loss [ASU and City of Phoenix, 2024]. Its city-wide water and energy extrapolations rest on a single day with 'a high level of uncertainty' and are not transferred to Dubai. Neither report gives a service life, re-application interval or unit cost; Year 1 defers life-cycle costing [ASU and City of Phoenix, 2021]. Grade: *measured-with-controls* for the field temperatures; *modelled* for the energy section.

Status and relevance. The programme continued across seven further neighbourhoods in 2022 [ASU and City of Phoenix, 2024]. Phoenix is the only controlled field evidence in the archive on a candidate intervention, and it cuts both ways: reflective coating lowers the surface and the night-time load but raises the pedestrian's radiant load in direct sun, so on an hourly comfort index it scores negatively by day on the walking line and positively for touch temperature, night conditions and pavement life; the sidewalk data and the authors' proposal to test coating under tree shade suggest a complement to a canopy, not a substitute. The abandoned before/after design is a direct lesson for Year 3 (evaluate against a simultaneous control) and the sensor accuracies (±0.2°C air, ±0.5°C surface) set the bar for the Dubai nodes.

3.2.9 Paris OASIS

Who, when, funder, scale, method. OASIS (Openness, Adaptation, Sensitisation, Innovation and Social ties) is the City of Paris schoolyard programme piloted under the EU's Urban Innovative Actions with an ERDF contribution of [4,995,793.16:20,982,331.27:5,828,425.35] (EUR:AED)(AED:USD) for 2018–2022, implemented 2019–2021, with CAUE 75, Météo-France and the Paris Interdisciplinary Energy Research Institute among the partners and Sciences Po for the social evaluation [Urban Innovative Actions, 2023][*EEA Climate-ADAPT, 2023*][15]. Ten pilot yards (four kindergartens, four elementary, two middle schools) were de-sealed and greened out of a stock with 73 hectares of sealed schoolyard surface in a city with about 6 m^2 of green space per central resident [*EEA Climate-ADAPT, 2023*][15]. The climate evaluation is a before–after–control–impact design: fixed stations at the site paired with a control station 200 m outside it, logging at one minute (explicit for the Maryse Hilsz site), mobile traverses at 1.5 m expressing heat stress as a UTCI offset from the control, and a GIS cooling-potential indicator computed before and after [OASIS research team, 2024a]; a companion laboratory study tested five paving structures under one 24-hour chamber cycle [OASIS research team, 2024b].

[15] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

Results and grade. No archived OASIS document reports a measured cooling effect of a renovated yard; what is reported is method stability (the UTCI offset constant within 2°C from 10:00 to 15:00 and drifting less than 1°C over a two-hour traverse) and a qualitative fall in cooling potential [OASIS research team, 2024a]. The laboratory results are quantitative: conventional asphalt reached 57°C and watered grass about 40°C (unwatered grass 58°C in an earlier test); light-coloured asphalt had the lowest daytime convective share (5 per cent of irradiance) but high radiosity, so sun-exposed comfort 'requires reducing the material's radiosity first and foremost' [OASIS research team, 2024b]. Scale-up: 75 yards transformed by 2022 per Climate-ADAPT, a further 60 on the city's own funds per the October 2022 UIA journal, about [9,000,000.00:37,800,000.00:10,500,000.00](EUR:AED)(AED:USD) a year for around 25 yards including surveillance and maintenance, lifetime above 50 years [Urban Innovative Actions, 2023] *[EEA Climate-ADAPT, 2023]*[15]. Grades: *case-report* and *policy-fact* for the programme documents; *method-reference* for the microclimate paper; *measured* (laboratory, single cycle, dry samples) for the paving paper.

Relevance. OASIS is the design template for the Dubai testbed and Year-3 campaign: paired case and control stations, native one-minute logging aggregated to the storyboard's ten-minute record, Al Fahidi untouched for three summers, a traverse protocol, and a comfort-anomaly metric that feeds Heat-Lost Activity Hours (HLAH) as hours above a threshold. Its instrument list (sheltered thermometer, hygrometer, black-globe thermometer, anemometer, net radiometer) exposes the storyboard's missing black globe, without which UTCI and WBGT cannot be computed at the node; the operating costs Climate-ADAPT records (guards, cleaning) are costs of keeping a cool place usable that the Dubai life-cycle model does not yet carry. For ABIG, the sequence of pilot, controlled evaluation with the national meteorological service, published method, own-funded scale-up and replication requests from more than 70 cities is the financing ladder in miniature.

Understanding: Urban heat island

TIER 1: IN PLAIN WORDS

A city is usually warmer than the countryside around it, especially at night, because roads and buildings soak up the sun by day and give the heat back after dark. How much warmer depends on the season, the hour and the street.

TIER 2: IN PROFESSIONAL TERMS

The urban heat island is the urban-minus-rural difference for a stated variable (air, surface or radiant temperature), reference station, averaging period and season; a daily mean can be near zero in summer while the night-time island and the daytime radiant load are both large.

TIER 3: IN TECHNICAL TERMS

Intensity is metric-specific: Abu Dhabi's daily-mean air-temperature island ranged from −0.23 to +3.21°C with a cool-season maximum [Manandhar et al., 2019]; Singapore quotes 'up to 7°C' without method *[Centre for Liveable Cities, 2024]*[4]; Arsht-Rock maps summer land-surface temperature at 30 m and reports hotspots 8°C hotter in New Delhi and 12°C in Los Angeles *[Arsht-Rock, 2022c][Arsht-Rock, 2022b]*. None is a pedestrian comfort index.

AS A MATHEMATICAL MODEL

$\mathrm{UHI}_{v,t} = T^{\mathrm{urban}}_{v,t} - T^{\mathrm{rural}}_{v,t}$ for variable v at time t; the daily mean $\overline{\mathrm{UHI}}_v$ can vanish while $\max_t \mathrm{UHI}_{v,t}$ does not, which is why HLAH is computed per place-hour.

3.3 Heat economics and policy

3.3.1 UNEP Beating the Heat

Beating the Heat: A Sustainable Cooling Handbook for Cities was launched on 3 November 2021 at COP26 by the Cool Coalition, UNEP, RMI, the Global Covenant of Mayors, Mission Innovation and the Clean Cooling Collaborative as 'an encyclopaedia of proven options to help cool cities' supported by 80 case studies [UNEP, Cool Coalition and RMI, 2021]. Only the landing page is archived; no case study can be cited. Grade: *policy-fact*. It is the compendium into which a validated corridor study would be reportable, and the source the Secretary-General's paper cites for cities heating at twice the global average rate [UN Secretary-General, 2024].

3.3.2 Singapore Heat Stress Advisory

On 24 July 2023 Singapore's Ministry of Sustainability and the Environment and the National Environment Agency launched a Heat Stress Advisory for the general population undertaking prolonged outdoor activity, developed with an eight-member Ministry of Health expert panel chaired by Professor Chia Kee Seng, after the warmest decade on record (2013–2022) and a 37°C reading on 13 May 2023 that equalled the 1983 record *[MSE Singapore, 2023]*[16]. It uses the wet bulb globe temperature, WBGT $= 0.7\,T_{\mathrm{nwb}} + 0.2\,T_g + 0.1\,T_d$, from nine island-wide sensors delivered through the myENV application from the sensor nearest the user, in three bands (low below 31°C, moderate from 31 to below 33°C, high at 33°C and above), each with actions on activity, hydration and attire; vulnerable groups include people recently arrived from cooler climates [NEA Singapore, 2023]*[MSE Singapore, 2023]*[16]. Grade: *policy-fact*. The bands are population guidance, not occupational limits, and no Singapore text mentions ISO 7243. The three-band scheme is a ready-made visitor-facing layer for the Dubai comfort map, the instruction to 'reduce' or 'minimise' outdoor activity in the upper bands is the qualitative basis for counting those hours as lost, and the formula makes the equipment gap concrete: WBGT needs natural wet-bulb and black-globe readings that the storyboard's nodes do not include [Iyyunni, 2026]. Nine sensors and a nearest-sensor rule serve a nation; the Dubai twin's advantage is to replace 'nearest sensor' with a modelled 1–5 m field.

3.3.3 Arsht-Rock, Hot Cities, Chilled Economies

Who, when, funder, method. *Hot Cities, Chilled Economies* is a 2022 study by the Adrienne Arsht-Rockefeller Foundation Resilience Center at the Atlantic Council with Vivid Economics, supported by Bank of America, with a methodology dated July 2022 [Arsht-Rock and Vivid Economics, 2022, Vivid Economics, 2022]. Across twelve cities (Abu Dhabi, Bangkok, Buenos Aires, Dhaka, Freetown, London, Los Angeles, Miami, Monterrey, New Delhi, Santiago and Sydney), a CMIP6 ensemble under SSP3-7.0 downscaled to 9 km^2 drives a simplified WBGT (no wind or radiation term) through workability functions for outdoor, shaded and sedentary work; O*NET exposure shares and a Cobb-Douglas production function convert lost labour to lost value added on a 1986–2005 climate and a 2019 economy, with 2050 at about 2°C of warming and fixed sectoral structure; Abu Dhabi alone is given hour-shifted 'adaptive' workability because heat regulation already exists [Vivid Economics, 2022].

Results and grade. Losses are USD 44 billion a year in 2020 rising to USD 84 billion by 2050 (derived: a factor of about 1.9), from 0.1 per cent of output in Santiago to 8.3 per cent in Dhaka, against about 0.3 per cent of world output lost to all natural catastrophes in 2021; Freetown loses 3.5 per cent, rising fivefold in dollars by 2050 [Arsht-Rock and Vivid Economics, 2022]. City pages:

[16] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

Bangkok 5 per cent of output, USD 8.6 billion around 2020, rising to USD 15.6 billion or 6 per cent by 2050, outdoor workers losing up to 40 per cent *[Arsht-Rock, 2022a]*[17]; Los Angeles almost USD 5 billion (0.5 per cent) rising to USD 11 billion, with a chief heat officer created in November 2021, a 2014 cool-roof ordinance and 250 miles of cool pavement planned *[Arsht-Rock, 2022b]*[18]; New Delhi 4 per cent of output, USD 3.9 billion in 2020, rising to 5 per cent or USD 6.1 billion by 2050, productivity losses of 20 per cent for shaded and 25 per cent for sun-exposed workers, construction producing 9 per cent of output but a third of the losses *[Arsht-Rock, 2022c]*[19]. No Abu Dhabi loss figure is archived. Grade: *modelled*, the 2050 values *projection*, the methodology *method-reference*; there is no validation against observed productivity.

Relevance. Arsht-Rock is the macro frame and the labour-hours analogue of HLAH: hours lost from WBGT through workability curves at sector-hour resolution over 9 km^2, where the Dubai metric is the same construct at place-hour resolution over metres, extended from workers to visitors. It models the leisure workforce, not the visitor, so it cannot be cited for Outdoor Tourism Capacity; its 30 m Landsat layer is where the best-known heat-economics study stops and the corridor twin starts. The UAE already encodes hours lost to heat as a rule (a midday break from 12:30 to 15:00 between 15 June and 15 September, introduced in 2004, with fines of [5,000.00:1,388.89](AED:USD) per breach per worker up to [50,000.00:13,888.89](AED:USD) *[The National, 2026]*[20], described by the Ministry of Human Resources and Emiratisation as its nineteenth consecutive year in 2023 *[MoHRE, 2023]*[21]) and Arsht-Rock's hour-shifting is the modelled form of that rule. The reference series behind these figures (the ILO's projection of 2.2 per cent of working hours, 80 million full-time-equivalent jobs and USD 2.40 trillion (purchasing-power parity) lost by 2030 assuming work in the shade [ILO, 2019], and the Lancet Countdown's observation-based model estimate of 512 billion potential work hours and USD 835 billion lost in 2023 [Romanello et al., 2024]) are treated in Chapter 4.

3.4 Tourism-climate indices

3.4.1 Holiday Climate Index and Tourism Climate Index

Scott, Rutty, Amelung and Tang published *An Inter-Comparison of the Holiday Climate Index (HCI) and the Tourism Climate Index (TCI) in Europe* in *Atmosphere* in June 2016 with SSHRC and IDRC funding [Scott et al., 2016]. The Tourism Climate Index (Mieczkowski, 1985) uses five weighted sub-indices with half its weight on thermal comfort and monthly data; the Holiday Climate Index replaces expert weights with tourists' stated preferences, HCI : Urban $= 4\,\mathrm{TC} + 2\,A + (3\,P + W)$, five daily variables (maximum temperature, mean relative humidity, cloud cover, precipitation, wind) in four weighted sub-indices, each rated 0–10 and summed to 0–100, with scores below

[17] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Bangkok, Thailand*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-bangkok/

[18] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Los Angeles, United States*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-los-angeles/

[19] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: New Delhi, India*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-new-delhi/

[20] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[21] Ministry of Human Resources and Emiratisation, UAE, *UAE to implement midday work break from 15 June to 15 September for 19th consecutive year*, MoHRE media centre, 2023. https://mohre.gov.ae/en/media-center/news/1/6/2023/uae-to-implement-midday-work-break-from-15-june-to-15-september-for-19th-consecutive-year

40 'unacceptable' and 50–59 'acceptable'; the thermal rating for 37–39°C effective temperature is 2 of 10 and no band above 39°C is printed [Scott et al., 2016]. Both indices were run for Barcelona, Istanbul, London, Paris, Rome and Stockholm on 25 km regional-climate output under A1B from 1961–1990 to the 2080s and checked against Paris hotel overnights for 2000–2010; the authors conclude that 'use of the TCI should be discontinued' because it rates shoulder and winter months unacceptable where demand is steady, and note that the HCI 'can utilize other measures of thermal comfort such as the humidex or universal thermal climate index' [Scott et al., 2016]. Grade: *method-reference* for the index; *modelled* for the comparison.

The report's own worked example, not a finding of the paper, exposes the structural gap. A dry, calm, clear Gulf afternoon at 37–39°C scores $4 \times 2 + 2 \times 8 + 3 \times 10 + 1 \times 10 = 64$, 'Good'; even with a thermal rating of zero it scores $0 + 16 + 30 + 10 = 56$, 'Acceptable', because only rain and wind can override the other facets. The HCI cannot classify a Dubai summer afternoon as unacceptable. That is the argument for an explicit heat-stress override (UTCI [Jendritzky et al., 2012] or WBGT), inside Outdoor Tourism Capacity, and for positioning OTC as the hourly, cell-level successor to a daily destination index: HCI-compatible in its preference layer (23–25°C → 10; 31–32°C → 6; 37–39°C → 2, with no Gulf sample behind the curve), physiologically bounded in its threshold layer, and validated against hourly footfall rather than monthly hotel arrivals. Index choice also determines the projected season (the TCI projects summer decline in Mediterranean cities and the HCI none), so any claim about Dubai's future tourism season must state its index.

3.5 Ocean and blue intelligence

Understanding: Integration layer

TIER 1: IN PLAIN WORDS

ABIG does not plan to launch satellites first. It plans to join up the observing stations, forecasts and satellite services Africa already has so that the people who make decisions (fisheries officers, port managers, coastal planners) get answers they can act on.

TIER 2: IN PROFESSIONAL TERMS

ABIG is positioned as an integration and application layer above three existing layers: observation (GOOS-Africa), forecasting (OceanPrediction Decade Collaborative Centre African Seas and OPERA) and Earth-observation services (GMES & Africa), with a federated data hub and a co-design chain from user decision to operational test [Terhile et al., 2026].

TIER 3: IN TECHNICAL TERMS

The posture matches the European Digital Twin Ocean, which combines EMODnet and Copernicus Marine into one framework and declares interoperability with Destination Earth *[Copernicus Marine Service, 2024]*, and Digital Earth Africa's chain from observations through analysis-ready data and products to decisions [WEF and Digital Earth Africa, 2021]. Its test is whether a decision improves per dollar, not whether a sensor is added.

3.5.1 GMES & Africa, MarCOSIO and MarCNoWA

GMES & Africa is the African Union–European Union programme that builds institutional, human and technical capacity for operational Earth-observation services; its marine and coastal service covers oceanographic monitoring and forecasting, coastal monitoring, ship-traffic monitoring, pollution monitoring and marine weather forecasting through institutional consortia [Terhile et al., 2026]; the programme's own page could not be captured *[GMES and Africa, 2025]*[1]. The southern consortium, MarCOSIO (Marine and Coastal Operations for Southern Africa and the Indian Ocean), is funded under the programme's Marine and Coastal Operations thematic area

(its e-Station at the CSIR was 'a gift from the African Union and European Union Commissions' under the 'GMES and Africa Program, 2017–2021') and lists twelve partners (CSIR, ABALOBI, the Benguela Current Convention, CORDIO East Africa, marine institutes and universities in Madagascar, Tanzania, Kenya, Mauritius, South Africa and Mozambique, and WIOMSA), implying South Africa, Tanzania, Kenya, Mozambique, Mauritius and Madagascar plus the Convention; no country list is printed *[MarCOSIO, 2025]*[22]. It runs three service lines (fisheries and aquaculture; ship traffic and safety at sea; water quality and coral bleaching), adopts the South African OCIMS method that identifies 'archetypal users, beneficiaries, policies, usage scenarios, as well as functional requirements', delivers sea-surface temperature and chlorophyll forecasts to small-scale fishers through the ABALOBI Fisher application, and had commissioned an Economic Benefit and Analysis Report only as terms of reference at capture *[MarCOSIO, 2025]*[22]. The northern and western consortium, MarCNoWA, is cited by title only; its reported eighteen-country coverage and phase dates are unverified *[GMES and Africa, 2023]*[2]. Grade: *policy-fact*; the ABALOBI item is a *case-report* without outcome data. No cost, uptake or benefit figure exists for any GMES & Africa marine service in the archive. It is at once ABIG's strongest asset (an operating consortium in the Western Indian Ocean and Benguela systems for the regional hubs to build on) and its principal duplication risk; the absence of published benefit measurement is the gap ABIG's 'decisions improved per dollar' indicator would fill.

3.5.2 Digital Earth Africa

Digital Earth Africa operates an open data cube of analysis-ready satellite data over more than 30 million km^2, 'the world's largest' [WEF and Digital Earth Africa, 2021]*[Digital Earth Africa, 2021]*[23]. The World Economic Forum's January 2021 report, prepared with the programme, estimated that its benefit 'could be higher than USD 2 billion a year' by 2024. Its Figure 1 totals USD 2.30 billion: USD 0.50 billion of extra Earth-observation industry sales (USD 385 million direct plus USD 117 million indirect, on a market growing from USD 1.32 billion in 2019 to USD 1.8 billion in 2024), USD 0.90 billion from regulating unregulated gold mining (10 per cent prevention of about USD 9 billion a year of damage scaled from Ghana and South Africa) and USD 0.90 billion in agriculture (17.6 billion m^3 of water saved worth USD 880 million at 10 per cent adoption and USD 0.05 per cubic metre, plus wheat sowing and insurance) [WEF and Digital Earth Africa, 2021]; the programme's release says 'up to USD 2 billion' and adds one unquantified anecdote, Tanzania's statistics bureau monitoring Lake Sulunga with the Water Observations from Space service *[Digital Earth Africa, 2021]*[23]. The method transfers single-site analogues continent-wide with assumed adoption rates and a twelve-observation regression whose coefficient was rounded by judgement; every impact figure is 'Team Analysis'. Grade: *projection*; the release is a *case-report*. No realised benefit and no cost side appear, so no benefit–cost ratio can be derived. For ABIG the programme is a continental asset to integrate, its readiness-index rubric (data infrastructure 20, policy 10, institutional capacity 20, user adoption 20, industry fabric 30 per cent) a capability-map template, and its unverified headline a caution to commit to ex-post measurement. For Dubai the benefit formula (physical change times unit value times adoption share) is a reusable structure for valuing restored activity-hours, and its weight on adoption supports the primacy of the written-usefulness gate condition.

[22] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[23] Digital Earth Africa, *Digital Earth Africa has the potential to contribute billions to the African economy using Earth observations*, news, 2021. https://digitalearthafrica.org/en_za/digital-earth-africa-has-the-potential-to-contribute-billions-to-the-african-economy-using-earth-observations/

3.5.3 Global Fishing Watch in Africa

Global Fishing Watch, founded by Oceana, SkyTruth and Google, frames its Africa programme with unsourced estimates (fisheries and aquaculture contribute USD 24 billion a year and employ some 12 million people; an estimated 40 per cent of catch is illegal, unreported or unregulated) and a second-hand projection that catch potential may fall by 30 per cent or more by 2050 in many tropical West African countries *[Global Fishing Watch, 2025a]*[24]. Its documented work: a port-inspection pilot from 2021 to 2023 with Senegal, Ghana, Côte d'Ivoire, Kenya and the Fisheries Committee for the West Central Gulf of Guinea, run with Trygg Mat Tracking, funded by the Moore and Oak foundations and Norad, announced in Dakar on 4 November 2021, which produced the Vessel Viewer tool in support of the Port State Measures Agreement in force since 2016 *[Global Fishing Watch, 2021]*[25]*[Global Fishing Watch, 2025a]*[24]; partnerships with three sub-regional fisheries bodies representing 24 countries; intelligence to the Gulf of Guinea Joint Analytical Cell; and training modules launched in July 2025 with the fisheries authorities of Senegal, Madagascar and Gabon *[Global Fishing Watch, 2025a]*[24]. The Senegal page was captured without article text *[Global Fishing Watch, 2025b]*[26]. The Gabon case is the archive's clearest product-to-action chain: after a workshop for 25 personnel from the fisheries directorate, the parks agency and the navy and a tailored intelligence report, Gabonese enforcement aboard Sea Shepherd's vessel intercepted three of five industrial trawlers (*Rong Chang 19*, *Rong Chang 26* and *Chang Hiu 01*) about 10 km inside Gabonese waters on 18 July 2026 without valid licences and escorted them 200 km to Libreville, where the owner later settled; Operation Albacore, run by Gabon with Sea Shepherd since 2016, has seized 16 vessels over the decade, a total that belongs to the operation, not to the satellite intelligence *[Global Fishing Watch, 2026]*[27]. Grade: *case-report* for Gabon; *policy-fact* for programme facts; the macro figures inherit no measured grade, and no cost, patrol-day denominator or counterfactual is given. For ABIG this is the proof that the capability-first sequence works without new hardware and the candidate first use case: illegal-fishing alerts with an authority already working with the provider. For Dubai it is the template of the Decision ID record (brief delivered, decision taken, outcome recorded) that gate condition 6.2.4 must prove, with the missing counterfactual as the gap the Year-3 paired-control design closes.

3.5.4 Great Blue Wall

The Great Blue Wall was launched at COP26 in Glasgow on 10 November 2021 by Western Indian Ocean countries with IUCN and partners as a movement to establish a connected network of conserved and restored seascapes, contribute to 30 per cent marine conserved-area coverage by 2030, achieve net gain of mangroves, seagrass and coral, and benefit at least 70 million people across ten countries of the region. It was endorsed by Kenya, Mozambique, Tanzania, Seychelles, the United States, France, Ireland, the United Kingdom, Sweden and Portugal, and Tanzania's Tanga-Pemba seascape was launched as the first *[IUCN, 2021]*[28]. By COP27 in November 2022 two seascapes had been designated (Tanga-Pemba, an IUCN Category VI area, and Quirimbas

[24] Global Fishing Watch, *Rising pressures on Africa's fisheries and marine resources*, 2025. https://globalfishingwatch.org/rising-pressures-on-africas-fisheries-and-marine-resources/

[25] Global Fishing Watch, *African Nations to Use New Technology in Tightening Port Controls, Fighting Illegal Fishing With Big Data*, press release, 2021. https://globalfishingwatch.org/press-release/africa-new-tech-tighten-port-control/

[26] Global Fishing Watch, *Our work in Senegal*, 2025. https://globalfishingwatch.org/our-work-in-senegal/

[27] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[28] International Union for Conservation of Nature, *Global launch of the Great Blue Wall*, IUCN news, November 2021. https://iucn.org/news/secretariat/202111/global-launch-great-blue-wall

in Mozambique, planned at 40,000 km^2) and the initiative aimed at 2 million km^2 of seascapes by 2030 (30 per cent of the region's ocean), 2 million ha of critical ecosystems, an estimated 100 million tonnes of carbon dioxide sequestered, and 500 ocean start-ups by 2030 from more than 20 already supported *[IUCN, 2022]*[29]. The first Impact Report (2021–2025) was launched on 17 June 2026 at the eleventh Our Ocean Conference in Mombasa, the first on African soil: partners 'have collectively contributed to supporting' more than 83,000 km^2 of marine and coastal areas, strengthened over 112,000 livelihoods, mobilised more than USD 80 million, and supported 'the protection, restoration, or commitment to protect' more than 175,000 ha *[IUCN, 2026]*[30]; the report itself was not reviewed [Great Blue Wall Initiative, 2026]. Grade: *case-report*, self-reported and contribution-based without baselines; the launch and COP27 statements are *policy-fact*. For ABIG's Eastern Africa hub the initiative supplies an operational user community, an accepted reporting vocabulary (area supported; hectares protected, restored or committed; livelihoods; money mobilised) and a convening cadence; it also shows the region is already convened, so ABIG must contribute products with attribution, not another forum.

3.5.5 European Digital Twin Ocean

The European Digital Twin Ocean (EU DTO, built by the EDITO projects) follows a commitment by Commission President von der Leyen at the One Ocean Summit in Brest in February 2022; its pre-operational platform was unveiled on 13 June 2024 at the third Digital Ocean Forum in Brussels as a fundamental action of the EU Mission 'Restore our Ocean and Waters' to 2030, with the public platform to enter operation in 2025 (a forward-looking statement as of July 2024) *[Copernicus Marine Service, 2024]*[31]. It combines EMODnet and the Copernicus Marine Service under Mercator Ocean International and the Flanders Marine Institute, draws applications from EDITO-Model Lab, Iliad, Blue-Cloud 2026 and the Mission Lighthouses, is 'fully compatible' with the Destination Earth climate twin, names the UN Decade of Ocean Science, and demonstrated seven use cases from marine litter and seagrass restoration to early warning and coastal hazards *[Copernicus Marine Service, 2024]*[31]. Grade: *case-report*; no users, outcomes, costs or African coverage. The timeline (28 months from commitment to demonstration (derived: February 2022 to June 2024), about three years to planned operation) is the reference class for ABIG's faster 3- to 36-month loops, which lack an equivalent of EuroHPC. It is the closest institutional precedent for ABIG's posture, and the addendum's capability map should record interoperability with the EU DTO and Destination Earth from the outset.

3.6 Finance

3.6.1 Seychelles blue bond

The first sovereign blue bond was announced by Seychelles on 29 October 2018 with World Bank support: USD 15 million over ten years, placed with Calvert Impact Capital, Nuveen and Prudential Financial through Standard Chartered, with Bank of New York Mellon as trustee; a USD 5 million IBRD partial credit guarantee lowered the borrowing cost by at least 2.0 per cent

[29] International Union for Conservation of Nature, *Leaders from Western Indian Ocean countries share progress on Great Blue Wall at COP27*, IUCN news, November 2022. https://iucn.org/news/202211/leaders-western-indian-ocean-countries-share-progress-great-blue-wall-cop27

[30] International Union for Conservation of Nature, *Great Blue Wall report showcases five years of impact on ocean regeneration across the Western Indian Ocean*, IUCN news, June 2026. https://iucn.org/news/202606/great-blue-wall-report-showcases-five-years-impact-ocean-regeneration-across-western

[31] Copernicus Marine Service, *The European Digital Twin Ocean Launched at the 2024 Digital Ocean Forum*, news. https://marine.copernicus.eu/news/european-digital-twin-ocean-launched-2024-digital-ocean-forum

a year and USD 5 million of GEF concessional financing by over 3 per cent, a combined 5 per cent, while a Rockefeller Foundation donation covered most transaction costs [World Bank, 2025]. Proceeds flow through a Blue Grants Fund managed by SeyCCAT and a Blue Investment Fund managed by the Development Bank of Seychelles to the third South West Indian Ocean Fisheries project; the World Bank case study of 27 June 2025 reports the expansion of marine-protected-area management coverage from 5 million to about 22 million hectares as a project outcome, records that the guarantee was not triggered and remains in place until 2028, and puts the global blue-bond market at USD 15.25 billion of cumulative issuance by June 2025, USD 14.76 billion of it from emerging-market issuers [World Bank, 2025]. The 2018 press release could not be captured *[World Bank, 2018]*[3], and the *Marine Policy* critique was not reviewed, so no claim about debt outcomes is made [Kilic, 2024]. Grade: *policy-fact* for the terms; *case-report* for the project outcomes. For ABIG's financing ladder it is a real African instance of concessional preparation followed by a small guaranteed instrument with a grants-plus-loans split; for the Dubai roadmap it is a precedent for matching instrument tenor to asset life and separating an evaluation window from asset capital. The report cites the terms only and does not extrapolate a 'heat bond'.

3.6.2 AU Blue Economy Strategy and the Luanda Declaration

The Africa Blue Economy Strategy was prepared by AU-IBAR in Nairobi with EU and member-state funding and Norwegian support under lead consultant Professor Pierre Failler, after visits to thirteen member states and a 125-delegate workshop on 30–31 August 2019, and was endorsed by the third Specialised Technical Committee on Agriculture, Rural Development, Water and Environment in October 2019; its five thematic annexes are not archived [AU-IBAR, 2019]. It counts 38 coastal states with an estimated 13 million km^2 of ocean territory and puts catch lost to illegal fishing at a conservative USD 10 billion a year. Its 2018 estimates are USD 296 billion and 49 million jobs from blue-economy sectors, projected to USD 405 billion and 57 million in 2030 and USD 576 billion and 78 million in 2063; fisheries USD 21 billion of value added (8.1 marine artisanal, 6.8 marine industrial, 6.3 inland) with 13 million employed and 10 million tonnes landed; aquaculture USD 2.77 billion, Egypt 1.37 of 1.98 million tonnes; coastal tourism USD 80 billion and 24 million jobs, above USD 100 billion and 28 million jobs by 2030 and USD 138 billion and 35 million by 2063; ecosystem services about USD 40 billion [AU-IBAR, 2019]. The strategy separately reports coastal tourism growing at 5.6 per cent a year against 3.9 per cent globally, and that rate is not the one its own projections carry: USD 80 billion compounding at 5.6 per cent a year reaches about USD 154 billion by 2030 (derived), not the USD 100 billion projected. The projected endpoints imply 1.88 per cent a year to 2030 and 1.22 per cent a year to 2063 (derived). The 5.6 per cent is read here as an observed historical growth rate and not as the forecast rate, and the two should not be compounded together. Absolute values are cited; the strategy's percentage shares are internally inconsistent and are not. Its objectives include a unified national accounting system, coordinated monitoring, control and surveillance, and an 'analytical and information support system at all levels for informed decision making' [AU-IBAR, 2019]. Grade: *projection* for 2030 and 2063; *policy-fact* for the institutional record.

The Luanda Declaration was adopted on 24 July 2026 at the Third Africa Blue Economy Week in Luanda (22–24 July by the release's body text), convened by the Government of Angola, the AU Commission and UNDP under the theme 'From Strategy to Action', with six commitments (accelerate the Strategy, strengthen governance and accountability, mobilise blue finance, protect ecosystems, deepen regional cooperation, expand value chains) and an emphasis on 'science, innovation, digital transformation and inclusive participation'; the *State of Africa's Blue Economy Report 2026* was launched alongside it *[African Union, 2026]*[32]. The figures the ABIG paper attributes

[32] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-a

to the Declaration (38 of 55 member states coastal, seven Large Marine Ecosystems, about 305,000 km^2 of inland waters, transboundary basins on almost 64 per cent of the land), appear only in that paper, the Declaration's text being unarchived [Terhile et al., 2026]. Grade: *policy-fact*. Together the Strategy and the Declaration are ABIG's political mandate two months before the Dubai Letter of Intent; every ABIG domain maps to a Strategy objective adopted in 2019, and the sector table is the natural denominator for a 'decisions improved per dollar' indicator. For Dubai, the coastal-tourism figures give Outdoor Tourism Capacity a named downstream market on the scale spiral of Chapter 7.

3.7 Global policy instruments

3.7.1 Global Cooling Pledge

The Global Cooling Pledge is an initiative of the United Arab Emirates as host of COP28, one of nine non-negotiated outcomes of the COP28 Presidential Action Agenda, with the UNEP-led Cool Coalition as partner and secretariat; its published targets are to cut cooling-related emissions by 68 per cent against the Pledge's 2023 baseline by 2050, to increase access to sustainable cooling significantly by 2030, and to raise the average efficiency of new air conditioners by 50 per cent, the emission targets drawn from UNEP's *Global Cooling Watch 2023* [UNEP, 2023]*[UN News, 2023]*[33]. Over 60 countries had signed at the launch on 5 December 2023 *[UN News, 2023]*[33]; by the COP29 ministerial round table of 19 November 2024 it united over 71 countries and 60 non-state actors around 14 actions, fifteen cities had joined as subnational signatories committing 'to integrate heat-resilience into urban design and nature-based cooling strategies', seventeen signatories had national cooling action plans, the UAE and Brazil were co-chairs for 2025, a 2030 Implementation Strategy was adopted and an Intergovernmental Committee on Cooling was announced for COP30 *[UNEP, 2024]*[34]. Grade: *policy-fact*. The published targets concern emissions, access and equipment efficiency; none is an adaptation or comfort indicator. That is the opening for a validated Heat-Lost Activity Hours metric as the Pledge's heat-adaptation indicator (a claim made about the published target summary, not the unarchived Pledge text) and the DEWA line in the Dubai optimiser (kilowatt-hours of district cooling avoided per restored activity-hour) is a Pledge-relevant quantity by construction.

3.7.2 UN Secretary-General's Call to Action on Extreme Heat

The Secretary-General's Call to Action on Extreme Heat was issued in July 2024 (the archived paper carries no printed date) by the Climate Action Team with inputs from FAO, ILO, UNDRR, UNEP, UNESCO, UN-Habitat, UNICEF, OCHA, WHO and WMO, in four pillars (care for the vulnerable, protect workers, boost resilience using data and science, limit warming to 1.5°C) with 23 actions [UN Secretary-General, 2024]. Every number is a secondary citation: about 489,000 heat-related deaths a year in 2000–2019 (modelled), potential income losses of USD 863 billion in 2022, more than 70 per cent of the global workforce (2.4 billion people) at high risk with 93 per cent affected in Africa and 84 per cent in the Arab states, productivity beginning to decline above 24–26°C and halving at 33–34°C, and the ILO's USD 2.40 trillion at purchasing-power parity and 80 million jobs by 2030 [UN Secretary-General, 2024]. The Call to Action carries that figure without its basis; it is the ILO projection at 1.5°C of warming and in purchasing-power-parity

ccelerate-africas-blue-economy

[33]UN News, *COP28: Over 60 countries pledge to slash cooling emissions amid rising temperatures*, December 2023. https://news.un.org/en/story/2023/12/1144382

[34]United Nations Environment Programme, *Cooling the Planet: COP29 affirms global climate action*, UNEP press release, November 2024. https://www.unep.org/news-and-stories/press-release/cooling-planet-cop29-affirms-global-climate-action

terms, and it is the same figure reported earlier in this chapter rather than a second, independent estimate [ILO, 2019]. It names WBGT, UTCI, the Heat Index and Humidex as recognised indices, mentions tourism once without a number, endorses 'reflective surfaces' generically within passive cooling, and asks for 'standardized reporting of extreme heat risk and impact data', heat action plans, and 'champions' or an office with a cross-sectoral heat-resilience mandate [UN Secretary-General, 2024]. Grade: *policy-fact*; each figure inherits its primary grade, and the generic endorsement of reflective surfaces does not override the Phoenix radiant-load finding. Its relevance is that hours lost to heat is already the accepted policy metric at national scale, that the UN asks for exactly the standardised reporting HLAH provides, and that no UN quantitative anchor exists for visitor outdoor activity: the gap Outdoor Tourism Capacity fills.

3.7.3 IPCC Special Report on Climate Change and Cities

The IPCC decided in April 2016 to produce a Special Report on Climate Change and Cities in its seventh cycle; a scoping meeting in Riga on 16–19 April 2024 produced the outline proposal submitted as IPCC-LXI/Doc. 2 [IPCC, 2024], which the Panel agreed at its 61st session in Sofia on 27 July to 2 August 2024 with more than 230 delegates from 114 governments, scheduling approval for March 2027 *[IPCC news, 2024]*[35]. The report has five chapters by 97 experts from 56 countries; lead author meetings were held in Osaka (March 2025), Mombasa (July 2025), Oslo (January 2026) and Bhopal (3–7 August 2026); the First Order Draft drew over 32,000 comments and the Second Order Draft's government and expert review, completed in July 2026, over 25,000 from 960 reviewers in 102 countries; it is the only special report of the cycle, and a revision of the 1994 technical guidelines on impacts and adaptation, including adaptation indicators, metrics and guidelines, is to be developed with the Working Group II report *[IPCC, 2026a]*[36]. The outline's indicative content includes 'urban observation and modelling tools for monitoring and evaluation', 'metrics for assessing mitigation and adaptation options' and 'climate and information services to enable action'; it never uses the words 'digital twin' [IPCC, 2024], and the report page itself is navigation only *[IPCC, 2026b]*[37]. Grade: *policy-fact*. With the Second Order Draft review closed on 3 July 2026 and approval on 15–19 March 2027, nothing produced after the Dubai grant starts on 15 March 2027 can enter the Special Report; the targets are the Working Group II contribution and the adaptation-indicator guidelines, for which Year-3 papers in 2029–2030 are plausible inputs, and the report's regional typology is where a hot-arid Gulf corridor and African coastal-city evidence would sit.

3.8 Comparison matrix and landscape

Table 3.1 condenses the line items into one matrix; Figure 3.1 places them by domain and spatial scale, and Figure 3.2 shows the resolution ladder from planetary twins to the Dubai street model. In the relevance column D denotes the Dubai Decision Twin and A denotes ABIG.

[35] Intergovernmental Panel on Climate Change, *IPCC approves outlines of the first two reports in the seventh assessment cycle*, IPCC news, 2 August 2024. https://www.ipcc.ch/2024/08/02/ipcc-approves-outlines-of-the-first-two-reports-in-the-seventh-assessment-cycle/

[36] Intergovernmental Panel on Climate Change, *Leading scientists gather in India for final meeting to work on the Special Report on Climate Change and Cities*, IPCC news, 3 August 2026. https://www.ipcc.ch/2026/08/03/srcities-lam4/

[37] Intergovernmental Panel on Climate Change, *Special Report on Climate Change and Cities*, IPCC Seventh Assessment Cycle; approval scheduled March 2027. https://www.ipcc.ch/report/special-report-on-climate-change-and-cities/

Table 3.1: Comparison matrix of similar efforts. Money figures carry the year of the source; grades use the report's seven labels; D = relevant to the Dubai Decision Twin, A = relevant to ABIG.

Effort	Lead, funder, dates	Scale and method	Headline result (grade)	Status	Rel.
Urban heat and decision twins					
Cooling Singapore, DUCT [NSCC Singapore, 2024, Cooling Singapore 2.0, 2024]*[Centre for Liveable Cities, 2024]*	Singapore-ETH Centre with SMU, NUS, SMART; NRF; since 2017	Island-wide meso- and micro-scale twin; >20 researchers, 4 years; 80 simulation-h per day on ASPIRE 2A	UHI up to 7°C (no method); 0.13°C island cooling on one modelled heat-wave day (modelled; case-report)	Explorer app in development	D, A
Destination Earth *[ECMWF, 2024]*	EU DG CNECT; ECMWF, ESA, EUMETSAT; Dec 2021; phase 2 from June 2024	Global 4.4 km, 4-day; regional 500–750 m, 2-day; climate 5–10 km hourly; EuroHPC	Design and resolutions only (method-reference)	Ramping up; target 2030	D, A
Dubai Digital Twin Platform *[Dubai Media Office, 2026]*	Dubai Municipality; launched 2 July 2026	>195,000 buildings; >280,000 assets; >1,500 layers; >100 applications	Geometry-and-data twin; rainfall simulation; no comfort layer (policy-fact)	Next phase; MoUs Al-Futtaim, Huawei	D
Dubai Walk, Future Loop *[Dezeen, 2024] [Khaleej Times, 2025]*	Government of Dubai; launched 7 Dec 2024	3,300 km new + 2,300 km renovated by 2040 (+900 km after); 160 areas; Future Loop 2 km, 30,000 m^2 AC + 30,000 m^2 shaded	Targets: soft mobility 13 to 25%; 80% within 20 min (policy-fact)	Future Loop under review Nov 2025; PPP	D
The Loop *[URB, 2025]*	URB (private); early 2023	93 km climate-controlled corridor; >3 M residents	Aspirations only (projection)	Unfunded concept	D
Green Riyadh *[Saudipedia, 2025][RCRC, 2025]*	Royal Commission for Riyadh City; launched 19 Mar 2019	>7.5 M trees by 2030; 1,350 km irrigation; 72 species	Claims −1.5 to −2°C city, −8 to −15°C afforested sites (projection); targets (policy-fact)	In progress; no reported outcomes	D
Makkah, Arafat cooled roads *[Arab News, 2024][Gulf News, 2025][Saudi Gazette, 2024]*	Saudi Roads General Authority; 2023–2025	Reflective surfaces >84,000 m^2 (2025); rubberised walkways; misting; Neem trees	Surface −12 to −15°C, agency claim, no protocol (case-report)	Expanding; Hajj leaves summer from 2026	D
Abu Dhabi LCZ [Manandhar et al., 2019]	UAE University, Khalifa University; no external funding; 2017–2019	26 stations; Landsat 30 m LCZ; ENVI-met 3 m; four seasons	Daily-mean UHI −0.23 to +3.21°C; corrected model RMSE 0.27–1.62°C (measured)	Completed; Dubai paper not reviewed [Bande et al., 2020]	D, A

continued overleaf

Table 3.1 (continued)

Effort	Lead, funder, dates	Scale and method	Headline result (grade)	Status	Rel.
Phoenix cool pavement [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]	City of Phoenix, ASU; 2020–2024	36 miles CoolSeal; MaRTy cart and vehicle traverses vs adjacent untreated streets	Surface −12.0°F noon; road-centre MRT +5.5°F (Year 1), +5.8°F (Phase II); reflectivity 33–38 to 19–30% in 10 months (measured-with-controls; modelled for energy)	Programme continues; no service life or cost	D
Paris OASIS [Urban Innovative Actions, 2023, OASIS research team, 2024a, OASIS research team, 2024b]*[EEA Climate-ADAPT, 2023]*	City of Paris; ERDF-UIA [4,995,793.16: 20,982,331.27: 5,828,425.35] (EUR:AED)(AED:USD) (2018–2022); implemented 2019–2021	10 pilot yards; BACI with control 200 m outside; lab paving tests	Method published, effect sizes not archived (case-report; method-reference; measured lab)	75 yards by 2022; [9,000,000.00: 37,800,000.00: 10,500,000.00] (EUR:AED)(AED:USD)/yr for 25	D, A
Heat economics and policy					
UNEP Beating the Heat [UNEP, Cool Coalition and RMI, 2021]	Cool Coalition, UNEP, RMI and partners; 3 Nov 2021	Handbook with 80 case studies	Compendium; content not archived (policy-fact)	Published	D
Singapore Heat Stress Advisory [NEA Singapore, 2023]*[MSE Singapore, 2023]*	MSE and NEA; 24 July 2023	9 WBGT sensors; myENV nearest-sensor rule	Bands <31, 31–<33, ≥33°C WBGT (policy-fact)	Operational; expansion planned	D, A
Arsht-Rock Hot Cities [Arsht-Rock and Vivid Economics, 2022, Vivid Economics, 2022] *[Arsht-Rock, 2022a]* *[Arsht-Rock, 2022b]* *[Arsht-Rock, 2022c]*	Atlantic Council, Vivid Economics; Bank of America; 2022	12 cities; CMIP6 SSP3-7.0 at 9 km^2; simplified WBGT; O*NET; Cobb-Douglas	USD 44 bn/yr (2020) to 84 bn (2050); Bangkok 5%, LA 0.5%, New Delhi 4% of output (modelled; projection)	Published; Heat Action Platform	D, A
Tourism-climate indices					
HCI and TCI [Scott et al., 2016]	Scott, Rutty, Amelung, Tang; SSHRC, IDRC; 2016	Daily destination index; six European cities; A1B at 25 km	HCI:Urban cannot rate a 37–39°C dry day unacceptable (method-reference; modelled)	Published; HCI in use	D, A
Ocean and blue intelligence					

continued overleaf

Table 3.1 (continued)

Effort	Lead, funder, dates	Scale and method	Headline result (grade)	Status	Rel.
GMES & Africa, MarCOSIO [Terhile et al., 2026] *[MarCOSIO, 2025]*	AU–EU; MarCOSIO 12 institutions; 2017–2021 programme period	Three service lines; OCIMS method; ABALOBI app	Services exist; no benefit measured (policy-fact); MarCNoWA not reviewed *[GMES and Africa, 2023]*	Economic report at ToR stage	A
Digital Earth Africa [WEF and Digital Earth Africa, 2021]*[Digital Earth Africa, 2021]*	DE Africa with WEF; Jan 2021	Open data cube >30 M km^2	'Higher than USD 2 bn/yr' by 2024 (USD 2.30 bn; projection); Lake Sulunga (case-report)	Operational; no realised benefit reported	A, D
Global Fishing Watch, Africa *[Global Fishing Watch, 2026]* *[Global Fishing Watch, 2025a]* *[Global Fishing Watch, 2021]*	GFW with TMT; Moore, Oak, Norad; 2021–2026	Port pilot in 4 countries; Vessel Viewer; 24-country partners; training	Gabon: 3 trawlers seized 18 July 2026 (case-report); 40% IUU (unsourced)	Active	A, D
Great Blue Wall *[IUCN, 2021]* *[IUCN, 2022]* *[IUCN, 2026]*	IUCN and WIO countries; launched COP26, 10 Nov 2021	Seascape network; 2 designated by 2022; 2 M km^2 by 2030 aim	>83,000 km^2 supported; >175,000 ha protected, restored or committed; >112,000 livelihoods; USD 80 M+ (case-report)	Impact report June 2026 [Great Blue Wall Initiative, 2026] not reviewed	A
European Digital Twin Ocean *[Copernicus Marine Service, 2024]*	EU Mission; Mercator Ocean, VLIZ; Brest Feb 2022; demo 13 June 2024	EMODnet + Copernicus Marine; 7 use cases; DestinE-compatible	Pre-operational platform (case-report)	Operation planned 2025	A
Finance					
Seychelles blue bond [World Bank, 2025]	Seychelles, World Bank, GEF; 29 Oct 2018	USD 15 M, 10 years; USD 5 M guarantee; USD 5 M concessional	Borrowing cost −5%/yr; MPA management 5 to 22 M ha (policy-fact; case-report)	Guarantee untriggered to 2028; critique not reviewed [Kilic, 2024]	A, D
AU Blue Economy Strategy, Luanda [AU-IBAR, 2019]*[African Union, 2026]*	AU-IBAR; EU and member states; Oct 2019; Declaration 24 July 2026	Continental; 38 coastal states; five thematic annexes (unarchived)	USD 296 bn and 49 M jobs (2018) to 576 bn and 78 M (2063); IUU loss USD 10 bn/yr (projection; policy-fact)	Declaration adopted; report 2026 launched	A
Global policy instruments					
Global Cooling Pledge [UNEP, 2023]*[UN News, 2023]*[UNEP, 2024]*	UAE as COP28 host; Cool Coalition; Dec 2023	>60 countries at launch; >71 by COP29; 14 actions	−68% cooling emissions by 2050; access by 2030; +50% AC efficiency; no adaptation metric (policy-fact)	2030 strategy adopted; committee at COP30	D, A

continued overleaf

Table 3.1 (continued)

Effort	Lead, funder, dates	Scale and method	Headline result (grade)	Status	Rel.
UNSG Call to Action [UN Secretary-General, 2024]	UN Secretary-General with 10 UN entities; July 2024	Four pillars; 23 actions	Secondary figures: 489,000 deaths/yr; USD 863 bn (2022); 84% Arab-states workforce (policy-fact)	Issued	D, A
IPCC SR Cities [IPCC, 2024] *[IPCC news, 2024][IPCC, 2026a]*	IPCC; outline agreed 2 Aug 2024	5 chapters; 97 authors, 56 countries	Approval 15–19 Mar 2027; no 'digital twin' in outline (policy-fact)	SOD review closed July 2026	D, A

	Spatial scale of the product			
	planet, global	**continent or region**	**city**	**district or corridor**
Global policy instruments	UNSG Call to Action (2024) Global Cooling Pledge (COP28) IPCC Special Report on Cities (2027)		UNEP Beating the Heat (2021) Pledge city signatories (15)	
Finance		AU Blue Economy Strategy (2019) Luanda Declaration (2026) Seychelles blue bond (2018)		
Ocean and blue intelligence	Destination Earth (climate twin, 5–10 km)	European Digital Twin Ocean GMES & Africa, MarCOSIO Digital Earth Africa Global Fishing Watch Great Blue Wall (WIO) **ABIG (integration layer)**		
Tourism-climate indices			HCI:Urban and TCI (destination, daily)	**Outdoor Tourism Capacity** (hourly, cell-level, with heat-stress override)
Heat economics and policy	ILO 2019 (50 km cells) Lancet Countdown 2024	UAE midday break (statutory hours)	Arsht-Rock 12 cities (9 km^2) Singapore Heat Stress Advisory (9 sensors)	**Heat-Lost Activity Hours** (place-hour)
Urban heat and decision twins	Destination Earth (extremes twin, 4.4 km; regional 500–750 m)		Cooling Singapore DUCT Dubai Digital Twin Platform Green Riyadh Abu Dhabi LCZ (30 m to 3 m) Dubai Walk Master Plan	Phoenix cool pavement Makkah, Arafat cooled roads Paris OASIS schoolyards Future Loop (2 km) **Dubai Decision Twin** (800 m corridor + control)

Figure 3.1: Landscape of similar efforts by domain (rows) and spatial scale of the product (columns). Entries in green are the two programmes assessed in this report and the metrics they introduce. The corridor column holds no prior effort that ties measured activity-hours to a built intervention validated against a twin's prediction.

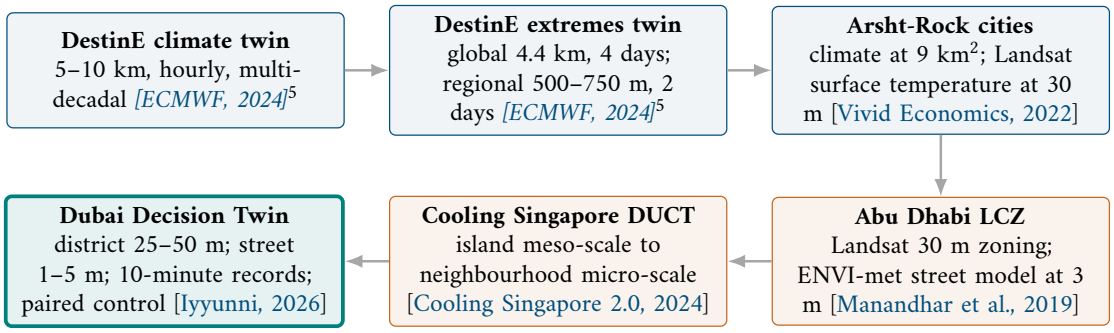


Figure 3.2: Resolution ladder from planetary twins to the Dubai street model. Each step is the finest published resolution of the effort named; the Dubai proposal sits two to three orders of magnitude below the continental twins in space and, at ten-minute cadence, below their hourly output in time.

What the landscape shows

Three facts stand out. First, no archived effort publishes activity-hours lost to heat at corridor scale, tied to a built intervention and validated against a twin's prediction; the corridor column of Figure 3.1 holds only interventions without twins (Phoenix, Makkah, OASIS, the Future Loop), and the one twin that runs what-if scenarios, Singapore's DUCT, sits a column to the left at city scale and tests no built intervention. Second, the two controlled field studies (Phoenix and OASIS) and the one measured UAE study (Abu Dhabi) agree on method (simultaneous treated-versus-control measurement, paired stations, radiant temperature as the pedestrian-relevant variable) and Phoenix shows that reflective pavement raises the pedestrian's radiant load in direct sun, so shade ranks first and cool pavement is conditional. Third, most headline numbers in the landscape are projections or self-reported outputs; the programmes that scaled (OASIS, the Seychelles bond, Global Fishing Watch in Gabon) did so after a validation step with a named operational user, which is what the Dubai month-12 gate and ABIG's capability-first sequencing are designed to reproduce.

3.9 Implications for the programmes

For the Dubai Decision Twin. The proposal should be written against the landscape, not beside it: it sits on the Dubai Digital Twin Platform and supplies the comfort layer the platform lacks *[Dubai Media Office, 2026]*[6]; it serves the demand Dubai Walk created (5,600 km of walkways by 2040 with no published method for deciding which need shade, where and when) through the Minimum Cooling Network *[Dezeen, 2024]*[8] *[Khaleej Times, 2025]*[7]; it extends Arsht-Rock's hours-lost construct from sector-hours over 9 km^2 to place-hours over metres and from workers to visitors [Vivid Economics, 2022]; and it gives the Holiday Climate Index the heat-stress override it lacks [Scott et al., 2016]. Four design consequences follow. The evaluation design is OASIS and Phoenix: paired case and control stations, native one-minute logging, a mobile traverse, Al Fahidi untouched for three summers, and the built canopy measured against a simultaneous unshaded reference rather than the previous summer [OASIS research team, 2024a, ASU and City of Phoenix, 2024]. The node specification must add a black-globe thermometer to at least the reference station and the paired nodes, or WBGT and UTCI cannot be computed and the comfort map will not be comparable with Singapore's bands or the ILO's thresholds [ILO, 2019]*[MSE Singapore, 2023]*[16]. The month-12 gate should state its error in the units of the Abu Dhabi precedent, roughly 1°C on bias-corrected hourly air temperature [Manandhar et al., 2019], and of the expected intervention effect. Cool pavement should enter the Year-2 candidate set only as a conditional measure or a complement under canopy [ASU and City of Phoenix, 2021, OASIS research team, 2024b]. The December 2026 site letters should include a written data-access confirmation from Dubai Municipality, and the decision briefs should be shaped for two vehicles that lack an adaptation metric: the Global Cooling Pledge's city

signatories and the IPCC's adaptation-indicator guidelines *[UNEP, 2024]*[34]*[IPCC, 2026a]*[36].

For ABIG. The concept paper's posture is confirmed and sharpened. The integration layer has three working precedents (the European Digital Twin Ocean, Digital Earth Africa's data cube and GMES & Africa's consortia) and a governance template in Destination Earth's separation of policy owner, modelling entity, platform and data federation [WEF and Digital Earth Africa, 2021] *[Copernicus Marine Service, 2024]*[31]*[MarCOSIO, 2025]*[22]*[ECMWF, 2024]*[5]. The first-year capability map should inventory MarCOSIO's service lines, MarCNoWA once documented, Digital Earth Africa's products, Global Fishing Watch's 24-country partnerships and the Great Blue Wall's seascapes as existing assets, and record interoperability with the EU DTO and Destination Earth as a requirement. The first use case should be illegal-fishing alerts with a fisheries authority already working with Global Fishing Watch, and the indicator set should adopt the Great Blue Wall's vocabulary while adding the attribution and denominators both lack: interdictions per targeted patrol, decisions improved per dollar *[Global Fishing Watch, 2026]*[27]*[IUCN, 2026]*[30]. The financing ladder has a real African rung in the Seychelles bond and the economic case a continental denominator in the AU-IBAR sector table [World Bank, 2025, AU-IBAR, 2019]; the Luanda Declaration is the political anchor, and the 93 per cent of Africa's workforce affected by excessive heat is the reason to keep heat in the climate-resilience domain [UN Secretary-General, 2024]*[African Union, 2026]*[32]. The landscape's two cautions are the absence of published benefit measurement for the GMES & Africa marine service and the unverified USD 2 billion headline of Digital Earth Africa: ABIG should be the programme that measures.

For both. The bridge between the proposals is already written into the ABIG paper: the UAE's 'tourism and climate applications' are to be 'demanding test environments for products intended for African markets' [Terhile et al., 2026]. The Dubai corridor is that test environment for a heat-adaptation metric no instrument in the landscape yet carries, and the same discipline (grade every number, pair every intervention with a control, put a named user at the gate) separates both programmes from the projections and announcements that fill most of Table 3.1. Chapter 4 asks what the measured studies prove; Chapter 5 turns the templates identified here into first steps.

٤ {4}

Feasibility Studies: Evidence, Results and Sources

Abstract

Feasibility studies do exist elsewhere. The harder questions are whether they carry real results, and where the studies and final reports may be found. Each archived document is opened in turn and recorded under identity, purpose, method, results with page references, evidence grade, limitations, and meaning for the Dubai testbed or for the Africa Blue Intelligence Grid. A seven-grade scheme separates what was measured from what was modelled, projected, announced or narrated, and a master evidence table condenses the reading. Three conclusions follow. No archived study publishes activity-hours lost to heat at corridor scale, tied to a built intervention and tested against a twin's prediction: that is the defensible gap. Cool pavement lowers surface temperature but raises the radiant load on a person standing on the coated surface, so shade ranks first and coatings are conditional. Most headline numbers are projections, and the few studies with controls supply the design for the Year-3 before/after campaign, specified here with paired control stations and radiant measurements. A closing section lists the sources that could not be archived and why; every source is registered with its grade and extent in Appendix A, and the address of each press release or programme page is in its footnote.

Asked whether feasibility studies exist elsewhere, a proposal writer is tempted to answer with a list of programme names. The stakeholders asked the harder question: do those studies have real results? Chapter 3 inventoried the landscape as line items; this chapter opens each archived document and asks what was measured, against what comparator, over what period, and what the authors themselves said their numbers do not show. The material is the source archive described in Chapter 1 and registered in Appendix A. Where the agreed baseline and a source disagreed, the source won: the Phoenix surface-temperature results belong to the Year-1 report and not to Phase II; the Phoenix radiant increase is a difference in degrees Fahrenheit and never a percentage; Paris OASIS published a controlled design and a measurement method but no archived effect size; the Makkah cooled-road figures are agency claims relayed by the press; and the Lancet Countdown's lost hours are an observation-based model estimate, not measured absence.

4.1 What counts as a feasibility study here

A feasibility study, in the sense used here, is a document that tests whether a proposed intervention, method or programme can be done, and what it delivers, by measuring, modelling or projecting an outcome. The archive holds three kinds: field or laboratory evaluations of a

built intervention against a comparator (the two Phoenix cool-pavement reports [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024] and the Paris OASIS schoolyard papers [OASIS research team, 2024a, OASIS research team, 2024b]); measurement and modelling studies with no intervention (the Abu Dhabi local climate zone study [Manandhar et al., 2019], the Cooling Singapore twin [Cooling Singapore 2.0, 2024] and the heat-economics models of Arsht-Rock, the International Labour Organization and the Lancet Countdown [Arsht-Rock and Vivid Economics, 2022, ILO, 2019, Romanello et al., 2024]); and programme reports that claim outcomes (the Makkah cooled roads, Green Riyadh, Digital Earth Africa, Global Fishing Watch in Gabon, the Great Blue Wall and the Seychelles blue bond), which are feasibility evidence only in so far as their claims can be graded [WEF and Digital Earth Africa, 2021, World Bank, 2025]*[Arab News, 2024]*[1] *[Saudipedia, 2025]*[2] *[Global Fishing Watch, 2026]*[3] *[IUCN, 2026]*[4]. The Holiday Climate Index paper and the UAE midday break are read alongside them because they define the incumbent metric and the incumbent policy against which the Dubai constructs are positioned [Scott et al., 2016]*[The National, 2026]*[5].

4.2 The evidence-grading scheme

Seven grades are used, in the exact words set out in the Preface and applied to line items in Chapter 3. This chapter applies them to individual quantities rather than to whole documents, because most documents carry several grades at once: the Phoenix Phase II report contains field measurements with a simultaneous control and, forty pages later, a city-scale dollar extrapolation from one October day. Table 4.1 states the test for each grade; Figure 4.1 gives the decision path.

Three rules of application matter. A secondary citation inherits the grade of its primary: the UN Secretary-General's Call to Action quotes the ILO's 2.2 per cent of working hours, so that quantity remains a projection wherever it appears [UN Secretary-General, 2024, ILO, 2019]. A self-reported programme output takes the case-report grade however large the number *[IUCN, 2026]*[4]. And a scenario result is modelled even when later quoted as an observation [Cooling Singapore 2.0, 2024].

Understanding: Evidence grade

TIER 1: IN PLAIN WORDS

A grade says how the number was produced, not how big it is. Numbers that come from somebody actually measuring what happened outrank numbers that come from a computer model, and both outrank numbers that come from a press release.

TIER 2: IN PROFESSIONAL TERMS

Grading separates observation from simulation, present from forecast, and independent evaluation from self-report. A programme may be impressive and still carry the weakest grade, because grade is a property of the method, not of the achievement.

[1] Arab News, *Road-cooling tech will provide Hajj pilgrims with respite from heat*, Saudi Arabia section, 2024. https://www.arabnews.com/node/2530731/saudi-arabia

[2] Saudipedia, *Green Riyadh Project*, encyclopaedia article (sources: Royal Commission for Riyadh City; Saudi Press Agency), 2025. https://saudipedia.com/en/green-riyadh-project

[3] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[4] International Union for Conservation of Nature, *Great Blue Wall report showcases five years of impact on ocean regeneration across the Western Indian Ocean*, IUCN news, June 2026. https://iucn.org/news/202606/great-blue-wall-report-showcases-five-years-impact-ocean-regeneration-across-western

[5] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

TIER 3: IN TECHNICAL TERMS

Each quantity is graded at its weakest supporting link. A same-day difference between a treated section and an adjacent untreated section is measured-with-controls; an instrument reading without a comparator is measured; a scenario output is modelled; a value for a future date or under an assumed adoption rate is a projection; a rule, date or target of the responsible authority is a policy-fact; an actor's account of an episode without a disclosed protocol is a case-report; a design, index or procedure reported without results is a method-reference.

AS A MATHEMATICAL MODEL

For a claim c with supporting sources $s \in \mathcal{S}(c)$ and grades $g(s)$, the reported grade is $g(c) = \min_{s \in \mathcal{S}(c)} g(s)$ under the ordering measured-with-controls $\succ$ measured $\succ$ modelled $\succ$ projection. The three categorical grades (policy-fact, case-report, method-reference) are carried as labels, not as ranks.

Table 4.1: The seven evidence grades and the test applied to each quantity.

Grade	Test	Example in this chapter
measured-with-controls	Observed with instruments against a simultaneous or paired comparator	Phoenix surface temperature, treated versus adjacent untreated street, same day [ASU and City of Phoenix, 2021]
measured	Observed with instruments, no intervention comparator	Abu Dhabi urban–rural air-temperature difference from a 26-station network [Manandhar et al., 2019]
modelled	Output of a simulation for a present or recent period	Singapore island-average cooling under a 2030 scenario [Cooling Singapore 2.0, 2024]
projection	Value for a future date, or under an assumed adoption or prevention rate	ILO 2030 working-hour losses [ILO, 2019]; Digital Earth Africa 2024 benefits [WEF and Digital Earth Africa, 2021]
policy-fact	A rule, date, budget or target of the responsible authority	UAE midday break hours and fines *[The National, 2026]*
case-report	An actor's or the press's account of an episode with no disclosed protocol	Makkah cooled-road temperature reduction *[Arab News, 2024]*; Gabon trawler seizure *[Global Fishing Watch, 2026]*
method-reference	A design, index or procedure published without outcome results	Holiday Climate Index specification [Scott et al., 2016]; OASIS microclimate protocol [OASIS research team, 2024a]

4.3 Urban-heat studies with field measurement

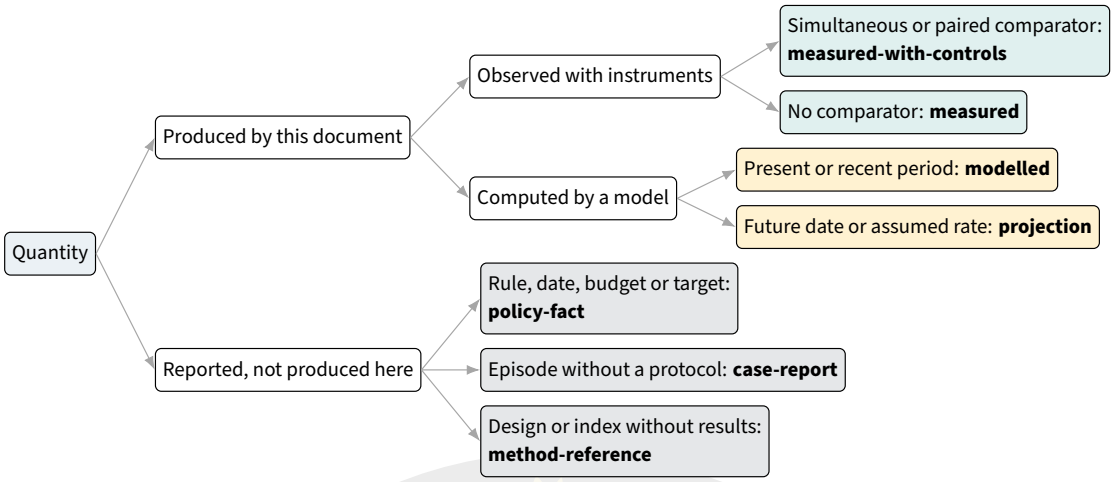


Figure 4.1: The grading decision path applied to every quantity cited in this chapter.

4.3.1 Phoenix cool pavement, Year 1

Identity. A joint study by Arizona State University and the City of Phoenix, reporting Year 1 of the Cool Pavement Pilot Program (project period 15 July 2020 to 14 July 2021; report September 2021) [ASU and City of Phoenix, 2021]. The archived file name and one bibliographic listing carry the year 2025; the document itself is dated September 2021, and this report cites it as the Year-1 report throughout.

Purpose and method. To evaluate the effectiveness, performance and community perception of a reflective coating applied to 36 miles of residential streets, one neighbourhood in each of eight council districts. Three one-day campaigns (18 August, 5 September and 20 September 2020) each ran four one-hour windows: pre-sunrise, solar noon, mid-afternoon and after sunset. Two platforms were used: a six-directional radiometer cart measuring mean radiant temperature on the street centre and on the adjacent footway, and a vehicle carrying shielded thermocouples at 2 m and a downward infrared radiometer, driven in two loops per window through the treated neighbourhood and through an *adjacent untreated reference neighbourhood*. Reflectance was tracked monthly against an untreated asphalt control road; twenty paired subsurface loggers recorded temperatures at half an inch and three inches over seven to ten months. Stated accuracies are ±0.2°C for air temperature and ±0.5°C for the infrared surface sensor (Appendix 4, p. 55).

Results. Surface temperature of the coated street averaged 12.0°F lower at noon and 10.5°F lower in the afternoon than adjacent asphalt concrete, with a range of 9–16°F and a maximum single-window difference of 16.0°F; at sunrise the reduction was 2.4°F (Findings; Table 6, p. 15). Two-metre air temperature fell by only 0.5°F on average after sunset (range −0.9 to +0.1°F) and 0.3°F during the day (range −1.2 to +0.2°F). Road-centre mean radiant temperature *rose* by approximately 5.5°F at noon and in the afternoon, ranging from 2.6 to 9.2°F, while the same measurements on the adjacent footway ranged from −5.2 to +1.9°F, showing no consistent penalty beside the coated road (Table 8, p. 19); after sunset it was 0.9–2.3°F cooler. Subsurface temperature was 4.8°F lower on average. Solar reflectance was 33–38 per cent at installation and 19–30 per cent ten months later across the eight neighbourhoods, against a consistent 12 per cent for the untreated control. Bitumen bond strength of the coating on asphalt cores was 34 psi against 230–276 psi for the reference binders, on a test the authors state was not designed for surface treatments (pp. 21–22). Figure 4.2 plots the mean differences together, which is the only honest way to read them; converted at 1.8°F per °C, the noon surface reduction is about 6.7°C and the radiant increase about 3.1°C (this report's conversion, indicative).

Evidence grade and limitations. Measured-with-controls for the field temperature results:

every comparison is same-day, same-instrument, treated against adjacent untreated, with paired subsurface cores and a control road for reflectance. The helicopter infrared overflights are a case-report and the 3–13 per cent rutting and fatigue improvement is modelled. Comparisons are against aged asphalt or chip-seal rather than a fresh dark sealant, which the authors call conservative; air-temperature differences are frequently smaller than the reported standard deviations and individual windows reverse sign; no confidence intervals or significance tests accompany the field differences; and nothing in the report measures a comfort index, pedestrian activity or dwell time. Critically, *no service life, re-application interval or unit cost is given anywhere*, and the life-cycle cost analysis is explicitly deferred because the coating's service life was unknown.

What it means for Dubai and ABIG. This is the strongest intervention evidence in the archive and it points the Minimum Cooling Network away from coatings on a walked surface. It sets the instrument bar for the twenty microclimate nodes and shows that the month-12 gate condition 'model within stated error' must state that error in the same units as the effect Year 3 hopes to resolve. For ABIG, the sequence (36 miles deployed in 2020, evaluation published in 2021 and again in 2024) documents what evaluation-after-deployment costs, and is the inverse of ABIG's principle that hardware follows evidence.

Understanding: Mean radiant temperature

TIER 1: IN PLAIN WORDS

It is the heat a person feels arriving from everything around them (sun overhead, sky, walls and the ground underfoot), as opposed to the temperature of the air. On a bright day it is what makes standing in the sun unbearable while a thermometer in the shade reads the same.

TIER 2: IN PROFESSIONAL TERMS

Mean radiant temperature is the uniform surrounding temperature that would give a person the same net radiant heat exchange as the actual, uneven surroundings. In sunlit arid streets it dominates outdoor thermal comfort, which is why a surface that reflects more sunlight can cool the pavement and still heat the pedestrian.

TIER 3: IN TECHNICAL TERMS

It is derived from six-directional short- and long-wave irradiance measured by a radiometer array, and enters comfort indices such as the Universal Thermal Climate Index as the radiative forcing term. A reflective coating lowers upwelling long-wave emission but raises upwelling short-wave reflection, with a net daytime increase at the surface.

AS A MATHEMATICAL MODEL

By definition, $\sigma T_{mrt}^4 = \sum_i F_i L_i + (a_k/\varepsilon_p) \sum_i F_i K_i$, where F_i are the angular weights of the six measured directions, K_i and L_i the short- and long-wave irradiances from each, and a_k/ε_p the ratio of the body's short-wave absorptivity to its long-wave emissivity. The archived Phoenix campaigns measure all six directions directly with a radiometer cart [ASU and City of Phoenix, 2021].

4.3.2 Phoenix cool pavement, Phase II

Identity and purpose. The Phase II report (October 2024) covers 2022 field data and extends Year 1 with a vertical air-temperature gradient, a same-day radiant comparison, a water and energy extrapolation, a pedestrian energy-balance assessment, a side-by-side test bed of two reflective products against a conventional dark sealant, and laboratory testing of recycling, wear, thermal properties and reflectance [ASU and City of Phoenix, 2024].

Method and results. The intended before/after design failed: the pre-coating and post-coating days differed in air temperature, the pre-coating data were withheld as misleading, and a *same-day*

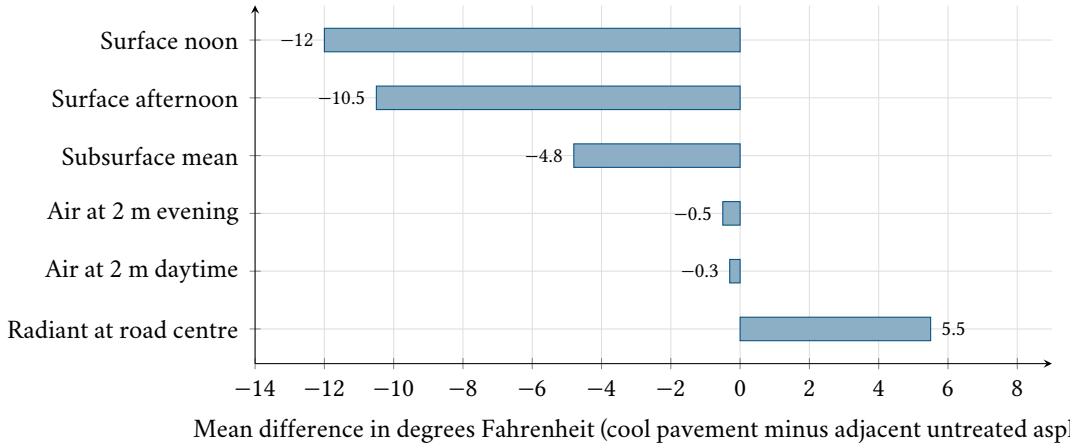


Figure 4.2: Phoenix Year-1 mean differences between cool pavement and adjacent untreated asphalt. The surface is much cooler; the air is barely cooler; the radiant load on a person standing on the coating is higher [ASU and City of Phoenix, 2021].

comparison against a stationary reference cart on untreated asphalt was substituted (p. 3). That substitution is itself a finding for Year 3. Road-centre mean radiant temperature over the coating was 5.8°F higher at noon and 4.5°F higher in the afternoon, and 0.4°F lower before sunrise and after sunset (Table 1, p. 3). Mean air-temperature differences were −0.13°F on the mobile transect and −0.9°F at the paired stationary arrays. A human energy-balance model gave 12 and 13 W m^{-2} of additional required heat loss for a person standing on the coating at noon and at 16:00 (p. 14). Laboratory solar reflectance was 38.62 and 30.29 per cent for the two coatings against 3.57 per cent for the dark sealant control (p. 34); over nine months of truck traffic the control lightened by 21 per cent while the coatings darkened by 17 and 13 per cent (p. 22).

Evidence grade and limitations. Measured-with-controls for the field and test-bed sections, measured for the laboratory work, modelled for the water and energy extrapolation. All radiant readings are at the road centre and are, in the authors' words, not representative of a pedestrian on an adjacent footway, and each residential result rests on a single measurement day. The water and energy figures (0.55°F city-wide cooling, ten to twenty million dollars a year, twenty-eight million gallons a month) are extrapolations from one October day through an assumed solar-scaling profile which the authors qualify as carrying a high level of uncertainty; they are specific to roughly 600,000 Phoenix households and do not transfer to Dubai.

What it means for Dubai and ABIG. Phase II supplies the comparison against a fresh dark sealant that Year 1 lacked, and the methodological warning that governs the Year-3 design: a before/after comparison without a simultaneous control is hostage to the weather. Its energy method (measured air-temperature difference, perturbed weather file, building archetypes, city total) is a reusable template for coupling the Dubai twin to the DEWA cooling load, though none of its dollar values transfers.

4.3.3 Paris OASIS

Identity and purpose. The OASIS programme de-sealed and greened ten pilot Paris schoolyards (four kindergartens, four elementary and two middle schools), as neighbourhood cool islands, with an instrumented evaluation. The archive holds two Urban Innovative Actions case-study reports, a Climate-ADAPT case study, a conference paper on the microclimate measurement protocol and a laboratory paper on paving materials [Urban Innovative Actions,

2023, OASIS research team, 2024a, OASIS research team, 2024b][EEA Climate-ADAPT, 2023][6].

Method. Implementation ran 2019–2021 within an ERDF funding window of 2018–2022, with measurements over three years. The microclimate paper specifies a before-after-control-impact design: fixed stations logging at one-minute resolution, a control station 200 m outside the Maryse Hilsz site, mobile traverses at 1.5 m with 15-second sampling and 10–20 minutes of dwell per point for black-globe stabilisation, a traverse window of 12:00 to 16:00, and a campaign-day filter of at most three oktas of cloud with daily maximum above 25°C and minimum above 16°C. Stated instrument uncertainties are 0.1°C for air temperature, 1.5 per cent for relative humidity, 0.15°C for black-globe temperature and 5 per cent daily for the net radiometer. The paving paper is a climate-chamber comparison of five surface structures over a single 24-hour cycle (8 hours day at 35°C, 16 hours night at 25°C) with conventional asphalt as the reference.

Results. The archived microclimate paper reports method-stability quantities only: the derived comfort offset was constant within 2°C between 10:00 and 15:00 in the 2019 campaign, and parameters fluctuated by less than 1°C within a traverse period of two hours or less. *No measured before/after cooling effect (no change in air temperature, mean radiant temperature or thermal-comfort index for a renovated yard) appears in any archived OASIS document.* The laboratory paper does give surface temperatures after the 8-hour day phase: conventional asphalt reached 57°C, wood chips plateaued near 55°C and watered grass reached about 40°C. Programme facts are firm: [4,995,793.16: 20,982,331.27:5,828,425.35](EUR:AED)(AED:USD) of ERDF contribution to the ten pilots; a city programme cost of about [9,000,000.00:37,800,000.00:10,500,000.00](EUR:AED)(AED:USD) a year for around twenty-five schoolyards including surveillance and maintenance; an expected lifetime above fifty years; and 73 hectares of asphalted, paved or cemented school surface in a city with about 6 m^2 of green space per central-Paris resident.

Evidence grade and limitations. The two UIA reports are case-reports; the Climate-ADAPT study is a case-report with policy-fact content; the microclimate paper is a method-reference; the paving paper is measured, in the laboratory. The paired control station is explicit only for the Maryse Hilsz site; elsewhere the paired design is implied rather than stated. The laboratory results are single-cycle, dry-sample chamber measurements, and the authors state that the total pedestrian effect would require site-specific simulation. Per-schoolyard costs (roughly [360,000.00:1,512,000.00:420,000.00](EUR:AED)(AED:USD) from the programme figure, or roughly [500,000.00:2,100,000.00:583,333.33](EUR:AED)(AED:USD) from the ERDF share) are derived here from different cost bases and are not comparable with each other.

What it means for Dubai and ABIG. OASIS is the design template, not the effect-size source. It supplies the control-station protocol, the campaign-day filter, the dwell time a black globe needs to stabilise and the instrument uncertainties that the Year-3 campaign should match or better. The honest statement for the proposal is that a controlled before/after evaluation was designed, instrumented and published as method, and that effect sizes are not in the archived documents.

4.3.4 The Abu Dhabi local climate zone study

Identity and purpose. A peer-reviewed study of local climate zones on Abu Dhabi island, combining a dense in-situ network with satellite classification and microscale simulation, published open access and carrying no external funding [Manandhar et al., 2019].

Method. Twenty-six urban weather stations were mounted on lamp posts with municipal approval, measuring air temperature at 3, 4, 5.5, 7 and 8.5 m at one-minute resolution, with a suburban field station as the rural reference. Four seasonal windows (June, September and December 2017 and March 2018) were each split into a calibration week and a held-out validation

[6] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

week. Landsat-8 imagery at 30 m multispectral and 100 m thermal resolution supported a nine-class, Abu Dhabi-specific zone scheme; the street-scale model ran in ENVI-met 4.0 at 3 m per pixel on a 100 by 100 node grid.

Results. Measured daily-mean urban-minus-rural differences across five districts and four seasons span −0.23 to +3.21°C: September −0.23 to +0.29°C, June +0.06 to +0.68°C, December +2.44 to +3.21°C and March +1.25 to +1.85°C (Table 2, p. 10). The five-district means are +0.07, +0.36, +2.72 and +1.51°C for those seasons (derived by this report from Table 2, arithmetic verified). The highest measured urban daily mean was 36.09°C, in June 2017, in the mixed-use traditional typology that covers most of the island. Simulation error against the in-situ series fell from 1.33–2.71°C root-mean-square uncorrected to 0.27–1.62°C after bias correction, and a model calibrated on one district predicted a second district of the same class within 0.24–0.68°C.

Evidence grade and limitations. Measured for the urban heat island values; modelled for the simulated district temperatures and the zone map. It is not measured-with-controls: no intervention was tested and no uncertainty statistics or significance tests are reported. Air temperature is the only quantity: no thermal-comfort index, no radiant measurement, no cost, no sensor-uptime figure. The near-zero summer daily means are an artefact of daily averaging, which hides the diurnal inversion, and must not be read as an absence of a summer heat problem. The paper's own statement that simulated values exceed measured ones by 0.88°C is contradicted by its Table 2 and is not cited here, and its zone numbers are local composites that do not correspond to the standard scheme [Stewart and Oke, 2012]. The companion Dubai study is not archived and was not reviewed [Bande et al., 2020].

What it means for Dubai and ABIG. This is the nearest regional precedent for the Dubai architecture and it validates the nested resolution the proposal adopts: satellite classification at tens of metres, street simulation at metres, both anchored to a reference station and validated on held-out data. The reduction of error from roughly 2.2 to roughly 0.7°C (derived from Table 4) is a defensible order of magnitude for the month-12 gate threshold, provided the threshold is stated in the same units as the intervention effect.

4.3.5 The Makkah and Arafat cooled roads

Identity and purpose. Two press reports relaying statements by the Saudi Roads General Authority about a heat-reducing road surface deployed on pilgrim routes, plus a third press report on the wider set of Hajj cooling measures *[Arab News, 2024]*[1] *[Gulf News, 2025]*[7] *[Saudi Gazette, 2024]*[8]. Neither of the first two is a study; no report, dataset or protocol is cited by either.

Results as claimed. The authority's spokesman gives a reduction in asphalt surface temperature of 12 to 15°C against untreated roads, and states that untreated surfaces retain heat up to 70°C after sunset *[Arab News, 2024]*[1]. The second report gives around 12°C, an increase of 30–40 per cent in reflected sunlight 'in the mornings', more than 84,000 m^2 of heat-reducing surface across the Arafat plains for Hajj 2025, an 82 per cent year-on-year expansion, and a 33 per cent expansion of the rubberised-walkway initiative *[Gulf News, 2025]*[7]; about 25,000 m^2 were treated near the Namira Mosque in 2024. The third adds misting columns, mist-covered walkways and the Neem trees of the Arafat plain, and relays a hospital study of forty years of data reporting a 74.6 per cent fall in heat-stroke cases and a 47.6 per cent fall in mortality over four decades *[Saudi Gazette, 2024]*[8].

Evidence grade and limitations. Case-report for all of it. The temperature effect is presented as measured ('data shows') but no instrument, site, period, sample or comparison design is disclosed,

[7] Gulf News, *Hajj 2025: Saudi Arabia expands cooled, rubberised roads*. https://gulfnews.com/world/gulf/saudi/hajj-2025-saudi-arabia-expands-cooled-rubberised-roads-1.500139040

[8] Saudi Gazette, *Hajj 2024: Saudi Arabia implements advanced cooling techniques in holy sites amid scorching heat*. https://saudigazette.com.sa/article/643620/SAUDI-ARABIA/Hajj-2024-Saudi-Arabia-implements-advanced-cooling-techniques-in-holy-sites-amid-scorching-heat

and no report is cited; the convergence of two outlets on about 12°C shows only that this is the authority's consistent public figure. Surface temperature is the sole quantity reported: there is no air temperature, no radiant temperature, no wet bulb globe temperature or comfort index, no cost, no durability and no re-application interval. The third report's 'about 20 degrees Celsius' for the white coating conflicts with the authority's 12 to 15°C and is not adopted here. The hospital study's four-decade improvement is confounded by every other change in Hajj management and cannot be attributed to the road surface, and the two area figures are not comparable, one being a single site and the other a regional total.

What it means for Dubai and ABIG. Makkah is the largest recurring heat-management operation in the world and the strongest argument that the Gulf is the right laboratory, but it is not evidence that cool pavement improves pedestrian comfort, because it reports the quantity Phoenix reports (surface temperature) and none of the quantity Phoenix shows moving the other way (radiant load). It also shows what Dubai could supply a regional partner: a published protocol for a claim an authority is already making.

4.3.6 Cooling Singapore and the Digital Urban Climate Twin

Identity and purpose. Cooling Singapore, launched in 2017 with national research funding, built a Digital Urban Climate Twin to test heat-mitigation strategies at city scale. The archive holds a conference poster, a supercomputing centre newsletter and a policy-centre box story, together with the national heat-stress advisory that the work sits beside [Cooling Singapore 2.0, 2024, NSCC Singapore, 2024, NEA Singapore, 2023]*[Centre for Liveable Cities, 2024]*[9] *[MSE Singapore, 2023]*[10].

Results. An urban heat island of up to 7°C is quoted as a maximum differential across the island, without a stated method or period, and projected additional warming of 1.4–4.6°C by 2100 is reported without a scenario or baseline *[Centre for Liveable Cities, 2024]*[9]. The poster's own modelled result is a whole-island average cooling of up to 0.13°C in 2 m air temperature under an approximate Green Plan 2030 scenario on one heat-wave day, peaking between 06:00 and 08:00, with effects that vary by district [Cooling Singapore 2.0, 2024]. More than twenty researchers developed the twin over four years, and the platform achieves up to eighty hours of climate-model simulation within twenty-four hours on the national supercomputer [NSCC Singapore, 2024]. The advisory divides risk by wet bulb globe temperature (low below 31°C, moderate to below 33°C, high at or above 33°C), computed as $0.7\,T_{nwb} + 0.2\,T_g + 0.1\,T_d$ from a nine-sensor network [NEA Singapore, 2023] *[MSE Singapore, 2023]*[10].

Evidence grade and limitations. Mixed: modelled for the 0.13°C scenario result, case-report for the programme descriptions, policy-fact for the advisory; nothing in the archived Singapore material reaches measured-with-controls. No model validation error, sensor specification, per-intervention pedestrian-scale effect size or cost figure exists in any archived Singapore document, so none of them can benchmark the Dubai gate threshold or the equipment line. The 2026 journal paper on the twin could not be archived and was not reviewed [Aydt et al., 2026]. The Singapore bands are population guidance, not occupational limits under the international standard [ISO, 2017b]. The acronym OTC collides: in the Singapore literature it means outdoor thermal comfort, whereas in this report it means Outdoor Tourism Capacity.

What it means for Dubai and ABIG. Singapore proves the institutional feasibility of a city-scale climate twin used by agencies, and it supplies the comparator index. It also exposes a concrete equipment gap: the Dubai node specification of air temperature, humidity, wind, radiation

[9] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

[10] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

and surface temperature omits the black-globe and natural-wet-bulb instruments that a wet bulb globe temperature requires, so comparability with Singapore and with the labour literature must be bought explicitly.

4.3.7 Green Riyadh

Identity, results and grade. Green Riyadh was launched on 19 March 2019 and aims to plant more than 7.5 million trees by 2030, raising green space from 1.5 to 9 per cent of the city's area, with 1,350 km of new irrigation networks and seventy-two selected species *[RCRC, 2025]*[11]. The claimed benefits are a city-level reduction of 1.5–2°C and 8–15°C in intensively afforested locations. The archived text is an encyclopaedia article naming the commissioning authority as its source, the programme's own page having proved uncapturable *[Saudipedia, 2025]*[2]. Grade: projection for the benefit figures, policy-fact for the scope items. The 8–15°C figure does not say whether it refers to air or surface temperature, and the 650 GWh energy saving is attributed by the source to green-building techniques rather than trees, so neither is evidence of tree cooling. No measured outcome, cost or progress figure appears in the archive: Green Riyadh is the scale ambition against which a corridor demonstration is positioned, not a source of effect sizes.

4.4 Heat-economics and heat-policy studies

4.4.1 Arsht-Rock, Hot Cities, Chilled Economies

Identity and method. A twelve-city assessment of heat-related worker-productivity loss, with a methodology document dated July 2022 [Arsht-Rock and Vivid Economics, 2022, Vivid Economics, 2022]. The chain is a downscaled multi-model climate ensemble at 9 km^2, a simplified wet bulb globe temperature, exposure-response damage functions, sector labour shares and a Cobb-Douglas production structure. Baseline climate is 1986–2005 and baseline economy 2019; the 2050 case is SSP3-7.0, about 2°C of warming. A twelve-hour working day is modelled in three four-hour blocks, the mid-point value being used above a daily maximum of 31°C to mimic hour-shifting. Abu Dhabi is one of the twelve study cities and the only one given adaptive hour-shifted workability.

Results. Losses across the twelve cities are put at USD 44 billion a year in 2020 and USD 84 billion by 2050 without emissions reduction, ranging from 0.1 to 8.3 per cent of city output (Dhaka 8.3, Bangkok 4.9, New Delhi 4.2, Freetown 3.5 per cent) against a comparator of 0.3 per cent of world output for all natural-catastrophe losses in 2021. The archived city pages give Los Angeles almost USD 5 billion a year, nearly half its USD 11.2 billion annual budget, rising to USD 11 billion by 2050 *[Arsht-Rock, 2022b]*[12]; Bangkok USD 8.6 billion, 5 per cent of output, rising to USD 15.6 billion and 6 per cent, with outdoor workers losing as much as 40 per cent of output *[Arsht-Rock, 2022a]*[13]; and New Delhi 4 per cent rising to 5 per cent, with construction producing 9 per cent of output but bearing a third of the losses because more than 60 per cent of its hours are outdoors *[Arsht-Rock, 2022c]*[14].

Evidence grade and limitations. Modelled for present-day values, projection for 2050;

[11] Royal Commission for Riyadh City, *Green Riyadh Program*, project page, 2025. https://www.rcrc.gov.sa/en/projects/green-riyadh-project/

[12] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Los Angeles, United States*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-los-angeles/

[13] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Bangkok, Thailand*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-bangkok/

[14] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: New Delhi, India*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-new-delhi/

the methodology document is a method-reference and reports no results. No productivity was measured and there are no controls; the model covers employed workers only, in normal years, with a simplified heat index that omits wind and radiation, and it is not validated. No Abu Dhabi loss figure exists in the archive, because that city page does not exist and the main report could not be captured. The framework cannot therefore be cited as evidence for Outdoor Tourism Capacity; it is the macroeconomic frame and the labour-hours analogue of Heat-Lost Activity Hours.

4.4.2 ILO, Working on a Warmer Planet

Identity, method and results. The ILO's global assessment combines gridded climate data at 67,420 cells of half a degree, exposure-response functions by metabolic class (200 W services, 300 W industry, 400 W agriculture and construction) and national employment data [ILO, 2019]. Working-hour loss rises from 1.4 per cent globally in 1995 to a projected 2.2 per cent in 2030 (about 80 million full-time-equivalent jobs and USD 2.40 trillion at purchasing-power parity) and to 3.8 per cent if agricultural and construction work is done in the sun rather than in shade. The exposure-response thresholds are the operative content for this report: productivity loss begins at 24–26°C wet bulb globe temperature, reaches 50 per cent at 33–34°C for a 300 W worker and becomes total at 38°C, against an annual denominator of 4,320 potential daylight working hours per worker. For the Arab States the loss rises from 0.4 to 1.0 per cent of hours, roughly 90,000 to 618,000 full-time equivalents, construction accounting for 40 per cent of the 2030 regional loss; Bahrain and the United Arab Emirates are each projected to lose more than 2 per cent of GDP by 2030.

Evidence grade and limitations. Projection throughout: even the 1995 values are hindcast estimates, and no confidence intervals are given. The central estimates assume all outdoor work is done in the shade and exclude both the urban heat island and heatwaves, which the ILO itself says means highly urbanised countries are under-estimated. The country row for the United Arab Emirates was reconstructed from a reflowed table and sum-checked against the printed regional totals, so the figures taken from it, including the projected loss of more than 2 per cent of GDP by 2030, are reported here as indicative.

What it means. The ILO method is the compatible skeleton for Heat-Lost Activity Hours: an hourly heat index, a metabolic class, an exposure-response curve and an explicit annual denominator. Adopting the same thresholds and denominator makes the Dubai metric comparable with the largest existing series, while the Dubai measurement supplies what the ILO excludes: the urban heat island, at street resolution, for a place people choose to visit rather than are employed to occupy.

4.4.3 Lancet Countdown 2024

The 2024 Lancet Countdown reports 512 billion potential work hours lost to heat globally in 2023, 49 per cent above the 1990–1999 average, with a potential income loss of USD 835 billion, or 0.82 per cent of gross world product; low and medium human-development groups lose 7.6 and 4.4 per cent of GDP, and agriculture accounts for 63 per cent of lost hours and 37 per cent of lost income, construction for 31 per cent of income [Romanello et al., 2024]. Exposure rose to 1,512 hours a year of at least moderate heat-stress risk for light outdoor activity, 328 hours above the 1990s baseline. Evidence grade: modelled, an observation-driven hindcast of *potential* hours and income lost, computed from reanalysis weather through exposure-response models and wage data, not measured absence or payroll. Group values for the very-high-development category must not be presented as UAE values, and the archived file is an accepted manuscript with typographical errors, so volume and page numbers must come from the journal record. The series is the benchmark against which a Dubai place-level metric would be reported, and it uses wet bulb globe temperature: a further argument for instrumenting the corridor to compute it.

4.4.4 The incumbent policy and the incumbent index

Two further documents are read as evidence because they define what already exists. The UAE midday break prohibits work under direct sunlight and in open spaces between 12:30 and 15:00 from 15 June to 15 September, under an Occupational Heat Stress Prevention Policy, with fines of [5,000.00:1,388.89](AED:USD) per breach per worker to a maximum of [50,000.00:13,888.89] (AED:USD); a national newspaper dates the rule's introduction to 2004 *[The National, 2026]*[5], while the labour ministry's own 2023 announcement described that year as the nineteenth consecutive year of the ban *[MoHRE, 2023]*[15]. Grade: policy-fact. Multiplying 2.5 hours by the 93 calendar days of the period gives about 232.5 hours: this report's own arithmetic, and only an indicative upper bound, since the ban covers direct-sun work, the calendar days include rest days and the rule reaches regulated workers only. The point is the precedent rather than the number: a government already encodes hours lost to heat as an instrument of policy.

The Holiday Climate Index rates five variables (daily maximum temperature, mean relative humidity, cloud cover, daily precipitation and wind speed) in four weighted sub-indices, with a score below 40 defined as unacceptable to the majority of tourists [Scott et al., 2016]. Grade: method-reference, its European inter-comparison results graded modelled. The thermal rating assigns a score of 2 to an effective temperature of 37–39°C, but because only rain and wind can override the other components, a dry, calm, clear Gulf afternoon at that temperature still scores in the 'Good' to 'Acceptable' band: 64, or 56 with a thermal rating of zero, on this report's own worked example from the paper's equation and tables. That is the argument for an explicit heat-stress override inside Outdoor Tourism Capacity and for reporting a Universal Thermal Climate Index value beside the tourism score [Jendritzky et al., 2012].

4.5 Blue-economy and Earth-observation studies

Digital Earth Africa and the World Economic Forum. A January-2021 assessment values the socioeconomic benefit of the continental open data cube at higher than USD 2 billion a year by 2024 (about USD 500 million from accelerated growth of the Earth-observation industry, at least USD 900 million from mining regulation and at least USD 900 million in agriculture) over a land area exceeding 30 million km^2 [WEF and Digital Earth Africa, 2021]. The agricultural water component pairs 17.6 billion m^3 a year of saved abstraction with USD 880 million at an assumed 10 per cent adoption rate; the assessment's own text mis-pairs the full-adoption volume with the 10 per cent value, and only the consistent pairing is used here. Grade: projection, ex-ante 2024 scenario estimates built from single-site analogues scaled continent-wide, with assumed adoption and prevention rates and a twelve-observation regression whose coefficient was overridden by judgement. There are no realised benefits, no cost side and therefore no benefit-cost ratio; the press release's 'up to USD 2 billion' is the weaker wording of the same estimate *[Digital Earth Africa, 2021]*[16]. For ABIG the lesson is methodological: an ex-ante benefit case built this way is a fund-raising instrument, not evidence.

Global Fishing Watch in Gabon. After a patrol-planning workshop that trained twenty-five personnel from the fisheries directorate, the parks agency and the navy, satellite intelligence identified five industrial trawlers about ten kilometres inside Gabonese waters; three were

[15] Ministry of Human Resources and Emiratisation, UAE, *UAE to implement midday work break from 15 June to 15 September for 19th consecutive year*, MoHRE media centre, 2023. https://mohre.gov.ae/en/media-center/news/1/6/2023/uae-to-implement-midday-work-break-from-15-june-to-15-september-for-19th-consecutive-year

[16] Digital Earth Africa, *Digital Earth Africa has the potential to contribute billions to the African economy using Earth observations*, news, 2021. https://digitalearthafrica.org/en_za/digital-earth-africa-has-the-potential-to-contribute-billions-to-the-african-economy-using-earth-observations/

intercepted on 18 July 2026 without valid licences and escorted 200 km to Libreville *[Global Fishing Watch, 2026]*[3]. Grade: case-report, a single enforcement episode narrated by the intelligence provider, with no counterfactual and no denominator; the sixteen vessels seized over the decade belong to a separate joint operation. The macro figures on the companion page (USD 24 billion a year and about twelve million jobs from African fisheries and aquaculture, and illegal, unreported and unregulated fishing at an estimated 40 per cent of catch) are uncited secondary estimates *[Global Fishing Watch, 2025a]*[17]. It is nonetheless the clearest available demonstration of the chain ABIG proposes: data, analysis, a named operational user, a decision, an action. The missing element (what the seizure changed) is the measurement ABIG should commit to producing.

The Great Blue Wall. An impact report covering 2021–2025 was launched on 17 June 2026 at the eleventh Our Ocean Conference in Mombasa. Partners collectively contributed to supporting more than 83,000 km^2 of marine and coastal areas, more than 175,000 hectares protected, restored *or committed to be protected*, more than 112,000 livelihoods strengthened and more than USD 80 million mobilised *[IUCN, 2026]*[4]. Grade: case-report, partner self-reported, contribution-based aggregates relayed by press release, with no baselines and no attribution method. The impact report itself could not be archived [Great Blue Wall Initiative, 2026], so every figure here is relayed at one remove.

The Seychelles blue bond. The world's first sovereign blue bond was announced on 29 October 2018: USD 15 million over a ten-year maturity, with a USD 5 million partial credit guarantee from the World Bank and USD 5 million of concessional financing from the Global Environment Facility, placed with three investors *[World Bank, 2018]*[18]. The World Bank case study of 27 June 2025 attributes at least 2.0 per cent a year of borrowing-cost reduction to the guarantee and more than 3 per cent to the concessional loan, and reports cumulative global blue-bond issuance of USD 15.25 billion by mid-June 2025 [World Bank, 2025]. Grade: policy-fact for the transaction terms, the market data being third-party. The published critique could not be archived and was not reviewed [Kilic, 2024], so no claim about the bond's effect on sovereign debt appears here.

The AU Blue Economy Strategy and the Luanda Declaration. The strategy, endorsed in October 2019, records 38 coastal states and about 13 million km^2 of collective ocean territory, USD 10 billion a year lost to illegal fishing in African exclusive economic zones, a 2018 baseline of USD 296 billion and 49 million jobs, USD 21 billion of fisheries gross value added and USD 80 billion from coastal tourism supporting 24 million jobs, with projections of USD 405 billion and 57 million jobs by 2030 [AU-IBAR, 2019]. Grade: projection for the forward values, whose method sits in unarchived annexes; only absolute values are cited, because the internal percentage shares are arithmetically inconsistent. The Luanda Declaration was adopted on 24 July 2026 at the third Africa Blue Economy Week *[African Union, 2026]*[19], and its scale figures reach this report only through the ABIG concept paper quoting it [Terhile et al., 2026]. No cost, uptake or benefit measurement exists anywhere in the archive for ABIG or for the regional marine services of GMES & Africa *[MarCOSIO, 2025]*[20].

[17] Global Fishing Watch, *Rising pressures on Africa's fisheries and marine resources*, 2025. https://globalfishingwatch.org/rising-pressures-on-africas-fisheries-and-marine-resources/

[18] World Bank, *Seychelles launches World's First Sovereign Blue Bond*, press release, 29 October 2018. https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond

[19] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

[20] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

4.6 Method references used as evidence about evidence

Several further documents are graded method-reference and are used in Chapters 7 and 9 rather than here, but they belong in the evidence register because they discipline the reading. The megaproject literature measures outcomes rather than prescribing them: nine of ten megaprojects overrun, and rail projects average 44.7 per cent cost overrun alongside a 51.4 per cent demand shortfall [Flyvbjerg, 2014]; the Olympic Games average 156 per cent overrun, median 90 per cent, none on or under budget [Flyvbjerg et al., 2016]. These concern projects of hundreds of millions to billions of US dollars and are reference-class context, not calibrated uplifts for a grant of [1,400,000.00:388,888.89](AED:USD) [Cantarelli et al., 2025]. The federal cost-estimating guide states that its risk techniques do not capture catastrophic events, which belong in force-majeure clauses rather than contingency percentages [GAO, 2020]; the life-cycle costing standard is archived only as a preview through clause 4.2.2, so no service life or discount rate may be cited to it [ISO, 2017a]. The spiral literature supplies the anchor points and incremental-commitment logic the month-12 gate implements [Boehm and Hansen, 2000], and the WinWin case study the sustainment finding: of five applications built in that programme's first year, only the one with a sustaining budget, people and facilities survived [Boehm et al., 1998].

4.7 Master evidence table

Table 4.2 gathers the reading in one place: what was tested, what came out, how strong it is, and where to find it. Every source is registered with its extent and evidence grade in Appendix A; the address of a press release or programme page is in its footnote.

Table 4.2: Master evidence table: archived feasibility documents, what each tested, the headline result, the evidence grade and the citation.

Study	What was tested	Result	Grade	Source
Phoenix Year 1 (2020 data)	Reflective coating on residential streets	Surface −12.0°F noon, −10.5°F afternoon (range 9–16°F); subsurface −4.8°F; air −0.3°F day, −0.5°F evening; road-centre radiant +5.5°F (2.6–9.2); reflectance 33–38% to 19–30% in 10 months	measured-with-controls	[ASU and City of Phoenix, 2021]
Phoenix Phase II (2022 data)	Same coating; two products against a dark sealant	Road-centre radiant +5.8°F noon, +4.5°F afternoon; +12–13 W m^{-2} required heat loss; lab reflectance 38.6/30.3% against 3.6% control	measured-with-controls (field); modelled (water, energy)	[ASU and City of Phoenix, 2024]
Paris OASIS	Ten de-sealed, greened schoolyards; paired control station	Controlled design instrumented and method published; no effect size in the archive; lifetime >50 years; [4,995,793.16: 20,982,331.27:5,828,425.35] (EUR:AED)(AED:USD) ERDF for ten pilots	case-report; method-reference	[Urban Innovative Actions, 2023, OASIS research team, 2024a]*[EEA Climate-ADAPT, 2023]*
OASIS paving laboratory	Five paving structures, 24-hour chamber cycle	Conventional asphalt 57°C surface; wood chips ≈55°C; watered grass ≈40°C	measured (laboratory)	[OASIS research team, 2024b]
Abu Dhabi local climate zones	26-station network; satellite classification; street simulation	Daily-mean urban–rural difference −0.23 to +3.21°C by district and season; simulation error 1.33–2.71 to 0.27–1.62°C after correction	measured; modelled (simulation)	[Manandhar et al., 2019]
Makkah and Arafat cooled roads	Reflective and rubberised pavement on pilgrim routes	Surface 12–15°C lower (agency claim); 30–40% more reflection in the mornings; >84,000 m^2 in Arafat; 82% expansion for 2025	case-report	*[Arab News, 2024]* *[Gulf News, 2025]*
Hajj cooling measures	Misting, coatings, shade, trees at the holy sites	Misting columns and mist-covered walkways deployed; hospital study reports 74.6% fall in heat-stroke cases over 40 years (confounded)	case-report	*[Saudi Gazette, 2024]*
Cooling Singapore / climate twin	City-scale twin; policy scenario	Up to 0.13°C island-average cooling on one modelled heat-wave day; urban heat island quoted as up to 7°C	modelled; case-report	[Cooling Singapore 2.0, 2024, NSCC Singapore, 2024] *[Centre for Liveable Cities, 2024]*
Singapore heat-stress advisory	Public heat guidance by wet bulb globe temperature	Bands below 31, 31 to below 33, and 33°C and above, from nine sensors	policy-fact	[NEA Singapore, 2023]*[MSE Singapore, 2023]*
Green Riyadh	Mass tree planting	Targets: >7.5 million trees by 2030; claimed 1.5–2°C city-wide, 8–15°C locally; no measured outcome	projection; policy-fact	*[RCRC, 2025]* *[Saudipedia, 2025]*

Study	What was tested	Result	Grade	Source
Arsht-Rock, twelve cities	Heat and humidity against labour productivity	USD 44 bn/yr in 2020 to USD 84 bn by 2050; 0.1–8.3% of city output; Los Angeles almost USD 5 bn; Bangkok USD 8.6 bn (5%)	modelled; projection (2050)	[Arsht-Rock and Vivid Economics, 2022]*[Arsht-Rock, 2022b][Arsht-Rock, 2022a][Arsht-Rock, 2022c]*
ILO, Working on a Warmer Planet	Global heat stress against working hours	1.4% (1995) to 2.2% (2030) of hours, ≈80 M full-time equivalents, USD 2,400 bn; thresholds 24–26, 33–34, 38°C	projection	[ILO, 2019]
Lancet Countdown 2024	Observed-weather hindcast of labour capacity	512 bn potential hours lost in 2023 (+49% on the 1990s); USD 835 bn potential income	modelled (hindcast)	[Romanello et al., 2024]
UAE midday break	Ban on direct-sun outdoor work	12:30–15:00, 15 June to 15 September; fines [5,000.00: 1,388.89](AED:USD) per breach per worker to [50,000.00: 13,888.89](AED:USD)	policy-fact	*[The National, 2026] [MoHRE, 2023]*
Holiday Climate Index	Tourism-climate rating	Five variables, four weighted sub-indices; below 40 unacceptable; no heat-stress override	method-reference	[Scott et al., 2016]
Digital Earth Africa / WEF	Value of Earth observation for Africa	>USD 2 bn a year by 2024: ≈500 M industry, ≥900 M mining, ≥900 M agriculture; water 17.6 bn m^3 for USD 880 M at 10% adoption	projection	[WEF and Digital Earth Africa, 2021] *[Digital Earth Africa, 2021]*
Global Fishing Watch, Gabon	Satellite intelligence into enforcement	25 personnel trained; three unlicensed trawlers seized 18 July 2026 and escorted 200 km	case-report	*[Global Fishing Watch, 2026][Global Fishing Watch, 2025a]*
Great Blue Wall	Regional seascape network	>83,000 km^2 supported; >175,000 ha protected, restored or committed; >112,000 livelihoods; >USD 80 M mobilised	case-report	*[IUCN, 2026]*
Seychelles blue bond	Sovereign blue finance	USD 15 M raised, ten-year maturity, USD 5 M guarantee and USD 5 M concessional loan; cumulative market USD 15.25 bn by June 2025	policy-fact	[World Bank, 2025] *[World Bank, 2018]*
AU Blue Economy Strategy	Continental sector valuation	USD 296 bn and 49 M jobs (2018) to USD 405 bn and 57 M (2030); USD 10 bn a year lost to illegal fishing	projection	[AU-IBAR, 2019] *[African Union, 2026]*
Megaproject and cost-estimating literature	Outturn against forecast	Nine of ten megaprojects overrun; Olympics average 156%, median 90%, none under budget; percentile-based contingency	measured; method-reference	[Flyvbjerg, 2014, Flyvbjerg et al., 2016, GAO, 2020, Cantarelli et al., 2025]
Spiral and WinWin	Risk-driven process with anchor points	Anchor points and per-cycle termination; only the sustained application survived year 1 of the case study	method-reference; case-report	[Boehm and Hansen, 2000, Boehm et al., 1998]

4.8 Three conclusions

4.8.1 The defensible gap

Conclusion 1

No archived study publishes hours of outdoor activity lost to heat, at corridor scale, tied to a specific built intervention, and tested against a twin's prior prediction. Each element exists somewhere; the intersection is empty.

The claim is falsifiable and the archive is the test. Phoenix measures microclimate around a built intervention with controls, but measures no activity and makes no prior prediction [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. Paris OASIS has the controlled design and the right spatial grain, but no archived effect size and no activity metric [OASIS research team, 2024a]. Abu Dhabi measures the urban heat island densely with model validation, but tests no intervention [Manandhar et al., 2019]. Cooling Singapore predicts scenario effects with a city twin, but reports no built intervention, no validation error and no activity outcome [Cooling Singapore 2.0, 2024]. Arsht-Rock, the ILO and the Lancet Countdown quantify lost labour hours at city, country and global scale, for employed workers, with no intervention and no place-level resolution [Arsht-Rock and Vivid Economics, 2022, ILO, 2019, Romanello et al., 2024]. The Holiday Climate Index scores destination climates at daily resolution with no heat-stress override [Scott et al., 2016]. The Dubai proposal occupies the empty cell (hourly, place-level, activity-denominated, intervention-tested, prediction-checked) and should say so in the Letter of Intent in those terms.

4.8.2 Cool pavement lowers the surface and raises the radiant load

Conclusion 2

On the only controlled field evidence in the archive, a reflective coating makes the pavement much cooler and the person standing on it hotter. Shade ranks first in the Minimum Cooling Network; coatings are conditional.

The Year-1 measurements give a surface reduction of 12.0°F at noon and 10.5°F in the afternoon, a subsurface reduction of 4.8°F and an air-temperature reduction of at most about 1°F, while road-centre mean radiant temperature rises by 5.5°F on average and up to 9.2°F [ASU and City of Phoenix, 2021]. Phase II reproduces the radiant increase at +5.8°F and +4.5°F and quantifies its effect on a body as 12 to 13 W m^{-2} of additional required heat loss [ASU and City of Phoenix, 2024]. In an arid, sunlit street the radiant term dominates the comfort of a standing or walking person, so an optimiser scoring hourly comfort ranks a bare reflective coating below shade. Three refinements follow from the same data rather than from principle. The penalty is to a person *on* the coated surface: footway measurements beside the coated road range from −5.2 to +1.9°F, with no consistent penalty. The coating is beneficial after sunset, when radiant temperature over it runs 0.9–2.3°F cooler, and for touch temperature, which follows from the measured 12.0°F surface reduction [ASU and City of Phoenix, 2021]; its effect on pavement life is modelled, not measured, at a 3–13 per cent rutting and fatigue improvement. The coating's own durability is unresolved: reflectance fell from 33–38 to 19–30 per cent within ten months, the coatings darkened 13–17 per cent under nine months of truck traffic, and no report gives a service life or unit cost. The Makkah claims do not change this, reporting only the quantity that moves favourably *[Arab News, 2024]*[1] *[Gulf News, 2025]*[7]. The Decision Twin should therefore carry cool pavement as a night-time and surface-durability measure and as a possible complement under canopy or trees, never as a stand-alone daytime comfort measure on the exposed 200 m.

4.8.3 Projections, measurements, and what informs Year 3

Conclusion 3

Most headline numbers in this field are projections. Exactly four archived documents carry a measured or measured-with-controls grade for a physical quantity, and three archived designs (Phoenix, OASIS and Abu Dhabi) supply the Year-3 campaign.

Counting Table 4.2, two documents are measured-with-controls for field results (the two Phoenix reports) and two are measured (the Abu Dhabi network and the OASIS paving laboratory); the rest are modelled, projections, policy-facts, case-reports or method-references. Every multi-billion figure in the chapter (USD 44 billion, USD 2.40 trillion, USD 2 billion a year, USD 296 billion) is modelled or projected, and the USD 2 billion rests on assumed adoption and prevention rates, including the 10 per cent adoption rate behind the agricultural water benefit. That is not a criticism of those studies but a statement about what a new project can add: measurement with controls at the scale at which a city actually decides. Three archived designs supply it. Phoenix gives the simultaneous paired comparator and the warning that an unpaired before/after collapses when the weather differs; OASIS gives the control station outside the treated site, the campaign-day filter and the instrument uncertainties; Abu Dhabi gives the reference station, the held-out validation week and the nested resolution. Year 3 is built from those three.

4.9 Running the Year-3 before/after study

The storyboard commits Year 3 to constructing one intervention (most likely a shade canopy over the most exposed 200 m of the corridor, between March and May 2029), then measuring the summer after construction against the baseline summers of 2027 and 2028 with the same sensor network, comparing the measured change with the twin's prediction, and repeating pedestrian counts and dwell-time measurement to October 2029 [Iyyunni, 2026]. The archive shows how that commitment can fail and how to protect it.

Understanding: Before-after-control-impact design

TIER 1: IN PLAIN WORDS

Measure the place you are about to change and a similar place you will not change, both before and after the work. The change that matters is the difference between how the two places moved, not how the treated place moved.

TIER 2: IN PROFESSIONAL TERMS

A before-after-control-impact design estimates an intervention effect as a difference of differences, so that season, weather and city-wide trend affect both arms and cancel. A single before/after comparison confounds the intervention with everything else that changed between the two periods.

TIER 3: IN TECHNICAL TERMS

Paired case and control stations recording simultaneously at one-minute resolution, with mobile traverses at pedestrian height on filtered campaign days, permit the treated-minus-control difference to be computed within the minute and aggregated to hourly comfort. The estimator is robust to slow instrument drift and to synoptic variation, and its residual variance sets the smallest effect the campaign can resolve.

AS A MATHEMATICAL MODEL

With Y the comfort variable, $\hat{\Delta} = \left(\bar{Y}^{T}_{\text{after}} - \bar{Y}^{T}_{\text{before}}\right) - \left(\bar{Y}^{C}_{\text{after}} - \bar{Y}^{C}_{\text{before}}\right)$, where T is the treated segment and C the paired control. The study is powered when $\hat{\Delta}$ exceeds the combined

instrument and sampling uncertainty by the margin declared before the build.

Five design rules follow from the evidence; Figure 4.3 shows the resulting structure and Table 4.3 specifies it. First, *pair every treated point with a simultaneous control*: Al Fahidi Historical Neighbourhood is already the calibration baseline, with no interventions, and serves as the district control, but an untreated corridor segment of comparable geometry must serve as the local control, because a district control cannot absorb street-level geometry. That is the OASIS structure [OASIS research team, 2024a], reinforced by Phoenix's abandoned unpaired comparison [ASU and City of Phoenix, 2024].

Second, *measure radiant load, not only air temperature*: Phoenix shows air temperature moving by tenths of a degree while the radiant environment moves by several degrees, so a six-directional radiometer cart operated under the canopy and at the paired unshaded reference within the same minute is not optional [ASU and City of Phoenix, 2021]. Third, *buy the instruments the index requires*: reporting a wet bulb globe temperature beside the Universal Thermal Climate Index needs a black globe and a natural wet bulb, which the node specification omits [NEA Singapore, 2023], and without them the corridor cannot be compared with the Singapore bands or the ILO and Lancet series [ILO, 2019, Romanello et al., 2024]. Fourth, *declare the resolvable effect before the build*: Phoenix reports air-temperature differences smaller than the standard deviations of the underlying series, so the protocol must state in advance the minimum effect it can detect in each variable, and the month-12 gate criterion on model error must use the same units [Iyyunni, 2026]. Fifth, *pre-register the twin's prediction*: the distinguishing claim is not that the canopy cools but that the twin said by how much, where and at which hours, so the prediction and its uncertainty must be lodged before construction and reported whether or not it verifies, on the precedent of Abu Dhabi's held-out validation week [Manandhar et al., 2019].

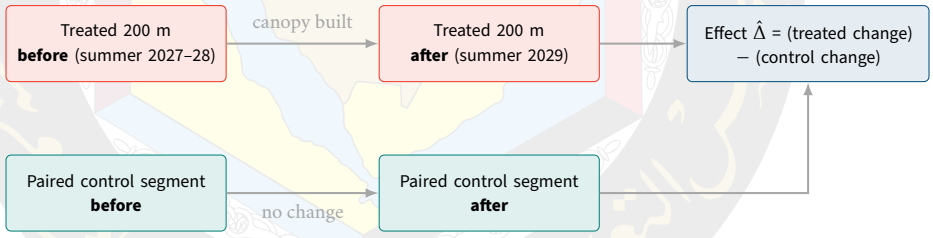


Figure 4.3: The Year-3 before-after-control-impact design. Every arm is measured with the same instruments in the same minutes, so weather and seasonal trend affect both and cancel in the difference of differences.

4.10 Sources that could not be archived, and why

The studies and the final reports are registered, with their extent and evidence grade, in Appendix A. A minority could not be captured from the network used, and those are listed in Table 4.4 so that no reader mistakes a title-only citation for a reviewed source. The rule applied throughout is simple: a source that was not archived is cited by title only, is stated not to have been reviewed, and carries no number into any table or conclusion.

Two further absences are absences of evidence rather than of documents. No archived source gives a service life, re-application interval or unit cost for cool pavement, so the life-cycle assumptions in Chapter 10 are this report's own and are labelled as such. And no archived source

Table 4.3: Specification of the Year-3 before/after campaign, with the archived design each element is drawn from.

Element	Specification	Precedent
Design	Before-after-control-impact; effect estimated as a difference of differences	Paris OASIS [OASIS research team, 2024a]
Controls	District control at Al Fahidi plus a paired untreated corridor segment of comparable geometry, recording simultaneously	Paris OASIS; Phoenix paired reference [OASIS research team, 2024a, ASU and City of Phoenix, 2021]
Fixed sensing	Existing nodes at pedestrian height with at least one vertical profile; one-minute logging during campaigns, ten-minute otherwise	Abu Dhabi profile heights [Manandhar et al., 2019]
Radiant sensing	Six-directional radiometer cart under the canopy and at the unshaded reference within the same minute; four windows per campaign day	Phoenix radiant cart [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]
Added instruments	Black globe and natural wet bulb, to report wet bulb globe temperature beside the comfort index	Singapore advisory [NEA Singapore, 2023]
Campaign days	Filtered on cloud cover and on daily maximum and minimum temperature, declared before the summer	Paris OASIS filter [OASIS research team, 2024a]
Behaviour	Repeat pedestrian counts and dwell time on the same segments and hours as the baseline summer	Storyboard [Iyyunni, 2026]
Prediction	Twin's predicted change, with uncertainty, lodged before construction and reported regardless of outcome	Abu Dhabi held-out validation [Manandhar et al., 2019]
Reporting	Restored Activity Hours per AED, per litre and per kWh, with the measured and predicted changes side by side	Chapter 10

Table 4.4: Sources that could not be archived, the reason, and how they are cited in this report.

Source	Reason	How cited here
Aydt et al. 2026, the Digital Urban Climate Twin paper	Publisher bot wall	Title only; not reviewed [Aydt et al., 2026]
Bande et al. 2020, Dubai local climate zones and ENVI-met	Publisher bot verification	Title only; not reviewed [Bande et al., 2020]
Kılıç 2024, critique of the Seychelles blue bond	Publisher bot wall; paywalled	Title only; no claim about the bond's debt effect [Kilic, 2024]
Hoffmann et al. 2023, Destination Earth climate services	Publisher bot wall	Title only; not reviewed [Hoffmann et al., 2023]
Great Blue Wall Impact Report 2021–2025	Hosted on file-sharing links	Figures relayed via the IUCN release [Great Blue Wall Initiative, 2026] *[IUCN, 2026]*
Arsht-Rock main report PDF and the Abu Dhabi city page	Main PDF not linked; the Abu Dhabi page does not exist	Methodology and three city pages archived instead; no Abu Dhabi loss figure cited [Vivid Economics, 2022]
Royal Commission for Riyadh City, Green Riyadh page	Connection reset	Encyclopaedia article archived instead *[Saudipedia, 2025]*
GMES & Africa programme pages, including the North and West Africa service	Connection reset	Title only; country counts and phase dates not asserted *[GMES and Africa, 2025][GMES and Africa, 2023]*
Ministry release on the UAE midday break, 2023	Connection reset	Newspaper report archived instead *[The National, 2026] [MoHRE, 2023]*
UNEP handbook and cooling-pledge briefing (document repository)	Access challenge on the repository	Landing pages archived; no content cited [UNEP, Cool Coalition and RMI, 2021, UNEP, 2023]
ISO 15686-5:2017, full standard	Paid standard	Preview through clause 4.2.2 only [ISO, 2017a]
Cooling Singapore twin summary (development partner PDF)	Access forbidden to the capture agent	Poster and newsletter archived instead [Cooling Singapore 2.0, 2024, NSCC Singapore, 2024]

gives any cost, uptake or benefit measurement for ABIG or the regional marine services it would build on, which is why Chapter 5 places a measured decision-improvement indicator in ABIG's first twelve months rather than its third year.

4.11 Implications for the programmes

Dubai. Four consequences follow. The Letter of Intent should state the gap in the falsifiable form of Conclusion 1 and cite the archive as the test, because a reviewer who knows the field will otherwise assume Phoenix or Singapore has already done this. The candidate-intervention set must rank shade first and carry cool pavement conditionally, with the radiant penalty encoded in the optimiser rather than mentioned in a footnote: on this evidence a coating on the exposed 200 m ranks below a canopy before cost is even considered. The equipment line must fund a black globe and a natural wet bulb alongside the twenty nodes, and access to a six-directional radiometer cart for campaign days, without which the corridor cannot resolve the radiant term on which Conclusion 2 turns. And the Year-3 protocol in Table 4.3 should be written into the proposal now rather than designed in 2029: the paired control, the campaign-day filter, the declared resolvable effect and the pre-registered prediction are what convert a demonstration into evidence, and they cost almost nothing if planned at the outset.

ABIG. The archive contains no measured benefit for any African Earth-observation or marine-intelligence programme (only ex-ante projections and self-reported aggregates [WEF and Digital Earth Africa, 2021][*IUCN, 2026*][4][*Global Fishing Watch, 2025a*][17]), so ABIG's comparative advantage is not a larger projection but the first credible measurement, and its first twelve months should commit to one: decisions improved per pilot, with the decision, the user and the counterfactual named in advance. The Gabon case is the template at the smallest useful scale (a named authority, a dated action, a traceable chain from data to decision) and the element it lacks, a stated outcome, is exactly what ABIG can add *[Global Fishing Watch, 2026]*[3]. The sustainment finding should then govern pilot selection: each pilot should carry a written sustainment commitment from its operational user before selection [Boehm et al., 1998], as the Dubai gate requires a written statement of usefulness from RTA or DEWA.

Both. The discipline that produced this chapter (grade every quantity, cite every number to a page, state what a source does not show, name what could not be archived) is transferable, and it is the cheapest credibility either programme can buy. Chapter 5 turns these findings into the next six months of work; Chapter 11 carries the unresolved items into the risk register.

{5}

Where to Start

Abstract

Where the two programmes should start is a question of order, not of ambition. One ordering rule (meaning before measurement, and the cheapest, most reversible and most informative step first) applies to the 180 days from this report of 22 September 2026 to 21 March 2027, a window that contains the storyboard's planned grant start of 15 March 2027. For the Dubai Decision Twin it sets out eight sequenced steps: write the semantic layer; define Heat-Lost Activity Hours and Restored Activity Hours per dirham, per litre and per kilowatt-hour, and give Outdoor Tourism Capacity an explicit heat-stress override; adopt the Universal Thermal Climate Index with wet-bulb globe temperature alongside it, and close the black-globe and natural-wet-bulb instrument gap; anchor the twin on the Dubai Digital Twin Platform; instrument a paired-control design; turn the storyboard into the written proposal; secure the December 2026 and February 2027 letters; and carry the undocumented third funding cycle as a dependency rather than an assumption. For the Africa Blue Intelligence Grid it sets out four: the alignment and mapping exercise, the capability map, two or three use cases with named operational users, and the ninety-day list placed in dependency order. A 180-day action table assigns owners among the five stakeholders, a dependency diagram shows what blocks what, and a closing section lists the defects in the source documents that must be fixed first.

Where to start is a question about order, not about ambition. Both papers already state what they want to build (Chapter 2), and the archived evidence already shows which designs hold and which numbers can be trusted (Chapters 3 and 4). What neither paper states is the order in which its first steps must be taken so that no later step is blocked by a missing earlier one. This chapter supplies that order. Its rule is drawn from decision analysis and from the spiral model: take first the step that is cheapest to take, cheapest to reverse and richest in information for the steps that follow [Howard, 1966, Boehm and Hansen, 2000]. For programmes whose money is not yet awarded, that rule points at paper before hardware, at shared meaning before shared data, and at written commitments before any sensor is bought.

5.1 The ordering rule

Three sourced ideas fix the order. The first is the value of information: an observation is worth acquiring in proportion to how far it can change a decision, so the steps that resolve the most decision-relevant uncertainty at least cost come first [Howard, 1966]. The second is the spiral model's third invariant, that the level of effort on any activity (how many sensors, how long to calibrate, how much to negotiate) is set where risk exposure, the probability of a loss multiplied by its size, reaches its interior minimum; underkill and overkill both cost [Boehm and Hansen,

2000]. The third is the WinWin lesson not to close negotiations before prototyping: the agencies that must confirm the Decision Twin's usefulness at the month-twelve gate cannot know what a useful comfort map looks like until they have seen one, so the definitions they endorse now must be light enough to revise then [Boehm et al., 1998].

Applied to the material in hand, the rule puts one step ahead of all others. The discussion notes filed with the storyboard distinguish six things that are routinely confused (data model, semantic model, knowledge graph, artificial-intelligence model, digital twin and decision system) and place the semantic layer between the data and everything that reasons over it [Iyyunni, 2026]. Every later step depends on it: the metrics need agreed meanings for *place*, *hour*, *activity* and *cohort*; the sensor specification must know which quantities the comfort index consumes; the data letters must name the fields each agency supplies; and the capability map of the African programme needs one vocabulary spanning many institutions. It is also the cheapest step, written on paper by people already in the team.

Understanding: Semantic layer

TIER 1: IN PLAIN WORDS

A short, agreed dictionary of what the words mean (what counts as a place, an hour, a visitor, a lost hour, a shade structure), so that every sensor, model and agency in the programme is talking about the same things.

TIER 2: IN PROFESSIONAL TERMS

A controlled vocabulary and concept map defining the entities of the domain (heat exposure, activity, route, place, intervention, resource, decision) and the relations among them, sitting between raw data and any model or dashboard, and distinct both from the database schema below it and from the knowledge graph, models and digital twin above it [Iyyunni, 2026].

TIER 3: IN TECHNICAL TERMS

A lightweight ontology whose classes and relations type every record the platform ingests, with provenance carried per record in the W3C PROV-O pattern of entity, activity and agent, so that observed, modelled and synthetic values can never be confused [W3C, 2013]; in WinWin practice every artefact points to at least one element of the domain taxonomy, which makes the ontology a completeness checklist for the proposal itself [Boehm et al., 1998].

AS A MATHEMATICAL MODEL

The ontology is a triple $\mathcal{O} = (E, R, \Sigma)$ in which the entity classes E are Place, TimeSlot, Cohort, Activity, Route, Intervention, Resource, Sensor, ModelRun and Decision; the typed relations satisfy $R \subseteq E \times E$; and Σ is the set of constraints. A record $m = (e, t, v, u, \pi)$ of entity, time slot, value, unit and provenance is admissible only if $e \in E$, only if u is the unit declared for e, and only if π names the activity that generated it.

5.2 Dubai: eight steps before the grant starts

The storyboard fixes the window: Letter of Intent in September 2026; site-access letters from Dubai Culture and Dubai Municipality by December 2026; Roads and Transport Authority and Dubai Electricity and Water Authority data letters, the Curtin University Dubai agreement and the final budget by February 2027; grant start on 15 March 2027; sensors live by May 2027 [Iyyunni, 2026]. The eight steps below are ordered so that each is finished before the milestone that needs it.

5.2.1 Step 1: write the semantic layer

The deliverable is two pages: a concept map of the chain the notes already draw (heat, pedestrian exposure, tourism route, visitor dwell time, attraction capacity, cooling intervention, energy and

water requirement, investment, outcome) and a data dictionary giving each entity an identifier, a unit and a provenance rule [Iyyunni, 2026]. The identifier scheme is that of Chapter 8 and Appendix B; what matters here is that it exists before the sensor procurement is specified, because the sensor identifier and its calibration version are exactly what the first two gate conditions will be audited against. The map should carry the agencies' own vocabulary, since Chapter 6 argues that what moves an agency from concern to influence is a metric computed on its own data.

5.2.2 Step 2: define the metrics

Chapter 2 stated the three constructs; this step fixes operational definitions compatible with the series that governments already publish.

HLAH, Heat-Lost Activity Hours per place-hour, should collapse (when summed over workers) to the International Labour Organization's working hours lost, and (when summed over visitors) to the Lancet Countdown's hours of at least moderate heat-stress risk for light outdoor activity. The ILO method supplies the scaffolding: metabolic classes of 200, 300 and 400 W; no loss of work capacity up to about 24–26°C on the wet-bulb globe temperature scale; about 50 per cent loss at 33–34°C for a 300 W worker; work becoming impossible above 38°C; and a denominator of 4,320 potential daylight working hours a year (grade: projection for the loss estimates, method-reference for the thresholds) [ILO, 2019]. The Lancet indicator is the visitor-side ancestor: a global mean of 1,512 such hours in 2023, an observation-based model estimate rather than measured absence [Romanello et al., 2024]. Arsht-Rock's daily-maximum threshold of 31°C, above which it substitutes a mid-point value to mimic the shifting of outdoor work away from the hottest hours, is a defensible default cut-off for a lost hour until the corridor's own calibration replaces it (grade: modelled) [Vivid Economics, 2022]. The UAE midday break (no work under direct sunlight or in open spaces from 12:30 to 15:00, 15 June to 15 September (grade: policy-fact) *[The National, 2026]*[1]) should be an explicit stratum, so that briefs report hours restored inside and outside the statutory window; the arithmetic of 2.5 hours across 93 calendar days, about 232.5 hours a year, is this report's own derivation and an indicative upper bound for regulated workers in direct sun only.

RAH, Restored Activity Hours, is the difference in HLAH with and without an intervention, reported in the storyboard's own denominators: per dirham of life-cycle cost, per litre of water and per kilowatt-hour of cooling load [Iyyunni, 2026]. The hourly Bur Dubai district cooling load requested from the water and electricity authority is the kilowatt-hour denominator; the litre denominator is why Green Riyadh's 1,350 km of irrigation network and its target of doubling treated-wastewater reuse (grade: projection and policy-fact) read as a warning rather than as a template *[RCRC, 2025]*[2].

OTC, Outdoor Tourism Capacity, must carry an explicit heat-stress override, for a structural reason. The Holiday Climate Index uses five variables in four weighted sub-indices, and only rain or wind can override an otherwise favourable score (grade: method-reference) [Scott et al., 2016]; this report's own worked example from the published equations scores a dry, calm, clear Gulf afternoon at 37–39°C as 'Good', and still 'Acceptable' when the thermal rating is set to zero. An index that cannot call such an afternoon unacceptable cannot measure what the corridor loses. The override is the acceptability function of Chapter 2 forced to zero above a cohort-specific limit, and Singapore's advisory names people who have recently travelled from cooler climates among the vulnerable, the visitor cohort by definition [NEA Singapore, 2023]. Note the collision of acronyms: in the Singapore sources OTC means outdoor thermal comfort, so the proposal should expand the term at every first use.

[1] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[2] Royal Commission for Riyadh City, *Green Riyadh Program*, project page, 2025. https://www.rcrc.gov.sa/en/projects/green-riyadh-project/

5.2.3 Step 3: fix the comfort index and close the instrument gap

The recommendation is the Universal Thermal Climate Index for pedestrians, with wet-bulb globe temperature computed and reported alongside it [Jendritzky et al., 2012, ISO, 2017b]. Both are needed, because the two global labour series and the only operational public heat product in the archive use the latter: the ILO through ISO 7243, the Lancet Countdown through a wet-bulb globe temperature model, and Singapore through a three-band public advisory [ILO, 2019, Romanello et al., 2024, NEA Singapore, 2023][*MSE Singapore, 2023*][3]. Those Singapore bands are population guidance, not occupational limits, and should not be quoted as though they were. The ILO itself observes that the Universal Thermal Climate Index lacks equivalent safety limits [ILO, 2019]; a corridor map expressed in that index alone would be incomparable with every series a ministry reads.

The equipment consequence is concrete. Singapore's published formula weights the natural wet-bulb, black-globe and dry-bulb temperatures at 0.7, 0.2 and 0.1 *[MSE Singapore, 2023]*[3]. The storyboard's nodes record air temperature, humidity, wind, radiation and surface temperature at ten-minute resolution [Iyyunni, 2026]: neither a natural wet-bulb nor a black-globe thermometer is listed, and without the globe there is no measured mean radiant temperature for the Universal Thermal Climate Index either. Paris OASIS gives the minimum comfort-capable node: a sheltered platinum resistance thermometer, a capacitive hygrometer, a black-globe thermometer to ISO 7726, an anemometer and a net radiometer, logged at one-minute resolution (grade: method-reference) [OASIS research team, 2024a]. The fix is to add a black globe and a natural wet-bulb to the reference station and to the paired case and control nodes at least, at pedestrian height (Phoenix found that its 2 m vehicle sensors 'may pick up less differences because air mixes more the further up' [ASU and City of Phoenix, 2021]) within the equipment line of about [250,000.00:69,444.44](AED:USD), and then to state which nodes measure the index and which estimate it.

5.2.4 Step 4: anchor the twin on the Dubai Digital Twin Platform

The Dubai Digital Twin Platform was launched on 2 July 2026 with three-dimensional models of more than 195,000 buildings, more than 280,000 infrastructure assets and more than 330,000 public facilities; the platform carries more than 1,500 geospatial layers and more than 100 applications; rainfall is the only physics-based simulation it names, and Dubai Municipality signed memoranda with Al-Futtaim Group and Huawei for the next phase (grade: policy-fact) *[Dubai Media Office, 2026]*[4]. The storyboard, written before that launch, plans to obtain municipal street and building geometry as a public baseline [Iyyunni, 2026]. Re-anchoring on the platform converts a platform-build risk into a data request: the district model at 25–50 m and the street model at 1–5 m take their geometry from the platform, and the Decision Twin's contribution can then be stated precisely as the hour-by-hour comfort layer the platform does not have. The launch text documents no access terms, update cadence or research interface, so the December 2026 letter from Dubai Municipality should include a written data-access confirmation naming the layers, not sensor permission alone.

5.2.5 Step 5: instrument a paired-control design

The storyboard already has a control by accident of geography: the heritage core at Al Fahidi is used only as the calibration baseline and receives no intervention [Iyyunni, 2026]. This step

[3] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

[4] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

makes that control deliberate and instruments it as Paris OASIS did. At the Maryse Hilsz site each fixed station is paired with a control station 200 m outside the schoolyard, logged at a one-minute timestep and analysed as a Before-After-Control-Impact design; the mobile protocol samples every 15 seconds at 1.5 m, dwells 10–20 minutes per point for the black globe to stabilise, runs between noon and 16:00 on radiative days only, and keeps each traverse under two hours, because microclimatic parameters then fluctuate by less than 1°C [OASIS research team, 2024a]. Phoenix shows the alternative: its intended before-and-after comparison was abandoned because the two days differed in weather, and every usable field result came from same-day treated-versus-untreated measurement [ASU and City of Phoenix, 2024, ASU and City of Phoenix, 2021]. Three rules follow. Al Fahidi stays untouched for all three summers. Every intervention segment on the Al Ghubaiba corridor carries a paired case and control node, not merely the single reference station. And the Year-3 canopy is measured against a simultaneous unshaded reference point, minute by minute, rather than against the previous summer.

Understanding: Paired-control design

TIER 1: IN PLAIN WORDS

Measure the treated stretch and an untreated stretch at the same time, before and after the canopy is built, so that a hot week or a dust storm cannot be mistaken for the effect of the canopy.

TIER 2: IN PROFESSIONAL TERMS

A Before-After-Control-Impact evaluation with fixed case and control stations logging continuously and mobile traverses on comparable days, so that the intervention effect is the change at the treated site net of the change at the control [OASIS research team, 2024a][*EEA Climate-ADAPT, 2023*][a].

TIER 3: IN TECHNICAL TERMS

Difference-in-differences on paired same-minute observations of the comfort index, with the control outside the intervention's influence but inside its climate; the noise floor is the observed stability of the case-minus-control offset (constant within about 2°C over a midday window and within 1°C over a two-hour traverse in Paris), so an effect of one to two degrees is at the edge of what traverses can resolve, and a full summer of fixed one-minute data is what resolves smaller effects [OASIS research team, 2024a].

AS A MATHEMATICAL MODEL

With U^I and U^C the comfort index at paired intervention and control points, $\hat{\Delta} = \left(\bar{U}^I_{\text{after}} - \bar{U}^I_{\text{before}}\right) - \left(\bar{U}^C_{\text{after}} - \bar{U}^C_{\text{before}}\right)$ over matched place-hours; the Year-3 proof requires $|\hat{\Delta} - \hat{\Delta}_{\text{twin}}| \leq \varepsilon_{\text{build}}$, and the design requires $\varepsilon_{\text{build}}$ to exceed the measured offset noise.

[a] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

The gate's second condition, that the model match sensor observations within a stated error, should be expressed in the units of the one UAE precedent. In Abu Dhabi an uncorrected street model reached a root-mean-square error of 1.33–2.71°C against a 26-station network, falling to 0.27–1.62°C after empirical bias correction, a mean of 0.69°C across twenty district-seasons (grade: measured; the mean is derived from the paper's table) [Manandhar et al., 2019]. A threshold of roughly 1°C on bias-corrected hourly air temperature across a summer is this report's indicative suggestion; the proposal should name the metric, the variable, the height and the averaging period as precisely as that paper's tables do.

5.2.6 Step 6: turn the storyboard into the written proposal

The storyboard is an outline rather than a document, and no drafted proposal yet exists for either programme, as Chapter 1 records. The drafting is not clerical. The call scores a Letter of Intent on five criteria (relevance to the priority areas, partnerships and capacity building, innovation in methodology and outcomes, potential for impact in strategic sectors, and budget adequacy) and a full application on ten, including deliverables with timelines and milestones, a management plan, risk and risk mitigation, and budget *[Dubai RDI Programme, 2025]*[5]. The storyboard has content for each, but not in that order or vocabulary; the call's own topic wording and its cross-cutting technology heading covering the internet of things, edge computing and digital twins should appear verbatim *[Dubai RDI Programme, 2025]*[5]. The month-twelve gate, with its four conditions and its fallback deliverable, answers the deliverables, management and risk criteria directly, and the semantic layer of Step 1 is the completeness check that every criterion has a section. The proposal, the Year-3 decision briefs and the method paper should draw on one bibliography of the archived sources, so that all three share one set of numbers.

5.2.7 Step 7: the letters of December 2026 and February 2027

December 2026: site access from Dubai Culture and Dubai Municipality, extended to data access for the platform's layers (Step 4) and to the paired-control site plan with marked node positions (Step 5). February 2027: the single dataset requested from the Roads and Transport Authority, namely Al Ghubaiba station exit counts with bus and abra ridership; the hourly Bur Dubai district cooling load for one summer from the Dubai Electricity and Water Authority; the Curtin University Dubai agreement with a signed budget share; and the final budget [Iyyunni, 2026]. The call also requires institutional support measures from the host, among them a letter of institutional commitment, a letter of collaboration, a reduced teaching load, fast-tracked procurement and expedited hiring *[Dubai RDI Programme, 2025]*[5]; procuring twenty nodes and hiring a postdoctoral researcher and a doctoral student within weeks of a March start is feasible only if those are signed in advance.

Three rules from Boehm apply. Record win conditions per stakeholder in a table of the WinWin kind (what each of the two authorities, the municipal and cultural bodies, the funder, the collaborating university and the host institution needs to see), together with the commitments made and the areas deliberately left flexible [Boehm et al., 1998, Boehm and Hansen, 2000]. Make sure each signatory is representative, empowered, knowledgeable, collaborative and committed, or the letter gives the appearance but not the reality of stakeholder participation [Boehm and Hansen, 2000]. And ask for sustainment: in the WinWin case only one of five first-year applications survived into use (the one with sufficient budget, people and facilities to sustain it) and sustainment commitment became the top selection criterion in the following year [Boehm et al., 1998]. Since the Year-3 intervention is partner-funded and platform operation after March 2030 is unfunded, a written sustainment commitment should be requested now and secured before the Year-2 intervention is chosen. Paris OASIS adds the tactic: a field visit by fifty stakeholders helped to clear doubts and scepticism before its pilots were built [Urban Innovative Actions, 2023]; a walk of the corridor with the four agencies before the December letters is the *nemawashi* the notes ask for [Iyyunni, 2026].

5.2.8 Step 8: carry the third funding cycle as a dependency

The archived sources document only the 2025 second cycle of the Dubai Future Foundation RDI Grant: a soft call on 15 May, the official call on 30 May, Letters of Intent due 13 June, shortlist

[5] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

notification on 15 July, full proposals on 15 August and awards announced on 10 September 2025, roughly four months from call to award (derived); funding of [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD) per project, an annual cap of [500,000.00:138,888.89] (AED:USD), durations of one to three years, and a requirement that at least half the grant be spent within Dubai (grade: policy-fact) *[Dubai RDI Programme, 2025]*[5] *[Dubai Future Foundation, 2025]*[6]. The initiative was launched in September 2024, and its first cycle funded 24 projects out of 374 proposals, about one in sixteen (derived) *[Dubai Future Foundation, 2025]*[6]. No archived text names a third cycle, its calendar, its envelope or its topics, and the [750,000,000.00:208,333,333.33](AED:USD) reported for the wider Dubai research and development programme is a phase allocation, not a grant envelope available to applicants *[Dubai Future Foundation, 2025]*[6]. The storyboard's Letter of Intent in September 2026 and start on 15 March 2027 therefore presuppose a call that appears in no source.

Dependency: the undocumented third cycle

Treat the cycle as an external dependency, not an assumption. Confirm the next call's calendar with the Dubai Future Foundation within the first thirty days; hold every milestone relative to the call date rather than to fixed months; and protect the summer. The first gate condition needs a full June-to-September season of observations, so if the start slips past the spring of 2027 the first summer moves to 2028 and the programme slips a year: the fixed-window penalty that the megaproject literature documents for schedules which cannot move (reference-class context drawn from projects far larger than this one, not a calibrated uplift) [Flyvbjerg et al., 2016, Flyvbjerg, 2014]. Chapter 11 carries the funding-timing risk; the annual cap of [500,000.00:138,888.89](AED:USD) set against a Year-1 equipment line of about [250,000.00:69,444.44](AED:USD) is the second constraint the budget profile must show [Iyyunni, 2026]*[Dubai RDI Programme, 2025]*[5].

5.3 ABIG: four first steps

The Africa Blue Intelligence Grid concept paper runs in horizons of 0–3, 3–6 and 6–12 months, then Year 2 and Year 3, with a twelve-item ninety-day list and an addendum recommending one formal exercise inside the first year [Terhile et al., 2026]. The four steps below order those items so that the addendum's exercise comes first and hardware comes last, as the paper's own capability-first sequencing requires.

5.3.1 Step 1: the alignment and mapping exercise

The addendum's alignment and mapping exercise is the first deliverable, because it settles the boundary on which every other decision rests: observation belongs to GOOS-Africa, forecasting to the OceanPrediction Decade Collaborative Centre's African Seas Regional Team and to OPERA, Earth-observation services to GMES & Africa, and the Grid is the integration and application layer above them [Terhile et al., 2026]. The addendum's immediate action is a small technical dialogue with those bodies around six questions: what already exists, what is being developed, where the genuine gaps are, where the Grid adds value, which existing programmes could host or test its use cases, and which one or two practical demonstrations could be developed jointly. The mapping scope has thirteen items, running from existing observation and forecasting systems through data repositories and operational users to gaps in digital infrastructure and opportunities

[6] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

for joint demonstration, and its output is the Africa Ocean Observation, Prediction and Intelligence Capability Map [Terhile et al., 2026]. Without it the programme is exposed to the first risk in its own register (becoming another coordination forum), which is the failure Boehm names as evolutionary development without a life-cycle architecture [Boehm and Hansen, 2000].

5.3.2 Step 2: build the capability map

The capability map is the African programme's semantic layer: one vocabulary across institutions before any data are federated. The concept paper supplies its rows and columns (ten institutional nodes, from the South African National Space Agency to the African Space Agency; ten capability layers, from consuming Earth-observation data to independent launch; and eight regional marine systems) and its closing note asks for a country-by-country capability database recording source, date, sensor, resolution, ownership, operational status, personnel, facilities, budgets and application domain [Terhile et al., 2026]. The archive supplies first entries and one rubric. Entries: MarCOSIO's twelve partner institutions and three service lines, with an eStation gifted under the 2017–2021 GMES & Africa programme *[MarCOSIO, 2025]*[7]; Digital Earth Africa's data cube over more than 30 million km^2, whose water-observation service Tanzania's national statistics bureau already uses on Lake Sulunga [WEF and Digital Earth Africa, 2021]*[Digital Earth Africa, 2021]*[8]; Global Fishing Watch's three sub-regional fisheries partners representing 24 countries, its 2021–2023 port-inspection pilot in Côte d'Ivoire, Ghana, Kenya and Senegal, and its July 2025 training modules with Senegal, Madagascar and Gabon *[Global Fishing Watch, 2025a]*[9]; and the two European twins, whose global products are candidate boundary conditions and whose mutual interoperability sets the standard the Grid should declare (grade: policy-fact for the programme facts and periods, case-report for the institutional self-reports) *[Copernicus Marine Service, 2024]*[10] *[ECMWF, 2024]*[11]. Rubric: the Countries Geospatial Readiness Index used in the World Economic Forum valuation weights data infrastructure at 20 per cent, policy framework at 10, institutional capacity at 20, user adoption at 20 and industry fabric at 30, and names Nigeria, South Africa and Ghana as African best practice on individual pillars (grade: projection for the valuation, method-reference for the rubric) [WEF and Digital Earth Africa, 2021].

Understanding: Capability map

TIER 1: IN PLAIN WORDS

A table of who in Africa can already observe, forecast, process and use ocean and Earth data, at what level and in which sea region, so that the programme connects what exists instead of building it twice.

[7] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[8] Digital Earth Africa, *Digital Earth Africa has the potential to contribute billions to the African economy using Earth observations*, news, 2021. https://digitalearthafrica.org/en_za/digital-earth-africa-has-the-potential-to-contribute-billions-to-the-african-economy-using-earth-observations/

[9] Global Fishing Watch, *Rising pressures on Africa's fisheries and marine resources*, 2025. https://globalfishingwatch.org/rising-pressures-on-africas-fisheries-and-marine-resources/

[10] Copernicus Marine Service, *The European Digital Twin Ocean Launched at the 2024 Digital Ocean Forum*, news. https://marine.copernicus.eu/news/european-digital-twin-ocean-launched-2024-digital-ocean-forum

[11] European Centre for Medium-Range Weather Forecasts, *Fact sheet: ECMWF and the EU's Destination Earth initiative*, ECMWF media centre, 2024. https://www.ecmwf.int/en/about/media-centre/focus/2024/fact-sheet-ecmwf-and-eus-destination-earth-initiative

TIER 2: IN PROFESSIONAL TERMS

An inventory of institutions against capability layers and regional marine systems, each cell scored and sourced, which becomes the programme's baseline dataset and the evidence on which any later hardware decision must rest [Terhile et al., 2026].

TIER 3: IN TECHNICAL TERMS

A scored matrix over institutions, the paper's ten capability layers and its eight marine systems, in which a gap is the difference between the observation, modelling and product requirements derived from a named user's decision and the capability recorded; because hardware follows evidence, the map is also the state of a sequential decision problem whose next action is the pilot, fellowship or sensor with the highest expected information value [Powell, 2019, Howard, 1966].

AS A MATHEMATICAL MODEL

$M_{i,k,r} \in \{0, 1, 2, 3\}$ for institution i, capability layer k and region r; for use case u with requirement $q_{u,k,r}$ the gap is $g_{u,k,r} = \max\left(0,\ q_{u,k,r} - \max_i M_{i,k,r}\right)$, and the first investments are the cells (k, r) that close the largest gap per unit cost for use cases which already have a named operational user.

5.3.3 Step 3: two or three use cases with named operational users

The paper lists ten applications, each with a class of potential user; the first year must turn two or three of them into a named agency, a named decision and a written sustainment commitment [Terhile et al., 2026]. The archive points to candidates. First, alerts on illegal, unreported and unregulated fishing with a fisheries authority already working with Global Fishing Watch: Gabon's General Directorate of Fisheries and Aquaculture, whose staff were among the 25 personnel trained before the patrol that intercepted three unlicensed trawlers on 18 July 2026; or Senegal's Directorate of Fisheries Protection and Surveillance and Madagascar's fisheries ministry, both partners in the July 2025 training (grade: case-report and policy-fact) *[Global Fishing Watch, 2026]*[12] *[Global Fishing Watch, 2025a]*[9]. The duplication risk is real, so the Grid's contribution must be the record that the Gabon account lacks: a denominator such as interdictions per targeted patrol, and a decision log running from brief to action to outcome. Second, inland-water monitoring with a national statistics bureau, where the Tanzanian use of the Digital Earth Africa water service is already operational and needs only a decision record and a benefit measure *[Digital Earth Africa, 2021]*[8]. Third, a heat use case in the climate-resilience domain: Africa has the highest share of workforce affected by excessive heat, at 93 per cent (grade: policy-fact; a secondary citation inheriting its primary grade) [UN Secretary-General, 2024]; Freetown, the one African city in the Arsht-Rock study set, loses an estimated 3.5 per cent of city output to heat (grade: modelled) [Arsht-Rock and Vivid Economics, 2022]; and the Dubai HLAH metric, once validated, is the product. Every use case needs an explicit objective (decisions improved per dollar, with the denominator stated) before any platform is chosen, since the sequential-decision framework admits no policy without a stated objective [Powell, 2019], and each needs the sustainment letter that WinWin made its leading criterion [Boehm et al., 1998].

5.3.4 Step 4: put the ninety-day list in dependency order

The paper's twelve ninety-day actions are sound but unordered [Terhile et al., 2026]. In dependency order they read: appoint the founding team and define the charter; prepare the two-page stakeholder proposition for the African Union Commission, the African Space Agency,

[12] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

GMES & Africa, the regional economic communities and priority national institutions; open the technical dialogue of Step 1; build the capability map and the data inventory of Step 2 together, since the inventory is a column of the map; select the use cases and the two or three demonstration geographies of Step 3, covering West, East, North or Southern Africa and at least one island or coastal state; adopt responsible-use principles for security analytics before any vessel data are touched; engage the BRICS space and ocean working groups, the UAE Space Agency and the New Development Bank; launch the challenge-based fellowship; publish the Africa Blue Intelligence Baseline; and convene the first technical workshop with actual users to agree the twelve-month demonstration portfolio. The political anchor is already fixed: the Luanda Declaration adopted on 24 July 2026 at the Third Africa Blue Economy Week *[African Union, 2026]*[13], and the African Union strategy's 2019 objective of enhancing analytical and information-support systems at all levels for informed decision making [AU-IBAR, 2019].

5.4 The 180-day action table

Table 5.1 places the twelve steps on a calendar of 180 days running from the date of this report, 22 September 2026, to 21 March 2027 (the day windows are derived by counting from the report date), spanning the storyboard's pre-award loop and its planned grant start. Owners are proposed from the five stakeholders by affiliation, and are this report's recommendation rather than an agreed allocation: Dr Chakradhar Iyyunni for the Dubai proposal and its partner letters; Mahesh Shastry for the semantic layer, the data dictionary and the drafting of the proposal, the role the discussion notes describe as knowledge architecture [Iyyunni, 2026]; Dr Uchi Dominic Terhile for the dialogue with GOOS-Africa, OceanPrediction and OPERA and for the mapping exercise, as the addendum's author [Terhile et al., 2026]; Abdullah Al Salmani for the UAE Space Agency, data-gateway and BRICS engagement; and Arti Ahluwalia for the stakeholder proposition, the operational-user council and the responsible-use principles.

[13] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

Table 5.1: The 180-day action table, 22 September 2026 to 21 March 2027; day windows derived by counting from the report date. Owners are proposed, not agreed. Every action is drawn from the storyboard, the concept paper, the funding-call rules or the design lessons of Chapters 3 and 4.

Days	Action	Owner	Depends on	Output
0–30	Dubai: draft the semantic layer (concept map, data dictionary, identifier and provenance rules) in the agencies' own vocabulary	Mahesh Shastry	n/a	Two-page semantic layer, version 0.1
0–30	Dubai: confirm the next funding call's calendar, envelope and topics with the Dubai Future Foundation; reset milestones relative to the call date	Dr Chakradhar Iyyunni	n/a	Call calendar on record; dependency closed or re-planned
0–30	ABIG: request the technical dialogue with GOOS-Africa, the OceanPrediction African Seas Regional Team and OPERA; circulate the addendum's six questions	Dr Uchi Dominic Terhile	n/a	Dialogue scheduled; participants named
0–30	Both: correct the source documents (Section 5.6); issue the scrubbed derivative of the discussion record	Mahesh Shastry	n/a	Defects closed; scrubbed copy issued
31–60	Dubai: fix the metric definitions (HLAH; RAH per AED, per litre, per kWh; OTC with heat-stress override) in a form compatible with the ILO and Lancet series	Dr Chakradhar Iyyunni; Mahesh Shastry	Semantic layer	Metric definition note with thresholds and strata
31–60	Dubai: adopt the comfort index; revise the node specification to add black-globe and natural-wet-bulb instruments; re-cost within the [250,000.00:69,444.44] (AED:USD) line	Dr Chakradhar Iyyunni	Metric definitions	Node specification version 1; equipment estimate
31–60	Dubai: draft the Digital Twin Platform data request (layers, cadence, interface) for Dubai Municipality	Dr Chakradhar Iyyunni	Semantic layer	Data-request letter, draft
31–60	Dubai: write the win-conditions table for the four agencies, the funder, the collaborating university and the host institution	Dr Chakradhar Iyyunni; Arti Ahluwalia	n/a	Win-conditions table
31–60	ABIG: build the capability-map template (institutions by layers by regions) and open the data inventory; prepare the two-page stakeholder proposition	Dr Uchi Dominic Terhile; Arti Ahluwalia	Dialogue scheduled	Map template; proposition circulated
61–90	Dubai: walk the corridor with the four agencies; agree the paired-control site plan with marked node positions; obtain site-access and data-access letters	Dr Chakradhar Iyyunni	Node specification; data request; win conditions	December 2026 letters (Dubai Culture, Dubai Municipality)
61–90	ABIG: hold the dialogue; select two or three use cases and demonstration geographies, each with a named agency and decision; adopt responsible-use principles	Dr Uchi Dominic Terhile; Arti Ahluwalia	Dialogue; map template	Use-case portfolio; principles adopted
61–90	ABIG: engage the UAE Space Agency and one BRICS counterpart on an Africa–UAE data and applications gateway	Abdullah Al Salmani	Proposition	Gateway engagement brief

continued on next page

Table 5.1: The 180-day action table (continued).

Days	Action	Owner	Depends on	Output
91–120	Dubai: draft the proposal against the five Letter-of-Intent and ten full-application criteria; add the gate, fallback and risk sections	Mahesh Shastry; Dr Chakradhar Iyyunni	Metrics; node specification; letters	Full proposal, draft 1
91–120	Dubai: request written sustainment commitments for the Year-3 intervention and for platform operation after March 2030	Dr Chakradhar Iyyunni	Win conditions	Sustainment requests on record
91–120	ABIG: complete the alignment and mapping exercise; publish the Africa Blue Intelligence Baseline; open New Development Bank project-preparation contacts	Dr Uchi Dominic Terhile; Abdullah Al Salmani	Dialogue; use cases	Capability Map version 1; Baseline; investment concept note
121–150	Dubai: secure the transport and utility data letters, the Curtin University Dubai agreement and the institutional support measures; finalise the budget against the [500,000.00:138,888.89](AED:USD) annual cap	Dr Chakradhar Iyyunni	Proposal draft; December letters	February 2027 letters; final budget
121–150	ABIG: launch the challenge-based fellowship on the first prototype; open sustainment discussions with each named user	Arti Ahluwalia; Dr Uchi Dominic Terhile	Use cases	Fellowship cohort; draft sustainment letters
151–180	Dubai: submit in the form the confirmed call requires; pre-position procurement and hiring so that sensors are live before June 2027	Dr Chakradhar Iyyunni	All Dubai steps	Submission; procurement plan
151–180	ABIG: first technical workshop with actual users; agree the twelve-month demonstration portfolio; record the objective and denominator of each use case	Dr Uchi Dominic Terhile; all stakeholders	Capability Map; use cases	Twelve-month portfolio; objective functions

5.5 Dependencies

Figure 5.1 draws the dependency structure behind the table. The Dubai chain is linear from the semantic layer to the grant start, with one external node (the third funding cycle) that no step within the team's control can close; it is drawn dashed because its calendar is undocumented. The African chain begins with the dialogue and the mapping exercise, and cannot reach its first workshop without named users. One dashed edge crosses the lanes: the vocabulary written for the Dubai corridor is the common data model that the ABIG inventory needs, which is the concrete form of the unifying thesis of Chapter 2.

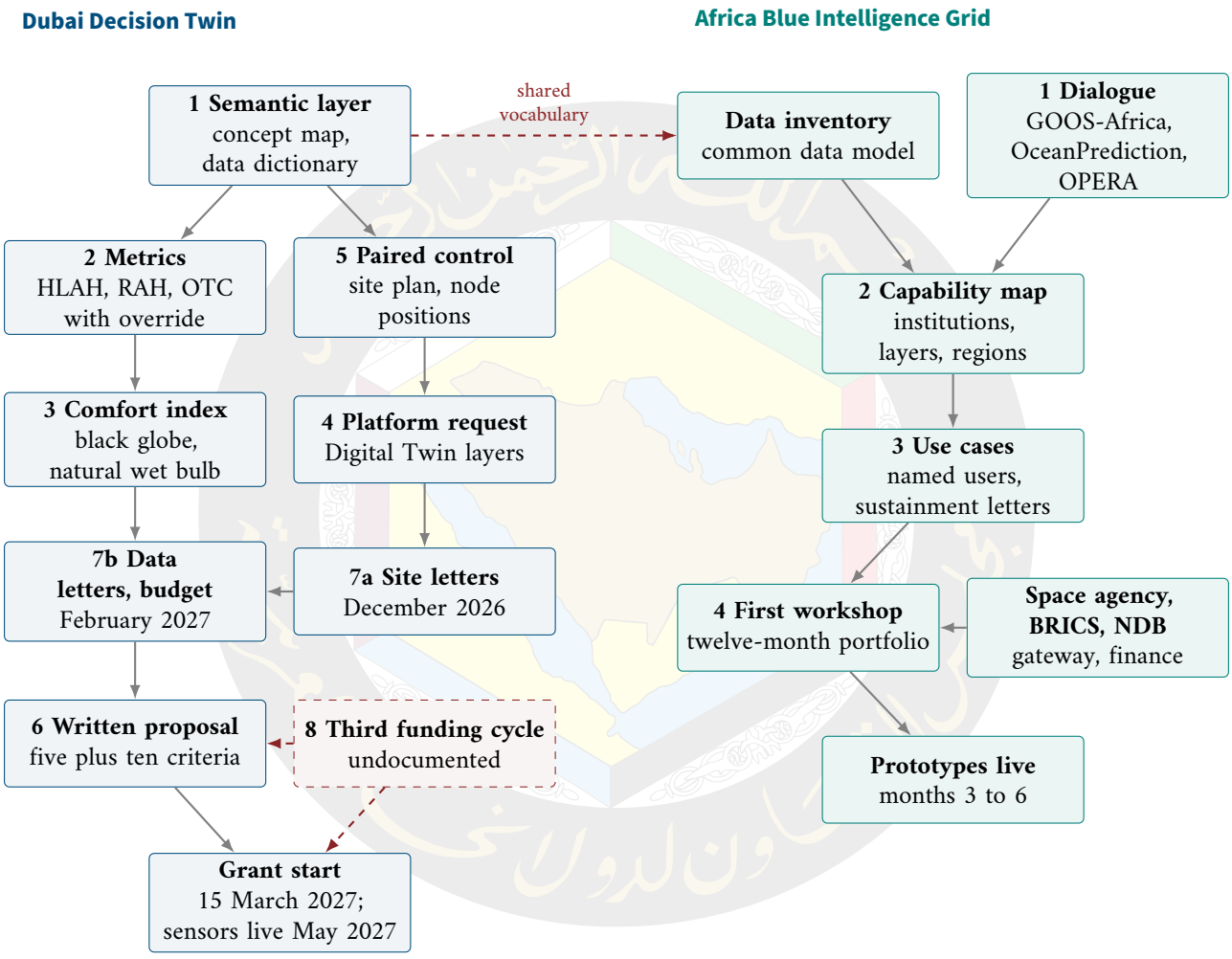


Figure 5.1: Dependency structure of the first steps. Solid arrows: a step that must be complete before the next can begin. Dashed arrows: dependencies outside the team's control, or shared across the two programmes. The third funding cycle is the only node with no documented calendar.

5.6 Putting the documents in order

The source documents are the first thing a funder or a partner will be shown, and four defects should be closed within thirty days. *Personal data*: the discussion record behind the storyboard contains a personal telephone number and a private letter to an investor (Chapter 1), and a scrubbed derivative should be produced before any wider circulation. *Unformalised proposals*: neither programme has yet been written up as a document a reviewer could assess (Chapter 1), and both should be drafted against the same evidence conventions, so that a change to a number is made once and once only. *Citation defects*: three items of publication metadata must be settled by one convention before anything is cited outside the team. The Holiday Climate Index paper renders its author list ambiguously, and its fourth author is Mantao Tang; the Seychelles case study is dated 27 June 2025 although it describes an issuance of October 2018; and the Abu Dhabi paper is dated 2019 but sits in a 2020 journal volume. *Missing evidence*: several sources named in the underlying analysis could not be obtained (among them the 2026 Cooling Singapore paper, the Dubai local climate zone paper, the Great Blue Wall impact report, the Seychelles critique and the Luanda Declaration itself), and each should either be secured and dated or cited by title only with a statement that it was not reviewed.

None of this is expensive, and all of it is the kind of defect that a reviewer notices first.

5.7 Implications for the programmes

For the Dubai Decision Twin. The next six months decide whether the storyboard becomes a proposal that can be scored, and the order of work is meaning, metrics, instruments, platform, control, document, letters. Two of those steps change the proposal's content and not merely its form. The node specification must add black-globe and natural-wet-bulb instruments, or the comfort map will be incomparable with the labour and advisory series that ministries actually read [ILO, 2019] *[MSE Singapore, 2023]*[3]; and the paired-control design means that the site *plan*, not only the site, must be agreed in the December letters [OASIS research team, 2024a]. Anchoring on the Dubai Digital Twin Platform makes the twin cheaper to build and easier to explain, because it supplies precisely the hour-by-hour comfort layer that platform lacks *[Dubai Media Office, 2026]*[4]. The third funding cycle is the one dependency the team cannot close by working harder, and the summer window makes its cost a year rather than a month.

For the Africa Blue Intelligence Grid. The concept paper's sequencing is right and its addendum names the first deliverable; the risk lies in order, not in content. The dialogue and mapping exercise must precede use-case selection, the use cases must precede any workshop, and each use case must arrive with a named agency, a stated objective and a sustainment letter, because the only surviving application in the WinWin case was the one with a budget to sustain it [Boehm et al., 1998]. The archive already names candidate users, in Gabon, Senegal and Madagascar for fishing alerts and in Tanzania for inland water, and it also names what the Grid must add that none of them publishes: decisions improved per dollar, with the denominator stated *[Global Fishing Watch, 2026]*[12] *[Global Fishing Watch, 2025a]*[9] *[Digital Earth Africa, 2021]*[8].

For both. The two programmes share their first step. The semantic layer written for the Dubai corridor is the common data model that the African inventory needs, and the same three disciplines (grade every number, pair every intervention with a control, put a named user at the gate) run through both action lists. The timing argument is external and fixed: the Special Report on Climate Change and Cities closed its second-order draft review on 3 July 2026 and is scheduled for approval in March 2027, so nothing from either programme can enter it, and the Working Group II assessment of the seventh cycle is the target instead [IPCC, 2024] *[IPCC, 2026a]*[14]. That is a reason

[14] Intergovernmental Panel on Climate Change, *Leading scientists gather in India for final meeting to work on the Special Report on Climate Change and Cities*, IPCC news, 3 August 2026. https://www.ipcc.ch/2026/08/03

to start now rather than later: a metric validated on one corridor by 2029 can be in the literature before that window closes, and Chapter 6 shows which actors must be moved, and in what order, for it to be adopted.

/srcities-lam4/

٦ {6}

Circles of Influence

Abstract

Influence in a programme of this kind runs in rings. It adapts the circles of concern and influence described in Covey's *The 7 Habits of Highly Effective People* (cited by title, since the book is not in the archive) into three rings, control, influence and concern, names the actors and levers in each, and draws the rings for the Dubai Decision Twin and for the Africa Blue Intelligence Grid. It sets out the mechanism that moves an actor inward, which this report calls the escalator: a metric the actor did not have, computed on the actor's own data, delivered as a decision brief and closed with a written confirmation. It then shows why gate condition 6.2.4 (one of two Dubai agencies confirming in writing that the output is useful for a decision it faces) is the one line in the storyboard the team cannot satisfy alone, and therefore the most important. It maps the five stakeholders to the institutions they can actually reach, and closes with an influence plan naming actor, lever, evidence, owner and date, which readers should use as the checklist for the letters due in December 2026 and February 2027 and for the Grid's first ninety days.

Every proposal lists what its authors will do; feasibility depends at least as much on the shorter list of what other people must do. The Dubai storyboard needs four agencies to sign letters before the grant begins, at least one of two to confirm in writing at month twelve that a map was useful, and two to agree which structure is built in Year 3 [Iyyunni, 2026]. The Africa Blue Intelligence Grid (ABIG) concept paper needs an observing community, a forecasting programme, an Earth-observation programme, a continental political body, national agencies, a space agency, a development bank and a diplomatic forum to act, in roughly that order, over three years [Terhile et al., 2026]. Not one of these actors reports to the stakeholders, so the question is practical: which decisions do the programmes take alone, which can they move with evidence, and which must they simply watch and plan around? The framing is Covey's circle of concern (everything a person cares about) with, inside it, a smaller circle of influence, read here as advice that effort spent on the inner circle enlarges it while effort spent on the outer circle does not [Covey, 1989]. Only those names and that reading are taken from the book, which was not reviewed for this report; following the baseline analysis, an innermost ring of control is added.

6.1 Three rings, and who decides

The rings are defined by who decides, not by how much something matters. The *control* ring holds decisions the team can take without permission: the design of the sensor network, the definitions of the metrics, the validation method, the wording of the gate, what is published. The *influence* ring holds decisions taken by other institutions that the team can move with evidence: a

data letter, a site permission, a grant award, a written confirmation, a pilot hosted by an agency. The *concern* ring holds conditions that shape the outcome but that no evidence the programme can produce will change within its lifetime: the climate trajectory, tourism demand, energy and water prices, the internal politics of a union of fifty-five states. The distinction earns its place because both programmes are small ([1,400,000.00:388,888.89](AED:USD) over 36 months for Dubai [Iyyunni, 2026], no budget at all yet for ABIG [Terhile et al., 2026]) and small programmes fail when they spend their attention on the outer ring.

Two cautions belong with the framing. Membership is per decision, not per institution: Dubai Municipality sits in the influence ring for a site permission and in the concern ring for the future of its own digital-twin platform. And the rings move: shifting named actors inward is the whole point, and evidence is the only instrument either programme has for doing it.

Understanding: Circle of influence

TIER 1: IN PLAIN WORDS

The people and organisations whose decisions the project can change by showing them something they did not know: as distinct from those it can only inform, and those it can only worry about.

TIER 2: IN PROFESSIONAL TERMS

A stakeholder segment defined by the existence of a lever: an institution belongs in the influence ring when the programme can produce an artefact (a dataset, a map, a brief, a validated method) that bears on a decision the institution actually faces and cannot resolve for itself.

TIER 3: IN TECHNICAL TERMS

In the multi-agent formulation of sequential decisions each agency is a separate agent with its own state, information and beliefs, and no agent observes the whole system [Powell, 2019]. Influence is the transmission of information from the programme's belief state into another agent's decision rule, observable only when that agent's decision changes, which is why the storyboard demands a written confirmation rather than a presentation [Iyyunni, 2026].

AS A MATHEMATICAL MODEL

For actor a facing decision d with options $x \in X_a$ and payoff u_a, the worth of a brief B is the classical value of information [Howard, 1966]: $V_a(B) = \max_x \mathbb{E}[u_a(x) \mid B] - \max_x \mathbb{E}[u_a(x)] \geq 0$. Actor a lies in the influence ring for d when $V_a(B) > 0$ *and* the programme can deliver B before d is taken.

6.2 Dubai: actors and levers by ring

Figure 6.1 draws the rings for the Climate Intelligence and Decision Twin. The control ring is larger than it first appears. The team (a principal investigator and four co-investigators at BITS Pilani Dubai, one postdoctoral researcher and one doctoral student) decides the node count and placement (twenty microclimate nodes and one reference station, about [250,000.00: 69,444.44](AED:USD)), the recording cadence, the platform architecture, the district and street model resolutions, the validation threshold, the definitions of Outdoor Tourism Capacity and the Minimum Cooling Network, the candidate intervention set and what is published [Iyyunni, 2026]. It also decides three things the storyboard leaves open and Chapter 5 asks it to settle first: the semantic layer, the comfort index and the identifier schema. The public baselines (National Center of Meteorology station data, Copernicus products, municipal geometry) require no agreement and so behave in practice as part of the control ring.

The influence ring has seven members, and the storyboard asks each for exactly one thing,



Figure 6.1: Circles of influence for the Dubai Decision Twin. Inner ring: decisions the team takes alone. Middle ring: institutions whose decisions the programme can move with evidence. Outer ring: conditions it can only watch, together with the second-stage actors it cannot yet approach. Actors and levers are taken from the storyboard [Iyyunni, 2026] and the baseline analysis; the arrow is the escalator of Section 6.4.

which is the right way to enter it. Table 6.2 sets out the levers row by row; two deserve comment here. Dubai Municipality is not only the permitting authority for the corridor's public realm: it also owns the Dubai Digital Twin Platform launched on 2 July 2026, holding three-dimensional models of more than 195,000 buildings and 280,000 infrastructure assets across more than 1,500 geospatial layers *[Dubai Media Office, 2026]*[1]. The geometry the Decision Twin needs already sits there and its access terms for third parties are undocumented, so the December 2026 letter should ask for it, converting a platform-build risk into a data request; that such permissions are granted to universities in the region is not speculation, since the Abu Dhabi municipality approved lamp-post mounting for a twenty-six-station urban weather network built for a university study [Manandhar et al., 2019]. The Dubai Future Foundation (DFF) and the Dubai Research, Development and Innovation (RDI) Programme control the money: [500,000.00:138,888.89] (AED:USD) to [1,500,000.00:416,666.67](AED:USD) per project, an annual ceiling of [500,000.00: 138,888.89](AED:USD), at least 50 per cent of the grant spent within Dubai, a principal investigator fully affiliated with a Dubai higher-education institution, and a rubric scoring partnerships and capacity building alongside deliverables and risk mitigation *[Dubai RDI Programme, 2025]*[2]. The only documented calendar is the 2025 second cycle *[Dubai Future Foundation, 2025]*[3]; the decision the storyboard most depends on (that a further cycle opens in time to receive a September 2026 letter) is undocumented, sits on the boundary between influence and concern, and is carried in Chapter 11 as a funding-timing risk. The Roads and Transport Authority (RTA) and the Dubai Electricity and Water Authority (DEWA) are treated in Section 6.5, because the gate turns on them; Dubai Culture, whose Al Fahidi Historical Neighbourhood serves only as a calibration baseline, Curtin University Dubai and the platform supplier complete the ring [Iyyunni, 2026].

Three actors sit just outside as second-stage targets, because the programme has nothing to offer them yet. The Department of Economy and Tourism (DET) controls destination policy, which an OTC figure for Al Fahidi would inform from mid-2028. The Dubai Walk master plan of December 2024 (3,300 km of new and 2,300 km of renovated walkways by 2040, with no owner named in the archived report *[Dezeen, 2024]*[4]) is precisely the programme the MCN is built for, and can be approached once the corridor's network exists in November 2028. The Ministry of Human Resources and Emiratisation (MoHRE) controls the midday break, a fixed-hours instrument running from 12:30 to 15:00 between 15 June and 15 September *[The National, 2026]*[5], whereas the International Labour Organization's method instead rests on temperature, humidity and workload thresholds [ILO, 2019]: an hour-by-hour map is the evidence for that conversation, but the conversation belongs after March 2030. The remaining concern items are exogenous inputs to the model of Chapter 7, not targets of persuasion.

[1] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

[3] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[4] Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

[5] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

6.3 ABIG: actors and levers by ring

ABIG's control ring is what a founding team can write without anyone's consent, and the concept paper lists it in the first ninety days: the charter; a capability map of African space agencies, Earth-observation centres, oceanographic institutions, universities, ports, fisheries and environmental agencies; a data inventory; responsible-use principles for security analytics; five to seven use cases and two to three demonstration geographies; and the common data model and provenance standards [Terhile et al., 2026]. The addendum adds the deliverable that matters most for the influence ring: an Africa Ocean Observation, Prediction and Intelligence Capability Map, produced by a formal alignment and mapping exercise with GOOS-Africa, the OceanPrediction Decade Collaborative Centre's African Seas Regional Team and OPERA within the first twelve months. Figure 6.2 draws the rings.



Figure 6.2: Circles of influence for the Africa Blue Intelligence Grid. Actors and ninety-day contacts are taken from the concept paper and its addendum [Terhile et al., 2026]; the outer-ring fishing pressures are Global Fishing Watch's uncited estimate and second-hand projection *[Global Fishing Watch, 2025a]*. AUC: African Union Commission; RECs: regional economic communities; NDB: New Development Bank; DCC: Decade Collaborative Centre.

The influence ring is deliberately institutional, because the addendum's entire argument is that ABIG connects what exists instead of building beside it [Terhile et al., 2026]. GOOS-Africa controls the observing community and its five working groups (in-situ observation, remote sensing, modelling, fundraising and end-user relations), which the addendum proposes as ABIG's explicit interface. The OceanPrediction African Seas Regional Team and OPERA control forecasting; the lever is ABIG's offer to be the downstream architecture carrying forecast products to fisheries, ports, pollution response and coastal resilience, and the addendum's first recommended action is not an endorsement but a small technical dialogue to establish what exists, what is under development, and where one or two joint demonstrations are possible. GMES & Africa controls Earth-observation services through consortia such as MarCOSIO, which runs three co-designed service lines (fisheries and aquaculture, marine and maritime, coastal monitoring) across institutions in South Africa, Tanzania, Kenya, Mozambique, Mauritius and Madagascar under the 2017–2021 programme *[MarCOSIO, 2025]*[6]. The African Union Commission, the African Space Agency and the regional economic communities control the political mandate, and the evidence that speaks to them is alignment with the Luanda Declaration of 24 July 2026 *[African Union, 2026]*[7] and with the 2019 Blue Economy Strategy's objective of an analytical and information support system for decision making [AU-IBAR, 2019]. National fisheries and environment agencies control the enforcement and management decisions the Grid exists to improve, and the paper's rule that satellite analytics support human decisions and lawful referral, never automated enforcement, is what makes them approachable. Finally the UAE Space Agency, the New Development Bank (NDB) and the BRICS space and ocean working groups each appear in the ninety-day list with a named first contact, which is what places them in the influence ring rather than the concern ring. Two further bodies belong in the ring on the same test: the Great Blue Wall community of practice convened by IUCN, which controls the regional reporting vocabulary ABIG's products must speak *[IUCN, 2026]*[8], and Digital Earth Africa, whose open data cube the capability map must integrate rather than duplicate [WEF and Digital Earth Africa, 2021].

The concern ring holds what the concept paper's own risk table treats as design constraints rather than as persuadable actors: data-sovereignty disputes, answered by African ownership and federated access; uneven capability; fragmented finance; and the two pressures Global Fishing Watch (GFW) reports without giving a source, an estimated 40 per cent of African catch that is illegal, unreported or unregulated, and a projected fall of 30 per cent or more in maximum catch potential by 2050 in many tropical West African countries *[Global Fishing Watch, 2025a]*[9]. BRICS membership belongs here too: the concept paper lists eleven members, a count this report does not repeat, because it has not been checked against the current membership.

6.4 The escalator

Nothing in the archive shows an agency entering a programme's influence ring because it was told the programme was important. Every archived case in which an agency acted shows the same shape: a metric computed on the agency's own data, put in front of the people who decide, followed

[6] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[7] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

[8] International Union for Conservation of Nature, *Great Blue Wall report showcases five years of impact on ocean regeneration across the Western Indian Ocean*, IUCN news, June 2026. https://iucn.org/news/202606/great-blue-wall-report-showcases-five-years-impact-ocean-regeneration-across-western

[9] Global Fishing Watch, *Rising pressures on Africa's fisheries and marine resources*, 2025. https://globalfishingwatch.org/rising-pressures-on-africas-fisheries-and-marine-resources/

by a commitment of the agency's own. The escalator adds the step none of those cases records: a written confirmation tied to a named decision.

Understanding: The escalator

TIER 1: IN PLAIN WORDS

Give an agency a number about its own problem, worked out from its own records, in a form its decision makers can use: then ask it to say in writing that the number helped.

TIER 2: IN PROFESSIONAL TERMS

A staged engagement in which the programme co-produces a metric on the partner's data (establishing relevance), converts it into a decision brief addressed to a named decision (establishing use), and records the partner's written acknowledgement (establishing adoption that a funder or a later partner can audit).

TIER 3: IN TECHNICAL TERMS

Three artefacts carrying identifiers from the schema of Chapter 8: a metric run keyed to the partner's dataset and a model-run identifier; a brief keyed to a decision identifier; and a confirmation keyed to that same identifier, its signatory and its date. It is the operational form of the WinWin spiral's front-end step of recording each stakeholder's win conditions and commitments before a cycle begins [Boehm and Hansen, 2000].

AS A MATHEMATICAL MODEL

With the partner's decision d at time t_d and the brief delivered at $t_B < t_d$, the escalator succeeds when a confirmation $C_a = 1$ is recorded for an actor with $V_a(B) > 0$. The adoption indicator is $\sum_a C_a$ over the decisions briefed and the value realised is $\sum_a C_a V_a(B)$, which is why realised value is bounded by adoption rather than by model accuracy.

The ocean case is Gabon. Global Fishing Watch trained twenty-five personnel from the fisheries directorate, the national parks agency and the navy in patrol planning, then produced a targeted intelligence report on a fleet of five trawlers about ten kilometres inside Gabonese waters; on 18 July 2026 the authorities, working with Sea Shepherd, intercepted three without valid licences and escorted them 200 km to Libreville *[Global Fishing Watch, 2026]*[10]. The chain is metric on their own waters, brief, action. What it lacks is a denominator (no comparison with untargeted patrols is offered), and a written confirmation tied to a recorded decision identifier is what makes one obtainable later. The urban case is Singapore, where the heat-stress advisory, built with a Ministry of Health expert panel, reaches the public through the environment agency's own application from the agency's own sensors *[MSE Singapore, 2023]*[11], and the Digital Urban Climate Twin's explorer application is being developed in close collaboration with agencies *[Centre for Liveable Cities, 2024]*[12]: because the agency owns the channel, research becomes an operational product. The public-realm case is Paris, where the city ran six workshops per school with pupils, three with staff and at least three working sessions with municipal departments at every site *[EEA Climate-ADAPT, 2023]*[13], and

[10] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[11] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

[12] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

[13] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

sent fifty heads of service, head teachers and technical staff on a field visit to Antwerp and Brussels that it credits with clearing doubts and scepticism, before extending the programme to a further sixty schoolyards on its own funds [Urban Innovative Actions, 2023]. The materials case is Phoenix, where a city–university partnership turned a first-year evaluation into numbered next steps that a second phase executed [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024].

The method literature explains why the sequence works. Boehm's Theory W holds that making winners of the key stakeholders is the condition for project success; in the WinWin case study five library clients wished to adopt or extend the application built for them, but only one was implemented, because it alone had sufficient budget, people and facilities to sustain the product. In the second year sustainment commitment became the top criterion for selecting what to build, and four of five selected applications transitioned to operations [Boehm et al., 1998]. That study's lesson not to finish negotiations before prototyping explains the storyboard's ordering (present the map first, obtain the confirmation afterwards) and Boehm's requirement that a team member be representative, empowered, knowledgeable, collaborative and committed, on pain of giving the appearance but not the reality of participation, is the test for who may sign [Boehm and Hansen, 2000]. That no single agent in a multi-agent system sees all the information [Powell, 2019] is the formal reason for briefs computed on each agency's own data rather than one system-wide dashboard. The economic reason is visible in the Digital Earth Africa valuation, where every projected benefit is scaled by an assumed 10 per cent adoption rate and the underlying readiness index gives user adoption a weight of 20 per cent [WEF and Digital Earth Africa, 2021]: adoption, not accuracy, sets realised value. The discussion notes name the social form of the practice *nemawashi* [Iyyunni, 2026]; the escalator is nemawashi with a paper trail.

6.5 Why gate condition 6.2.4 matters

The month-twelve gate has four conditions. Three sit squarely inside the control ring: at least 90 per cent sensor uptime across the summer, a model matching sensor observations within the stated error, and an hour-by-hour comfort map of both zones. The fourth does not: at least one of the Roads and Transport Authority or the Dubai Electricity and Water Authority must confirm in writing that the output is useful for a decision it faces [Iyyunni, 2026]. Each of those two supplies a single dataset (station exit counts with bus and abra ridership in the first case, one summer of hourly Bur Dubai district cooling load in the second) and each is asked in return for a judgement about its own decisions. Condition 6.2.4 is the only one whose satisfaction depends on somebody outside the team, and therefore the only one that tests whether the programme has an influence ring at all. A network can run, a model can validate and a map can be delivered without a single decision changing anywhere in Dubai; the condition is the storyboard's escalator written as a stop condition, and if the gate fails the project closes with the dataset and platform as its deliverables.

Understanding: Written confirmation as evidence

TIER 1: IN PLAIN WORDS

A short letter from one of the two agencies saying that the comfort map helped it with a real decision. Without that letter the project stops at the end of Year 1 and hands over what it has built.

TIER 2: IN PROFESSIONAL TERMS

An acceptance criterion for adoption: the outcome a funder cannot read off technical deliverables. It corresponds to the partnerships, capacity-building and impact criteria of the Dubai RDI rubric *[Dubai RDI Programme, 2025]*[2] and to the stakeholder concurrence Boehm requires at the Life-Cycle Architecture anchor point [Boehm and Hansen, 2000].

TIER 3: IN TECHNICAL TERMS

A dated document naming the decision and the agency dataset on which the metric was computed, signed by a person meeting the five criteria for an empowered team member, and recorded against a decision identifier so that the outcome can later be attached to it. The Gabon case shows an action without a counterfactual *[Global Fishing Watch, 2026]*[10]; the recorded identifier is what lets the Year-3 before-and-after study supply one.

AS A MATHEMATICAL MODEL

The gate passes on the fourth condition when $\max\{C_{\text{RTA}}, C_{\text{DEWA}}\} = 1$, where $C_a = 1$ requires a tuple $(a, d_a, \text{data}_a, s_a, t)$ with $t \leq$ March 2028, d_a a named decision and s_a a qualified signatory. For the Year-2 choice of intervention this report recommends the stronger criterion $C_a = 1$ together with a written sustainment commitment $S_a = 1$ from the agency that will own the structure.

Three refinements follow. The confirmation must be specified now, in the February 2027 data letters, rather than improvised in March 2028: each letter should name the decision the agency expects the Year-1 map to inform, so the brief has an address before it is written. The criterion of usefulness, by contrast, should be settled after the first map has been seen, because agencies cannot know what a useful comfort map looks like until one exists (Boehm's 'I'll know it when I see it' [Boehm and Hansen, 2000]) and the storyboard's sequence of presenting in February and confirming in March 2028 allows for it. And the confirmation is necessary but not sufficient for Year 3: what survives is what somebody can sustain, so the choice of intervention between January and March 2029 should require a written commitment to operate and maintain whatever is built [Boehm et al., 1998]. The United Nations Secretary-General's Call to Action on extreme heat asks governments to appoint a champion or an office with a cross-sectoral heat-resilience mandate [UN Secretary-General, 2024]; the storyboard names none, and the agency that signs the first confirmation is the natural candidate.

6.6 The five stakeholders and their reach

Table 6.1 maps the five stakeholders to the institutions they can actually reach, using only what the working papers and the archived sources state. Reach is recorded by documented role or affiliation; where the archive documents no relationship, the table says so rather than assume one.

Two observations follow. The Dubai influence ring is reached through one stakeholder and one document, which is efficient and fragile in equal measure, while the ABIG influence ring is reached through one stakeholder's membership of the two bodies the addendum most needs: the single strongest lever either programme holds. And neither programme has a documented route to the funders that matter in the outer years, the New Development Bank for ABIG and a Dubai agency's procurement budget after March 2030 for the corridor, so the plan below treats both as escalator targets rather than as assumptions.

6.7 The influence plan

Table 6.2 is the working instrument of this chapter: for every actor in either influence ring, what that actor controls, what the programme needs from it, the evidence expected to move it, the proposed owner and the timing. Owners are this report's proposals, drawn from roles the documents name; dates are the storyboard's milestones and the concept paper's horizons.

Stakeholder	Documented role	Institutional reach (ring)	Instrument
Dr Uchi Dominic Terhile, Federal University Dutse, Nigeria	Lead author of the ABIG concept paper and of the addendum, signed for the OceanPrediction DCC African Seas Regional Team and GOOS-Africa [Terhile et al., 2026]	Core of the ABIG influence ring: the GOOS-Africa working groups, the African Seas Regional Team and OPERA	The alignment and mapping exercise; the small technical dialogue the addendum recommends
Dr Chakradhar Iyyunni, BITS Pilani Dubai Campus, UAE	Author of the Dubai storyboard [Iyyunni, 2026]; co-author of the ABIG paper	Dubai control ring as lead institution, and through the Letter of Intent the Dubai influence ring; eligible principal investigator under the RDI rules *[Dubai RDI Programme, 2025]*	Letter of Intent, September 2026; site and data letters; the month-twelve gate
Abdullah Al Salmani, SpacePoint, UAE	Co-author of the ABIG paper [Terhile et al., 2026]	The UAE space-sector ecosystem the paper names as the enabling hub; reach by affiliation, with no relationship documented in the archive	The ninety-day action to engage the UAE Space Agency on an Africa–UAE gateway
Arti Ahluwalia, Country Chair, Ecocivilisation.earth, India	Co-author of the ABIG paper; contributor to the civil-society documents cited in this report [Commons Cluster, 2024, Commons Cluster, 2023]	Civil-society networks convened at the UN High-Level Political Forum; India, listed by the paper among the BRICS members	The inclusive-participation strand; the data and ethics board; a BRICS technical counterpart
Mahesh Shastry, Kalman Systems Pty Ltd, Sydney, Australia	Convening stakeholder; one architecture and one evidence-grading scheme across both programmes	Control ring of both programmes (architecture, metric definitions, evidence grading, identifier schema) and no institution in either influence ring	This report, its source archive, and the influence plan of Table 6.2

Table 6.1: The five stakeholders and their documented institutional reach. Affiliations use the report's standard wording; reach is asserted only where a working paper or an archived document supports it.

Table 6.2: Influence plan for the two programmes. Levers and dates come from the storyboard [Iyyunni, 2026], the Dubai RDI call rules *[Dubai RDI Programme, 2025]* and the ABIG concept paper [Terhile et al., 2026]; evidence entries cite the archived case that supports them. Owners are proposed by this report.

Actor	What they control	What we need	Evidence that moves them	Owner	Timing
Dubai: Climate Intelligence and Decision Twin					
RTA	Al Ghubaiba exit counts; bus and abra ridership	Data letter; written confirmation under 6.2.4	Hour-by-hour comfort map of the first- and last-mile link computed on its own counts; dwell time and route abandonment in Year 2	PI; Co-I for mobility	Feb 2027; Feb–Mar 2028
DEWA	Bur Dubai district cooling load, hourly, one summer	Data letter; written confirmation under 6.2.4	Comfort hours gained per kilowatt-hour and per litre; options that add comfort without adding load	PI; Co-I for systems integration	Feb 2027; Mar 2028; MCN Nov 2028
Dubai Culture	Access to the heritage core; sensor permissions	Site letter; agreement on the intervention	Heritage core used only as a calibration baseline, with no intervention placed in it	PI; Co-I for built environment	Dec 2026; Jan–Mar 2029
Dubai Municipality	Public-realm permissions; municipal geometry; the Digital Twin Platform *[Dubai Media Office, 2026]*	Site letter; access to geometry layers; sustainment commitment	The 1–5 m comfort layer the platform does not hold; the Abu Dhabi precedent of municipal approval for 26 stations [Manandhar et al., 2019]	PI; Co-I for edge sensing	Dec 2026; nodes live May 2027; Jan–Mar 2029
DFF and Dubai RDI	Grant envelope and rules; cycle calendar; Phase-2 release *[Dubai RDI Programme, 2025]*	A cycle open to a Sep 2026 letter (undocumented); release of Phase 2 at the gate	Gate design with an explicit stop condition; letters of institutional commitment; rubric-aligned deliverables and risk plan	PI; university research office	LoI Sep 2026 (contingent); gate Mar 2028
Curtin University Dubai	Green AI, downscaling and optimisation capacity	Signed agreement and budget share	A named role and the letter of collaboration the rubric requires	PI	Feb 2027
DataAlt Dynamics	The Twin platform, or replacement in house	A decision kept open through Year 1	Year-1 platform performance against an in-house dashboard	Co-I for systems integration	By Mar 2028

continued overleaf

Table 6.2, continued

Actor	What they control	What we need	Evidence that moves them	Owner	Timing
DET (second stage)	Destination policy	Adoption of OTC as a destination indicator	OTC for Al Fahidi computed from RTA counts and the comfort index	PI; doctoral student	From mid-2028
Dubai Walk programme (second stage)	3,300 km new and 2,300 km renovated walkways by 2040 *[Dezeen, 2024]*	Adoption of the MCN as a per-kilometre shade specification	Corridor MCN; Year-3 measured against predicted change	Co-I for built environment	Nov 2028; Sep 2029
MoHRE (second stage)	The midday break, 12:30–15:00, 15 Jun–15 Sep *[The National, 2026]*	Nothing within the grant period	Hour-by-hour maps as a basis for threshold-based rules of the ILO kind [ILO, 2019]	PI	After Mar 2030
Africa Blue Intelligence Grid					
GOOS-Africa	The observing community and its five working groups	A formal interface on the scientific and technical council; the joint mapping exercise	A capability map showing what exists and where the gaps are	Dr Terhile	0–12 months
Ocean-Prediction DCC; OPERA	Operational forecasting	A small technical dialogue; one or two joint demonstrations	ABIG as downstream architecture for forecast products, with no duplication	Dr Terhile	0–3 months; 6–12 months
GMES & Africa; MarCO-SIO	Earth-observation services; three co-designed service lines *[MarCOSIO, 2025]*	An interoperable gateway; hosting of prototypes	Two or three prototypes built from data that already exists	Programme office	3–12 months

continued overleaf

Table 6.2, continued

Actor	What they control	What we need	Evidence that moves them	Owner	Timing
AUC; African Space Agency; RECs	Political mandate and council seats	A two-page stakeholder proposition; a political council	Alignment with the Luanda Declaration *[African Union, 2026]* and the Blue Economy Strategy's information-support objective [AU-IBAR, 2019]	Dr Terhile; Dr Iyyunni	0–3 months
A national fisheries authority already working with GFW	Enforcement and management decisions	A named operational user for the illegal-fishing use case; a sustainment budget	The Gabon precedent of 18 July 2026 *[Global Fishing Watch, 2026]*; a brief on its own patrol data, with a denominator	Regional hub; Dr Terhile	6–12 months
UAE Space Agency	Space Data Center; Geo-Spatial Analytics platform	An Africa–UAE access and analytics gateway	Use cases with named African users attached to each	UAE-based stakeholders	90-day list; Year 2
New Development Bank	Project finance for digital, water and environmental infrastructure	An initial investment concept; a project-preparation pipeline	Independent validation of a pilot before blended finance; the small, de-risked Seychelles blue bond precedent [World Bank, 2025]	Programme office; UAE-based stakeholders	90-day list; Year 2
BRICS space and ocean working groups	Virtual Constellation; polar and ocean science cooperation	One technical counterpart per relevant member	A use-case portfolio with explicit data requirements	Stakeholders in member states	90-day list
IUCN and Great Blue Wall partners	A regional community of practice and its reporting vocabulary *[IUCN, 2026]*	Products rather than another forum; attribution reported alongside contribution	Regional-hub products that add a counterfactual to programme-reported indicators	Eastern Africa hub	Year 2

continued overleaf

Table 6.2, continued

Actor	What they control	What we need	Evidence that moves them	Owner	Timing
Digital Earth Africa	An open data cube of more than 30 million km^2 [WEF and Digital Earth Africa, 2021]	Integration in the capability map, not duplication of it	The data inventory; a first knowledge product built on analysis-ready data	Programme office	0–3 months

6.8 Implications for the programmes

Key finding

Both programmes have small control rings and depend on influence rings they have not yet entered. Every archived case in which an agency acted (Gabon, Singapore, Paris, Phoenix) shares one mechanism: a metric computed on the agency's own data, delivered as a brief, followed by the agency's own commitment. Gate condition 6.2.4 writes that mechanism into the Dubai storyboard as a stop condition, and ABIG's ninety-day list writes it as a set of named first contacts. Neither yet demands the sustainment commitment that the WinWin evidence shows decides what survives.

For Dubai, five actions follow. Treat the December 2026 and February 2027 letters as win conditions in Boehm's sense and record them: each should name the dataset, the person empowered to sign, and the decision the agency expects the Year-1 map to inform [Boehm and Hansen, 2000, Iyyunni, 2026]. Ask that Dubai Municipality's letter cover geometry held in the Digital Twin Platform *[Dubai Media Office, 2026]*[1]. Specify condition 6.2.4 as a tuple (decision, dataset, signatory, date, decision identifier) and settle the criterion of usefulness after the first map is presented rather than before. Make a written sustainment commitment a condition of the Year-2 intervention selection, because the only WinWin application that survived was the one its client could sustain [Boehm et al., 1998]. And keep the Department of Economy and Tourism, the Dubai Walk programme and the labour ministry in view as second-stage targets whose escalator cannot start until OTC and the MCN exist, in mid-2028 and November 2028 respectively.

For ABIG, five likewise. Begin with the addendum's small technical dialogue with GOOS-Africa, the OceanPrediction African Seas Regional Team and OPERA, conducted through the stakeholder who already belongs to two of them, and make the capability map the first product [Terhile et al., 2026]. Choose the two or three pilots by the sustainment rule (a named operational user with a pathway to an operational budget), which is the paper's own answer to pilot fatigue; the fisheries authorities already working with Global Fishing Watch are the obvious candidates *[Global Fishing Watch, 2025a]*[9]. Give every pilot a denominator from the first day, so that decisions improved per dollar can be reported where the Gabon case could not *[Global Fishing Watch, 2026]*[10]. Route the financing conversation through the UAE Space Agency and the New Development Bank as the paper proposes, but only after a pilot has been independently validated, keeping to the order of the financing ladder. And verify the BRICS membership count before repeating it anywhere.

The implication common to both is the one Chapter 7 formalises and Chapter 10 costs: no loop of either spiral should begin without the written commitments of the actors whose decisions that loop depends on. Table 6.2 lists those commitments in the order they fall due; read the other way, it is a register of single points of failure (one signature at the Dubai gate, one alignment exercise for ABIG) and Chapter 11 treats it as such.

V {7}

System Boundaries: Spiral Development and Stochastic Dynamic Programming

Abstract

Where the boundaries of the two programmes should be drawn, first under Boehm's risk-driven spiral model and then under stochastic dynamic programming in Powell's five-element form. Part A maps the Dubai storyboard's pre-award, Year-1, Year-2 and Year-3 loops onto the anchor points Life-Cycle Objectives, Life-Cycle Architecture and Initial Operational Capability, states the exit criterion of each loop, applies the same instrument to the 3-, 6-, 12-, 24- and 36-month horizons of the Africa Blue Intelligence Grid, and follows the scale spiral from one corridor to the arid world. Part B writes the Decision Twin as a sequential decision problem (state, decision, exogenous information, transition, objective) with the Bellman recursion, decisions at two timescales, the information boundary set by twenty sensor nodes, the expected value of information as the justification for Year 1, the four policy classes, and the water and energy couplings shared with the utility. Part C unifies both views in a taxonomy of eight boundaries. The method sources carry no effect sizes; the chapter therefore produces boundary clauses, not performance claims.

Every feasibility question is, in the end, a boundary question: what lies inside the programme and what lies outside it, what is decided and what is merely observed, what is committed now and what is deferred until more is known. The two proposals draw their boundaries in prose (one corridor, named partners and a hard gate at month twelve for Dubai [Iyyunni, 2026]; an integration and application layer above the observing and forecasting programmes that Africa already runs, for the Africa Blue Intelligence Grid (ABIG) [Terhile et al., 2026]) but neither gives its boundaries a method. This chapter supplies two. The spiral model treats a programme as a sequence of risk-driven loops, each ending at a milestone where the stakeholders commit further resources or stop, so that the boundary of a loop is the set of things the next commitment depends upon [Boehm and Hansen, 2000, Boehm et al., 1998]. Stochastic dynamic programming treats the same programme as a sequential decision problem, so that the boundary is the line between the state a decision-maker carries forward and the exogenous information that arrives regardless [Powell, 2019, Powell, 2022, Bellman, 1957].

The two methods draw the same line from opposite sides. The spiral asks what must be *true* before the next loop may begin; the decision model asks what must be *known* before the next decision can be a good one. Where they agree, a boundary is defensible and can be written into a proposal as a clause. Where they disagree, the disagreement is itself a finding, and is carried forward to the risk register of Chapter 11. The evidence base is deliberately thin: the Software Engineering Institute report that defines the spiral model is graded method-reference; the two-year university case study

of the WinWin variant is graded case-report, its year-on-year improvements confounded by several simultaneous process changes and unsupported by any control group; Powell's unified framework contains no data of its own; and Bellman's monograph is reachable here only through Powell's citations [Boehm and Hansen, 2000, Boehm et al., 1998, Powell, 2019, Bellman, 1957]. None of these texts measures anything about heat, oceans or cities. They are used for what they are (instruments for drawing boundaries) and not as evidence that a boundary drawn with them will hold.

7.1 Part A: boundaries under the spiral model

7.1.1 The spiral as a boundary-drawing instrument

Boehm defines the spiral model as a risk-driven process model *generator* rather than a fixed sequence of phases, and sets out six invariants that a process must exhibit to count as spiral [Boehm and Hansen, 2000]. In plain words: artefacts are determined concurrently rather than one after another; every cycle considers objectives, constraints, alternatives, risks, a review and a commitment; the level of effort is set by risk; the degree of detail is set by risk; the cycles are punctuated by three anchor-point milestones; and the emphasis falls on the whole system over its whole life rather than on the software alone. The same report catalogues six hazardous look-alikes that violate those invariants: among them sequential phases with key stakeholders excluded, evolutionary development with no life-cycle architecture, and impeccable spiral plans carrying no commitment to manage the risks they identify [Boehm and Hansen, 2000]. Part C shows that these are precisely the failure modes the two programmes most need to name.

The three anchor points give each loop its exit criterion. Life-Cycle Objectives (LCO) is the stakeholders' commitment to support architecting: at least one feasible architecture and a business case must exist. Life-Cycle Architecture (LCA) is the commitment to the full life cycle: one architecture has been chosen, and every major risk has been resolved or is covered by an element of the risk-management plan. Boehm calls LCA the point of significant escalation of staff levels and resource commitments. Initial Operational Capability (IOC) is the commitment to operations and has three elements: software preparation, site preparation, and user, operator and maintainer preparation. Six artefacts are reviewed at LCO and at LCA (operational concept, prototyping results, requirements description, architecture description, life-cycle plan and feasibility rationale), the last of which asks whether a product built with the stated architecture and processes will support the concept, realise the prototyping results, satisfy the requirements and finish within the budgets and schedules of the plan [Boehm and Hansen, 2000].

Understanding: Spiral anchor point

TIER 1: IN PLAIN WORDS

A point in a project where everyone who matters looks at what has been learned so far and decides together whether to put in more money and effort or to stop.

TIER 2: IN PROFESSIONAL TERMS

A stakeholder commitment milestone with defined entry artefacts and pass or fail criteria (LCO for architecting, LCA for the full life cycle, IOC for operations) at which the feasibility rationale is reviewed and the level of resource commitment changes [Boehm and Hansen, 2000].

TIER 3: IN TECHNICAL TERMS

An anchor point is the boundary between two risk-driven cycles: simultaneously the exit criterion of cycle k and the entry condition of cycle $k+1$. Boehm's analogy for the underlying discipline is stud poker: a small stake buys one more round of information, after which the

player bets or folds. Committing the full funding of a programme on very incomplete early information is the error the anchor points exist to prevent [Boehm and Hansen, 2000].

AS A MATHEMATICAL MODEL

Let $\mathscr{R}_k$ be the major risks identified in cycle k, $\mathscr{M}_k$ those covered by the risk-management plan, and $\mathrm{RE}(r) = \Pr(\text{loss} \mid r) \times \text{size}(\text{loss} \mid r)$ the risk exposure of risk r [Boehm and Hansen, 2000]. The LCA condition is $\forall\, r \in \mathscr{R}_k \;:\; \mathrm{RE}(r) \le \tau$ or $r \in \mathscr{M}_k$, where τ is the tolerance the stakeholders have written down.

7.1.2 The Dubai timeline as four loops

The Dubai storyboard already has a loop structure: a pre-award period, three annual phases, and a stop-or-go decision at month twelve at which the Phase 2 funds are released [Iyyunni, 2026]. Table 7.1 maps that structure onto the anchor points and states, for each loop, what lies inside the boundary and what must be true for the next loop to begin. Figure 7.1 draws the same mapping. Three features of the mapping are not obvious from the storyboard.

First, the pre-award loop (Loop 0, September 2026 to March 2027) is the LCO cycle, and it is the loop in which the boundary is actually drawn: one corridor of roughly 800 m from Al Ghubaiba metro station to the heritage core, the Al Fahidi Historical Neighbourhood retained untreated as the calibration baseline, named partners, named tools, funded posts and one built intervention [Iyyunni, 2026]. Its exit artefacts are letters (site access from Dubai Culture and Dubai Municipality by December 2026, data letters from the Roads and Transport Authority (RTA) and the Dubai Electricity and Water Authority (DEWA) by February 2027), together with a business case, which the storyboard does not yet contain and which Chapters 9 and 10 supply. In Boehm's terms the storyboard is an LCO package that is part complete: it holds an operational concept, a life-cycle plan and a budget, but not the prototyping results, a requirements description, an architecture description or a feasibility rationale.

Second, the month-12 gate is the LCA anchor point, and the correspondence is exact in three respects. The gate precedes the release of Phase 2 funds, which is Boehm's escalation of resource commitment. Its four conditions (a sensor network live with at least 90 per cent uptime across the summer; model output matching sensor observations within the stated error; an hour-by-hour comfort map of both zones delivered; and at least one of RTA or DEWA confirming in writing that the output is useful for a decision it faces) map onto three of the six anchor-point artefacts (prototyping results, the operational concept realised and the feasibility rationale) together with the stakeholders' concurrence that the anchor-point review itself requires [Iyyunni, 2026, Boehm and Hansen, 2000]. And the storyboard's rule that a failed gate closes the project with the dataset and the platform as deliverables is the fold in the poker analogy. Boehm's own supports are the per-cycle plan to terminate a project that proves too risky or infeasible and the design-to-schedule rule that separates a feasible core capability from optional features [Boehm et al., 1998, Boehm and Hansen, 2000]; treating a dataset as the fallback *deliverable* is this report's inference from that guidance rather than Boehm's phrase, and is labelled as such.

Third, Year 3 is the IOC cycle, and Boehm's three IOC elements name what the storyboard leaves implicit. Software preparation is Decision Twin version 1 and the method document. Site preparation is the built intervention (most probably a shade canopy over the most exposed 200 m) with its water and power connections and its maintenance access. User, operator and maintainer preparation is the element with no owner after March 2030. The WinWin case study is the warning here: of five applications that clients wished to adopt in its first year, only the one with a sustaining budget, people and facilities was ever implemented, and in the second year a sustainment commitment became the top criterion for selecting what to build [Boehm et al., 1998]. This report therefore adds an exit criterion that the storyboard does not have: the intervention selected during

Loop 2 must carry a written sustainment commitment from the agency that will own it.

Table 7.1: The Dubai storyboard as four risk-driven loops, with a fifth beyond the grant, and their anchor points, boundaries and exit criteria. Dates, scope and gate conditions are from the storyboard [Iyyunni, 2026]; anchor-point definitions from Boehm [Boehm and Hansen, 2000]; the sustainment criterion in Loop 2 is this report's addition after the WinWin case study [Boehm et al., 1998].

Loop, period, anchor	Inside the boundary	Exit criterion
0 Pre-award Sep 2026 – Mar 2027 *LCO*	One corridor of about 800 m plus the heritage core as untreated baseline; named partners; named tools; funded posts; draft ontology; ethics design	Site-access letters (Dec 2026); RTA and DEWA data letters, Curtin agreement and final budget (Feb 2027); written win conditions per partner; a business case
1 Observe and Model Mar 2027 – Mar 2028 *LCA, at the gate*	Measurement and validation only: 20 nodes plus one reference station; data platform; one summer of records; district model at 25–50 m and street model at 1–5 m; first comfort map	The four gate conditions: uptime of at least 90 per cent; model within the stated error; hour-by-hour comfort map of both zones; one written confirmation of usefulness from RTA or DEWA
2 Predict and Optimise Mar 2028 – Mar 2029 *(no anchor)*	One corridor's intervention set only (shade canopy, tree planting, cool pavement, misting rest point, revised route); OTC metric; multi-objective optimiser; Decision Twin v1; second summer	MCN delivered (Nov 2028); Twin v1 (Feb 2029); one intervention selected with Dubai Culture and Dubai Municipality *and* a written sustainment commitment
3 Build and Prove Mar 2029 – Mar 2030 *IOC*	One built intervention; before-and-after over one summer against both the twin's prediction and the untreated zone; repeated counts and dwell times; decision briefs; method document. The citywide 100 m product is outside the boundary	Measured change compared with predicted change; behaviour change tested; briefs delivered to a named recipient; follow-on proposal lodged
4 Scale 2030 onward *a new LCO*	A second corridor; procurement by a Dubai agency; platform operation, retraining and standards work	A new business case and feasibility rationale for the enlarged boundary

7.1.3 How much is enough: the risk boundary

The third and fourth invariants make risk exposure the arbiter of how much of any activity is enough: too little testing leaves defects, too much delays the market, and total risk exposure therefore has an interior minimum [Boehm and Hansen, 2000]. This converts three open questions in the storyboard from matters of taste into matters of argument. The twenty nodes plus one reference station should be justified as the number at which the exposure from an unrepresentative sample stops falling faster than the exposure from spending the equipment budget; the calibration and first-campaign duration should be set the same way; and the threshold $\varepsilon_{\text{gate}}$ in gate condition 6.2.2 should be fixed *after* the first summer's data rather than written into the proposal. Boehm's own illustration is an early-1980s information system in which freezing a one-second response-time requirement before prototyping implied an architecture costing nearly US$100 million, while a prototype showing that four seconds satisfied users 90 per cent of the time

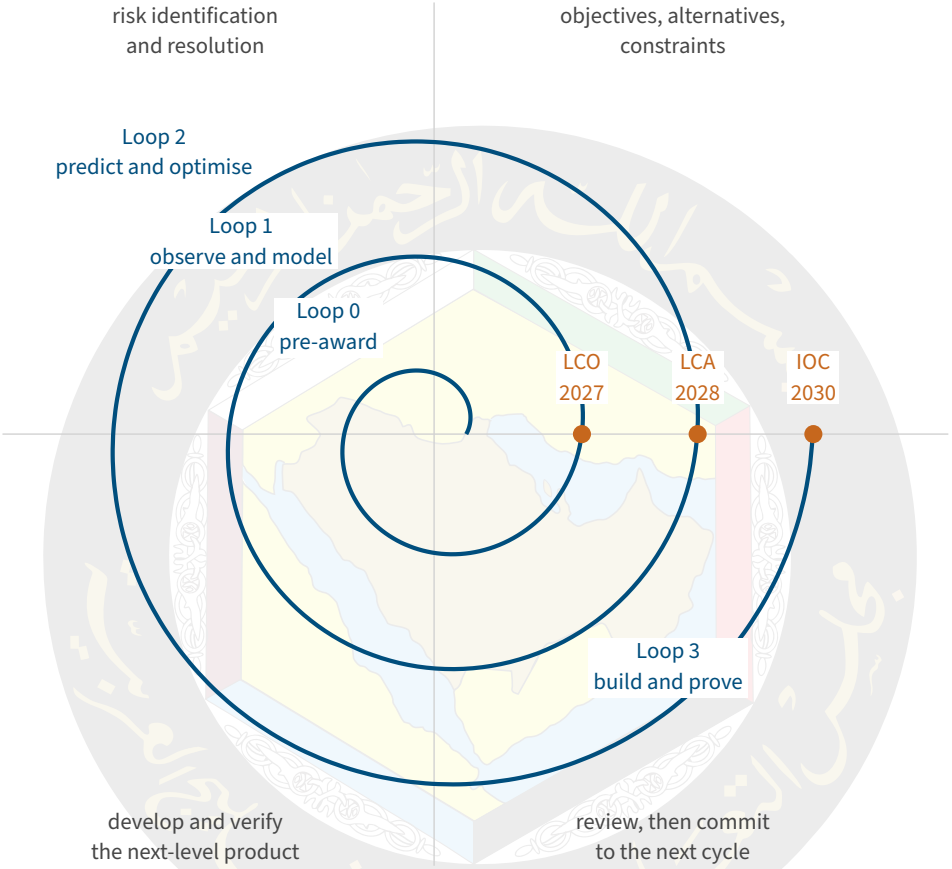


Figure 7.1: The Dubai programme as a risk-driven spiral. Each turn passes through the four per-cycle elements of the spiral model; the radius grows with cumulative commitment; the three marked points on the ray are Boehm's anchor points, at which the stakeholders either commit further resources or fold [Boehm and Hansen, 2000]: Life-Cycle Objectives at the award in March 2027, Life-Cycle Architecture at the month-12 gate in March 2028, and Initial Operational Capability at the close of the grant in March 2030. Loop dates are from the storyboard [Iyyunni, 2026].

permitted a different architecture at US$30 million [Boehm and Hansen, 2000]. That is a single anecdote about one system and no saving ratio may be derived from it; what transfers is the structure of the error (freezing a numerical performance requirement before the prototype exists), which is exactly the error a model-error threshold written into a proposal in 2026 would commit.

The fourth invariant also determines where *not* to specify. Interfaces whose failure is expensive should be specified precisely: node data format, calibration version, sampling interval, the provenance fields of Chapter 8. Choices whose premature fixing is expensive should be left open: the dashboard, and the decision between a platform partner and an in-house build. The same invariant supports the nested spatial resolutions. A district layer at 25–50 m and a street layer at 1–5 m follow the precedent set in Abu Dhabi, where 30 m multispectral imagery (100 m thermal) was used for zone classification and a 3 m street model for the microclimate [Manandhar et al., 2019]; each layer is specified to the precision its decisions require and no further.

7.1.4 Win conditions: the commitment boundary

The WinWin variant adds a negotiation front end to each cycle, in which the stakeholders' win conditions are elicited, conflicts surfaced as issues, options explored and agreements recorded, with the commitments and the areas deliberately left flexible both written into the project's design record [Boehm and Hansen, 2000, Boehm et al., 1998]. Two consequences bind the Dubai boundary. First, no loop should begin without written win conditions for every partner, which is why the letters of Loop 0 are exit artefacts and not courtesies. Second, the person who signs on behalf of an agency must meet Boehm's criteria for a qualified team member (representative, empowered to make commitments the organisation will honour, knowledgeable, collaborative and committed), because an unqualified contact gives the appearance but not the reality of stakeholder participation [Boehm and Hansen, 2000]. Gate condition 6.2.4 is worth exactly as much as the standing of the person who signs it.

A third consequence is procedural. The WinWin study's blunt lesson was not to finish negotiations before prototyping: clients do not know what they want until they see something [Boehm et al., 1998]. The meaning of 'useful' in gate condition 6.2.4 should therefore be negotiated with RTA and DEWA *after* the first hour-by-hour comfort map exists and before the gate, not written in advance.

7.1.5 ABIG as five loops

The ABIG concept paper sets out a roadmap over five horizons (0–3 months, 3–6 months, 6–12 months, Year 2 and Year 3) with goals and processes for each, a 90-day action list, a governance stack of seven layers, nine principal risks with design responses, and a financing ladder running from a public policy mandate through project preparation, pilot, independent validation and blended finance to a commercial service [Terhile et al., 2026]. What the roadmap lacks is exit criteria: goals are stated as outcomes rather than as conditions that must hold before the next horizon may be entered. Without them the roadmap risks becoming one of Boehm's hazardous look-alikes: an impeccable spiral plan with no commitment to managing the risks it identifies [Boehm and Hansen, 2000]. Table 7.2 supplies the missing column.

One boundary dominates all five loops. The addendum places ABIG as an integration and application layer above GOOS-Africa for observation, OceanPrediction and OPERA for forecasting, and GMES and Africa for Earth-observation services [Terhile et al., 2026]. Holding that line is what keeps ABIG from the failure its own risk table names first, becoming another coordination forum; in spiral terms, letting observation or forecasting drift inside the boundary would be evolutionary development with no life-cycle architecture [Boehm and Hansen, 2000]. The boundary must therefore be fixed in writing at the LCO of month twelve and re-tested at LCA, not defended informally meeting by meeting.

Table 7.2: The ABIG roadmap as five risk-driven loops. Horizons, goals, governance layers, risks and the financing ladder are from the concept paper [Terhile et al., 2026]; anchor points and exit criteria are this report's application of the spiral model [Boehm and Hansen, 2000, Boehm et al., 1998].

Horizon	Anchor	Inside the boundary	Exit criterion
0–3 months	pre-LCO	Charter; capability and data inventory; use-case scoring; ethics and responsible-use principles; pilot-geography selection	Charter adopted; 5–7 use cases scored; named focal points per institution; data-sharing negotiations opened
3–6 months	n/a	First prototypes from existing Earth-observation and data sources; fellowship design; common metadata and provenance standards	At least two prototypes live with a named operational user each; metadata and provenance standard published
6–12 months	LCO	The capability map; 2–3 operational pilots; independent validation; user-acceptance tests; investment-ready programme definition	Capability map published; at least one feasible federated architecture with a business case; benefits measured with the user, not for the user
Year 2	LCA, then IOC	Regional hubs; shared data infrastructure; procurement standards; operational service agreements; first financed infrastructure	All nine concept-paper risks resolved or covered by the risk-management plan; one use case in routine operational use by a named agency, with software, site and operator preparation complete
Year 3	new LCO	Repeatable continental services; recurring budgets; service-level agreements; independent impact evaluation	Recurring budgets secured for services that survive the programme; a new business case for the next enlargement

7.1.6 The scale spiral

There is a second spiral, orthogonal to the first. Each turn of the programme spiral enlarges the spatial boundary while reusing the method and the metric: corridor, district, city, other Gulf Co-operation Council cities, arid world (Figure 7.2). The turns are not equal. Corridor to district is an enlargement of the same physics and the same instruments. District to city crosses a computational boundary treated in Section 7.2.7. City to region crosses an institutional boundary: a different regulator, a different data-protection regime, a different procurement route. Region to arid world crosses an evidential boundary, because a method validated in one climate and one urban form is a hypothesis elsewhere until replicated. The policy argument of Chapter 12 (that a metric with an open method, replicated in three cities, becomes a norm) is the outermost turn of this spiral, and it depends on the innermost one being completed first.

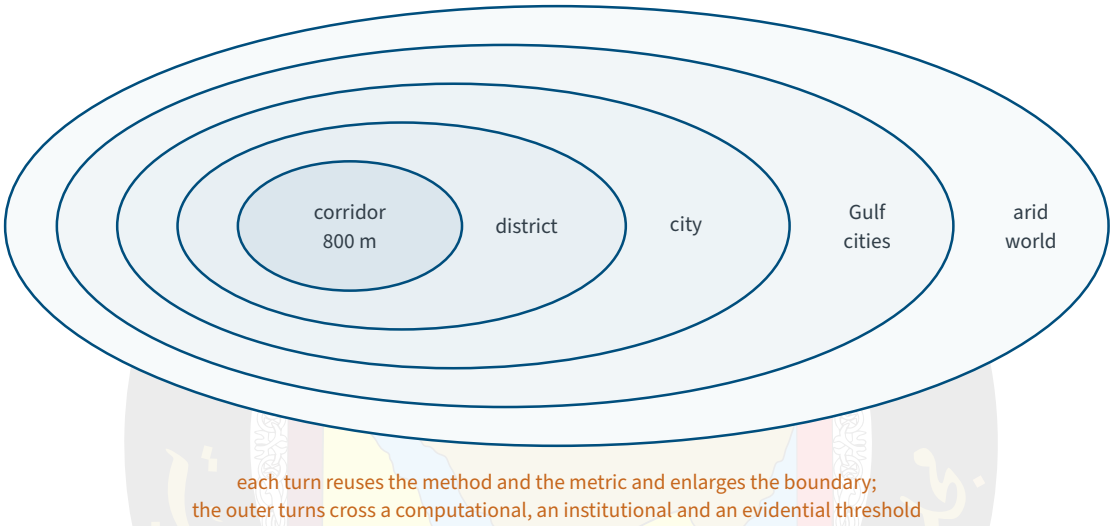


Figure 7.2: The scale spiral: the boundary of the Dubai method enlarged one turn at a time. The corridor and the district share physics and instruments; the city turn crosses a computational threshold (Section 7.2.7); the regional turn crosses an institutional one; the outermost turn crosses an evidential one and requires replication.

7.2 Part B: boundaries under stochastic dynamic programming

7.2.1 The five elements

Powell's framework models any sequential decision problem with five elements: the state variable, the decision variable, the exogenous information, the transition function and the objective function [Powell, 2019, Powell, 2022]. Its discipline is to model the problem first and to choose the solution method afterwards, which is exactly the discipline a proposal needs: writing the five elements down forces every boundary to be declared before any algorithm is promised.

The state is everything at epoch t that is needed to compute the decision, the transition and the contribution. Powell decomposes it into a physical state R_t, other information I_t and a belief state B_t [Powell, 2019]. For the Decision Twin this decomposition is a boundary in itself. The physical state holds the interventions installed and their condition, and the remaining capital, water and energy budgets. The information state holds the comfort field per place-hour, footfall and dwell, and the forecasts in hand. The belief state holds the distribution of model error and of the behavioural-response parameters, so gate condition 6.2.2, a statement about model error, is formally a statement about B_t, and Year 1 is a programme for narrowing it.

Understanding: State variable

TIER 1: IN PLAIN WORDS

Everything you need to know right now to make the next choice well. If a piece of information changes what you would do, it belongs in the state; if it does not, leaving it out costs nothing.

TIER 2: IN PROFESSIONAL TERMS

The minimal sufficient description of the system at a decision epoch: what is physically in place, what has been observed, and what is currently believed about the quantities that are not observed directly [Powell, 2019].

TIER 3: IN TECHNICAL TERMS

Bellman's formulation characterises a physical system at any stage by a small set of parameters, the state variables [Bellman, 1957, Powell, 2019]. Powell's decomposition $S_t = (R_t, I_t, B_t)$ adds the belief state explicitly, which is what makes a partially observable problem expressible without leaving the framework. Any forecast used by a policy must itself be carried in the state; otherwise the model is incorrect in Powell's sense [Powell, 2019].

AS A MATHEMATICAL MODEL

$S_t = (R_t, I_t, B_t)$, and $S_{t+1} = S^M(S_t, x_t, W_{t+1})$, where S^M is the transition function and W_{t+1} the exogenous information revealed after the decision.

Figure 7.3 draws the five elements as one epoch of the decision loop and marks the boundary they impose. Everything in the shaded band is modelled and, in principle, influenced; the exogenous information enters from outside it and can only be forecast, hedged or insured against. Deciding on which side of that band a quantity sits is the single most consequential act of modelling in the whole programme, because it determines what the team is accountable for.

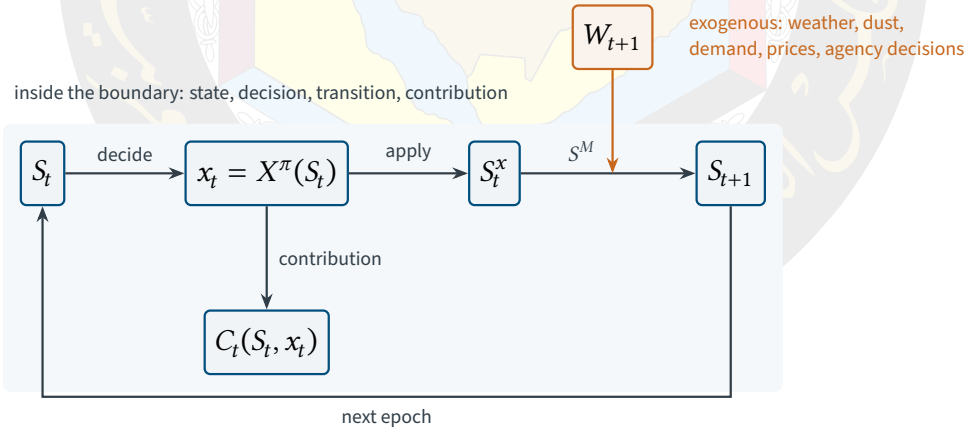


Figure 7.3: One epoch of the sequential decision model [Powell, 2019]. The shaded band is the endogenous system: state, decision, post-decision state, transition and contribution; the exogenous information enters from outside it after the decision has been taken and can only be forecast or hedged, never chosen.

7.2.2 The four metrics as functions of the state

Powell's insistence on an explicit objective over policies imposes a requirement the storyboard has not yet met: a metric that cannot be written as a function of the state and the decision cannot enter the optimiser. The four named metrics of this report are therefore fixed here as such functions. Two conventions apply throughout. The comfort index $U_{p,h}$ is the Universal Thermal Climate Index

for visitors and residents and the wet-bulb globe temperature (WBGT) for workers, reported side by side: the ILO's global series and its exposure-response thresholds (loss beginning at 24–26 °C WBGT and reaching 50 per cent at 33–34 °C for work at 300 W) are stated in WBGT, and the ILO itself notes that the Universal Thermal Climate Index lacks equivalent safety limits [ILO, 2019]. The place set $\mathscr{P}$ is indexed on H3 cells, whose resolution 11 (about 24.9 m edge) and resolution 13 (about 3.6 m edge) align with the district and street layers respectively *[Uber, 2025]*[1].

Definition 7.2.1 Heat-Lost Activity Hours (HLAH)

The person-hours of outdoor activity that would have taken place in the absence of heat and do not take place because of it, accumulated over places, hours and cohorts.

$$\mathrm{HLAH} = \sum_{p \in \mathscr{P}} \sum_{h \in H} \sum_{c} A_{p,h,c} \left(1 - \phi_c(U_{p,h})\right) \Delta h. \tag{7.1}$$

$A_{p,h,c}$ is the counterfactual activity of cohort c and ϕ_c its acceptability function, falling from one to zero as the comfort index worsens. Both are estimated, not observed; the estimate's uncertainty belongs in the belief state B_t and is what the Year-1 campaign narrows. The unit is person-hours.

Definition 7.2.2 Restored Activity Hours (RAH)

The reduction in HLAH attributable to an intervention, and the reporting unit of the whole programme once divided by the resources consumed.

$$\mathrm{RAH}(x) = \mathrm{HLAH}^{\,0} - \mathrm{HLAH}^{\,x}, \qquad \frac{\mathrm{RAH}(x)}{\mathrm{AED}}, \quad \frac{\mathrm{RAH}(x)}{\mathrm{litre}}, \quad \frac{\mathrm{RAH}(x)}{\mathrm{kWh}}. \tag{7.2}$$

The superscripts denote the untreated and treated configurations. Because the untreated term is counterfactual, RAH is only measurable under a design with a control, which is why the untreated heritage core is inside the Year-3 boundary.

Definition 7.2.3 Outdoor Tourism Capacity (OTC)

How many people a place can comfortably hold in a given hour: its physical capacity discounted by the acceptability of the conditions.

$$\mathrm{OTC}_{p,h,c} = \kappa_p \, \phi_c(U_{p,h}), \qquad \mathrm{OTC}_p = \left|\{\, h \in H \,:\, U_{p,h} \le U^{*} \,\}\right|. \tag{7.3}$$

The first form is an hourly capacity in persons, the second the count of usable hours. The acronym collides with Outdoor Thermal Comfort in the Singapore literature; in this report OTC always means Outdoor Tourism Capacity. Both forms require an explicit heat-stress threshold: on this report's own worked example of the Holiday Climate Index, an index built on daily aggregates in which only rain and wind can override the thermal rating will still score a dry, calm, clear Gulf summer afternoon as acceptable [Scott et al., 2016].

[1] Uber Technologies, *H3 Geospatial Indexing System: Table of Cell Areas and Edge Lengths by Resolution*, H3 documentation, 2025. https://h3geo.org/docs/core-library/restable

Definition 7.2.4 Minimum Cooling Network (MCN)

The smallest and cheapest set of interventions that makes the priority routes comfortable, rather than the largest set the budget allows.

$$x^\star = \arg\min_{x \subseteq \mathcal{I}} \left[c^{\text{cap}}(x) + \sum_{h \in H} c_h^{\text{om}}(x) \right] \quad \text{subject to} \quad U_{p,h}(x) \le U^* \quad \text{for all } (p,h) \in \mathcal{P}^{\text{pri}} \times H^{\text{pri}}. \tag{7.4}$$

$\mathcal{I}$ is the candidate intervention set, $\mathcal{P}^{\text{pri}}$ and H^{pri} the priority places and hours. The constrained form makes the comfort requirement explicit and auditable; the dual form (maximise RAH subject to budget, water and energy caps) is the one the Year-2 optimiser will actually solve.

7.2.3 Decisions at two timescales

The decision variable is not one thing. Operating decisions x_t^{op} are taken hourly or in 15-minute bins (misting on or off, a route advisory, a district-cooling load schedule agreed with DEWA) and are reversible. Capital decisions x_y^{cap} are taken once a programme year (which intervention, and where) and are not. The two are coupled in both directions: a capital decision changes the physics that the operating policy must work within, and the operating policy determines how much of a capital asset's benefit is actually realised, so evaluating a canopy while ignoring how the misting is run understates or overstates it.

This two-timescale structure is the temporal boundary, and it dictates the shape of the model: an hourly inner loop within a seasonal evaluation window, within annual capital epochs bounded by March 2027, 2028, 2029 and 2030. It also sets the granularity of every identifier in Chapter 8: a time-slot identifier binned at 15 minutes and aggregated upward is what makes both timescales addressable in one dataset.

7.2.4 Exogenous information and the transition

The exogenous information W_{t+1} is everything that arrives after a decision and cannot be influenced by it: weather and dust events, tourism demand, energy and water prices, and the decisions of agencies outside the consortium. Drawing this line is the most consequential boundary in Part B, because a variable inside the state must be modelled and a variable in W must only be forecast and hedged. Three placements matter. The district layer at 25–50 m is exogenous context to a corridor decision, not a decision variable, which is the formal reason the Dubai Digital Twin Platform, with its geospatial layers and infrastructure assets, should be consumed as a data source rather than rebuilt *[Dubai Media Office, 2026]*[2]. The street layer at 1–5 m is state, because interventions change it. And the behaviour of visitors and workers is part state and part exogenous: the number of people who would have come is exogenous, the fraction that stays given the conditions is a behavioural response inside the model, with its parameters in B_t.

The transition function S^M therefore has three components: a physics component, in which the district and street models propagate the comfort field given the weather and the installed interventions; an asset component, in which condition degrades and maintenance restores it; and a behavioural component, in which the acceptability functions ϕ_c convert conditions into activity. The gate tests the first, Year 2 estimates the third, and the second is an assumption until the roadmap

[2] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

of Chapter 10 supplies service lives, which are this report's own indicative assumptions and are cited to no source.

7.2.5 The objective and the Bellman recursion

With the elements defined, the objective is a search over policies rather than over decisions [Powell, 2019]:

$$\max_{\pi \in \Pi} \mathbb{E}\Big\{ \sum_{t=0}^{T} \gamma^{t}\, C_t\big(S_t, X^{\pi}(S_t)\big) \;\Big|\; S_0 \Big\}, \qquad S_{t+1} = S^{M}\big(S_t, X^{\pi}(S_t), W_{t+1}\big), \tag{7.5}$$

with the contribution function written in the report's own units,

$$C_t(S_t, x_t) = \sum_{c} v_c\, \mathrm{RAH}_{t,c}(x_t) \;-\; c^{\mathrm{cap}}\big(x^{\mathrm{cap}}_{y(t)}\big) \;-\; c^{\mathrm{om}}_t(x_t), \tag{7.6}$$

subject to the comfort constraints of equation (7.4) and the resource constraints of Section 7.2.8. The classical solution is Bellman's recursion, which replaces the search over whole policies with a search over one decision at a time against a value function [Bellman, 1957, Powell, 2019].

Understanding: Bellman equation

TIER 1: IN PLAIN WORDS

Instead of planning the whole future at once, work out what each situation is worth, then in every situation pick the choice that gives the best pay-off now plus the best worth afterwards.

TIER 2: IN PROFESSIONAL TERMS

A recursive statement of optimality: the value of being in a state equals the best immediate contribution available in it plus the discounted expected value of the state it leads to [Bellman, 1957, Powell, 2019].

TIER 3: IN TECHNICAL TERMS

Solving the recursion exactly by backward induction over a lookup table is possible only for very small state spaces. Powell names three curses of dimensionality (the state space, the action space and the outcome space) and gives the rule of thumb that one-dimensional problems solve in under a minute while four-dimensional ones can take a year, a figure offered as illustration and not as a benchmark [Powell, 2019].

AS A MATHEMATICAL MODEL

$$V_t(S_t) = \max_{x_t \in \mathcal{X}_t} \Big(C_t(S_t, x_t) + \gamma\, \mathbb{E}\big[V_{t+1}\big(S^{M}(S_t, x_t, W_{t+1})\big) \;\big|\; S_t \big] \Big).$$

The terminal condition matters more here than the recursion. A value function set to zero at March 2030 would make the canopy worthless on the day the grant ends, which no owning agency would accept. V_{T+1} is therefore supplied by the life-cycle roadmap of Chapter 10, and the choice of terminal value is itself a boundary: it is the assumption that carries the programme across the funding cliff.

7.2.6 The information boundary and the value of Year 1

Nothing in the corridor is fully observed. Twenty nodes and one reference station sample a field that exists at every point; footfall is counted at a station exit and estimated elsewhere; the acceptability functions are inferred from behaviour. The problem is therefore partially observable, and its state carries a belief B_t that the observation programme Z updates [Powell, 2019]. The instrument list *is* the information boundary, and it has a known gap: the specified nodes measure air temperature, humidity, wind, radiation and surface temperature, and so lack the black-globe and

natural wet-bulb instruments that a directly measured WBGT requires. Either the instruments are added in Loop 0 or WBGT is estimated and the estimation error declared; either way the decision belongs in the proposal, because comparability with the international labour series depends on it [ILO, 2019].

This gives Year 1 its economic justification. Year 1 builds no intervention and restores no activity-hour; what it buys is information, and the standard measure of the worth of information is the improvement in the expected value of the subsequent decision [Howard, 1966]:

$$\mathrm{EVPI} = \mathbb{E}_W\Big[\max_x C(x, W)\Big] - \max_x \mathbb{E}_W\big[C(x, W)\big], \tag{7.7}$$

$$\mathrm{EVSI}(Z) = \mathbb{E}_Z\Big[\max_x \mathbb{E}\big[C(x, W) \mid Z\big]\Big] - \max_x \mathbb{E}_W\big[C(x, W)\big]. \tag{7.8}$$

The first is the value of perfect knowledge and bounds the second, the value of the imperfect sample a sensor network actually provides. Year 1 should therefore be presented as the purchase of $\mathrm{EVSI}(Z)$ at a price equal to the equipment line and the first year of the two funded posts; the gate is the test of whether the purchase delivered. The terminology is decision analysis, not Powell, whose nearest constructs are active learning and the knowledge gradient; the report uses both and keeps the attributions distinct [Howard, 1966, Powell, 2019].

Understanding: Value of information

TIER 1: IN PLAIN WORDS

What it is worth paying to find something out, measured by how much better the decision becomes once you know it: not by how interesting the finding is.

TIER 2: IN PROFESSIONAL TERMS

The difference between the expected pay-off of the best decision taken with an observation in hand and the expected pay-off of the best decision taken without it [Howard, 1966].

TIER 3: IN TECHNICAL TERMS

Perfect information gives an upper bound, EVPI; a real instrument network yields sample information, $\mathrm{EVSI}(Z) \leq \mathrm{EVPI}$, and the gap is the price of measurement error and sparse sampling. Within the sequential framework the same idea appears as active learning: decisions taken partly in order to learn, evaluated by the knowledge gradient [Powell, 2019].

AS A MATHEMATICAL MODEL

Equations (7.7) and (7.8). Instrumenting is justified when $\mathrm{EVSI}(Z)$ exceeds the cost of Z; the same test decides an ABIG sensor or payload.

7.2.7 Policy classes and the curse of dimensionality

Powell's second claim is that every method for making sequential decisions falls into four classes, arrived at by two strategies: searching over policies, and constructing lookahead approximations [Powell, 2019, Powell, 2022]. Table 7.3 maps them onto the Dubai decisions. The mapping matters for what the proposal promises: since each of the four classes was found to win on a different problem in the same published comparison, the honest commitment is to compare policy classes against a common objective, not to promise that 'AI' will solve the corridor [Powell, 2022, Powell, 2019].

The classes also explain why the citywide 100 m product is a stretch goal and not merely a bigger run. The corridor has a small decision set and can be enumerated near-exactly; a city multiplies places, hours and candidate interventions until exact dynamic programming is unavailable and approximate policies are required. Powell's caution is pointed: value function approximations tend to work well only on a fairly small set of problems where structure such as convexity can

Table 7.3: The four policy classes [Powell, 2019, Powell, 2022] mapped onto the decisions of the Decision Twin. Powell's observation that a policy with two to four tunable parameters is far easier to tune than a time-dependent vector with one component per hour is the reason the forecast belongs in the state rather than in the policy.

Class	What it is	Where it fits in Dubai
Policy function approximation (PFA)	An analytic rule mapping state to decision, with a few tunable parameters θ	Corridor operating rules: mist when the comfort index passes a threshold; issue a shaded-route advisory in given hours
Cost function approximation (CFA)	An optimisation problem solved each period, with a modified or parameterised objective or constraint	The Year-2 multi-objective MCN optimiser over comfort hours gained, capital cost, energy, water and maintenance
Value function approximation (VFA)	An approximation of V_{t+1} used inside the Bellman recursion	Capital sequencing across epochs; viable only where exploitable structure exists, which Powell warns is a narrow class
Direct lookahead (DLA)	An explicit forward model solved over a horizon, then rolled forward	Day-ahead district-cooling scheduling with DEWA against a forecast; hybrid DLA–CFA when forecast constraints are parameterised

be exploited, and neural-network policies may need tens of millions of training iterations without any guarantee of quality [Powell, 2019]. The citywide product therefore belongs outside the Loop-3 boundary and inside the research-risk register of Chapter 11.

7.2.8 Resource coupling: water, energy and the annual cap

Three constraints bind the objective and none of them is owned by the consortium:

$$\sum_{t \in H_y} c_t^{\mathrm{om}}(x_t) + c^{\mathrm{cap}}(x_y^{\mathrm{cap}}) \le \bar{B}_y, \qquad \sum_{t \in H_y} w_t(x_t) \le \bar{W}_y, \qquad \sum_{t \in H_y} e_t(x_t) \le \bar{E}_y. \tag{7.9}$$

Water and electricity are DEWA's, which is why they must appear as constraints with shadow prices rather than as footnotes, and why RAH is reported per litre and per kilowatt-hour as well as per dirham: an intervention that restores activity-hours by consuming water the utility must supply has moved a cost rather than removed one. The budget constraint is the funder's. The archived rules of the 2025 grant cycle set a per-project envelope of [500,000.00:138,888.89] (AED:USD) to [1,500,000.00:416,666.67](AED:USD) over one to three years with an annual cap of [500,000.00:138,888.89](AED:USD) *[Dubai RDI Programme, 2025]*[3]; a storyboard budget of [1,400,000.00:388,888.89](AED:USD) over 36 months averages [466,667.00:129,629.72](AED:USD) per year (this report's own arithmetic, and indicative only), which fits, but a Year-1 equipment line of about [250,000.00:69,444.44](AED:USD) alongside two funded posts must be phased to respect the cap in the year it falls [Iyyunni, 2026]. That the storyboard presupposes a third cycle whose calendar and envelope are documented nowhere is a funding-timing risk carried to Chapter 11, not a boundary that can be drawn here.

[3] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

7.2.9 ABIG as a sequential decision problem

The same five elements write ABIG. The state is the capability map: R_t the institutions, datasets, sensors, trained people and operational services in place; I_t the decisions pending at fisheries, port, environment and maritime-security agencies and the forecasts available to them; B_t the beliefs about where an observation or forecasting gap actually changes a decision. The decision is which pilot, fellowship, standard, sensor or mission to fund next. The exogenous information is continental politics, donor behaviour, climate, and the conduct of the fleets and coasts observed. The transition is that capability grows where a pilot reaches an operational budget and decays where trained people leave. The objective is the one the concept paper already names in its utilisation workstream (decisions improved, measured with the operational user) per dollar spent [Terhile et al., 2026].

Written this way, the concept paper's sequencing rule, that hardware follows evidence of need, is a value-of-information policy: a sensor or payload is commissioned when the expected value of the information it yields for a named decision exceeds its cost [Howard, 1966], which is the active-learning family applied to a programme rather than to a corridor [Powell, 2019]. The nine risks are the constraint set on $\mathcal{X}_t$: no pilot without a pathway to an operational budget, no security analytic without a human in the loop and an audit trail, no platform without open standards [Terhile et al., 2026]. And the federated data hub of the addendum has a formal counterpart. Powell's multiagent extension observes that a central agent is simply another agent whose decisions are communicated to field agents, and that a single system state is meaningless because no agent ever sees all of it [Powell, 2019]. Observing nodes and forecasting centres are learning agents, national agencies are controlling agents, and ABIG is a combined agent that connects them: the sourced argument against a centralised continental data lake and for the addendum's federated model. The information boundary for ABIG is consequently not a list of instruments but a set of agreements about what each agent will share, which is why the data-sharing negotiations of months 0–6 sit on the critical path of every later loop. A unit of the objective already exists in the archive: an enforcement action in which satellite intelligence led to three unlicensed trawlers being intercepted on 18 July 2026 (grade: case-report, narrated by the provider, no counterfactual) is precisely the kind of decision improved that the measure should count *[Global Fishing Watch, 2026]*[4].

7.3 Part C: a unified boundary taxonomy

The two methods name the same boundaries in different vocabularies. Table 7.4 collects them into one taxonomy of eight and states each for both programmes. The first three are the spiral's and the last five the decision model's, and each can be read from the other side: the scope boundary of a loop is the decision set $\mathcal{X}_t$ during that loop; the risk boundary is the tolerance τ below which a risk may be carried rather than resolved, which is the LCA condition; the commitment boundary is the set of agents whose win conditions are written, which is the set of agents whose information the model may rely upon; the endogenous–exogenous boundary is the line between S_t and W_{t+1}, which in spiral terms separates what a loop can change from what it must plan around.

Two of the eight are where the methods disagree, and the disagreements are informative rather than embarrassing. On the temporal boundary the spiral stops at IOC and the decision model does not: the spiral is content to hand over a working capability, whereas the objective function keeps accruing value and cost for as long as the asset stands, so the roadmap must supply what the spiral does not. On the information boundary the spiral is satisfied when prototyping results have been reviewed, while the decision model asks whether those results changed a decision. Gate condition 6.2.4 already bridges the two by requiring a written statement of usefulness from an

[4] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

Table 7.4: A unified taxonomy of eight boundaries, drawn from the spiral model [Boehm and Hansen, 2000, Boehm et al., 1998] and the sequential decision model [Powell, 2019, Howard, 1966], with the content of each for the two programmes [Iyyunni, 2026, Terhile et al., 2026].

Boundary	Dubai Decision Twin	ABIG
Scope *the decision set in a loop*	Loop 1 measures and validates only; Loop 2 optimises one corridor's intervention set; Loop 3 builds one intervention; the citywide product is outside	Integration and application layer only; observation, forecasting and Earth-observation services remain with the existing programmes
Risk *the LCA condition*	Four gate conditions; fold to dataset and platform if unmet; error threshold set after the first summer, not before	All nine concept-paper risks resolved or covered before Year 2; every pilot with a pathway to an operational budget
Commitment *whose word the model relies on*	Site letters (Dec 2026), data letters (Feb 2027), one gate letter (Mar 2028), one sustainment commitment (Mar 2029)	Named focal points per institution; data-sharing agreements in months 0–6; sustainment before any pilot is funded
Endogenous / exogenous S_t *versus* W_{t+1}	Street layer at 1–5 m is state; district layer at 25–50 m is context; weather, demand, prices and agency decisions are exogenous	Capability map is state; continental politics, donor decisions and fleet behaviour are exogenous
Spatial *the place set* $\mathscr{P}$	Corridor of about 800 m with the heritage core as baseline; H3 resolutions 11 and 13; city is a stretch goal	Regional marine systems as hubs; continental scope only as a federation of them
Temporal *horizon and epochs*	Hourly operations in 15-minute bins; one summer per evaluation; capital epochs Mar 2027–2030; terminal value from the roadmap	0–3, 3–6, 6–12 months, Year 2, Year 3; recurring budgets and service agreements as the terminal condition
Information *contents of* I_t*, width of* B_t	20 nodes and one reference station; RTA counts; one summer of DEWA load; WBGT instruments absent; Year 1 valued as EVSI	Federated hub; agreements on what each agent shares; the capability map as first dataset; no central data lake
Resource *the binding constraints*	[1,400,000.00:388,888.89](AED:USD) over 36 months against an annual cap; water and district-cooling load owned by DEWA	Six-stage financing ladder; hardware only once the value of its information is shown

agency, and it should be preserved in exactly that form; it is the one clause in the storyboard that both methods endorse.

Where the two methods agree

Both instruments place the same three lines. The scope of Loop 1 is measurement and validation only; the commitment boundary is a set of signed letters rather than a set of intentions; and the citywide product lies outside the funded boundary: for the spiral because no loop should promise what its risk analysis cannot cover, and for the decision model because the corridor is solvable while the city requires approximate policies whose success is not assured on the framework's own evidence [Boehm and Hansen, 2000, Powell, 2019].

7.4 Implications for the programmes

Dubai.

1. **Write the loop table into the full proposal.** Table 7.1 is the boundary clause the storyboard lacks: each loop's scope, anchor point and exit criterion, with Loop 0 completed by a business case and a feasibility rationale in Boehm's sense [Boehm and Hansen, 2000].
2. **Treat the month-12 gate as the LCA milestone and keep its four conditions as written**, adding two procedural rules: fix the model-error threshold after the first summer's data, and negotiate the meaning of 'useful' with RTA and DEWA after the first comfort map and before the gate [Boehm and Hansen, 2000, Boehm et al., 1998].
3. **Add a fifth exit criterion to Loop 2:** a written sustainment commitment for the chosen intervention from the agency that will own it. In the only archived case study, adoption followed sustainment and nothing without it survived [Boehm et al., 1998].
4. **Fix the four metrics as functions of the state** (Definitions 7.2.1 to 7.2.4), report WBGT alongside the Universal Thermal Climate Index, and decide in Loop 0 whether to add the black-globe and wet-bulb instruments or to estimate WBGT and declare the error [ILO, 2019].
5. **Present Year 1 as the purchase of information** whose value is EVSI (equation (7.8)) and whose price is the equipment line plus the first year of two posts [Howard, 1966, Iyyunni, 2026]. The proposal cannot yet quantify it; it can and should make the argument in that form.
6. **Keep the city a stretch goal and label it a research risk**, and promise a comparison of policy classes against one objective rather than the deployment of 'AI' [Powell, 2019, Powell, 2022].
7. **Phase Loop 1 against the annual cap.** If the 2025 rules carry over, the equipment line and the first-year posts must fit [500,000.00:138,888.89](AED:USD) in the year they fall; the undocumented third grant cycle is a funding-timing risk for Chapter 11 [Iyyunni, 2026]*[Dubai RDI Programme, 2025]*[3].

ABIG.

1. **Give the roadmap anchor points** (Table 7.2): LCO at month twelve with the capability map published and at least one feasible federated architecture carrying a business case; LCA at the end of Year 2 with the federated architecture chosen and all nine risks resolved or covered; IOC when one use case is in routine operational use by a named agency, with software, site and operator preparation complete [Boehm and Hansen, 2000, Terhile et al., 2026].

2. **Hold the integration boundary in every loop.** No observation pipeline, sensor procurement or forecasting centre inside ABIG; those belong to the existing programmes. The alternative is evolutionary development with no life-cycle architecture, which is the formal name for the coordination forum the concept paper fears [Boehm and Hansen, 2000, Terhile et al., 2026].
3. **Make sustainment the selection criterion for pilots**, applied before funding rather than after, as the WinWin clients were eventually required to do [Boehm et al., 1998]; this is the operational form of the answer to pilot fatigue.
4. **State the objective as decisions improved per dollar**, defined per use case with the operational user, so that every hardware decision becomes a value-of-information decision [Howard, 1966, Terhile et al., 2026].
5. **Put data-sharing agreements on the critical path of months 0–6.** In a federated architecture the information boundary is a set of agreements, and every later loop inherits whatever is agreed there [Powell, 2019, Terhile et al., 2026].

The boundaries drawn in this chapter are the inputs to the three that follow: the identifiers that make the state observable and auditable (Chapter 8), the costs and values that give the objective its units (Chapter 9), and the roadmap that supplies the terminal value beyond the grant (Chapter 10).

{8}

A Taxonomy of Synthetic Data Across the Systems Architecture

Abstract

Which synthetic-data identifiers prove critical to measuring the life cycle of the project? The answer treats both proposals as one layered architecture and explores it recursively. Three families of diagrams do the work: a master tree of the ten architecture layers, from the physical environment through sensing, data, semantics, the knowledge graph, models, the Decision Twin, the decision and the action to learning; one sub-tree per layer whose leaves are the classes of synthetic data that the layer produces or consumes, each annotated with the keys it must carry; and a tree of the identifier schema itself, twelve canonical keys and seven synthetic-record flags. Three results follow. Identifiers inherit upward, so a key omitted at the sensing layer cannot be recovered at the decision layer. One sticky flag, `is_synthetic`, is the firewall that lets synthetic data train and stress-test the twin while keeping it out of the month-twelve gate. And every life-cycle indicator the programmes promise is a join over these keys. The same recursion is then shown for the ocean twin of the Africa Blue Intelligence Grid.

Every number these two programmes promise to report is a join. Heat-Lost Activity Hours are hours joined to a place, an activity and a group of people; Restored Activity Hours per dirham are those hours joined to an intervention, its version and the cost elements charged against it; decision uptake is a brief joined to the decision it produced. Not one of these quantities is observed by any instrument. They are computed late in the life cycle and high in the architecture, from records created early and low, and the only thing that travels with a record from the sensor that made it to the brief that uses it is the set of identifiers it carries. The eleventh stakeholder question therefore has a precise form: which identifiers must be attached to synthetic data (and to the observed data it stands in for), so that the life cycle of the project can be measured at all? Answering it requires a taxonomy of the architecture itself, explored layer by layer, because each layer inherits the keys of the layer beneath it and adds its own. This chapter builds that taxonomy for the Dubai Decision Twin, states the identifier schema and the synthetic-record flags, explains the firewall that separates synthetic from observed records at the gate, shows how each life-cycle indicator is computed through the schema, and then demonstrates that the same recursion describes the ocean twin of the Africa Blue Intelligence Grid.

8.1 Why identifiers decide what can be measured

The storyboard's unanswered question has three parts: which activities, routes and places are lost to heat and for how many hours a year; what economic and social value attaches to restoring

them; and which single intervention restores the most activity per dirham, litre of water and kilowatt-hour [Iyyunni, 2026]. Each part names the keys on which its answer depends. Hours lost are counted per place and per time slot; the activity lost belongs to a category, and the storyboard supplies its own list: walk, wait, visit and work; the people who lose it belong to a cohort; the value attaches to a decision an agency takes; and the restoring intervention has an identity, a version and a set of cost elements. The discussion notes filed with the storyboard set out the architecture in which these keys live (data, semantic layer, knowledge graph, intelligence layer, Decision Twin, agentic workflow, executive interface) and insist that a data model, a semantic model, a knowledge graph, an artificial-intelligence model, a digital twin and a decision system are six different things [Iyyunni, 2026]. Six different things need one thing in common, and that thing is the key set. A record produced at the data layer is meaningless at the decision layer unless it still says where, when, for whom and by which instrument it was made.

The discipline has a precedent in the method literature. In the WinWin spiral case study the domain taxonomy was used as a checklist, under the rule that every artefact should point to at least one taxonomy element [Boehm et al., 1998]; the artefacts reviewed at the Life-Cycle Objectives and Life-Cycle Architecture anchor points, from the operational concept and the prototyping results to the architecture, the life-cycle plan and the feasibility rationale, are all traceable to it [Boehm and Hansen, 2000]. The equivalent rule for a data-driven programme is that every record should point to at least one identifier in a published schema, and every indicator should be computable as a join across those identifiers. Chapter 7 describes the state of the Decision Twin in the three parts of Powell's framework, the physical state, the other information and the belief state [Powell, 2019]; the identifier schema is what allows a record to be assigned to one of those parts, and to be revised when the belief changes.

Understanding: Identifier

TIER 1: IN PLAIN WORDS

A label that stays attached to a piece of information wherever it goes, so that two pieces of information about the same street corner, the same quarter-hour or the same person's journey can be put side by side.

TIER 2: IN PROFESSIONAL TERMS

A key: a field whose value uniquely names an entity in an agreed vocabulary, on which records from different systems can be joined without guesswork or manual matching.

TIER 3: IN TECHNICAL TERMS

A canonical key drawn from a controlled scheme with a stated version (an H3 cell index, an ISO 8601 interval, a term from the project ontology), carried by every record of every layer and inherited by every record derived from it.

AS A MATHEMATICAL MODEL

Let N_i be the set of keys born at layer i and K_i the set carried by records at that layer. Then $K_i = K_{i-1} \cup N_i$, with $K_0 = \varnothing$; no join at layer $j > i$ can use a key absent from K_i.

Understanding: Synthetic data

TIER 1: IN PLAIN WORDS

Numbers that were made by a computer rather than measured by an instrument, but that are meant to behave like the real thing: filled-in gaps, imagined crowds, invented weather years.

TIER 2: IN PROFESSIONAL TERMS

Records generated by a model, a simulator or a statistical procedure, used where observation is impossible, too expensive, too sparse or too revealing of individuals, and always

distinguishable from observation.

TIER 3: IN TECHNICAL TERMS

Any record whose value is the output of a generator rather than a transducer: simulated microclimate fields, gap-filled and imputed series, agent-based populations, scenario draws, privacy-preserving surrogates. Its epistemic status is set by its generator, its seed, its source dataset and its fidelity assessment, not by its format.

AS A MATHEMATICAL MODEL

Writing $\mathrm{syn}(r)$ for the synthetic flag of record r and $\mathrm{gen}(r)$ for the generator that produced it, $\mathrm{syn}(r) = \mathbf{1}[\mathrm{gen}(r) \neq \emptyset] \vee \bigvee_{q \in \mathrm{parents}(r)} \mathrm{syn}(q)$: the flag is sticky under derivation.

8.2 The master tree of the architecture

Figure 8.1 is the master tree. Its root is the single chain the discussion notes propose for both twins; its ten children are the layers, ordered from the physical environment to learning, and each is annotated with the identifiers born there. The order matters because it is also the order of inheritance. Place and time are born with the physical environment, because nothing can be observed without a where and a when. The sensor identity is born at sensing. The synthetic flags are born at the data layer, because that is where a record first exists as a record. Activity, route and cohort are born in the semantic layer, because they are meanings rather than measurements. The model-run identity is born with the models. The intervention identity and its cost elements are born in the Decision Twin, where alternatives are first compared. The decision identity is born when a brief is delivered. The as-built version and the validity window are born with the action. New calibration, generator and model versions are born in learning, from which the recursion restarts.

Two consequences follow from the inheritance rule of the tiered box above. First, a key omitted at a low layer cannot be recovered at a high one. If the node firmware does not stamp a calibration version on each ten-minute record, no comfort map, no Restored Activity Hours figure and no decision brief can ever state which calibration it rests on, and the second gate condition (that the model matches sensor observations within the stated error [Iyyunni, 2026]) becomes unauditable. Second, the synthetic flag is sticky: if any ancestor of a record is synthetic, so is the record. That is the mechanism of the firewall in Section 8.5, and it is why the flag belongs to the data layer rather than to the models. Table 8.1 summarises the master tree with the observed records and the synthetic classes each layer will hold in the Dubai case.

8.3 Sub-trees: the synthetic-data classes of each layer

The recursion is applied one level down in Figures 8.2 to 8.4. Each sub-tree takes one layer of the master tree as its root, divides it into the sources or functions it contains, and terminates in leaves that are classes of synthetic data. Every leaf is annotated with the keys that a record of that class must carry over and above the keys it inherits: a leaf annotation is therefore a specification, not a description, and it is what Appendix B turns into field definitions.

The sensing sub-tree contains the chapter's sharpest equipment finding. The storyboard's node specification records air temperature, humidity, wind, radiation and surface temperature every ten minutes [Iyyunni, 2026]. It contains no black-globe thermometer and no natural wet-bulb instrument, so any wet-bulb globe temperature the programme reports will be a modelled quantity, not an observation: a synthetic record in the strict sense of this chapter. Such a record must carry a model-run identifier and a fidelity score if it is ever to be compared with the exposure-response thresholds the international labour literature uses, where productivity loss begins at a wet-bulb globe temperature of 24 to 26°C and reaches half at 33 to 34°C for work at 300 W [ILO, 2019]. Gap

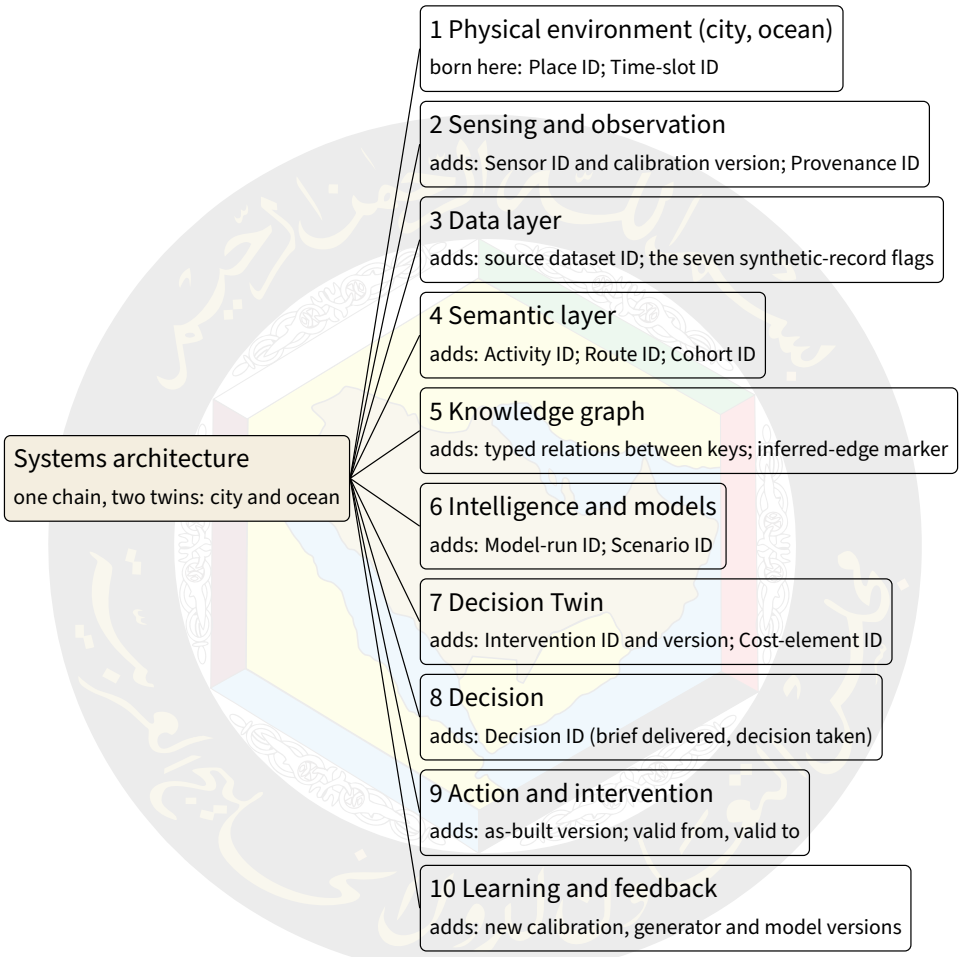


Figure 8.1: Master tree of the systems architecture. The layers are those of the chain set out in the discussion notes filed with the storyboard [Iyyunni, 2026], with the Decision Twin and the decision separated so that the identifier born at each is visible. Every layer inherits each identifier born at the layers below it.

Table 8.1: Layers of the architecture, the identifiers born at each, and the observed and synthetic records each will hold in the Dubai Decision Twin. The observed records are those named in the storyboard [Iyyunni, 2026]; the synthetic classes are this report's taxonomy.

Layer	Keys born here	Observed records	Synthetic classes
1 Physical environment	Place ID; Time-slot ID	NCM station records; Copernicus land-surface and vegetation products; municipal street and building geometry	Scenario weather years; extreme-day sets; dust-event scenarios
2 Sensing and observation	Sensor ID and calibration; Provenance ID	Twenty nodes and one reference station at ten-minute cadence; RTA exit and ridership counts; DEWA hourly district cooling load	Gap-filled sensor series; virtual sensors at unsensed cells; modelled globe temperature; simulated calibration drift
3 Data layer	Source dataset ID; the seven flags	Ingested, quality-checked records in one spatial–temporal store behind one interface	Imputed ridership; synthetic pedestrian traces; placeholder geometry versions
4 Semantic layer	Activity ID; Route ID; Cohort ID	Ontology of heat exposure, activity, route, place, intervention, resource and decision; the data dictionary	Test fixtures for the ontology; inferred activity labels; cohort assignment rules
5 Knowledge graph	Typed relations; inferred-edge marker	Links between places, routes, sensors, activities, agencies and assets	Inferred edges for adjacency, ownership and exposure; entity-resolution candidates
6 Intelligence and models	Model-run ID; Scenario ID	Validated district fields at 25–50 m and street fields at 1–5 m	Simulated microclimate fields; downscaled and bias-corrected fields; ensemble members; coarse-to-fine training pairs
7 Decision Twin	Intervention ID and version; Cost-element ID	The candidate set: shade canopy, tree planting, cool pavement, misting rest point, revised route	Scenario outputs; agent-based visitor populations; synthetic cost draws; stress tests
8 Decision	Decision ID	Briefs delivered to RTA, DEWA, Dubai Culture and Dubai Municipality	Do-nothing baselines; option sets; what-if variants of a brief
9 Action and intervention	As-built version; validity window	The single intervention built in Year 3, most likely a canopy over the most exposed 200 m	As-designed geometry before the build; predicted before-and-after fields
10 Learning and feedback	New versions	Measured before and after across one summer; repeated counts and dwell times	Hindcast replays under a new calibration; retrained generators; posterior belief updates

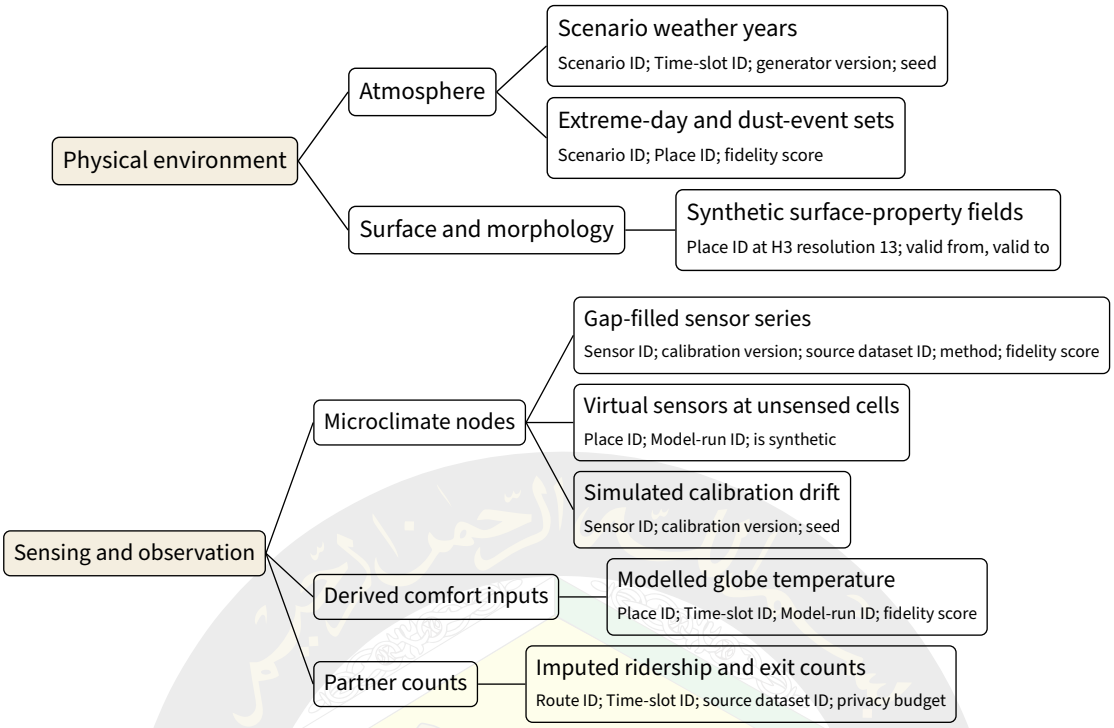


Figure 8.2: Sub-trees of the physical-environment and sensing layers. Leaves are classes of synthetic data; the annotation on each leaf lists the keys that class must carry in addition to those it inherits from the layers below.

filling is the second class here, and it is not hypothetical: gate condition 6.2.1 requires at least 90 per cent uptime across the summer [Iyyunni, 2026], which concedes that up to a tenth of the record may be missing. The flag on a gap-filled value is what prevents an uptime statistic from being computed over a series that has been silently completed.

The lower tree of Figure 8.3 carries the layer at which synthetic data is most defensible, because there its fidelity can be measured. The nearest archived precedent is the Abu Dhabi local climate zone study, which coupled Landsat-8 imagery at 30 m, with a thermal band at 100 m, to an ENVI-met street model at 3 m per pixel and validated it against a 26-station in-situ network: the uncorrected model showed root-mean-square errors of 1.33 to 2.71°C across twenty district-seasons, and an empirical correction brought these to 0.27 to 1.62°C, with a model calibrated on one district predicting another district of the same class at 0.24 to 0.68°C (grade: measured) [Manandhar et al., 2019]. Those numbers are what a fidelity score looks like when it is done properly: a stated variable, a stated averaging period, a held-out window and a range rather than a point. The study's classes are its own composites and do not correspond to the standard local climate zone numbering [Stewart and Oke, 2012], which is precisely why a scheme version belongs in the key.

8.4 The identifier schema

Figure 8.5 is the third tree: the schema itself, twelve canonical keys in four families and seven flags that mark a record as synthetic. Table 8.2 gives each key its scheme, its granularity and the gate or indicator that depends on it.

Two keys deserve explanation because they resolve an argument already open between the two papers. The place key is an H3 cell index at two resolutions: resolution 11, whose average edge

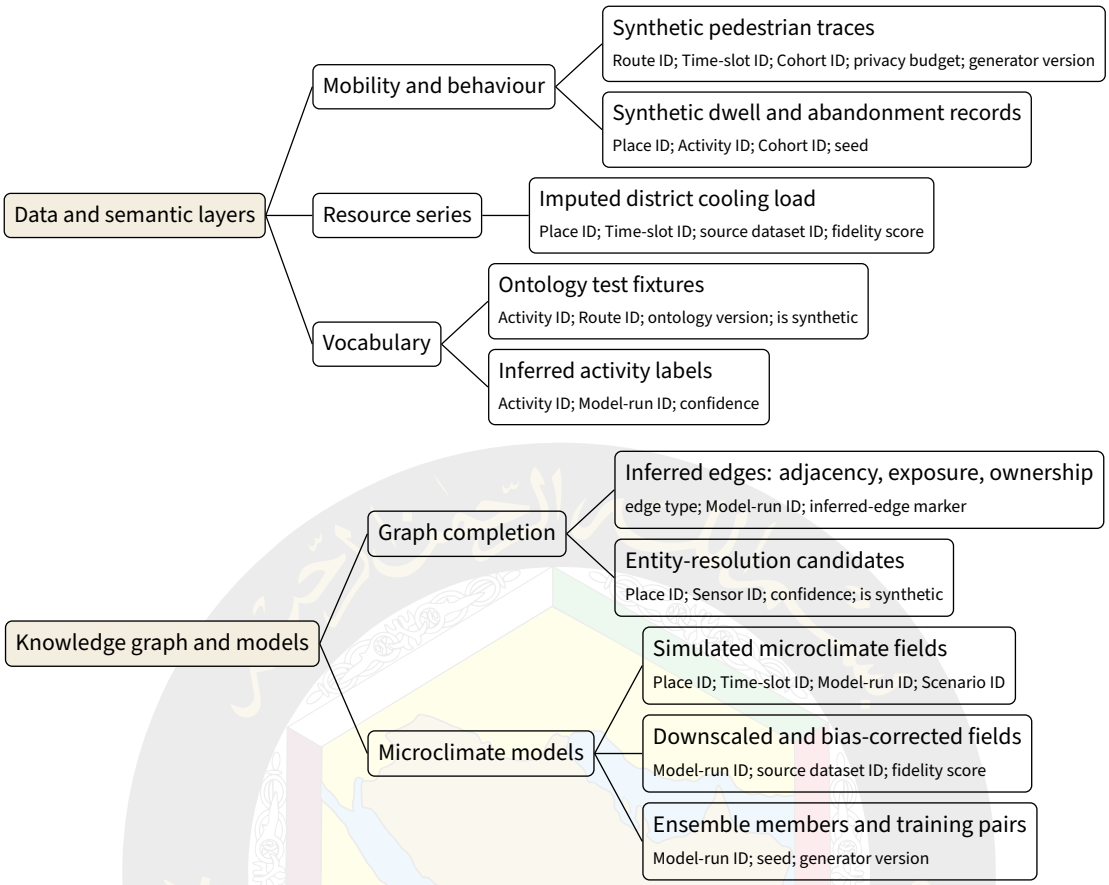


Figure 8.3: Sub-trees of the data, semantic, knowledge-graph and model layers. The leaves of the model layer are the largest synthetic population in the programme by volume, and the ones whose fidelity can be checked most directly against the sensor network.

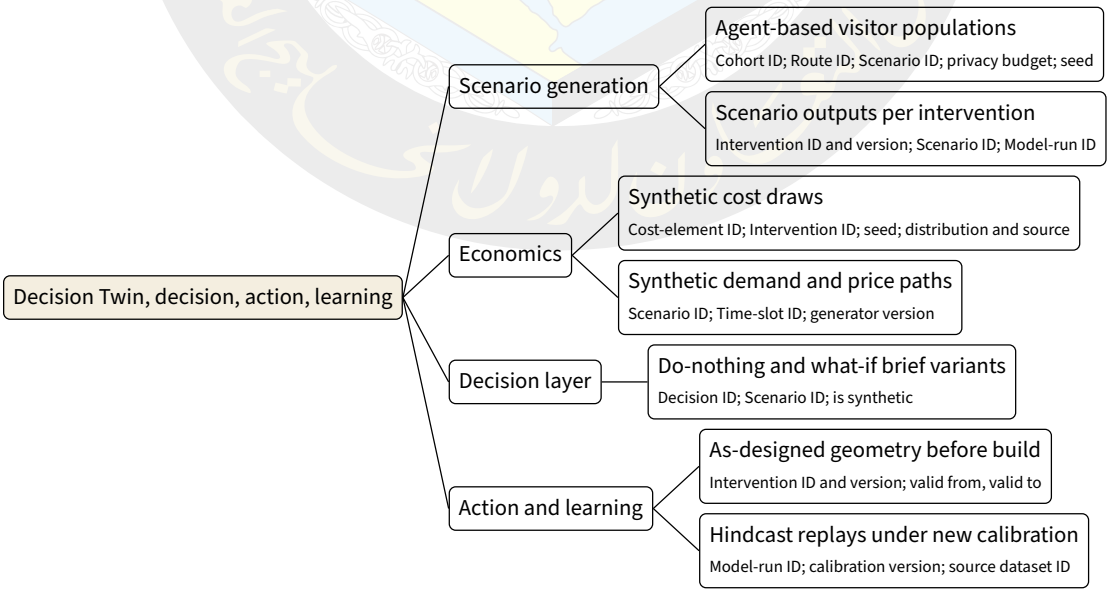


Figure 8.4: Sub-tree of the upper layers. Synthetic cost draws and agent-based visitor populations are the two classes that carry the largest economic weight, because they feed the multi-objective optimiser that produces the Minimum Cooling Network.

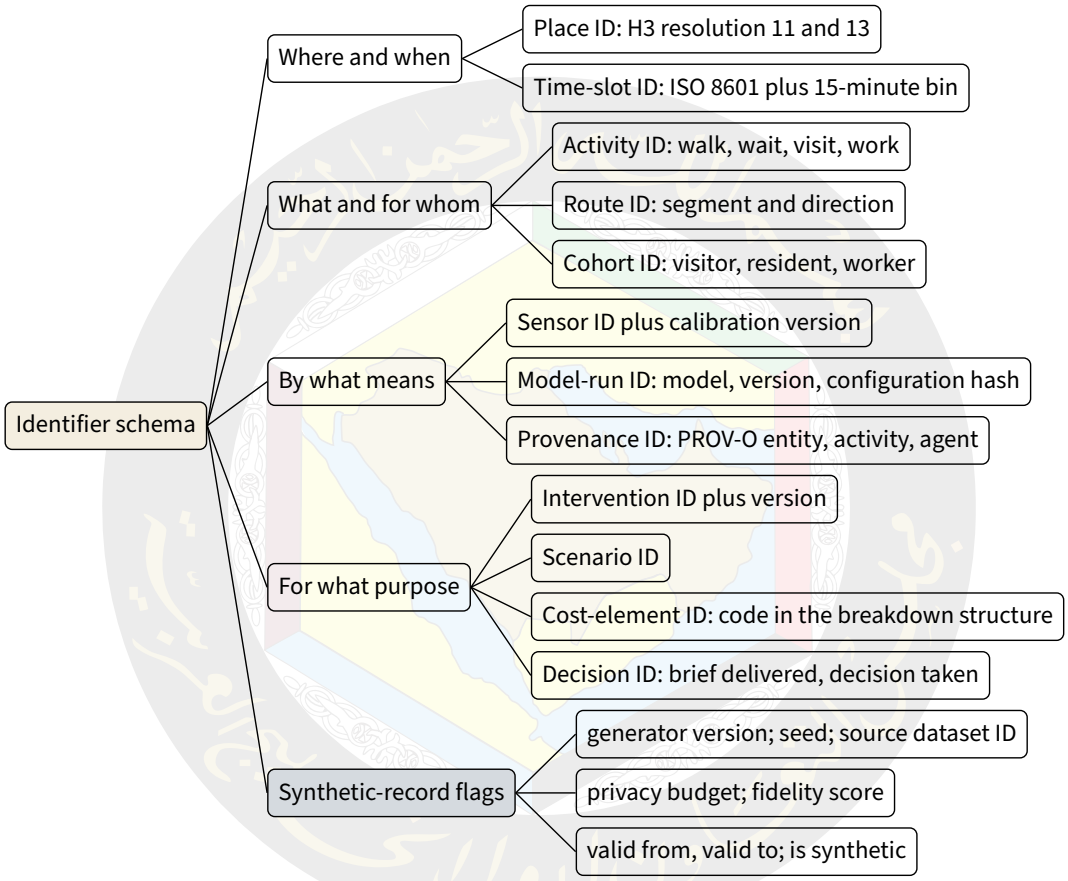


Figure 8.5: The identifier schema: twelve canonical keys in four families, and the seven flags that every synthetic record carries. The place key follows the H3 resolution table *[Uber, 2025]* and the provenance key follows PROV-O [W3C, 2013]; the remaining schemes are the project's own and are defined in Appendix B.

length is about 24.9 m, and resolution 13, whose average edge length is about 3.6 m *[Uber, 2025]*[1]. Those two resolutions are not chosen for elegance. They are the nearest hexagonal equivalents of the two model grids the storyboard commits to, the district model at 25 to 50 m and the street model at 1 to 5 m [Iyyunni, 2026], and of the nesting the Abu Dhabi precedent demonstrated between 30 m classification and 3 m street simulation [Manandhar et al., 2019]. Because resolution 13 cells nest inside resolution 11 cells, a street-scale result can be aggregated to the district layer without a spatial join, and a citywide product can be produced later without re-keying the corridor archive. The time key settles the question the discussion notes raise twice, whether hourly updates suffice or whether fifteen-minute updates are needed [Iyyunni, 2026]. The schema stores a fifteen-minute bin and aggregates upward, so that the hourly product remains available while the finer question stays answerable. Ten-minute sensor records map into fifteen-minute bins by an explicit, versioned aggregation rule rather than by convention.

Understanding: Provenance

TIER 1: IN PLAIN WORDS

A record of where a number came from: which instrument or program made it, from what, and when, so that anyone can retrace the steps.

TIER 2: IN PROFESSIONAL TERMS

An auditable chain of custody for data: the inputs, the process and the responsible party for every derived value, retained for the life of the programme.

TIER 3: IN TECHNICAL TERMS

A graph of entities, activities and agents in the sense of the PROV-O ontology, in which every derived entity is linked by `wasDerivedFrom` to its inputs, by `wasGeneratedBy` to the activity that produced it, and by `wasAttributedTo` to a responsible agent [W3C, 2013].

AS A MATHEMATICAL MODEL

The lineage of a record is the transitive closure of `wasDerivedFrom`; a record is admissible at the gate only if that closure contains no entity whose `is_synthetic` flag is true.

8.5 Synthetic-record flags and the `is_synthetic` firewall

Seven flags travel with every synthetic record. `generator_version` names the program and version that produced it, so that a change of generator is visible as a change of value. `seed` makes a stochastic draw reproducible; without it, an agent-based visitor population cannot be regenerated and a scenario cannot be re-examined after a challenge. `source_dataset_id` names the observed data the generator was fitted or conditioned on, which is what makes leakage detectable: a synthetic series fitted on the same summer against which a model is then validated is not an independent test. `privacy_budget` records the differential-privacy parameter ε spent in producing the record, so that a cumulative budget can be enforced across releases; it is the field that lets the programme publish synthetic pedestrian traces without publishing anyone's journey, and it is the technical expression of the storyboard's ethics, data-governance and publication lines [Iyyunni, 2026]. `fidelity_score` states how well the record reproduces the statistic it is meant to stand in for, with the variable, the metric and the held-out window named, in the manner of the Abu Dhabi validation design [Manandhar et al., 2019]. `valid_from` and `valid_to` bound the period for which the record is meant to be true, which is what lets an as-designed geometry be superseded by an as-built one without either being deleted. And `is_synthetic` is a single boolean that is true

[1] Uber Technologies, *H3 Geospatial Indexing System: Table of Cell Areas and Edge Lengths by Resolution*, H3 documentation, 2025. https://h3geo.org/docs/core-library/restable

Table 8.2: The identifier schema in detail. Gate conditions are those of the storyboard's month-twelve gate [Iyyunni, 2026]. Schemes marked "project" are defined in Appendix B.

Key	Scheme and granularity	What depends on it
Place ID	H3 resolutions 11 and 13; edge about 24.9 m and 3.6 m *[Uber, 2025]*	Every spatial join; the comfort map; aggregation from street to district and later to city
Time-slot ID	ISO 8601 instant plus 15-minute bin	Hour-by-hour comfort map (gate 6.2.3); uptime; all hours-based indicators
Activity ID	Project: walk, wait, visit, work	Heat-Lost and Restored Activity Hours; Outdoor Tourism Capacity
Route ID	Project: segment and direction	Minimum Cooling Network; first and last mile analysis; ridership joins
Cohort ID	Project: visitor, resident, worker; no personal data	Distributional analysis; the privacy budget; the ethics line of the programme budget
Sensor ID and calibration	Project: node serial plus calibration version and date	Uptime (gate 6.2.1); model error (gate 6.2.2); drift correction
Model-run ID	Project: model, version, configuration hash	Reproducibility of every field; fidelity scores; hindcast replay
Provenance ID	PROV-O entity, activity, agent [W3C, 2013]	The lineage closure; the firewall; audit of any published number
Intervention ID and version	Project: type, site, version	Restored Activity Hours per intervention; before-and-after comparison
Scenario ID	Project: weather year, demand path, price path	Scenario comparison in the twin; stress tests; option sets
Cost-element ID	Project: code in a breakdown structure of at least three levels [GAO, 2020]	Life-cycle cost; the dirham denominator of every efficiency ratio
Decision ID	Project: brief, recipient, date, outcome	Decision uptake; gate 6.2.4; the influence ring of Chapter 6

if the record was generated or if any of its ancestors was.

Understanding: Firewall

TIER 1: IN PLAIN WORDS

Made-up data may be used to practise on and to test what might go wrong, but it must never be used to prove that the system works.

TIER 2: IN PROFESSIONAL TERMS

A governance rule enforced in the data layer: generated records may enter training, sensitivity analysis and stress testing, and are excluded by construction from any validation set, gate evidence or published measurement.

TIER 3: IN TECHNICAL TERMS

A lineage predicate evaluated before any gate computation: a query that contributes to gate evidence is executed against the view in which `is_synthetic` is false throughout the transitive closure of `wasDerivedFrom` [W3C, 2013]; violations are refusals, not warnings.

AS A MATHEMATICAL MODEL

Let V be the validation set, $L(r)$ the lineage closure of record r and $\mathrm{syn}(\cdot)$ the synthetic flag. The gate is computed on $\{\, r \in V \;:\; \mathrm{syn}(q) = 0 \text{ for all } q \in L(r) \,\}$; a model error, an uptime figure or a comfort map computed on any other set is not gate evidence.

The firewall exists because all four gate conditions can be made to look satisfied by synthetic data. Uptime can be made to look like 95 per cent by gap filling; model error can be made small by validating against the very fields the model produced; an hour-by-hour comfort map can be complete because the holes were filled; and an agency can be shown a brief whose numbers rest on an imputed ridership series. None of these would be fraud; each is the ordinary consequence of a convenient default. The remedy is that the four gate computations run against the observed-only view, and that this is a property of the query, not of the analyst's memory. Three operating rules follow, and they belong in the data-governance document written before the first node is installed. Synthetic records may train, tune, stress-test and illustrate. They may be published, provided the privacy budget and fidelity score are published with them. They may never appear in the lineage of a number presented as a measurement: in the gate evidence, in the before-and-after comparison of Year 3, or in any figure a partner agency is asked to act on.

This is not an argument against synthetic data; it is what makes synthetic data usable. The value of the twin's scenario work lies precisely in asking what would happen under a weather year that has not occurred, a visitor population that has not been counted and a cost draw that has not been tendered. The distinction between exploring a possibility and reporting a measurement is old, and Boehm's risk-exposure formulation (probability of loss multiplied by size of loss [Boehm and Hansen, 2000]) names the risk that is being managed here: the loss from a gate passed on unsound evidence is the whole of Phase 2 and the programme's standing with four agencies, which is why the control belongs in the schema rather than in a reviewer's diligence.

8.6 Data lineage

Figure 8.6 traces one indicator from instrument to brief and shows where the firewall cuts. The upper path is observed throughout and ends in gate evidence; the lower path passes through a generator, acquires the synthetic flag, and is admitted to training and stress testing but refused at the gate. Both paths carry the same place, time and provenance keys, which is what allows them to be compared at all.

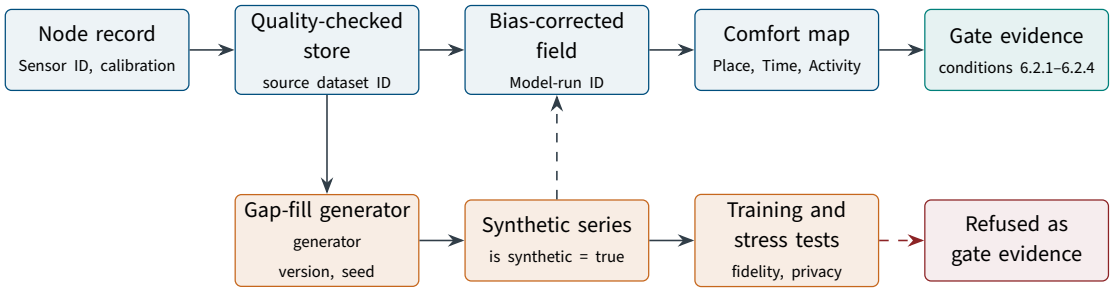


Figure 8.6: Data lineage for one indicator. Every box is a PROV-O entity or activity carrying the keys shown [W3C, 2013]; the synthetic flag propagates along every derivation, so the lower path can train the model but cannot supply gate evidence.

8.7 Life-cycle indicators measured through the identifiers

Table 8.3 states each life-cycle indicator as a join. The point of the table is negative as much as positive: it shows which single missing key makes each indicator uncomputable, and therefore which fields cannot be deferred to a later release of the platform.

Table 8.3: Life-cycle indicators as joins over the identifier schema. "Breaks without" names the key whose absence makes the indicator uncomputable. All indicators are computed on the observed-only view unless the row says otherwise.

Indicator	Join	Breaks without
Heat-Lost Activity Hours	Place × Time-slot × Activity × Cohort, over slots whose comfort index exceeds the threshold	Time-slot ID at a bin finer than the decision interval
Restored Activity Hours	The same join differenced across Intervention ID and version, before and after	Intervention version
RAH per AED, per litre, per kWh	Restored Activity Hours divided by life-cycle cost, water and energy summed over Cost-element ID	Cost-element ID
Outdoor Tourism Capacity	Place × Time-slot × Cohort, counting people who can be accommodated within comfort limits	Cohort ID
Model error	Model-run ID joined to Sensor ID and calibration over held-out slots	Calibration version
Sensor uptime	Sensor ID × Time-slot, counting received against expected records	The synthetic flag on gap-filled values
Decision uptake	Decision ID joined to the brief, the recipient and the outcome	Decision ID
Fidelity of a synthetic class	Model-run or generator version joined to the held-out observed window	Source dataset ID

Two of the rows carry a definitional commitment worth stating explicitly. Heat-Lost Activity Hours is an hours-count over place-slots that fail a comfort threshold, weighted by the number of people who would otherwise have used the place:

$$\mathrm{HLAH}(p,d) \;=\; \sum_{t \in T_d} \Delta t \; n(p,t) \; \mathbf{1}\!\left[\, C(p,t) > c^{*} \,\right],$$

where p is a place key, T_d the set of time slots in day d, Δt the bin width, $n(p,t)$ the exposed population and $C(p,t)$ the comfort index against threshold c^{*}. Three of the five symbols are identifiers, and the fourth, $n(p,t)$, is the quantity most likely to be synthetic in the early years,

because the RTA series covers exits and ridership rather than street occupancy [Iyyunni, 2026]. The fifth, $C(p,t)$, is itself a synthetic quantity wherever the comfort index rests on the modelled globe temperature of Section 8.3, so the health threshold and the exposure term both have to carry the flag. The consequence is that Heat-Lost Activity Hours will initially be reported in two versions, one restricted to observed counts and one using imputed occupancy, and the schema is what keeps them apart. The efficiency ratios are then simple quotients over the same keys,

$$\mathrm{RAH}_{\mathrm{AED}}(v) \;=\; \frac{\mathrm{RAH}(v)}{\mathrm{LCC}(v)},$$

for intervention version v, with the denominator assembled from cost elements in the manner of Chapter 9; the requirement that a breakdown structure have at least three levels, with children summing to their parent [GAO, 2020], is what makes the denominator auditable. Where the numerator or the denominator is drawn from a distribution rather than observed (synthetic cost draws are the obvious case), the result is a percentile statement rather than a point, and the cost-estimating literature's 50th, 70th and 80th percentile language is the right vocabulary for reporting it [GAO, 2020].

A note on the value of the identifier work itself. Chapter 7 treats the first year as the purchase of information about a poorly known state, in the sense of information value theory [Howard, 1966]. Identifiers are what convert that purchase into an asset: a summer of records keyed to place, time, sensor and calibration remains usable if the gate fails and the project closes with the dataset and platform as deliverables [Iyyunni, 2026], whereas the same summer stored without calibration versions is a collection of numbers nobody can defend. The schema is therefore the cheapest insurance in the programme, and it must exist before procurement, not after.

8.8 The same recursion for the ocean twin

The Africa Blue Intelligence Grid describes the same chain in ocean vocabulary. Its stated design sequence (user decision, information requirement, observation requirement, modelling or prediction requirement, knowledge product, operational testing, user feedback, service improvement) is the master tree read from the top down rather than the bottom up, and its federated model explicitly names discovery, metadata interoperability, common standards, controlled access, provenance, quality assurance and reproducibility as the properties that African institutions must retain when they share data [Terhile et al., 2026]. Those seven properties are the ocean statement of this chapter's schema. The concept paper also states the ethical requirement in the language of provenance: models should surface patterns for trained analysts, preserve uncertainty, document provenance, minimise unnecessary personal data, and establish audit trails that an outside party can follow [Terhile et al., 2026]. Figure 8.7 applies the recursion.

Two differences from the city case matter. The first is that synthetic vessel tracks are not an optional convenience but a structural necessity: the concept paper's own maritime-security pipeline relies on synthetic-aperture radar and optical Earth observation for vessel detection precisely where the signal of the automatic identification system is absent [Terhile et al., 2026], so a detection without an identification signal is by construction a partly generated record, and the gap between what was seen and what was inferred must be carried in the record rather than resolved in it. The macro scale is large (one estimate relayed by the intelligence provider puts illegal, unreported and unregulated fishing at around 40 per cent of the total catch in Africa, an uncited secondary estimate that should be handled as such *[Global Fishing Watch, 2025a]*[2]) and the operational consequence is visible in the Gabon case of 18 July 2026, where an intelligence report on a fleet operating inside national waters

[2]Global Fishing Watch, *Rising pressures on Africa's fisheries and marine resources*, 2025. https://globalfishingwatch.org/rising-pressures-on-africas-fisheries-and-marine-resources/

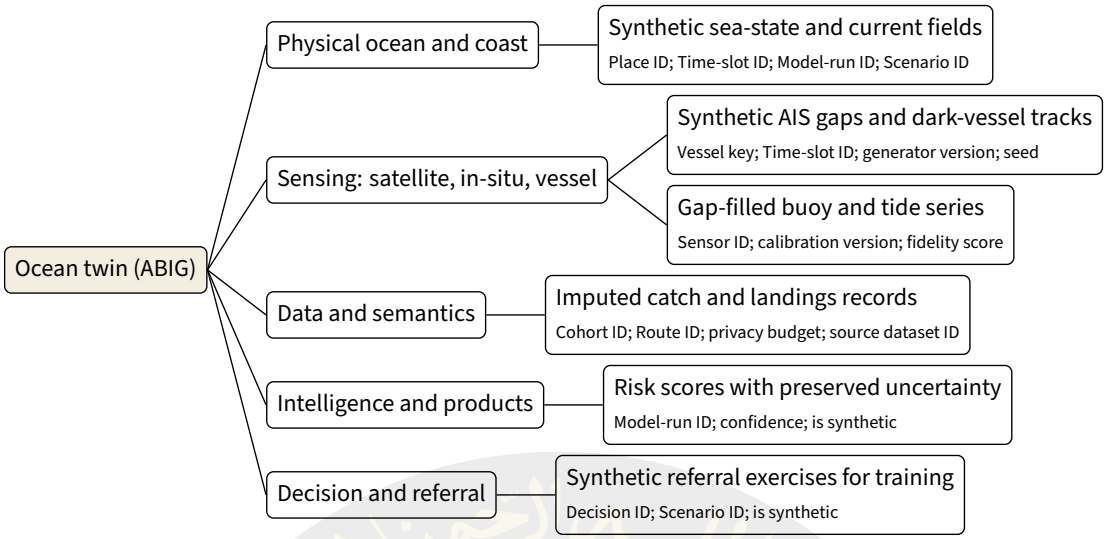


Figure 8.7: The recursion applied to the ocean twin. The leaves are the ABIG analogues of the Dubai synthetic classes; the flags are identical, and the firewall applies with greater force because the decisions at the top of this tree are enforcement referrals [Terhile et al., 2026].

preceded the interception of three trawlers without valid licences (grade: case-report) *[Global Fishing Watch, 2026]*[3]. A referral of that kind is a Decision ID whose lineage must be producible in a court, which is the strongest possible argument for the firewall.

The second difference is that the ocean twin has no month-twelve gate to protect, and therefore needs an equivalent. The concept paper's first-year deliverable is an Africa Ocean Observation, Prediction and Intelligence Capability Map [Terhile et al., 2026], and that map is itself a keyed artefact: an inventory of institutions, datasets and products in which every entry carries a provenance identifier and a synthetic flag on any coverage estimated rather than confirmed. Regional programmes give the geography its keys: the marine service for the southern and eastern African region ran within the GMES and Africa programme over 2017 to 2021, with a partner consortium spanning several coastal states and the Benguela Current Convention (grade: policy-fact) *[MarCOSIO, 2025]*[4]. The continental economic frame gives the cost-element family its scale: African fisheries were valued at about USD 21 billion and coastal tourism at about USD 80 billion in 2018 (grade: projection for later years, compiled secondary estimates for the baseline) [AU-IBAR, 2019]. What ABIG must not do is report a coverage figure or a benefit estimate whose lineage contains an unflagged model output; the archive already shows how easily that happens, since no cost, uptake or benefit measurement exists for any of the continental marine programmes reviewed in Chapter 4.

[3]Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[4]MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

8.9 Implications for the programmes

What the taxonomy decides

Twelve keys and seven flags are sufficient to compute every life-cycle indicator the two programmes promise, and they must be fixed before procurement because identifiers inherit upward and cannot be added retrospectively. The single most consequential field is `is_synthetic`, because it is what allows generated data to train and stress-test the Decision Twin while keeping it out of the evidence on which the month-twelve gate turns.

For Dubai, four things follow and all of them are pre-award work. The identifier schema and the data dictionary belong in the semantic-layer deliverable that Chapter 5 places first, before the sensor specification is written, because the sensor identifier and the calibration version are the fields against which gate conditions 6.2.1 and 6.2.2 will be audited [Iyyunni, 2026]. The node procurement specification must require per-record stamping of sensor identity, calibration version and firmware version, and must state whether a black-globe instrument is being bought; if it is not, every wet-bulb globe temperature the programme reports is a synthetic record and must be flagged as one, which is a material qualification when the programme's labour-relevant thresholds come from a literature that measures that quantity directly [ILO, 2019]. The data-governance document must contain the three firewall rules of Section 8.5 and must express them as a view definition rather than a policy statement. And the request to the Dubai Digital Twin Platform, which held more than 1,500 geospatial layers, more than 280,000 infrastructure assets and more than 195,000 buildings at its launch on 2 July 2026 (grade: policy-fact) *[Dubai Media Office, 2026]*[5], should ask for layer identifiers and version stamps, not only for geometry, since a twin keyed to an unversioned base map cannot support a before-and-after comparison across three summers.

For the Africa Blue Intelligence Grid, the same schema is the cheapest form of the integration layer the concept paper describes. Adopting H3 place keys *[Uber, 2025]*[1] and PROV-O provenance [W3C, 2013] as the two non-negotiable common standards would let the capability map be assembled from federated national holdings without renegotiating each interface, and would give the data and ethics board something concrete to audit: the flag, the privacy budget and the lineage closure. The firewall rule needs strengthening for the maritime-security domain, where a referral is a legal act: no risk score whose lineage contains a synthetic record should be transmitted to a competent authority without the synthetic component named in the transmission, which is the operational form of the concept paper's own insistence that analysis be decision support rather than automated accusation [Terhile et al., 2026].

Two cautions close the chapter. The schema is a design, not a finding: no archived source in this report tests it, and its cost (in firmware, in platform engineering and in the discipline of the analysts) has not been estimated and belongs in the cost elements of Chapter 9. And the schema does not make synthetic data true. A fidelity score is a statement about a held-out window, and the Phoenix reflectivity series is the standing reminder of how fast a calibrated property can decay in a dusty, low-rainfall climate: solar reflectivity measured at 33 to 38 per cent at installation had fallen to 19 to 30 per cent within ten months (grade: measured-with-controls) [ASU and City of Phoenix, 2021]. A generator fitted on the first summer will be wrong about the third for the same kind of reason, which is why `valid_from` and `valid_to` are fields and not documentation.

[5] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

٩ {9}

Forecasting Life-Cycle Costs

Abstract

In what ways can the life-cycle costs of the two programmes be forecast? It sets out ten methods, deterministic life-cycle costing with net present value in the vocabulary of ISO 15686-5, parametric estimating, analogy and engineering build-up, reference class forecasting, Monte Carlo simulation reported as percentiles and S-curves, service-life and degradation curves, the month-12 gate valued as a real option, total cost of ownership for the digital platform, learning curves, and the value function of the decision model of Chapter 7 read as expected life-cycle cost. Each is graded by what the archive can support: the ISO standard is held only as a preview of its first four clauses, the GAO guide's figures are notional illustrations, and the megaproject overrun statistics describe projects two to four orders of magnitude larger than an [1,400,000.00:388,888.89](AED:USD) grant. A method-selection matrix maps method to cost element and programme phase; worked examples use only storyboard figures with the arithmetic shown and labelled indicative; a discount-rate discussion and a nine-point uncertainty reporting standard close the chapter. Readers should use it to decide which forecast to demand for which budget line, and how to read the number they are given.

Forecasting a cost is easy; forecasting the right cost, with the right uncertainty, for the right decision is not. Chapter 2 recorded what the Dubai storyboard asks for ([1,400,000.00:388,888.89](AED:USD) over 36 months from 15 March 2027, released in two phases with a hard gate at month twelve, of which about [250,000.00:69,444.44](AED:USD) is equipment for twenty microclimate nodes, one reference station and one field workstation [Iyyunni, 2026]) and that the Africa Blue Intelligence Grid (ABIG) concept paper carries no budget at all [Terhile et al., 2026]. Between those two states, a single point figure and no figure, lies the question answered here: in what ways can a life-cycle cost be forecast, which way fits which cost element at which phase, and how should the result be reported so that a funder or a partner agency can tell a measured cost from an assumed one? Chapter 10 applies these methods to a breakdown by phase; Chapter 11 treats the events no cost method captures.

The evidence base is thin in an instructive way. The archive holds the international life-cycle costing standard only as a preview of its front matter and first four clauses [ISO, 2017a]; the United States Government Accountability Office cost guide as a best-practice compendium whose numbers are worked illustrations [GAO, 2020]; and the overrun literature as measured statistics on projects costing hundreds of millions to billions of US dollars [Flyvbjerg, 2014, Flyvbjerg et al., 2016]. Not one archived urban-heat or blue-economy programme publishes a unit cost, a service life or a benefit–cost ratio. What follows therefore shows the machinery in full and labels every input.

9.1 What is being forecast

The vocabulary comes from ISO 15686-5:2017, the standard for life-cycle costing of buildings and constructed assets, second edition dated July 2017, of which this archive holds only clauses 1 to 4.2.2 (grade: method-reference) [ISO, 2017a]. Its definitions are usable; its procedures, discounting clause and worked annexes are known by heading alone and are not cited as requirements. A life-cycle cost is the 'cost of an asset or its parts throughout its life cycle, while fulfilling the performance requirements' (clause 3.1.7); life-cycle costing is the 'methodology for systematic economic evaluation of life-cycle costs over a period of analysis, as defined in the agreed scope' (clause 3.1.8); net present value is the 'sum of the discounted future cash flows' and 'the standard criterion for deciding whether an alternative can be justified on economic principles' (clause 3.2.2 and its note).

Four further definitions do real work here. The period of analysis 'is determined by the client' (clause 3.3.6, note 1), making the length of the appraisal a governance decision rather than an analytical one; sunk costs 'are ignored in an appraisal' (clause 3.1.13); residual value is the asset's worth at the horizon (clause 3.3.8); and 'project costs before commencement of design for construction (e.g. feasibility studies) form part of WLCs, not LCCs' (clause 4.2.1), where whole-life cost is the wider aggregate adding externalities, non-construction costs and income. That last rule has a direct consequence: the [1,400,000.00:388,888.89](AED:USD) grant is, in ISO terms, part of the whole-life cost of an eventual cooling asset and not part of that asset's life-cycle cost, and the value of restored activity-hours is a whole-life benefit sitting beside the asset's cost rather than inside it. The four cost categories the standard names (acquisition; operation, including energy, insurance, inspection and taxes (clause 3.1.11); maintenance, including cleaning, servicing, repair and replacement of parts (clause 3.1.9); and end of life, including decommissioning, deconstruction, disposal and regulatory costs (clause 3.1.5)) are the categories onto which every line of the nine-category cost breakdown structure of Chapter 10 maps.

Understanding: Life-cycle cost

TIER 1: IN PLAIN WORDS

Everything a thing costs from the day it is bought to the day it is taken away: buying it, running it, cleaning and mending it, replacing worn parts, and finally removing it.

TIER 2: IN PROFESSIONAL TERMS

The sum of acquisition, operation, maintenance (including replacement of parts) and end-of-life costs of an asset over a period of analysis agreed with the client, normally expressed as a present value so that money spent in different years can be compared; the grant that funds the research and the benefits the asset produces are accounted for separately, as whole-life items [ISO, 2017a].

TIER 3: IN TECHNICAL TERMS

LCC is the net present cost of the asset's cost streams over the agreed period of analysis at a stated real discount rate, with residual value credited at the horizon, sunk costs excluded, and externalities, income and pre-design project costs carried instead in whole-life cost [ISO, 2017a].

AS A MATHEMATICAL MODEL

$\mathrm{LCC} = C_0 + \sum_{t=1}^{N} \frac{C_t^{\mathrm{op}} + C_t^{\mathrm{mt}} + C_t^{\mathrm{rp}}}{(1+r)^t} + \frac{C_N^{\mathrm{eol}} - R_N}{(1+r)^N}$, with N the period of analysis, r the real discount rate and R_N the residual value.

9.2 Ten ways to forecast

Table 9.1 summarises the ten methods, what each needs, what each yields and how far the archive supports it.

Table 9.1: Ten ways to forecast a life-cycle cost: input required, output produced, and the support available in this report's archive. 'Preview only' marks the standard held in truncated form; 'notional' marks figures that are worked illustrations rather than measured outcomes.

Method	Needs	Yields	Archive support
Deterministic LCC with NPV	Agreed scope and period of analysis; cost streams; a real discount rate	A present value per option, comparable across options	Definitions only; preview [ISO, 2017a]
Parametric estimating	A cost estimating relationship fitted to historical data, with its valid range	Cost as a function of a driver; fast rescaling	Method and notional example [GAO, 2020]
Analogy and build-up	One similar completed item, or a decomposed task list with quotations	A quick benchmark, or an auditable bottom-up total	Method and notional examples [GAO, 2020]
Reference class forecasting	A class of completed comparable projects and their outturn-to-estimate ratios	An empirical distribution of overrun; an uplift at a chosen risk level	Measured base rates for megaprojects [Flyvbjerg, 2014, Flyvbjerg et al., 2016]; critique [Cantarelli et al., 2025]
Monte Carlo, percentiles, S-curves	Distributions per element and a correlation assumption	A cumulative distribution of total cost; contingency at a chosen percentile	Method and notional example [GAO, 2020]
Service-life and degradation curves	Service life or a performance-decay function per asset	Timing and size of replacement cost	No archived service life; assumptions are this report's own
Real options	A gate with pass conditions, a fallback value and a pass probability	The value of deferring commitment; expected commitment	Gate design [Iyyunni, 2026]; incremental commitment [Boehm and Hansen, 2000]
Total cost of ownership	Every recurring cost of a platform over its useful life	Whole-of-life platform cost, including exit	Platform facts only *[Dubai Media Office, 2026]*
Learning curves	A slope and a first-unit cost	Declining unit cost with cumulative output	Listed as a sensitivity factor [GAO, 2020]
Value function of the decision model	State, decision, exogenous information, transition, objective	Expected discounted cost under a policy, not a schedule	Framework only [Powell, 2019, Bellman, 1957]

9.2.1 Deterministic life-cycle costing with net present value

Agree the scope and the period of analysis with the client, lay the cost streams out year by year in constant prices, discount at a real rate, and compare options on net present value. ISO 15686-5 requires that the scope of costs 'should be defined and agreed with the client at the outset' (clause 4.1) and allows allowances 'for foreseeable changes, such as future occupancy levels or changing legislative or regulatory parameters' (clause 4.2.1) [ISO, 2017a]. This is the method for the

grant's own cash-flow schedule and for the Year-2 ranking of candidate interventions, and it forces two decisions into the open that the storyboard leaves implicit: who signs the period of analysis for a shade canopy, and at what rate a cost in 2033 is compared with one in 2029 (Section 9.5). Its weakness is that a single present value conceals its own uncertainty, which is why every other method here exists.

9.2.2 Parametric estimating

A parametric estimate uses a cost estimating relationship fitted to historical data to predict cost from a driver such as area, node count or model-run hours. The GAO illustration is a support-cost relationship SA = USD 82,800 + (USD 26,500 × NW) fitted to eleven data points covering seven to forty-seven workstations, giving USD 1,142,800 at forty; the guide is explicit that extrapolating to two or two hundred workstations is inappropriate, and that a software relationship built on ten thousand to two hundred and fifty thousand lines of code cannot be used for a million [GAO, 2020]. The method is quick and makes cost drivers visible, but lacks detail and demands that the estimator understand the model's behaviour. The Dubai programme's natural parameters (cost per node, per square metre of canopy, per street-model run-hour, per instrumented place-hour) cannot be fitted from this archive, because no analogue publishes a unit cost. The honest use of the method is therefore inverted: record the programme's own unit costs from month one, against the identifiers of Chapter 8, so that a relationship exists for the second corridor and for ABIG's regional hubs.

9.2.3 Analogy and engineering build-up

Analogy scales the cost of one similar completed item: the GAO illustration takes a 12,000-pound-thrust engine at USD 5.2 million to a 16,000-pound engine at USD 6.9 million, noting that accuracy depends entirely on similarity. Its cautionary case suits any programme claiming a large capability jump cheaply: a programme office estimated USD 6 billion assuming ten times a predecessor's capacity at half its cost and weight, while an independent estimate of USD 8.7 billion predicted a USD 2.7 billion overrun [GAO, 2020]. Engineering build-up is the opposite: decompose to task level, price from quotations and labour rates, and sum; it is 'easily audited' but 'slow and laborious' [GAO, 2020]. Both rest on a work breakdown structure of at least three levels, standardised at level three so data can be shared across organisations, with programme management always a level-two element and each parent equal to the sum of its children [GAO, 2020]. For Dubai the available analogies are structural rather than monetary: Abu Dhabi's twenty-six-station network, Paris's ten instrumented schoolyards and Singapore's city twin say what such a network contains, not what it costs. The one archived cost analogy with a published figure is the Paris programme, at approximately [9,000,000.00: 37,800,000.00:10,500,000.00](EUR:AED)(AED:USD) a year for around twenty-five schoolyards including surveillance and maintenance, as reported in 2022 *[EEA Climate-ADAPT, 2023]*[1]: an intervention analogy rather than a sensing one, in a different climate, labour market and currency.

9.2.4 Reference class forecasting

Reference class forecasting replaces the inside view of one project with the outside view of a class of completed ones. The measured base rates are stark: nine of ten megaprojects overrun; overruns up to 50 per cent in real terms are common and above 50 per cent not uncommon; the rates have been high and constant across the seventy years for which comparable data exist; rail averages 44.7 per cent cost overrun with a 51.4 per cent demand shortfall and roads 20.4 per cent;

[1] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

one information-technology project in six becomes an outlier averaging 200 per cent overrun; and a one-year extension of the implementation phase correlates with 4.64 percentage points of additional overrun [Flyvbjerg, 2014]. The Olympic Games, the most extreme documented class, average 156 per cent overrun with a median of 90 per cent across the nineteen Games with bid-to-outturn data, 79 per cent of them above 50 per cent and none at all on or under budget [Flyvbjerg et al., 2016].

The critique matters as much as the base rates. A systematic review of sixty-one articles published between 2001 and 2025 finds substantial evidence of improved accuracy, but also studies where the method did not help, evidence from fifty-seven World Bank hydropower projects that screening by uplift can reject projects with positive net present value, a manipulation risk in which sponsors lower initial estimates knowing an uplift will follow, and a 'contingency on a contingency' objection. It reports the rail uplifts as arithmetic (40 per cent for a fifty-fifty chance of staying within budget, 68 per cent to cut the chance of overrun to one in ten), questions whether a modest 20 to 30 per cent uplift is 'the worst of both worlds', and finds the method least applicable to small organisations and routine projects; the class must be neither too large nor too small [Cantarelli et al., 2025]. None of the reviewed studies concerns a research grant or an urban-climate sensor network. The conclusion is firm: megaproject rates are reference-class *context* for these programmes, never calibrated uplifts, and the class actually needed (small urban sensing and twin pilots) does not yet exist and must be built by publishing outturns.

Understanding: Reference class forecasting

TIER 1: IN PLAIN WORDS

Instead of asking what this job should cost, look at what a set of similar finished jobs actually ended up costing, and assume this one will behave like them.

TIER 2: IN PROFESSIONAL TERMS

An outside-view technique that places a project in a class of completed comparable projects, derives the empirical distribution of outturn against original estimate for that class, and applies an uplift matching the overrun risk the client will accept [Flyvbjerg, 2014, Cantarelli et al., 2025].

TIER 3: IN TECHNICAL TERMS

A distributional correction for optimism bias and strategic misrepresentation, valid only where the class is sufficiently similar and sufficiently populated; vulnerable to gaming when uplifts are known in advance, capable of rejecting positive-value projects when used as a screen, and with no published class for research grants or urban sensing networks [Cantarelli et al., 2025].

AS A MATHEMATICAL MODEL

$\hat{C}_{\mathrm{RCF}} = \hat{C}_0\,(1 + u_\alpha)$, where u_α is the class's empirical uplift at acceptable overrun probability α; for rail, $u_{0.5} = 0.40$ and $u_{0.1} = 0.68$ [Cantarelli et al., 2025].

9.2.5 Monte Carlo simulation, percentiles and S-curves

Where reference classes are absent, uncertainty is built from the bottom up: give each cost element a distribution (triangular, lognormal, beta, uniform and normal are the commonest), sample the elements together thousands of times with a stated correlation, and plot the total as a cumulative curve. Because cost inputs are right-skewed the total tends towards a lognormal shape and the mean lies above the median. The guide's own illustration makes the mechanics concrete: a point estimate of USD 825,000 sits at roughly the 40th percentile, the 50th percentile is USD 908,000 (a contingency of USD 83,000) and the 70th percentile USD 1,096,000 [GAO, 2020].

Three cautions govern how such numbers must be read. Expert-elicited ranges cover only

60 to 85 per cent of possible outcomes ('at best' 70 per cent), so a correction treats an expert's stated minimum and maximum as the 15th and 85th percentiles. No confidence level is itself a best practice: budgeting to the mean is common, the mean of a cost distribution typically falls between the 55th and 65th percentiles, a higher level of 70 or 80 per cent is reserved for high-risk programmes, and budgeting every project in a portfolio at its 80th percentile can put the portfolio above its 95th. And the techniques 'are designed to capture general uncertainties about the future, not unforeseen catastrophic events such as major earthquakes and hurricanes' [GAO, 2020], which belong to the force-majeure clauses and gate fallbacks of Chapter 11, not to a contingency percentage. The shorthand P50 and P80 used below is exactly that: the guide's own language is the 50th, 70th and 80th percentile.

Understanding: P50/P80 (percentile) estimate

TIER 1: IN PLAIN WORDS

A P80 figure is the number you would not expect to exceed in four years out of five. A P50 figure is the coin-toss number: as likely to be beaten as missed.

TIER 2: IN PROFESSIONAL TERMS

The value of total cost at a stated cumulative probability on the S-curve produced by simulating the uncertain cost elements together; the gap between that percentile and the point estimate is the contingency the chosen confidence level requires, held separately from the management reserve for known unknowns [GAO, 2020].

TIER 3: IN TECHNICAL TERMS

The p-quantile of the simulated total-cost distribution, approximately lognormal for right-skewed inputs so that the mean exceeds the median; the choice of p is a management decision, element-level percentiles do not sum to the portfolio percentile, and catastrophic events lie outside the distribution entirely [GAO, 2020].

AS A MATHEMATICAL MODEL

$C_p = \inf\{c : \Pr(\tilde{C} \le c) \ge p\}$; for $\ln \tilde{C} \sim \mathcal{N}(\mu, \sigma^2)$, $C_p = \exp(\mu + \sigma z_p)$ with z_p the standard normal quantile, and contingency at confidence p is $C_p - \hat{C}$.

Figure 9.1 draws the shape with clearly labelled illustrative data: a lognormal distribution with median [1,460,000.00:405,555.56](AED:USD) and log-standard-deviation 0.18, parameters chosen by this report solely to display the method. No archived source supports that spread and the curve is not a forecast. On the illustration the storyboard's [1,400,000.00:388,888.89](AED:USD) point estimate sits near the 41st percentile, the mean is [1,480,000.00:411,111.11](AED:USD) near the 54th, the 80th percentile is about [1,700,000.00:472,222.22](AED:USD), and the [1,500,000.00:416,666.67](AED:USD) ceiling of the funding range lies near the 56th *[Dubai RDI Programme, 2025]*[2]. The lesson is structural: when the top of a funding envelope sits close to the median of an honest cost distribution, contingency cannot be held as money and must be held as scope, phase and partner funding (Section 9.4).

9.2.6 Service-life and degradation curves

Replacement cost is driven by service life, and service life is what the archive most conspicuously lacks. The Phoenix Year-1 report defers its life-cycle cost analysis explicitly because the coating's service life is unknown, while measuring that solar reflectivity fell from 33–38 per cent at installation to 19–30 per cent within ten months (grade: measured-with-controls) [ASU and City of Phoenix, 2021]; the Phase II test bed records the reflective products darkening by 13

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

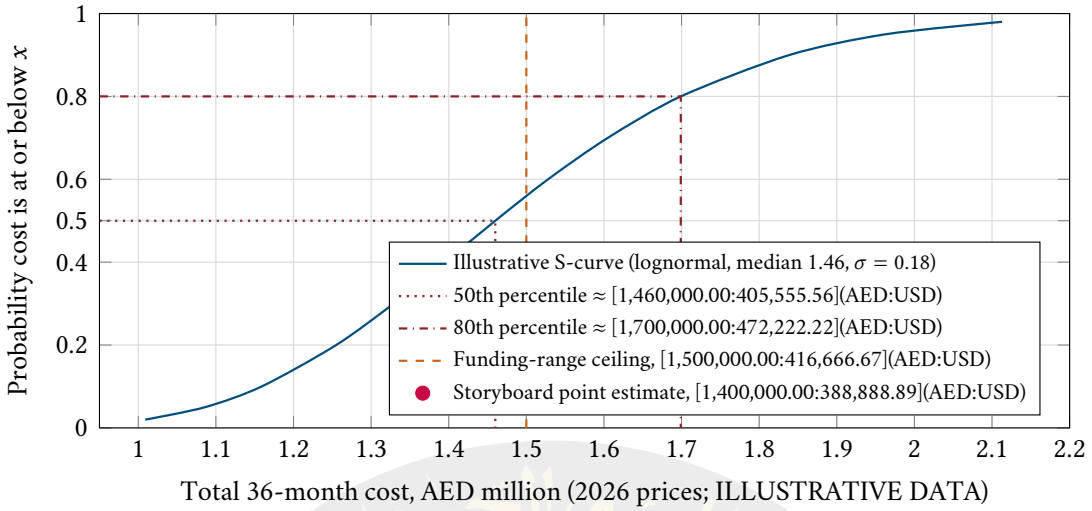


Figure 9.1: Cumulative cost S-curve with clearly labelled illustrative data. The curve is generated by this report from a lognormal distribution (median [1,460,000.00:405,555.56](AED:USD), log-standard-deviation 0.18) purely to show how a point estimate, the 50th and 80th percentiles and a funding ceiling relate; it is not a forecast, and no archived source supports the spread. The real curve must be built at the pre-award loop from three-point ranges per cost element with a documented correlation assumption [GAO, 2020]. Only the point estimate and the ceiling are taken from sources [Iyyunni, 2026]*[Dubai RDI Programme, 2025]*.

and 17 per cent in seven months under truck traffic [ASU and City of Phoenix, 2024]. Paris, at the other extreme, reports an expected lifetime above fifty years for its adopted schoolyard solutions with regular maintenance *[EEA Climate-ADAPT, 2023]*[1]. Between those poles this report adopts its own indicative service-life assumptions, which no archived source states and which must never be attributed to Phoenix, to Paris or to ISO 15686-5: reflective coatings renewed at about five-year intervals, sensor nodes about five years, canopy structures about fifteen years. A degradation curve turns such an assumption into a cost stream by asking not when an asset fails but when its remaining benefit falls below the cost of renewal, and the Phoenix reflectivity series suggests that for a coating in a dusty, low-rainfall climate the benefit half-life is measured in months rather than years. The sensor and calibration identifiers of Chapter 8 are what would convert these assumptions into measured curves over three summers.

9.2.7 The month-12 gate as a real option

The storyboard's gate releases Phase 2 funds only if all four of its conditions are met, and closes the project with the dataset and platform as deliverables if they are not [Iyyunni, 2026]. That has a cost-forecasting reading: the programme does not commit [1,400,000.00:388,888.89](AED:USD) in March 2027; it commits Phase 1 and buys the right, not the obligation, to commit Phase 2 a year later on better information. The real-option framing is this report's own. It is consistent with, but not drawn from, Boehm's incremental-commitment analogy in which a small stake buys one more round of information before the player bets or folds, and his risk-exposure rule that the right level of effort minimises the summed losses from doing too little and too much [Boehm and Hansen, 2000]; the value of what the gate buys is expected value of information in the standard decision-analytic sense [Howard, 1966]; and the requirement that each cycle carry a termination plan comes from the WinWin case study [Boehm et al., 1998]. None of those sources uses the term 'real option', and none should be cited as if it did. Figure 9.2 draws the resulting commitment structure.

Understanding: Real option

TIER 1: IN PLAIN WORDS

Paying a little now to keep the choice open, instead of committing all the money before you know whether the idea works.

TIER 2: IN PROFESSIONAL TERMS

A staged commitment in which a first tranche of expenditure buys the right (but not the obligation) to make a larger, later commitment, exercised only if defined conditions are met; its value rises with the uncertainty resolved during the first stage and with the value retained if the option is abandoned [Iyyunni, 2026].

TIER 3: IN TECHNICAL TERMS

A deferral option over a capital commitment whose exercise condition is the gate's four-part pass test; expected commitment is the first-stage cost plus the probability-weighted second stage, and the fallback deliverable sets the floor value on abandonment. The framing is this report's; the underlying discipline is incremental commitment with a per-cycle termination plan [Boehm and Hansen, 2000, Boehm et al., 1998] and value of information [Howard, 1966].

AS A MATHEMATICAL MODEL

$\mathbb{E}[C] = c_1 + \Pr(G)\,c_2$ and $\mathbb{E}[V] = \Pr(G)\,V_{\text{go}} + \big(1 - \Pr(G)\big)V_{\text{fallback}}$, where G is the event that all four gate conditions hold at month twelve.

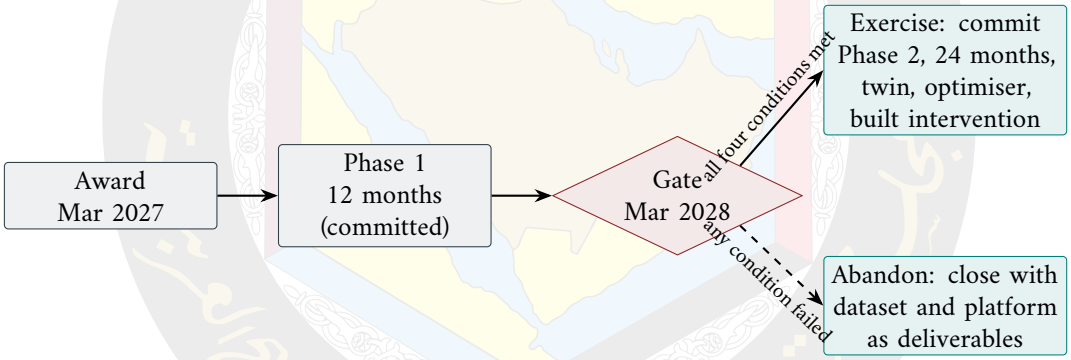


Figure 9.2: The month-12 gate read as a commitment structure. Phase 1 is sunk on exercise or abandonment; Phase 2 is committed only on a pass. The four conditions (sensor uptime of at least 90 per cent across the summer, model error within the stated threshold, an hour-by-hour comfort map of both zones, and written confirmation from RTA or DEWA that the output is useful) and the fallback deliverable are the storyboard's [Iyyunni, 2026]; the option reading is this report's framing, consistent with incremental commitment and a per-cycle termination plan [Boehm and Hansen, 2000, Boehm et al., 1998].

9.2.8 Total cost of ownership for the digital platform

The Decision Twin is software, and software is the cost element with the worst documented record: one project in six becomes an outlier averaging 200 per cent overrun, against an estimated USD 55 billion a year of waste from failed and underperforming information-technology projects in the United States alone [Flyvbjerg, 2014]. A total cost of ownership model is the discipline that answers it, and must carry every recurring line the storyboard names only in outline: compute for two summers of street-scale simulation, storage and egress, data licences, model retraining as sensors and land cover change, security and access administration, support for whichever platform path is chosen, the technical debt of code written to a grant deadline, and the cost of exit (migration

or archival) at the end of the period of analysis. The Dubai Digital Twin Platform, launched on 2 July 2026 with more than 1,500 geospatial layers and more than 100 applications across some 195,000 buildings and 280,000 infrastructure assets (grade: policy-fact) *[Dubai Media Office, 2026]*[3], changes the shape of this line materially: requesting geometry and layers through an existing municipal platform converts a platform-build cost into an integration cost, and a platform-build risk into a dependency. It removes none of the recurring lines, and the archive gives no figure for any of them.

9.2.9 Learning curves

Where a task repeats, unit cost falls with cumulative output. The GAO guide lists learning curves among the factors a valid sensitivity analysis varies [GAO, 2020]; the slope itself must come from the programme's own data. The repeating tasks here are node installation and calibration, the summer field campaign and the production of a decision brief. The method's value at this scale is not the saving it predicts but the distinction it enforces: first-unit effort, which the pre-award budget must carry in full, is a different thing from recurring effort, and a budget that prices the twentieth node like the first is wrong in both directions at once.

9.2.10 The value function as expected life-cycle cost

The last method reads the decision model of Chapter 7 as a cost forecast. In the five-element framework (state, decision, exogenous information, transition, objective), the solution of a sequential decision problem is a value function in Bellman's sense [Powell, 2019, Powell, 2022, Bellman, 1957]. If the contribution at each epoch is the value of restored activity-hours minus that epoch's life-cycle cost, the expected discounted sum of costs under a given policy is itself a life-cycle cost forecast, differing from every other method here in being conditional on a decision rule rather than a fixed schedule: the expected cost of 'mist only when the comfort index crosses a threshold' is not the expected cost of 'mist every afternoon', and that difference is what the Year-2 optimiser trades against comfort hours when it assembles the Minimum Cooling Network. Formally $\mathrm{LCC}^{\pi} = \mathbb{E}\big[\sum_{t} \gamma^{t} c_{t}(S_t, X_t^{\pi}(S_t), W_{t+1})\big]$, with the optimal policy satisfying $V_t(S_t) = \max_{x_t}\{C_t(S_t, x_t) + \gamma\, \mathbb{E}[V_{t+1}(S_{t+1}) \mid S_t, x_t]\}$. Two consequences follow. Weather-driven operating costs (water for misting, district-cooling load, sensor replacement after dust events) are distributions rather than lines, and Monte Carlo simulation is the numerical route to them. And the gate is a decision inside the same model, its expected commitment being Phase 1 plus the pass probability times Phase 2, which is why the option and value-function readings agree.

9.3 Choosing a method

No single method serves every line. Table 9.2 is the method-selection matrix: for each cost element and phase it names the primary method and, in brackets, the cross-check, following the GAO practice of testing one method with a second [GAO, 2020]. Three rules generate it. Where a vendor quotation exists, build-up from the quotation is primary and parametric unit cost the cross-check. Where a cost recurs with the weather, Monte Carlo is primary and deterministic net present value the cross-check. Where a line is a commitment a later decision may cancel, the option value is reported alongside the expected cost. Reference class forecasting appears only as a schedule cross-check and a plausibility test on the total, never as a money uplift.

[3] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

Table 9.2: Method-selection matrix: primary method with cross-check in brackets, by cost element and programme phase. Phases follow Chapter 10: 0 pre-award, 1 observe and model, 2 predict and optimise, 3 build and prove, 4–5 scale, operate and renew. Codes: D deterministic LCC/NPV; P parametric; B engineering build-up; A analogy; R reference class; M Monte Carlo and percentiles; S service-life curve; O real option; T total cost of ownership; L learning curve; V value function.

Cost element	0	1	2	3	4–5	Reporting unit
Proposal and pre-award effort	A	n/a	n/a	n/a	n/a	AED, 2026 prices
Sensing equipment	B (P)	B (P, L)	S (P)	S	S (P, M)	AED per node
Personnel	D (A)	D	D	D	A (R)	AED per person-month
Computing and platform	A (T)	T (M)	T (M)	T (M)	T (A)	AED per run-hour
Travel, ethics, governance, publication	D	D	D	D	D	AED per year
Phase-2 commitment as a whole	O (V)	O (V)	n/a	n/a	n/a	AED, expected
Intervention capital	A (R)	A (P)	P (M, R)	B (R)	P (L)	AED per m^2
Intervention operation, water and energy	A	A	M (V)	M (V)	M (V, D)	AED per litre, per kWh
Replacement and end of life	S	S	S (M)	S (M)	S (M, D)	AED per renewal
Schedule contingency	R	R	R	R	R	months
Programme total	M (R)	M (R)	M (R)	M (R)	V (M, R)	AED at stated percentile
ABIG hubs, data hub and pilots	A (R)	A (R, T)	P (M)	P (M, O)	V (M, R)	USD per decision improved

9.4 Worked indicative examples

Every figure below is a storyboard figure or arithmetic performed on one, and each result is labelled indicative [Iyyunni, 2026].

A parametric upper bound per node. The equipment line of about [250,000.00:69,444.44] (AED:USD) covers twenty nodes, one reference station and one field workstation. If station and workstation were free, the upper bound on one node would be 250,000/20, or [12,500.00:3,472.22] (AED:USD) (indicative); the true figure is lower by their shares, which the storyboard does not give. A vendor quotation at the pre-award loop replaces this bound with a build-up, and must also price a black-globe thermometer, absent from the storyboard's node specification and required if wet-bulb globe temperature is to be computed at a node *[MSE Singapore, 2023]*[4].

The annual profile against the funder's cap. The average annual cost is 1,400,000/3, or [466,667.00:129,629.72](AED:USD), inside the [500,000.00:138,888.89](AED:USD) annual cap of the archived call *[Dubai RDI Programme, 2025]*[2]. But equipment is a Year-1 purchase: if the cap binds per project year, [250,000.00:69,444.44](AED:USD) of equipment leaves at most 500,000 − 250,000, or [250,000.00:69,444.44](AED:USD), for everything else in Year 1, including the postdoctoral and doctoral posts, computing and the platform. The archived rules give a funding range of [500,000.00: 138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD), a duration of one to three years and a minimum share of the grant to be spent within Dubai, but do not say whether the cap applies per calendar or per project year, or whether carry-over is allowed *[Dubai RDI Programme, 2025]*[2]. The schedule of Section 9.2.1 must therefore be a cash-flow profile by project year, and the cap's interpretation is a question to put to the funder before submission.

Reference-class uplifts against the envelope. Applying the illustrative uplifts of Section 9.2.4 to the point estimate gives $1.4 \times 1.20 = [1.68 : 0.47](AED : USD)$ million, $1.4 \times 1.40 = [1.96 : 0.54](AED : USD)$ million and $1.4 \times 1.68 = [2.35 : 0.65](AED : USD)$ million (indicative). Every one breaches the [1,500,000.00:416,666.67](AED:USD) ceiling *[Dubai RDI Programme, 2025]*[2]. The conclusion is not that the programme is under-budgeted by a megaproject multiplier (the classes are not transferable) but that a money uplift is unavailable as an instrument, so contingency must take the three forms the storyboard already implies: scope reserve (the citywide 100 m product as a stretch goal, the corridor map as the core), phase reserve (the gate) and partner-funded capital (the built intervention).

Delay as reference-class context. At 4.64 percentage points of overrun per year of delay, a one-year slip would imply 0.0464 × 1,400,000, or approximately [65,000.00:18,055.56](AED:USD) (indicative; the underlying regression concerns projects of hundreds of millions of US dollars and is quoted as context only) [Flyvbjerg, 2014]. The Olympic finding that a fixed opening date removes the schedule–cost trade-off [Flyvbjerg et al., 2016] applies to the two summer windows: sensors live by May 2027 and a canopy built by May 2029 are fixed by the climate, so missing either is a full-year slip, not a month's.

Expected commitment through the gate. Taking Phase 1 at the cap, so that c_1 is [500,000.00: 138,888.89](AED:USD), and $c_2 = 1{,}400{,}000 - 500{,}000$, or [900,000.00:250,000.00](AED:USD),

[4] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

expected commitment is $\mathbb{E}[C] = 500{,}000 + \Pr(G) \times 900{,}000$: [950,000.00:263,888.89](AED:USD) at a pass probability of one half, [1,220,000.00:338,888.89](AED:USD) at 0.8, and the full [1,400,000.00:388,888.89](AED:USD) only if passing is certain (indicative). $\Pr(G)$ appears in no source; it is a quantity the pre-award loop should elicit from the team, with the expert-range correction of Section 9.2.5 applied. The option is worth holding not only for the [900,000.00:250,000.00](AED:USD) not committed on a fold but for the dataset and platform retained, which is why the fallback must itself be costed as a deliverable.

A learning curve on installation. On an assumed 90 per cent slope (this report's illustration), the twentieth node installation costs $20^{\log_2 0.9} \approx 0.63$, about 63 per cent of the first, with the twenty-unit average between the two.

Replacing the nodes at the indicative service life. Under this report's indicative five-year node life, nodes bought in 2027 fall due for replacement in 2032, after the grant ends. Taking the whole equipment line of [250,000.00:69,444.44](AED:USD) in 2026 prices as an upper bound on that replacement, Table 9.3 gives its present value at four illustrative real discount rates by $250{,}000/(1+r)^5$. The rates are this report's illustrations; the ISO preview prescribes none and none may be attributed to it.

Table 9.3: Present value of an indicative [250,000.00:69,444.44](AED:USD) node replacement falling due in year five, at four illustrative real discount rates (arithmetic: $250{,}000/(1+r)^5$; 2026 prices). The rates are illustrations chosen by this report; ISO 15686-5 prescribes no rate and none is cited to it [ISO, 2017a].

Real discount rate r	0 per cent	3 per cent	5 per cent	8 per cent
$(1+r)^5$	1.0000	1.1593	1.2763	1.4693
Present value (AED, rounded)	250,000	215,600	195,900	170,100
Share of undiscounted cost	100 per cent	86 per cent	78 per cent	68 per cent

9.5 Discount rates and the period of analysis

ISO 15686-5 distinguishes the real discount rate, which relates present and future money without taking general or specific inflation into account (clause 3.3.7), from the nominal rate, which includes it (clause 3.3.5), and both from the escalation rate, the differential price movement of a particular commodity (clause 3.3.2) [ISO, 2017a]. The preview gives no value for any of them. The overrun literature is stated throughout in real terms [Flyvbjerg, 2014, Flyvbjerg et al., 2016], and the GAO guide lists inflation indices among the economic sources of cost risk [GAO, 2020]. Four rules follow.

First, state every cost in constant prices with the year named (2026 prices for the Dubai grant, the year of each figure for every international analogue), so that real discounting is the default and no comparison silently mixes price years. Second, treat the water and energy tariffs that enter the optimiser's objective through DEWA as commodities with their own escalation rate, separate from the general discount rate, because comfort hours per litre and per kilowatt-hour are among the storyboard's own reporting units [Iyyunni, 2026]. Third, report net present value over a range of real rates and undiscounted, as Table 9.3 does, since the choice of rate is a client decision, not an analyst's. Fourth, agree the period of analysis in writing before any ranking is done: 36 months for the grant's cash-flow schedule, and for a built canopy a period signed by Dubai Municipality and Dubai Culture

before the Year-2 optimiser ranks candidates, because a fifteen-year assumption and a fifty-year assumption rank a coating and a canopy in different orders. ABIG faces the same choice at hub level, and the Seychelles precedent shows what matching looks like: a USD 15 million sovereign blue bond announced on 29 October 2018 with a custom ten-year maturity schedule aligned to project funding needs and the issuer's liability profile, de-risked by a USD 5 million World Bank partial credit guarantee and USD 5 million of concessional financing from the Global Environment Facility (grade: policy-fact) [World Bank, 2025][*World Bank, 2018*][5]. Instrument tenor should follow the period of analysis of the asset financed.

9.6 Reporting uncertainty

A cost forecast is only as useful as a reader's ability to tell what kind of number it is. This report adopts a nine-point reporting standard, drawn from the GAO's four characteristics of a reliable estimate (comprehensive, well documented, accurate, credible [GAO, 2020]) and from the evidence-grading scheme of the Preface. (1) Every figure carries its currency and price year. (2) Every figure is labelled measured, derived, indicative or illustrative, and every derived figure shows its arithmetic. (3) Every figure names the method that produced it and the evidence grade of its source. (4) Point estimates are placed on an S-curve with their percentile stated in the guide's own language, and the budgeting percentile declared a management decision. (5) Contingency and management reserve are separated and each held at a named level: contingency for unknown unknowns above the programme office, management reserve for known unknowns at delivery level. (6) Sensitivity analysis follows the guide's five steps and never 'simply var[ies] factors by a subjective plus or minus percentage', which it says 'does not constitute a valid sensitivity analysis'; a parameter is sensitive when a change of 10 to 50 per cent alters the preferred alternative and very sensitive when a change below 10 per cent does [GAO, 2020]. (7) Catastrophic events are declared outside the distribution and referred to Chapter 11. (8) The Letter of Intent budget remains the permanent baseline, because rebaselining 'resets the measurement of cost growth to zero' [GAO, 2020], and the cost risk analysis is re-run before each key decision point, of which the month-12 gate is the first; the cone of uncertainty narrows as actuals replace estimates, and the gate is where the first summer's actuals arrive. (9) An audit trail links every figure to its source, in the spirit of the standard's reporting clause, known here by heading only [ISO, 2017a]. Two cautionary cases justify the effort: a census life-cycle estimate whose risk analysis covered only about 37 per cent of the estimated cost, and an icebreaker baseline whose risk modelling excluded operating, support and disposal costs amounting to about 75 per cent of the life cycle [GAO, 2020].

What the methods show

The archive supports the machinery of life-cycle cost forecasting in full and its inputs only in part. Definitions come from ISO 15686-5 and procedure from the GAO guide; measured overrun classes exist only for projects thousands of times larger than the Dubai grant; and no archived cooling or ocean programme publishes a unit cost, a service life or a benefit–cost ratio. Three findings follow. First, money contingency is not available: every plausible reference-class uplift breaches the [1,500,000.00:416,666.67](AED:USD) ceiling, so contingency must be held as scope, phase and partner capital. Second, the month-12 gate is the programme's most valuable cost instrument, converting some [900,000.00:250,000.00] (AED:USD) of Phase-2 commitment into an option exercised on evidence, with a costed

[5] World Bank, *Seychelles launches World's First Sovereign Blue Bond*, press release, 29 October 2018. https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond

fallback setting its floor. Third, the largest unforecast lines are recurring rather than capital: platform operation after March 2030, and the supervision, cleaning, water and energy of a built cooling asset. The reporting standard of Section 9.6 is the condition under which any of these numbers may appear in a proposal.

9.7 Implications for the programmes

For the Dubai Decision Twin. The full proposal should present its budget as a cash-flow schedule by project year against the [500,000.00:138,888.89](AED:USD) annual cap, with the equipment purchase profiled and the cap's interpretation confirmed with the Dubai Future Foundation in advance, since budget adequacy is scored at the Letter of Intent, while budget, deliverables, management plan and risk mitigation are all scored at full application *[Dubai RDI Programme, 2025]*[2]; the first cycle funded 24 of 374 proposals, so a budget that invites a question is an expensive luxury *[Dubai Future Foundation, 2025]*[6]. Each cost element should name its method from Table 9.2: build-up from vendor quotations for the nodes, with the black-globe instrument added to the specification; total cost of ownership for the platform, priced both ways for the contracted-partner and in-house paths, with a written sustainment commitment from the host agency made a condition of the Year-2 intervention choice, after the WinWin finding that unsustained applications do not survive [Boehm et al., 1998]; and three-point ranges with a Monte Carlo S-curve for the total, the point estimate's percentile stated. Contingency should be declared openly as scope, phase and partner capital rather than money, and schedule contingency argued against a small documented class of sensor-network and urban-twin pilots rather than megaproject rates. The optimiser's cost inputs (capital per unit, operating cost per unit-year, an indicative service life per intervention, water and energy at DEWA tariffs with their own escalation) must be tabulated and labelled before any candidate set is ranked, and the period of analysis for the canopy agreed in writing before the Minimum Cooling Network is assembled. The programme should also record its own unit costs from month one against the identifiers of Chapter 8: the reference class this work most needs is the one it is in a position to create.

For the Africa Blue Intelligence Grid. ABIG has no budget, and the archived analogues cannot supply one: the African Union strategy's sector projections rest on annexes that are not archived [AU-IBAR, 2019], the continental Earth-observation valuation has no cost side at all [WEF and Digital Earth Africa, 2021], and the regional marine service's economic assessment exists only as a terms of reference *[MarCOSIO, 2025]*[7]. The cost structure must therefore be built by analogy and parametrics on a programme work breakdown structure with one common element set (programme office, scientific and technical council, operational user council, data and ethics board, regional hubs, federated data hub, pilots, training), standardised at level three so that every hub reports against the same codes [GAO, 2020], which is how a continental reference class comes to exist by Year 3 where none exists now. The federated data hub and analytics platform belong to the information-technology class, the class with the fattest tails [Flyvbjerg, 2014], and should be forecast with a total cost of ownership model and budgeted at the higher confidence level reserved for high-risk programmes; any sensor, ground-station or vessel hardware admitted by a later 'hardware follows evidence' decision belongs to the heavy-infrastructure classes with their measured delays.

[6] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[7] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

The pilot portfolio should be risk-adjusted at portfolio level, since budgeting each pilot at its own 80th percentile over-budgets the whole [GAO, 2020]. The concept paper's financing ladder (policy mandate, project preparation, pilot, independent validation, blended finance, commercial service [Terhile et al., 2026]) is a sequence of options in exactly the incremental-commitment sense, and each rung should carry pre-agreed pass conditions and a costed fallback as the Dubai gate does. And the capability map should add the line Chapter 10 identifies as missing: a cost breakdown per regional hub and a cost per decision improved, reported in constant US dollars with the year named, so that the outcome indicators the concept paper promises acquire a denominator.

١٠ {10}

Life-Cycle Cost Breakdown and Strategic Roadmap

Abstract

The life-cycle costs of both programmes are broken down inside one strategic roadmap: a cost breakdown structure of nine categories (people; sensing; data and compute; models and software; physical interventions; governance, ethics and legal; evaluation with controls; dissemination and standards; and contingency), crossed with six phases and an end-of-life stage, from pre-award in September 2026 to operation and renewal to 2050. Each line carries an owner and a funding source, so that the reader can see which costs the [1,400,000.00: 388,888.89](AED:USD) grant carries and which it does not: about [250,000.00:69,444.44] (AED:USD) of equipment does sit inside it, the built intervention of Year 3 does not. Three stage gates in the vocabulary of the spiral anchor points (life-cycle objectives, life-cycle architecture and initial operational capability) fix what must be true before money moves. The same structure is then built for the Africa Blue Intelligence Grid against its financing ladder and its regional hubs, with one reporting unit for each programme: dirhams per restored activity-hour, and cost per decision improved.

Budgets list things to buy; a cost breakdown structure is a claim about what a programme is made of, and a roadmap is that claim laid along a calendar with the moments named at which somebody may say no. Chapter 9 set out ten ways to forecast a life-cycle cost and found the machinery well documented and the inputs largely absent: no archived urban-heat or blue-economy programme publishes a unit cost, a service life or a benefit–cost ratio. This chapter takes the next step. It does not produce a forecast. It produces the structure into which a forecast must be poured (the rows, the columns, the owner of each cell, the source of the money in each cell, and the gates between the columns) for the Dubai Decision Twin and for the Africa Blue Intelligence Grid (ABIG). The structure is the deliverable, because a structure can be agreed with partners in 2026 while numbers cannot.

Two disciplines govern what follows. The first is accounting: the four cost categories of the international life-cycle costing standard (acquisition; operation, including energy, insurance, inspection and taxes; maintenance, including cleaning, servicing, repair and replacement of parts; and end of life, including decommissioning, deconstruction, disposal and regulatory costs) with the standard's own boundary rule that project costs incurred before commencement of design, feasibility studies among them, belong to whole-life cost and not to the life-cycle cost of the asset (grade: method-reference; the archive holds only clauses 1 to 4.2.2) [ISO, 2017a]. The second is programme structure: a breakdown that has at least three levels, is standardised at level three so that data can be shared across organisations, carries programme management as a level-two element,

obeys the rule that the sum of a parent's children must equal the parent, and is accompanied by a dictionary defining every element (grade: method-reference) [GAO, 2020]. Neither discipline supplies a single dirham. Both supply the shape of the answer.

10.1 The cost breakdown structure

Understanding: Cost breakdown structure

TIER 1: IN PLAIN WORDS

A list of the kinds of things a programme spends money on, arranged so that every cost has exactly one place to go and nothing is left out.

TIER 2: IN PROFESSIONAL TERMS

A product- and service-oriented hierarchy of cost elements, at least three levels deep and standardised at the third, in which each child rolls up into exactly one parent, programme management appears as an element in its own right rather than being spread invisibly across the others, and a written dictionary defines the boundary of every element so that two organisations reporting against it report the same thing [GAO, 2020].

TIER 3: IN TECHNICAL TERMS

A partition of the whole-life cost space into mutually exclusive and collectively exhaustive elements, each mapped to the standard's acquisition, operation, maintenance and end-of-life categories, with the period of analysis agreed with the client, sunk costs excluded from appraisal, residual value credited at the horizon, and pre-design project costs carried in whole-life cost rather than in the asset's life-cycle cost [ISO, 2017a, GAO, 2020].

AS A MATHEMATICAL MODEL

$C^{\text{total}} = \sum_{i \in I} \sum_{t \in T} C_{i,t}$ with I the level-three element set and T the phase set, subject to $C_{\text{parent}} = \sum_{\text{children}} C_{\text{child}}$ at every level and I a partition, so that no cost is double-counted and none is homeless.

Nine level-two categories carry every cost of both programmes. They are proposed by this report, not taken from any source, and are chosen so that the boundary of each is decidable by a person reading a single invoice. Table 10.1 gives them with their principal contents, their owner and their funding source for the Dubai case.

Three of the nine deserve a note. Category 2 contains a known gap: the node specification records air temperature, humidity, wind, radiation and surface temperature [Iyyunni, 2026], which does not include the black-globe and natural-wet-bulb instruments that wet-bulb globe temperature requires, and comparability with the labour-hours literature depends on reporting that index [ILO, 2019]. Closing the gap is a purchase, and a purchase belongs in a cost line before it belongs in a discussion. Category 5 is the line the grant does not carry and the line that dominates the life cycle after 2030. Category 9 is not money at all, for reasons Section 10.4 sets out.

10.2 Phases and stage gates

Understanding: Stage gate

TIER 1: IN PLAIN WORDS

A point in the calendar where the work stops, somebody checks whether the agreed things are true, and only then is the next tranche of money released.

Table 10.1: The nine-category cost breakdown structure for the Dubai Decision Twin, with the owner accountable for each line and the source of the money. Categories and the owner/funding assignment are this report's proposal; the budget facts they organise are the storyboard's [Iyyunni, 2026] and the funding rules the programme page's *[Dubai RDI Programme, 2025]*.

Category	Principal contents	Owner	Funding source
1 People	Principal Investigator and four Co-Investigators; one postdoctoral researcher and one doctoral student for 36 months; the collaborating entity's staff share; named advisers	Lead institution	Grant; institutional in kind; collaborator budget line
2 Sensing	Twenty microclimate nodes, one reference station, one field workstation; mounting, power and connectivity; calibration; spares; the missing black-globe and natural-wet-bulb instruments	Lead institution	Grant (about [250,000.00: 69,444.44] (AED:USD))
3 Data and compute	Ingestion, quality control, spatial-temporal store and one interface; two summers of street-scale simulation; archive and egress	Lead institution; platform partner	Grant; agency platform in kind
4 Models and software	District and street models; machine-learning downscaling and bias correction; the multi-objective optimiser; Decision Twin version 1; licences	Lead institution; collaborating entity	Grant; collaborator budget line
5 Physical interventions	The single built intervention; its foundations, supervision, cleaning, water, energy, repair, replacement and eventual removal	Municipality and heritage authority	**Not the grant:** partner capital and operating budgets
6 Governance, ethics, legal	Ethics review; data-governance instruments; site permissions; data agreements; the collaborating-entity agreement; insurance	Lead institution with partners	Grant (ethics and governance lines); partner legal in kind
7 Evaluation with controls	Paired control instrumentation; before-and-after campaigns; repeat pedestrian counts and dwell-time measurement; independent review	Lead institution	Grant; partner data in kind
8 Dissemination and standards	Decision briefs; the transferable method document; publications; standards and indicator work	All stakeholders	Grant (publication line); institutional in kind
9 Contingency and reserve	Contingency for unknown unknowns held above the project; management reserve for known unknowns at delivery level [GAO, 2020]	Funder; lead institution	Scope, phase and partner capital, not money (Section 10.4)

TIER 2: IN PROFESSIONAL TERMS

A milestone at which a defined set of artefacts is reviewed against pass conditions written before the work began, with a costed fallback deliverable prepared in advance so that failing the gate closes the programme tidily rather than expensively; six artefacts (operational concept, prototyping results, requirements, architecture, life-cycle plan and feasibility rationale) are the review set at the first two anchor points [Boehm and Hansen, 2000].

TIER 3: IN TECHNICAL TERMS

An incremental-commitment decision point in a risk-driven process: the life-cycle objectives anchor point commits to architecting on at least one feasible architecture and a business case; the life-cycle architecture anchor point, 'the point of significant escalation of staff level and resource commitments', requires all major risks resolved or covered by a risk-management plan; initial operational capability commits to operations through software, site, and user, operator and maintainer preparation [Boehm and Hansen, 2000].

AS A MATHEMATICAL MODEL

Release Phase $k{+}1$ funds if and only if $\bigwedge_j g_j = \text{true}$ for the pre-agreed conditions g_j; otherwise pay the fallback cost F_k and stop. Expected commitment is $C_k + \Pr(\text{pass})\, C_{k+1} + (1-\Pr(\text{pass}))\, F_k$, and $\text{RE} = \Pr(\text{Loss}) \times \text{Size}(\text{Loss})$ sets how much verification is worth buying [Boehm and Hansen, 2000].

The spiral model's three anchor points map onto the Dubai calendar without strain, and the mapping is this report's, not Boehm's. The pre-award loop from September 2026 to March 2027 ends at a life-cycle objectives milestone: a feasible architecture (one corridor, twenty nodes, named models, named partners) and a business case in the form of the Letter of Intent. The month-twelve gate of March 2028 is a life-cycle architecture milestone: the storyboard's four conditions are exactly the requirement that the major technical risks be resolved rather than merely listed, and it is the point at which the remaining two-thirds of the grant escalate into the field [Iyyunni, 2026, Boehm and Hansen, 2000]. The built intervention of May 2029, measured across the summer and compared with the twin's prediction, is initial operational capability: something exists that a member of the public walks under. Table 10.2 records the phases, gates and conditions.

One lesson from the archive belongs at the Phase-2 gate rather than anywhere else. In the WinWin spiral case study, of five applications carried forward after the first year only one was actually implemented, and it was the only one with sufficient budget, people and facilities to sustain the product; in the second year a sustainment commitment became a selection criterion (grade: case-report, no control group, confounded process changes) [Boehm et al., 1998]. The Dubai analogue is exact: the intervention chosen in early 2029 will need supervision, cleaning, water, energy and repair for as long as it stands, and none of that is in the grant. A written commitment from the owning authority to carry those costs should therefore be a condition of selection, not a hope entertained after construction.

10.3 The breakdown crossed with the phases

Table 10.3 crosses the nine categories with the six phases and end of life, marking where each category is the principal cost, where it is present but minor, and where it is dormant. It is a map of attention, not of amounts.

Four features of the pattern matter more than any individual mark. Sensing is a single large acquisition, about [250,000.00:69,444.44](AED:USD) for twenty nodes, one reference station and one field workstation [Iyyunni, 2026], that recurs only when the hardware is renewed: on this report's own indicative assumption of roughly five years for a field node, once inside the grant and again early in Phase 4. That assumption is the report's and is cited to nobody: no

Table 10.2: Phases, anchor points and gate conditions for the Dubai Decision Twin. Dates and the four month-twelve conditions are the storyboard's [Iyyunni, 2026]; the anchor-point mapping and the Phase-4 and Phase-5 horizons are this report's [Boehm and Hansen, 2000]. Phases 4 and 5 lie outside the grant and outside any archived source.

Phase	Window	Anchor point and pass condition
0 Pre-award	Sep 2026 – Mar 2027	Life-cycle objectives. Site letters from the heritage and municipal authorities by December 2026; data letters and the collaborator agreement by February 2027; one feasible architecture and a written win condition per partner
1 Observe and model	Mar 2027 – Mar 2028	Life-cycle architecture (the month-twelve gate). Sensor network live with at least 90 per cent uptime across the summer; model matching observations within the stated error; hour-by-hour comfort map of both zones delivered; at least one of the transport or the utility authority confirming in writing that the output is useful for a decision it faces. Failure closes the project with dataset and platform as deliverables
2 Predict and optimise	Mar 2028 – Mar 2029	Minimum Cooling Network for the corridor (November 2028); Decision Twin version 1 (February 2029); one intervention selected with the municipality and the heritage authority, with a written sustainment commitment attached
3 Build and prove	Mar 2029 – Mar 2030	Initial operational capability. Intervention built (May 2029); before-and-after measurement complete (September 2029); measured change compared with predicted change; decision briefs and transferable method document (February 2030)
4 Scale	2030 – 2033	A new life-cycle objectives milestone for a second corridor: agency procurement, platform operation transferred, restored activity-hours per dirham demonstrated at district scale
5 Operate and renew	2033 – 2050	Asset operation, maintenance, replacement and renewal; cost per restored activity-hour trending downwards; periodic reassessment against the agreed period of analysis [ISO, 2017a]
End of life	As each asset falls due	Decommissioning, deconstruction, disposal and regulatory costs; residual value credited; data archived or destroyed under the governance instruments [ISO, 2017a]

Table 10.3: Cost breakdown structure against phase for the Dubai Decision Twin. • principal cost driver in that phase; ◦ present but minor; – not active. Phases as in Table 10.2. The pattern, not the amounts, is the claim: sensing is a large acquisition twice, once inside the grant and once at renewal, while people, compute and physical operation recur for as long as the programme lives.

Category	P0	P1	P2	P3	P4	P5	EoL
1 People	◦	•	•	•	•	◦	◦
2 Sensing	–	•	◦	◦	•	◦	◦
3 Data and compute	–	•	•	◦	•	•	◦
4 Models and software	◦	•	•	◦	•	◦	–
5 Physical interventions	–	–	–	•	•	•	•
6 Governance, ethics, legal	•	◦	◦	◦	◦	◦	◦
7 Evaluation with controls	◦	◦	◦	•	•	◦	–
8 Dissemination and standards	◦	◦	◦	•	•	◦	–
9 Contingency and reserve	◦	◦	◦	◦	•	•	◦

archived source states a service life for a microclimate node. Data and compute never stop; the continental analogue makes the scale legible, since the Singapore urban-climate twin was developed by more than twenty researchers over four years and its high-performance runs deliver up to eighty hours of climate-model simulation within twenty-four hours of wall-clock time on a national supercomputer (grade: case-report) [NSCC Singapore, 2024]. Physical intervention costs begin only in Phase 3 and then never end, which is why the Paris schoolyard programme is the honest comparator: its solutions are expected to last more than fifty years with regular maintenance, and the city's own programme cost is given as approximately nine million euros per year for around twenty-five schoolyards, a figure the case study states as including surveillance and maintenance and as co-funded by the city, the state and the water agency (grade: case-report with policy-fact content) *[EEA Climate-ADAPT, 2023]*[1]. Dividing that figure gives about 360,000 euros per yard per year; this report derives that number, labels it derived, and does not compare it with the pilot's European funding of 4,995,793.16 euros for ten schoolyards over the 2018–2022 window, because the two figures have different cost bases *[EEA Climate-ADAPT, 2023]*[1].

The fourth feature is renewal. Where a surface coating is a candidate intervention, its cost is not a purchase but a subscription: the Phoenix residential trial measured solar reflectivity falling from 33–38 per cent at installation to 19–30 per cent within ten months across eight neighbourhoods, against untreated asphalt concrete at about 12 per cent (grade: measured-with-controls) [ASU and City of Phoenix, 2021]. Neither Phoenix report gives a service life, a re-application interval or a unit cost [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024], and the Year-1 report explicitly defers its life-cycle cost analysis because the service life is unknown [ASU and City of Phoenix, 2021]. The report's own indicative assumptions (roughly five years for a coating, roughly five for a node, roughly fifteen for a canopy fabric) are therefore placeholders whose only function is to force the replacement rows of Phase 5 to exist. They must be replaced by supplier warranties before any candidate set is ranked.

[1] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

10.4 What the grant covers and what it does not

The Dubai Future Foundation programme page states grants of [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD) per project, an annual budget per project that must not exceed [500,000.00:138,888.89](AED:USD), a requirement that at least 50 per cent of the grant funding be utilised within Dubai, and a Principal Investigator fully affiliated with a Dubai-based higher-education institution (grade: policy-fact, documenting the 2025 second cycle) *[Dubai RDI Programme, 2025]*[2]. The storyboard asks for [1,400,000.00:388,888.89](AED:USD) over 36 months, of which about [250,000.00:69,444.44](AED:USD) is equipment [Iyyunni, 2026]. Dividing the ask by the three project years gives an average of about [466,667.00:129,629.72](AED:USD) per project year; that is this report's arithmetic (derived), and it sits just under the annual cap with roughly [33,000.00:9,166.67](AED:USD) of headroom *[Dubai RDI Programme, 2025]*[2]. The headroom is thin enough that the equipment purchase must be profiled deliberately: [250,000.00:69,444.44](AED:USD) of nodes bought in the first months of Phase 1 consumes half of one year's ceiling on its own, leaving about [250,000.00:69,444.44](AED:USD) for twelve months of salaries, compute, ethics and travel in the same year. Either the nodes are bought inside the first project year and the rest of that year's cap is planned around them, with only spares and later instruments deferred into the following project year, or the cap's interpretation is confirmed with the funder in advance. What no profile can do is defer the network past the first summer: the month-twelve gate requires it live at 90 per cent uptime across that summer (Table 10.2), so nodes bought after it buy budget headroom at the cost of the baseline on which every heat and health claim in the programme rests.

The boundary of the grant

The [1,400,000.00:388,888.89](AED:USD) covers people, sensing, data and compute, models and software, governance and ethics, evaluation and dissemination for 36 months. It does not cover the physical intervention. The storyboard is explicit that the Year-3 structure is built 'with a partner' and that the municipal and heritage authorities are asked for written permission to build, not for a construction budget [Iyyunni, 2026]. Nor does the grant cover anything after 14 March 2030: platform operation, sensor renewal, model retraining, and the supervision, cleaning, water, energy and repair of whatever was built are Phase-4 and Phase-5 costs belonging to an agency that has not yet been asked to carry them. In life-cycle costing terms the grant is a whole-life cost of a future cooling asset, incurred before design, and is therefore not part of that asset's life-cycle cost at all [ISO, 2017a]. Two consequences follow. First, money contingency is unavailable: the top of the funding envelope, [1,500,000.00:416,666.67](AED:USD), is only about seven per cent above the ask (derived), so the reserve must be held as scope (the citywide 100 m product is already a stretch goal [Iyyunni, 2026]), as phase, through the month-twelve gate, and as partner capital. Second, the single most consequential number in the whole life cycle, the capital and recurrent cost of the intervention, is one the proposal does not control: the proposal must therefore name its owner before Phase 2 closes.

The prudence of this boundary is worth stating plainly rather than apologising for. The measured overrun literature concerns projects costing hundreds of millions to billions of US dollars (nine of ten megaprojects overrun, with overruns up to 50 per cent common in real terms and above 50 per cent not uncommon, a pattern constant over seventy years [Flyvbjerg, 2014]) and those base rates are reference-class context for how a built asset behaves, not a calibrated uplift for a research

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

grant three orders of magnitude smaller. What transfers is the direction of the error, not its size: the construction line is the line that moves, and it is the line the grant does not hold.

10.5 One reporting unit

Understanding: Cost per restored hour

TIER 1: IN PLAIN WORDS

How much it costs to give people back one hour in which they can comfortably be outside somewhere they could not before.

TIER 2: IN PROFESSIONAL TERMS

The whole-life cost of an intervention divided by the activity-hours it restores over its life: capital, operation, water, energy, maintenance and replacement in the numerator; the measured difference in usable hours, with and without the intervention (Restored Activity Hours, RAH) in the denominator, reported alongside the same quantity per litre of water and per kilowatt-hour of cooling load, because comfort bought with resources is not free [Iyyunni, 2026].

TIER 3: IN TECHNICAL TERMS

Discounted whole-life cost per unit of Restored Activity Hours at a stated comfort threshold, aggregated over place-hours and cohorts, with the counterfactual supplied by a paired untreated control rather than by a model, and the comfort index reported as both the Universal Thermal Climate Index and wet-bulb globe temperature so that the denominator remains comparable with the international labour-hours series [ILO, 2019].

AS A MATHEMATICAL MODEL

$u = \dfrac{\mathrm{LCC}}{\sum_{c,h}\left(H^{\mathrm{base}}_{c,h} - H^{\mathrm{int}}_{c,h}\right)} = \dfrac{\mathrm{LCC}}{R}$, where H is Heat-Lost Activity Hours per cell-hour and R is Restored Activity Hours, giving AED per restored activity-hour, with parallel denominators in litres and kilowatt-hours.

Both programmes need a single number that survives translation between a laboratory, a municipality and a bank. For Dubai it is dirhams per restored activity-hour, computed from Heat-Lost Activity Hours (HLAH) with and without the intervention and reported for every candidate in the Minimum Cooling Network (MCN); for ABIG it is cost per decision improved. Neither exists in any archived source, and that absence is the finding. The continental Earth-observation valuation projects benefits higher than two billion US dollars a year by 2024 with no cost side at all and no realised outcomes, every figure labelled as the authors' own team analysis (grade: projection) [WEF and Digital Earth Africa, 2021]; the African Union strategy gives sector values (about 21 billion US dollars for fisheries and about 80 billion for coastal tourism in 2018), whose projection method lies in unarchived annexes [AU-IBAR, 2019]; the regional marine service's economic benefit assessment exists only as a terms of reference *[MarCOSIO, 2025]*[3]. A denominator is exactly what all three lack.

What a decision improved looks like in practice is, however, documented. A satellite-intelligence provider trained twenty-five personnel drawn from a fisheries directorate, a national parks agency and a navy at a patrol-planning workshop, then supplied an intelligence report on a fleet operating inside national waters; three trawlers without valid licences were intercepted on 18 July 2026 and escorted to port (grade: case-report, narrated by the provider, no counterfactual)

[3] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[Global Fishing Watch, 2026][4]. The countable unit is the enforcement action that would not otherwise have been taken. The cost of producing it (the workshop, the analysts, the data, the patrol) is not published, and would complete the ratio. ABIG's outcome indicators promise response-time and case-support metrics at Year 3 [Terhile et al., 2026]; attaching a cost denominator to them is the change this chapter recommends.

10.6 The roadmap

Figure 10.1 places both programmes on one calendar with their gates marked. The Dubai bars and milestones are the storyboard's dates [Iyyunni, 2026]; the Phase-4 and Phase-5 bars are this report's horizons and rest on no source. The ABIG bars are the concept paper's horizons of three, six and twelve months and Years 2 and 3, anchored by this report to the paper's own date of 20 September 2026 [Terhile et al., 2026].

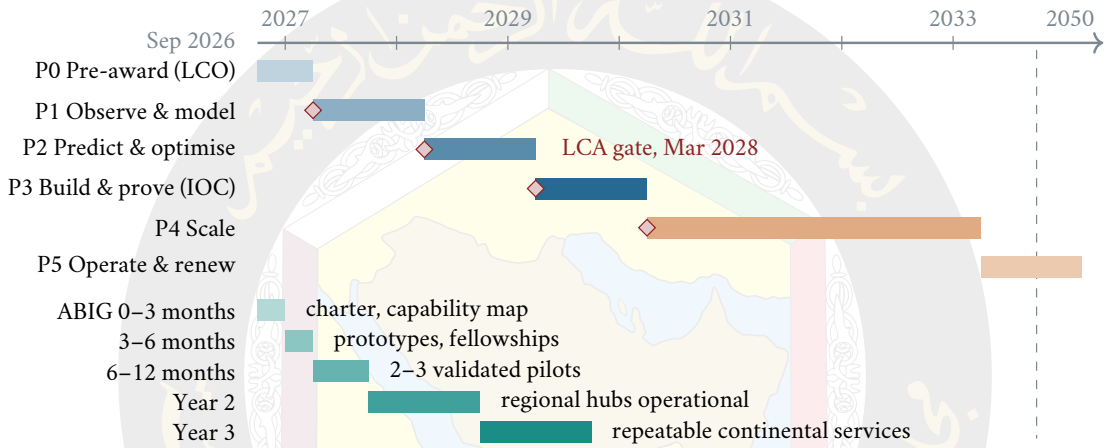


Figure 10.1: Strategic roadmap for the two programmes on one calendar. Diamonds are stage gates at which funds are released only against written pass conditions. Dubai phases and milestones are the storyboard's [Iyyunni, 2026] for phases 0–3; phases 4 and 5 are this report's horizons and are supported by no source. The dashed rule marks the break to the long operating tail to 2050. The Africa Blue Intelligence Grid horizons are the concept paper's, anchored by this report to its date of 20 September 2026 [Terhile et al., 2026].

Two calendar constraints deserve to be drawn on the chart before it is drawn on a wall. The first is the construction window. The national midday-break rule prohibits work under direct sunlight and in open spaces between 12:30 and 15:00 from 15 June to 15 September (grade: policy-fact) *[The National, 2026]*[5]; the storyboard's build window of March to May 2029 sits deliberately ahead of it, and any slippage into June pushes labour productivity, and therefore construction cost, into the season the programme exists to study. The second is the funding calendar. The archived sources document only the 2025 second cycle of the grant programme, with its Letter of Intent deadline of 13 June 2025 *[Dubai Future Foundation, 2025]*[6]*[Dubai RDI Programme, 2025]*[2]; the storyboard's

[4] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[5] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[6] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

submission in September 2026 and start in March 2027 presuppose a third cycle whose calendar, envelope and topics appear in no source. Chapter 11 carries that as a funding-timing risk; the roadmap carries it as a dependency with a named fallback rather than as an assumption.

10.7 The same structure for ABIG

ABIG has no budget, so its cost breakdown structure is not a division of a known sum but a declaration of what will have to be paid for. Its natural columns are not calendar phases but the rungs of the concept paper's financing architecture: public policy mandate, project-preparation grant, pilot or proof of value, independent validation, blended finance, and commercial service or infrastructure financing [Terhile et al., 2026]. Read as a sequence of gates rather than as a diagram, that ladder is incremental commitment in the strictest sense (a couple of chips in the pot, then the choice to fold or to buy one more round, committing incrementally rather than all at once [Boehm and Hansen, 2000]) and each rung should therefore carry pre-agreed pass conditions and a costed fallback exactly as the Dubai month-twelve gate does.

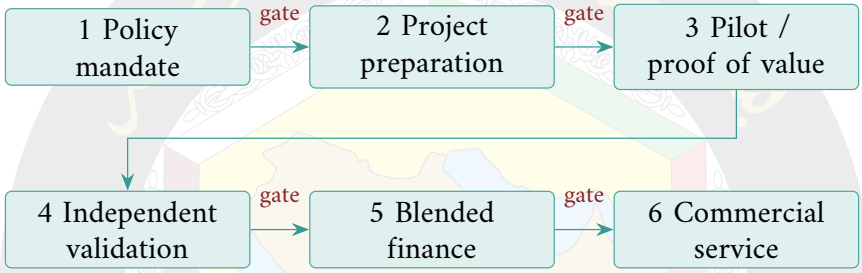


Figure 10.2: The Africa Blue Intelligence Grid financing ladder read as a gated commitment structure. The six rungs are the concept paper's financing architecture [Terhile et al., 2026]; reading each transition as a stage gate with pre-agreed pass conditions and a costed fallback is this report's proposal, on the incremental-commitment principle [Boehm and Hansen, 2000]. No amount attaches to any rung in any archived source.

The rows of the ABIG structure are the same nine categories, with three read differently. Sensing is deferred by design: the concept paper's own sequencing rule is demand to information to sensing to mission, and its stated first objective is not to build more satellites but to build the capability to ask better questions [Terhile et al., 2026]. Physical interventions become ecosystem and coastal works rather than shade structures. Governance is heavier, because the paper specifies seven governance layers (political and strategic council, scientific and technical council, operational user council, data and ethics board, programme office, regional hubs, and industry and investment forum [Terhile et al., 2026]) and a governance layer that meets is a cost line, whether or not anybody books it. Table 10.4 sets the ladder against the structure with owner and funding source per line.

Two requirements follow that the concept paper does not yet contain. The first is a cost breakdown structure per regional hub. The paper proposes a regional architecture by marine system (West Africa and the Gulf of Guinea, North Africa and the Mediterranean, the Canary Current, eastern Africa and the western Indian Ocean, Benguela and the southern Atlantic, Agulhas and the southern Indian Ocean, the Red Sea and adjacent waters, and island systems [Terhile et al., 2026]). Eight hubs reporting against eight different chart-of-account structures would make the continental total uncomputable. Standardising the structure at level three so that data can be shared across organisations is the specific remedy, and it is also how a continental reference class comes to exist by Year 3 where none exists today [GAO, 2020]. The second requirement is a sustainment condition on every pilot. The concept paper already names pilot fatigue as a risk and answers it

Table 10.4: The Africa Blue Intelligence Grid financing ladder against the cost breakdown structure, with the owner and funding source of each rung. Rungs, governance layers, candidate financiers and the roadmap horizons are the concept paper's [Terhile et al., 2026]; the assignment of cost categories, evidence conditions and the per-hub reporting requirement are this report's proposal. No monetary amount is attached to any rung because no archived source supplies one.

Rung	Dominant cost categories and evidence required to pass	Owner	Candidate funding source
1 Policy mandate	Governance, ethics, legal; dissemination. Pass: charter adopted; responsible-use principles for security analytics agreed; continental baseline published	Political and strategic council	Member-state and continental institutional budgets
2 Project preparation	People; data and compute; governance. Pass: capability map and data inventory complete; 5–7 use cases identified; 2–3 demonstration geographies selected	Programme office	Project-preparation grant; convening-hub capital
3 Pilot	People; data and compute; models and software; evaluation. Pass: 2–3 pilots running with a named operational user each, and a measured decision outcome per pilot	Regional hub with the user agency	Project-preparation facility; development finance
4 Independent validation	Evaluation with controls; dissemination. Pass: user-acceptance tests passed; benefits measured against a stated counterfactual; cost per decision improved reported	Scientific and technical council	Programme budget; philanthropy
5 Blended finance	Physical interventions; sensing (only if the evidence demands it); contingency. Pass: a bankable pipeline with an operational budget line behind each service	Industry and investment forum	Development bank; sovereign, climate and private capital
6 Commercial service	Operation, maintenance, renewal; people. Pass: service-level agreements and recurring national budgets in place	Regional hubs and national agencies	Service revenue; recurrent national budgets

with the rule that every pilot needs a pathway to operational budget, procurement or investment [Terhile et al., 2026]; the WinWin finding gives that rule its evidence, and rung 3 above turns it into a gate condition rather than an aspiration [Boehm et al., 1998].

Finally, the ladder's upper rungs have one archived precedent worth naming for structure rather than for scale. A sovereign blue bond of 15 million US dollars was issued on 29 October 2018, de-risked by a 5 million dollar partial credit guarantee from the World Bank and 5 million dollars of concessional financing from the Global Environment Facility, and placed with three investors (grade: policy-fact) [World Bank, 2025]*[World Bank, 2018]*[7]. That is what a rung-5 instrument looks like: a small principal made investable by two layers of public risk absorption. Separately, a regional seascape programme reports having mobilised more than 80 million US dollars and supported 83,000 square kilometres of marine and coastal areas, with more than 175,000 hectares under protection, restoration or commitment to protect and over 112,000 livelihoods strengthened (grade: case-report, partner self-reported, no baselines) *[IUCN, 2026]*[8]. Neither figure is a cost, and neither can be divided by anything the sources provide, which is the point of insisting on the denominator now.

10.8 Implications for the programmes

For the Dubai Decision Twin. Publish the nine-category cost breakdown structure with its dictionary as an annex to the full proposal, standardise it at the third level, and make programme management a visible element rather than an overhead spread across the other eight [GAO, 2020]. Present the budget as a cash-flow schedule by project year against the [500,000.00:138,888.89](AED:USD) annual cap, profile the [250,000.00:69,444.44](AED:USD) equipment purchase explicitly across the cap boundary, and confirm the cap's interpretation with the funder before submission [Iyyunni, 2026]*[Dubai RDI Programme, 2025]*[2]. State in the proposal itself, in one sentence, that the grant does not fund the built intervention and does not fund anything after 14 March 2030, and name the authority expected to carry each. Attach a written sustainment commitment (supervision, cleaning, water, energy, repair) to the Phase-2 intervention choice as a pass condition, on the evidence that unsustained applications do not survive [Boehm et al., 1998]. Hold contingency as scope, phase and partner capital rather than as money, and say so. Add the black-globe and natural-wet-bulb instruments to the sensing line so that the denominator of the reporting unit remains comparable with the international labour-hours series [ILO, 2019]. Record actual unit costs from month one against the identifiers of Chapter 8, so that the reference class this work lacks begins to exist. And carry the undocumented third funding cycle as a dependency with a fallback, not as an assumption *[Dubai Future Foundation, 2025]*[6].

For the Africa Blue Intelligence Grid. Adopt the same nine categories, publish them as a level-three standard before the first hub spends anything, and require every regional hub to report against them so that a continental total and a continental reference class both become computable [GAO, 2020, Terhile et al., 2026]. Read the financing ladder as six gates rather than six stages, and write pass conditions and a costed fallback for each, with the hardest condition at rung 3: a named operational user and a measured decision outcome per pilot [Boehm and Hansen, 2000, Terhile et al., 2026]. Add the two missing lines to the capability map (a cost breakdown per hub, and cost

[7] World Bank, *Seychelles launches World's First Sovereign Blue Bond*, press release, 29 October 2018. https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond

[8] International Union for Conservation of Nature, *Great Blue Wall report showcases five years of impact on ocean regeneration across the Western Indian Ocean*, IUCN news, June 2026. https://iucn.org/news/202606/great-blue-wall-report-showcases-five-years-impact-ocean-regeneration-across-western

per decision improved in constant US dollars with the year named), so that the Year-3 outcome indicators acquire a denominator that the archived continental valuations conspicuously lack [WEF and Digital Earth Africa, 2021, AU-IBAR, 2019]. Keep sensing below the line until the evidence demands it, exactly as the concept paper's own sequencing rule requires [Terhile et al., 2026]. And treat the governance layers as a funded cost category from the first month: a political council, a scientific council, a user council, a data and ethics board, a programme office, eight regional hubs and an investment forum are the programme's standing overhead, and a structure that does not book them will discover them anyway.

١١ {11}

Risk Modelling, SWOT Analysis, Loss Remediation and Recovery, and Force Majeure

Abstract

Neither proposal carries a risk apparatus, and this chapter builds one. It sets out a risk-management process and a taxonomy, then compiles a full risk register for the Dubai Decision Twin (sensor uptime, model error, data access, site and partner letters, the Curtin agreement, the platform partner, the undocumented third grant cycle, heat and dust damage, ethics and privacy, and failure of the month-12 gate itself) and for the Africa Blue Intelligence Grid, extending its concept-paper risk table with cross-border data law, the misuse of a vessel-anomaly score and geopolitical exposure. Risks are plotted on a likelihood-by-impact heat map and then modelled quantitatively: Monte Carlo on schedule and cost, a Bayesian update of the probability that the gate is passed, and a sensitivity analysis drawn as a tornado. Key risk indicators convert the register into monthly numbers. A SWOT analysis for each programme and for the combined portfolio is turned into a TOWS strategy matrix. Loss remediation and recovery follows: six playbooks, recovery time and point objectives for the data platform, insurance and warranty options, and the gate-failure fallback. The chapter closes on force majeure: what it is, what a clause must contain, which regional events belong in it, and which questions must go to counsel.

Risk is the only chapter of this report that the two source documents do not attempt. The Dubai storyboard names a gate and a fallback but no register, no likelihoods and no owners [Iyyunni, 2026]; the Africa Blue Intelligence Grid (ABIG) concept paper carries a nine-row table of principal risks with a design response against each, which is a strong start and an incomplete one, since it gives no likelihood, no impact, no indicator and no recovery plan [Terhile et al., 2026]. Chapter 7 established that both programmes are risk-driven in Boehm's sense, and Chapter 9 showed that money contingency is not available to a grant whose point estimate already sits near the top of its funding envelope *[Dubai RDI Programme, 2025]*[1]. What follows therefore does three things at once: it identifies and scores the risks, it models the ones that can be modelled, and (because the cost guide is explicit that simulation does not capture catastrophic events [GAO, 2020]), it specifies in contract language what happens to the events that no model reaches.

A caution about sources governs the whole chapter. The three standards and model clauses

[1] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

named below (the international risk-management guidelines, the business-continuity requirements standard and the International Chamber of Commerce's force-majeure and hardship clauses of March 2020) are *not* held in this report's archive. They are cited by title, they were not reviewed, and every process step, definition and clause element attributed to them here is this report's own formulation of common practice, offered as a drafting checklist and to be replaced by the standard's own wording once the documents are obtained [ISO, 2018, ISO, 2019, ICC, 2020]. Likewise, every likelihood, every impact score, every recovery objective and every threshold in this chapter is an indicative assessment made by this report, not a measurement; they are the stakeholders' to revise, and the discipline that matters is that they are written down, dated and owned.

11.1 A process, and what counts as a risk

The sequence used here is the one conventionally associated with the international risk-management guidelines: establish the scope, context and risk criteria; identify risks; analyse them for likelihood and consequence; evaluate them against the criteria; treat the ones that exceed tolerance; monitor, review and report; and run communication and consultation alongside the whole cycle rather than at the end of it [ISO, 2018]. Two features of that sequence do real work in this report. The first is that criteria come before identification: unless the programme has said in advance what it is unwilling to lose, every risk looks equally alarming and none gets treated. The second is that monitoring is part of the process, not an afterthought, which is what Section 11.7 operationalises as key risk indicators.

The report's risk criteria are stated plainly. For Dubai, the intolerable outcomes are: losing the summer measurement window, because the window is annual and a slip of weeks costs a year; failing the month-12 gate for a reason inside the team's control; and any breach of privacy or data governance, because a single such breach ends the agency relationships on which everything else depends [Iyyunni, 2026]. For ABIG, the intolerable outcomes are: becoming a coordination forum with no operational user; any use of the maritime security layer that harms a person without lawful process; and any arrangement in which an African institution loses ownership of African data [Terhile et al., 2026]. Everything else is tolerable at some price.

The cost guide's distinction between risk and uncertainty is retained throughout: a risk is quantifiable and may or may not occur, while uncertainty is always present and ambiguous [GAO, 2020]; the life-cycle costing standard's definition of risk as the probability of an event multiplied by its consequences is the same arithmetic in another vocabulary [ISO, 2017a]. Boehm gives the operational form of that arithmetic, and it is the one used in this chapter: risk exposure is the probability of an unsatisfactory outcome multiplied by the size of the loss if it occurs, and the level of effort spent on reducing it has an interior minimum: underkill and overkill both cost [Boehm and Hansen, 2000]:

$$\mathrm{RE}_j \;=\; \Pr(L_j)\times S(L_j), \qquad \min_{e}\Big[\,e \;+\; \textstyle\sum_j \mathrm{RE}_j(e)\,\Big], \tag{11.1}$$

where e is the effort devoted to risk resolution. Equation (11.1) is why the answer to 'how many sensors, how long a calibration campaign, how much testing' is neither the maximum nor the minimum, and why the register below carries a response column rather than a worry column.

Understanding: Risk register

TIER 1: IN PLAIN WORDS

A list of the things that could go wrong, how likely each one is, how badly it would hurt, who is responsible for it, what is being done about it, and the number that will warn you it is happening.

TIER 2: IN PROFESSIONAL TERMS

A controlled document maintained across the life of a programme in which each identified risk carries an identifier, a description written as cause–event–consequence, an assessed likelihood and impact, an owner, a treatment (avoid, reduce, transfer, accept), a residual assessment after treatment, and a monitored indicator with a trigger threshold [ISO, 2018].

TIER 3: IN TECHNICAL TERMS

A register is the programme's estimate of the joint distribution of its own failure modes, decomposed into marginals for tractability; entries feed the risk-driver terms of the cost simulation, the branch probabilities of the gate model, and the trigger conditions of the continuity playbooks, and it is reviewed at every anchor point rather than annually [GAO, 2020, Boehm and Hansen, 2000].

AS A MATHEMATICAL MODEL

$\mathrm{RE}_j = \Pr(L_j) \times S(L_j)$; residual exposure after treatment t_j is $\mathrm{RE}_j^{\mathrm{res}} = \Pr(L_j \mid t_j) \times S(L_j \mid t_j)$, and the treatment is justified when $\mathrm{RE}_j - \mathrm{RE}_j^{\mathrm{res}} > c(t_j)$ [Boehm and Hansen, 2000].

11.2 A taxonomy for two programmes

Identification is more reliable when it is prompted by categories than when it is left to imagination. The cost guide's own list of the sources of risk and uncertainty (economic, cost-estimating, programmatic, requirements, schedule and technology [GAO, 2020]) covers a project's inside view but not the institutional and ethical exposures that dominate both programmes here, so Figure 11.1 extends it to eight branches. The branches are the prompt list used to build the two registers; nothing enters a register that cannot be placed on one of them.

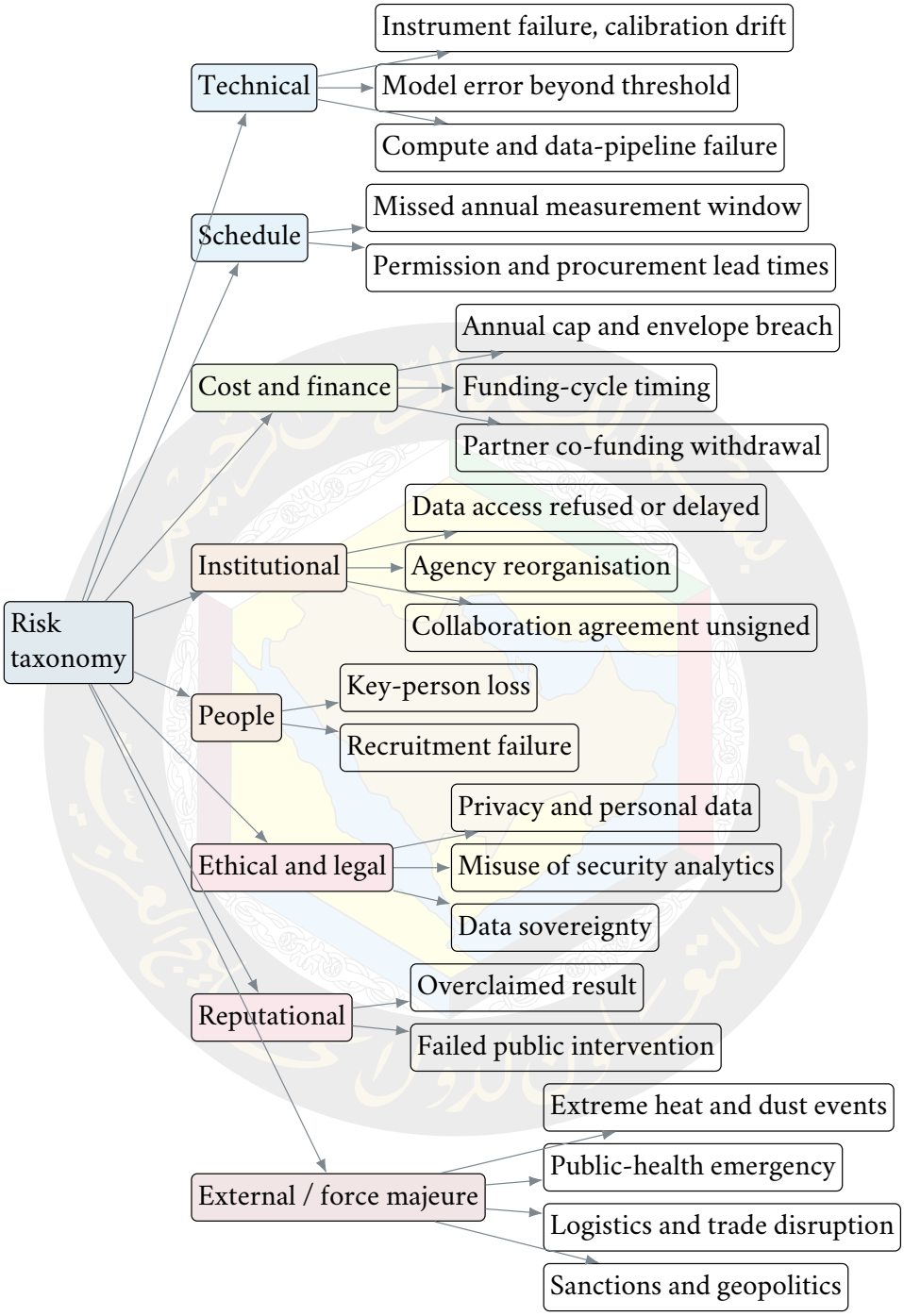


Figure 11.1: The eight-branch risk taxonomy used to build both registers. The first seven branches extend the cost guide's list of the sources of risk and uncertainty (economic, cost-estimating, programmatic, requirements, schedule and technology [GAO, 2020]) with the institutional, ethical and reputational exposures that dominate a programme built on agency data and public space. The eighth branch is the one that quantitative techniques are explicitly not designed to capture [GAO, 2020] and is therefore handled by contract in Section 11.11.

11.3 The Dubai risk register

Table 11.1 carries the risks that bear directly on the four gate conditions of March 2028: at least 90 per cent sensor uptime across the summer, model agreement with the sensors within the stated error, delivery of an hour-by-hour comfort map of both zones, and a written confirmation from the Roads and Transport Authority (RTA) or the Dubai Electricity and Water Authority (DEWA) that the output is useful for a decision they face [Iyyunni, 2026]. Table 11.2 carries the programme, resource, institutional and external risks. Likelihood and impact are scored on a one-to-five scale; both scores and the resulting exposure are this report's indicative assessment, made from the storyboard and the archived programme rules, and are offered for the stakeholders to revise at the first risk review.

Table 11.1: Dubai Decision Twin: gate-critical risks. L = likelihood, I = impact, each scored 1–5 by this report; RE = L×I is the indicative risk exposure of equation (11.1). All four gate conditions are as written in the storyboard [Iyyunni, 2026].

ID	Risk (cause → event → consequence)	L	I	RE	Treatment and owner
D1	Heat, dust, power or connectivity faults reduce node availability below 90 per cent across the summer → gate condition 6.2.1 fails	4	5	20	Reduce: over-provision nodes and spares; local buffering so a link outage is not a data loss; weekly uptime reporting from May 2027; pre-agreed replacement stock. Owner: systems-integration Co-I
D2	District and street models disagree with sensor observations beyond the stated error → condition 6.2.2 fails	3	5	15	Reduce: fix the error threshold after the first summer rather than before it; hold out a validation week as the Abu Dhabi precedent did [Manandhar et al., 2019]; report error by hour and by zone, not as one mean. Owner: principal investigator
D3	Pipeline or staffing delay → the hour-by-hour comfort map of both zones is late or covers one zone → condition 6.2.3 fails	2	4	8	Reduce: make the map a Year-1 deliverable in its own right, produced monthly from July 2027 in draft form. Owner: postdoctoral researcher
D4	No decision owner is identified inside RTA or DEWA → no written confirmation of usefulness → condition 6.2.4 fails	4	5	20	Reduce: negotiate the meaning of 'useful' after the first comfort map and before the gate; name an individual decision owner in each agency in Loop 0; deliver a decision brief on a decision they already face. Owner: principal investigator
D5	Data letters from RTA and DEWA are not signed by February 2027 → Year-1 behavioural analysis has no flow data	3	4	12	Reduce and accept: request one dataset only from each, as the storyboard already scopes; carry a public-baseline fallback (station data, Copernicus products, municipal geometry). Owner: principal investigator
D6	Site-access letters from Dubai Culture and Dubai Municipality are not obtained by December 2026 → sensors cannot be installed	2	5	10	Avoid: treat as a pre-award gate; no full proposal without them; route the request through the municipal digital-twin platform where possible *[Dubai Media Office, 2026]*. Owner: lead institution
D15	Any of D1–D4 → the month-12 gate fails → Phase 2 funds are not released and the project closes	3	5	15	Accept with a costed fallback: dataset and platform as deliverables, published and citable (Section 11.10.4). Owner: steering group

Three entries deserve comment. D9 is the risk this report discovered rather than inherited: the

Table 11.2: Dubai Decision Twin: programme, resource, institutional and external risks. Scores are this report's indicative assessment. D9 arises because the archived grant sources document only the 2025 second cycle *[Dubai RDI Programme, 2025][Dubai Future Foundation, 2025]* while the storyboard presupposes a third [Iyyunni, 2026].

ID	Risk	L	I	RE	Treatment and owner
D7	The Curtin University Dubai collaboration agreement and budget share are unsigned at submission → the downscaling and optimisation workstream has no contracted owner	3	4	12	Reduce: sign before the full proposal; state the budget share and the deliverable list in the agreement, not in correspondence. Owner: lead institution
D8	The platform partner does not contract, fails commercially, or delivers late → no Decision Twin front end	3	3	9	Transfer and reduce: keep the storyboard's own alternative (a simple in-house dashboard) as a costed parallel path; own the data model and the application interface in every case, so the partner is replaceable. Owner: systems-integration Co-I
D9	A third grant cycle is not opened, or is opened on a calendar incompatible with a September 2026 letter of intent and a March 2027 start → the whole schedule shifts	4	5	20	Reduce: confirm the cycle calendar with the funder before committing partners to dates; write date-relative rather than date-absolute milestones into the letters; prepare one alternative funding route. Owner: principal investigator
D10	Equipment-heavy Year 1 breaches the annual budget ceiling of [500,000.00: 138,888.89](AED:USD) per project → the budget is rejected or rescoped late	3	4	12	Avoid: profile the purchase across project years and present a cash-flow schedule against the cap; confirm the cap's interpretation in writing in advance *[Dubai RDI Programme, 2025]*. Owner: lead institution
D11	Summer heat, dust deposition and solar load degrade nodes, radiation shields, power systems and any coated surface → data loss and unrepresentative readings	4	3	12	Reduce: specify for the environment, not the catalogue; scheduled cleaning and recalibration between June and September; treat performance decay as measurable, as the only archived degradation evidence requires: reflectivity of a cool-pavement coating fell from 33–38 per cent at installation to 19–30 per cent within ten months [ASU and City of Phoenix, 2021]. Owner: systems-integration Co-I
D12	Footfall, dwell and cohort data are collected or published in a way that identifies individuals → ethics breach, agency relationships end	2	5	10	Avoid: cohort identifiers only and no personal data, as the storyboard's ethics line already requires; provenance recorded for every derived record [W3C, 2013]; ethics approval before the first campaign. Owner: data and ethics lead
D13	Loss of the principal investigator, the postdoctoral researcher or the doctoral student → a workstream stops	3	4	12	Reduce: named deputy for each role; code, models and configurations in a shared, access-controlled store from month one; no single-person knowledge of the calibration procedure. Owner: lead institution
D14	Procurement, customs, permissions or recruitment slip past May 2027 → the summer window is missed → a full year is lost	4	5	20	Reduce: start procurement at award; treat the window as a fixed deadline with scope as the variable, since a fixed deadline with fixed scope removes the trade-off entirely [Flyvbjerg et al., 2016]. Owner: lead institution
D16	Permission to build the Year-3 intervention is withdrawn or delayed → no before-and-after study, the distinctive contribution is lost	3	5	15	Reduce: obtain the build permission in principle with the December 2026 site letters, not in 2029; require a written sustainment commitment from the owning agency before the intervention is chosen [Boehm et al., 1998]. Owner: principal investigator

archived funder sources document a second cycle with a letter-of-intent deadline of 13 June 2025 and the published rules of that cycle, grants of [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD) per project, an annual budget per project not exceeding [500,000.00:138,888.89](AED:USD), at least 50 per cent of the funding to be spent within Dubai, and a principal investigator fully affiliated with a Dubai-based higher-education institution *[Dubai RDI Programme, 2025]*[1]*[Dubai Future Foundation, 2025]*[2]. The storyboard's September 2026 letter of intent and March 2027 start presuppose a third cycle whose calendar, envelope and priorities appear in no archived source [Iyyunni, 2026]. That is not a reason to abandon the schedule; it is a reason to write every partner letter in relative time and to confirm the calendar before any partner is asked for a date.

D14 is the schedule risk that the measurement design creates. Because the observation window is a summer, the schedule has no slack of the ordinary kind: a four-week slip in May is a four-week slip, but a ten-week slip is a fifty-two-week slip. The Olympic study's finding that a fixed deadline removes the schedule–cost trade-off and invites a blank cheque is the relevant warning [Flyvbjerg et al., 2016], and the megaproject literature's measured association between a one-year extension of the implementation phase and 4.64 additional percentage points of cost overrun is reference-class context for how expensive a lost year is: context only, since those statistics concern projects of hundreds of millions to billions of US dollars [Flyvbjerg, 2014, Cantarelli et al., 2025].

D11 is where the archive is thinnest and the engineering judgement heaviest. No archived source measures the survival of microclimate instruments in a Gulf summer. The nearest evidence of environmental degradation of any deployed surface is the decline of cool-pavement reflectivity from 33–38 per cent at installation to 19–30 per cent ten months later across eight Phoenix neighbourhoods [ASU and City of Phoenix, 2021]; it is not a sensor result, and it is used here only to make the general point that installed performance decays and must be re-measured rather than assumed. The instrument specification itself carries a known gap recorded in Chapter 5: the node set (air temperature, humidity, wind, radiation and surface temperature) lacks the black-globe and natural wet-bulb instruments that a wet-bulb globe temperature measurement requires, and wet-bulb globe temperature is the index in which the international labour projections are stated [ILO, 2019].

11.4 The ABIG risk register

The concept paper's own table of principal risks and design responses is reproduced faithfully in the first nine rows of Table 11.3, because it is a genuinely good list and because nothing is gained by paraphrasing a document the stakeholders wrote [Terhile et al., 2026]. What this report adds is the scoring, the owner, the indicator, and three rows: two that sharpen entries the paper states only in general terms (cross-border data-sharing law as a legal rather than a governance question, and the specific harm of treating a vessel-anomaly score as proof), and one the paper does not raise, geopolitical exposure.

The pattern in the ABIG register differs from Dubai's. Dubai's highest exposures are operational and time-bound: a window, a threshold, a letter. ABIG's highest exposures are structural and slow: the programme fails not by missing a summer but by producing documents for three years while no fisheries authority, port or environment agency changes a decision. That is why the single most valuable line in the ABIG risk treatment is A1's indicator, and why the archived case that ABIG most resembles operationally (an intelligence provider's targeting report leading to the interception of three unlicensed trawlers off Gabon on 18 July 2026, after twenty-five personnel from the fisheries

[2] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

Table 11.3: Africa Blue Intelligence Grid risk register. Rows A1–A9 are the concept paper's own principal risks and design responses [Terhile et al., 2026]; rows A10–A12 are added by this report. Scores are this report's indicative assessment.

ID	Risk	L	I	RE	Response (A1–A9 as written in the concept paper; additions marked)
A1	Becoming another coordination forum	4	5	20	Start with operational pilots and measurable user outcomes. *Added indicator:* number of named operational users with a signed use-case brief by month 12
A2	Hardware-first programme	3	4	12	Use demand → information → sensing → mission sequencing. *Added:* no payload commitment before an application requirement is written and valued [Howard, 1966]
A3	Data sovereignty concerns	4	5	20	African ownership, explicit data-governance rules, federated access where appropriate
A4	Security misuse	3	5	15	Human-in-the-loop, legal review, purpose limitation, audit trails and privacy safeguards
A5	Uneven capability across countries	4	3	12	Regional hubs, fellowships, shared infrastructure and train-the-trainer programmes
A6	Vendor lock-in	3	3	9	Open standards, interoperable application interfaces and multiple-source architectures
A7	Research disconnected from users	4	4	16	Embed researchers in operational agencies and require user acceptance tests
A8	Pilot fatigue	4	4	16	Every pilot needs a pathway to operational budget, procurement or investment
A9	Fragmented finance	4	4	16	Project-preparation facility, blended-finance pipeline, and co-financing
A10	*Added.* Cross-border data-sharing law diverges between member states → the federated hub cannot lawfully serve a regional product	4	4	16	Legal mapping in months 0–6 alongside the capability map; product design that keeps data in national custody and moves only derived indicators
A11	*Added.* A vessel-anomaly score is treated as proof and a person is harmed without lawful process → programme legitimacy lost	2	5	10	Model outputs carry uncertainty and provenance [W3C, 2013]; analyst review and lawful referral are mandatory stages; independent audit of every referral pathway
A12	*Added.* Geopolitical exposure: sanctions, export controls or a partnership rupture make a partner's contribution unavailable	3	4	12	No single-partner dependency on the critical path; open standards and portable data so any contributor can be substituted; a documented substitution plan per pilot

directorate, the national parks agency and the navy had been trained in patrol planning) is a case report of exactly that decision-changing kind, and not a measurement of programme effect *[Global Fishing Watch, 2026]*[3].

11.5 The likelihood and impact heat map

Figure 11.2 plots both registers on one five-by-five grid. The point of the picture is not precision (the scores are judgements) but triage: the upper-right cells are the risks that must have a named owner and a dated treatment before the Dubai letter of intent is submitted and before the ABIG charter is signed, and, for the entries of Table 11.4, a monitored indicator with a trigger threshold.

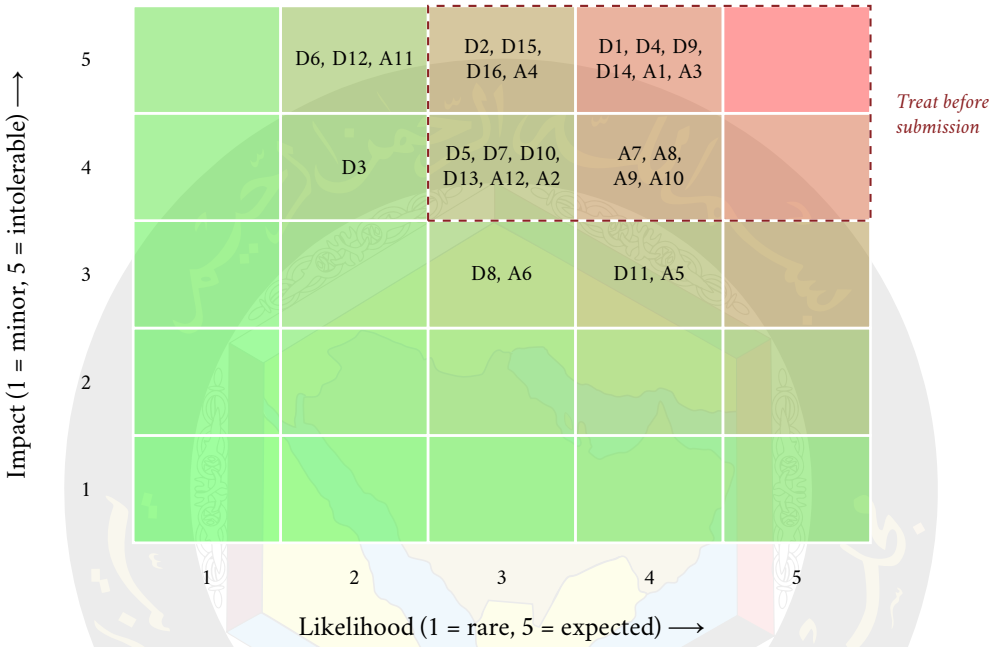


Figure 11.2: Likelihood-by-impact heat map for both registers. Identifiers are those of Tables 11.1, 11.2 and 11.3; D marks an entry of the Dubai registers and A an entry of the ABIG register. The dashed boundary encloses the exposures that this report judges must carry a named owner and a dated treatment before the Dubai letter of intent is submitted and before the ABIG charter is signed, and, for the entries of Table 11.4, a monitored indicator with a trigger threshold. All positions are this report's indicative judgement, not measurements.

11.6 Quantitative risk modelling

Three quantitative treatments are worth the effort at this scale: a simulation of schedule and cost, a Bayesian model of the gate, and a sensitivity analysis. Each is set out with its equations and a worked illustration whose inputs are labelled as this report's own.

11.6.1 Monte Carlo on schedule and cost

Give each uncertain element a three-point range (optimistic a, most likely m, pessimistic b) and a distribution over that range. Two summaries are used throughout: the triangular mean and the

[3] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

beta (PERT) mean and standard deviation,

$$\mathbb{E}[X]_{\triangle} = \frac{a + m + b}{3}, \qquad \mathbb{E}[X]_{\mathrm{PERT}} = \frac{a + 4m + b}{6}, \qquad \sigma_{\mathrm{PERT}} \approx \frac{b - a}{6}. \tag{11.2}$$

The total is then sampled thousands of times by Monte Carlo simulation with a stated correlation between related elements, and the result is read as a cumulative distribution rather than a number [GAO, 2020]. Chapter 9 drew that curve for cost and set out how to read it; what this chapter adds is the risk-driver form, in which named register entries, rather than anonymous spreads, generate the tail:

$$\tilde{C} \;=\; \hat{C}_0 \cdot U \;+\; \sum_{j} \mathbb{1}\{L_j\}\,\Delta_j, \qquad \mathbb{1}\{L_j\} \sim \mathrm{Bernoulli}\big(\Pr(L_j)\big), \tag{11.3}$$

where U is a multiplicative uncertainty factor applied to the point estimate and Δ_j is the cost consequence of register entry j if it occurs. The guide's own illustration of this method (likelihoods of 25, 20 and 15 per cent with cost-effect multipliers, alongside always-present uncertainties such as labour productivity at 97–105 per cent and software sizing at 76–140 per cent of the most likely value) produces a wider spread than three-point ranges alone, which is the point: risk drivers put the tail where the register says it is [GAO, 2020].

Two cautions from the same source govern the inputs. Expert-elicited ranges cover only 60 to 85 per cent of possible outcomes, 'at best' 70 per cent, so a correction treats an expert's stated minimum and maximum as the 15th and 85th percentiles before sampling; and when no correlation data exist, a nominal coefficient of 0.3 is better than assuming independence, because correlation barely moves the median but moves the 80th percentile materially [GAO, 2020].

Schedule is modelled the same way but read differently, because of the annual window. Let d be the cumulative delay in weeks to the 'sensors live' milestone of May 2027. The effective loss is not d but

$$\Lambda(d) \;=\; \begin{cases} d & \text{if the network is live before the summer window opens,} \\ 52 + d' & \text{otherwise,} \end{cases} \tag{11.4}$$

with d' the residual delay into the following year. Equation (11.4) is a step function, not a smooth one, and it is the formal reason that schedule contingency for this programme must be bought in the first eight weeks after award or not at all.

11.6.2 A Bayesian model of the month-12 gate

The gate is a conjunction: Phase 2 funds are released only if all four conditions are met [Iyyunni, 2026]. Writing p_i for the probability that condition i holds and assuming (generously) that the four are independent,

$$\Pr(\text{gate passed}) \;=\; \prod_{i=1}^{4} p_i. \tag{11.5}$$

With this report's indicative pre-award judgements of $p_1 = 0.85$ for uptime, $p_2 = 0.70$ for model error, $p_3 = 0.95$ for the comfort map and $p_4 = 0.60$ for the written confirmation, equation (11.5) gives $0.85 \times 0.70 \times 0.95 \times 0.60 = 0.339$, or about a one-in-three chance of passing. The arithmetic is shown because the conclusion is uncomfortable and instructive: the binding constraint is not the technology but the letter, and the independence assumption flatters the result, since a late comfort map and an unpersuaded agency share a common cause. Every point of probability bought on p_4 (by naming a decision owner early, by delivering a brief on a decision the agency already faces, by agreeing in advance what 'useful' means) is worth more than the same effort spent on the model.

The uptime condition is the one that data updates fastest. Treat the availability of a node-day as a Bernoulli trial with unknown probability θ, place a Beta(α_0, β_0) prior on it, and observe s successes in n node-days. Conjugacy gives

$$\theta \mid s, n \;\sim\; \mathrm{Beta}(\alpha_0 + s,\; \beta_0 + n - s), \qquad \mathbb{E}[\theta \mid s, n] = \frac{\alpha_0 + s}{\alpha_0 + \beta_0 + n}. \tag{11.6}$$

Two illustrations follow, both with a uniform prior $\alpha_0 = \beta_0 = 1$ and a first month of twenty nodes over thirty days, that is $n = 600$ node-days. Both are this report's own arithmetic, shown in full, and neither is a forecast. If $s = 570$, the posterior is Beta$(571, 31)$ with mean $571/602 = 0.949$ and standard deviation $\sqrt{0.949 \times 0.051/603} = 0.009$, placing the 0.90 threshold more than five standard deviations below the mean: the condition is all but secured, and management attention should move elsewhere. If instead $s = 540$, the posterior is Beta$(541, 61)$ with mean $541/602 = 0.899$ and standard deviation $\sqrt{0.899 \times 0.101/603} = 0.012$, so the threshold sits essentially at the mean and the condition is a coin toss. The practical consequence is that one month of operation converts an assumption into a decision: at $s = 540$, with the first month of node-days closing at the end of May 2027, the programme has the ten weeks to the August peak in which spares, cleaning intervals and power redesign can move the posterior (derived).

The same update applies to the gate as a whole. Writing G for the event that the gate is passed and E for the evidence accumulated by month nine,

$$\Pr(G \mid E) \;=\; \frac{\Pr(E \mid G)\,\Pr(G)}{\Pr(E \mid G)\,\Pr(G) + \Pr(E \mid \neg G)\,\big(1 - \Pr(G)\big)}, \tag{11.7}$$

which is the formal statement of the incremental-commitment logic that Chapter 7 took from the spiral model: commit in stages and let each stage's evidence price the next, rather than committing everything on the most incomplete information the programme will ever have [Boehm and Hansen, 2000]. Equation (11.7) is also the reason the gate should be rehearsed at month nine rather than encountered at month twelve.

11.6.3 Sensitivity and the tornado

Sensitivity analysis asks which assumption, varied alone across its plausible range, moves the answer most. The procedure is five steps (identify the factors, re-estimate varying one factor between two set amounts, document the result, repeat until all factors are tested, and evaluate) and the guide is explicit that simply varying everything by a subjective plus or minus percentage is not a valid sensitivity analysis. In an analysis of alternatives a parameter is 'sensitive' if a change of 10 to 50 per cent produces a different preferred alternative, and 'very sensitive' if a change of less than 10 per cent does [GAO, 2020]. The swing of factor j is

$$\mathrm{swing}_j \;=\; C\big(x_j^{\mathrm{high}},\, \mathbf{x}_{-j}^{\mathrm{base}}\big) - C\big(x_j^{\mathrm{low}},\, \mathbf{x}_{-j}^{\mathrm{base}}\big), \tag{11.8}$$

and ordering the factors by $|\mathrm{swing}_j|$ produces the tornado diagram of Figure 11.3.

11.7 Key risk indicators

A register that is not measured is a document. Table 11.4 converts the highest exposures into key risk indicators with trigger thresholds, reporting frequencies and named escalation routes. The thresholds are this report's indicative proposals; what is not negotiable is that each has a number, a frequency and an owner, because an indicator whose trigger is 'if it looks bad' triggers nothing.

Table 11.4: Key risk indicators for the highest exposures in both registers. Thresholds are this report's indicative proposals, to be fixed by the stakeholders at the first risk review; the gate conditions themselves are as written in the storyboard [Iyyunni, 2026].

Risk	Indicator	Trigger threshold (indicative)	Frequency	Escalation
D1	Rolling 30-day node availability, reported per node and for the network	Any node below 95 per cent, or the network below 93 per cent	Weekly from May 2027	Systems Co-I, then steering group
D2	Mean and 90th-percentile model error against held-out sensor hours, by zone and hour	Error outside the agreed threshold for two consecutive validation runs	Monthly from July 2027	Principal investigator
D4	Number of named agency decision owners who have received a decision brief and responded	Fewer than two by month nine	Monthly	Steering group
D9	Days since the funder's cycle calendar was last confirmed in writing	More than 90 days	Quarterly	Lead institution
D11	Calibration drift per node since last service; days since last cleaning	Drift beyond the instrument specification, or more than 30 days uncleaned in summer	Fortnightly, June–September	Systems Co-I
D12	Derived records released without a provenance entry, and re-identification tests run on each published cohort series	Any record released without provenance, or any test not run before release	Every release	Data and ethics lead, then steering group
D14	Float in weeks to the 'sensors live' milestone	Below four weeks	Weekly to May 2027	Lead institution
A1	Operational users with a signed use-case brief; decisions demonstrably changed	Fewer than three users by month 12	Quarterly	Programme office
A4, A11	Referrals	Any referral without	Monthly	Data and ethics

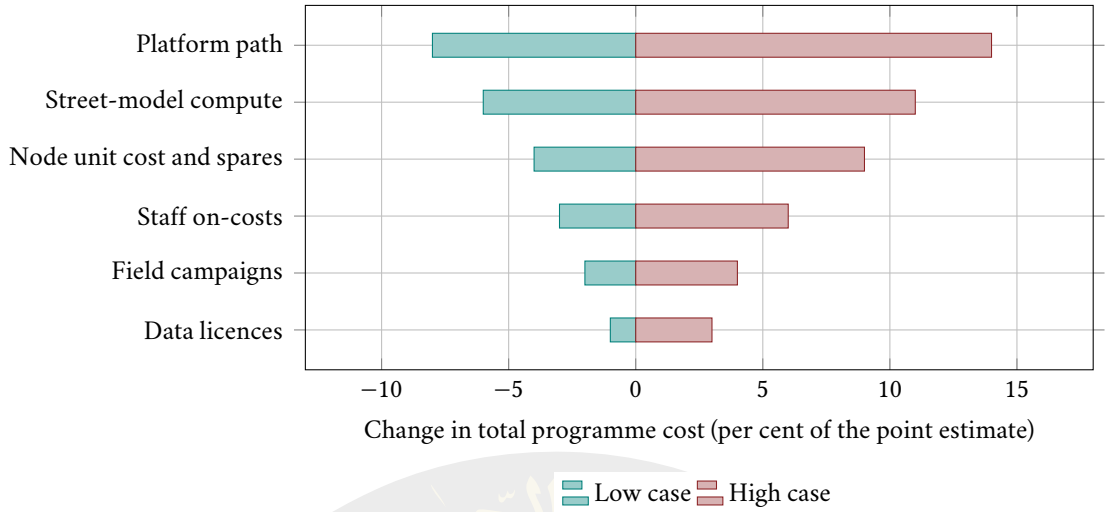


Figure 11.3: Tornado diagram of the Dubai cost estimate's sensitivity to six factors, each varied alone between a low and a high case and expressed as a percentage of the point estimate. **The percentages are this report's own illustrative values, chosen solely to display the method; no archived source supports them and this is not a forecast.** The ordering is the substantive claim: the choice between a contracted platform and an in-house dashboard, and the number of street-scale model-run hours, dominate the equipment line that the storyboard discusses most [Iyyunni, 2026, GAO, 2020].

11.8 What the models do not capture

The most important sentence in the archived cost-estimating literature, for the purposes of this chapter, is a footnote: the risk and uncertainty techniques are 'designed to capture general uncertainties about the future, not unforeseen catastrophic events such as major earthquakes and hurricanes' [GAO, 2020]. Simulation samples from a distribution the programme has already imagined. It cannot price an event outside that distribution, and adding contingency percentages in the hope of covering one is expensive and ineffective, which is the same conclusion the reference-class literature reaches when it observes that it is prohibitively expensive to carry an uplift large enough to cover a fat tail, and that the correct response is to cut the tail off through mitigation and contractual allocation rather than to fund it [Cantarelli et al., 2025]. Three consequences follow, and they structure the rest of this chapter: catastrophic and systemic events belong in force-majeure clauses (Section 11.11); recoverable but severe events belong in continuity playbooks with objectives and rehearsals (Section 11.10); and the residual belongs in the gate's fallback, which converts an unaffordable downside into a defined and publishable one (Section 11.10.4).

The five exposures that decide whether either programme survives

D4: no written confirmation of usefulness. A gate condition whose owner is an agency, not the team; the lowest of the four condition probabilities and the one most responsive to early effort [Iyyunni, 2026].

D14: the missed summer window. Because the observation window is annual, moderate delay is catastrophic delay; contingency must be bought in the first eight weeks after award [Flyvbjerg et al., 2016].

D9: the undocumented third grant cycle. The archived sources document only the 2025 second cycle; the storyboard's dates presuppose a third [Iyyunni, 2026][*Dubai RDI Programme,*

2025][1] *[Dubai Future Foundation, 2025]*[2].

A1: becoming another coordination forum. The concept paper's own first risk, and the failure mode that consumes three years without an operational user [Terhile et al., 2026].

A3 and A11: sovereignty and misuse. A single mishandled referral or a single sovereignty dispute ends the legitimacy on which a continental federation depends [Terhile et al., 2026].

11.9 SWOT analysis

A SWOT analysis sorts what is true about a programme into four boxes: strengths and weaknesses, which are internal and present, and opportunities and threats, which are external and prospective. Its value is not the sorting but what follows it: a TOWS matrix pairs the boxes to generate strategies, so that a strength is matched to an opportunity, a weakness to a threat, and so on. Tables 11.5 to 11.7 give the three analyses; Table 11.8 gives the strategies.

Understanding: SWOT and TOWS

TIER 1: IN PLAIN WORDS

First list what you are good at, what you are bad at, what is out there to be taken and what is out there that could hurt you. Then pair them up: use each strength on an opportunity, and fix each weakness that a threat could exploit.

TIER 2: IN PROFESSIONAL TERMS

A structured internal–external appraisal in which strengths and weaknesses describe present capability and opportunities and threats describe the environment; the paired form converts the four inventories into four families of strategy (offensive, defensive, adaptive and contingent), so that the analysis produces actions rather than lists.

TIER 3: IN TECHNICAL TERMS

SWOT is an unweighted qualitative appraisal with no inferential content: it does not establish causality, does not rank, and is vulnerable to the same optimism bias documented in project appraisal [Flyvbjerg, 2014]. It is used here as an elicitation frame whose outputs are disciplined downstream: every threat must map to a register entry, and every strength-based strategy must name the evidence that supports it.

AS A MATHEMATICAL MODEL

Strategy set $= (S \times O) \cup (S \times T) \cup (W \times O) \cup (W \times T)$, with each generated pair admitted only if it names an owner and a date.

11.10 Loss remediation and recovery

Risk treatment reduces the chance of a loss. Remediation and recovery decide what the loss costs once it has happened, and the difference between a programme that recovers and one that does not is almost never the severity of the event: it is whether somebody had written down, in advance, what to do in the first forty-eight hours. The vocabulary used here is that of business-continuity management: a business impact analysis that establishes how long each activity can be unavailable, continuity plans that say who does what, recovery objectives expressed as time and as data, and exercises that test the plans before they are needed [ISO, 2019]. As with the risk guidelines, the standard is cited by title and was not reviewed for this report; the formulations here are the report's own and should be replaced by the standard's on acquisition.

Table 11.5: SWOT analysis: Climate Intelligence and Decision Twin for Dubai. Internal items are drawn from the storyboard [Iyyunni, 2026]; external items from the archived programme, policy and comparator sources cited in each cell.

Strengths	A single named testbed with a heritage calibration zone and an intervention corridor, rather than a citywide promise; a hard month-12 gate with four testable conditions; a built intervention with a before-and-after design in Year 3; two funded posts; a distinctive metric set (HLAH, OTC, MCN) that no archived programme publishes [Iyyunni, 2026]
Weaknesses	The gate's fourth condition depends on a party outside the team for a written confirmation, and its first depends on site-access letters from two further agencies before a sensor can be installed; the instrument set omits the black-globe and wet-bulb sensors that wet-bulb globe temperature requires [ILO, 2019]; no unit cost, service life or reference class exists for the work [GAO, 2020]; the platform partner is not contracted; the schedule presupposes an undocumented funding cycle *[Dubai RDI Programme, 2025]*
Opportunities	A municipal digital-twin platform launched on 2 July 2026 carrying more than 1,500 geospatial layers, more than 280,000 infrastructure assets and more than 195,000 buildings, which converts a platform-build risk into a data request *[Dubai Media Office, 2026]*; a city walking-network programme of 3,300 km of new and 2,300 km of renovated walkways by 2040 that needs exactly this evidence *[Dezeen, 2024]*; a labour regulation that already encodes hours lost to heat *[The National, 2026][MoHRE, 2023]*; an international policy agenda seeking heat-adaptation indicators [UN Secretary-General, 2024]
Threats	Agency reorganisation or a change of decision owner; the pedestrian radiant-load penalty that makes cool pavement a trap for a walking corridor [ASU and City of Phoenix, 2021]; comparator programmes that publish neither cost nor effect size, leaving no benchmark *[EEA Climate-ADAPT, 2023]*; summer heat and dust as operational hazards; competition for the same agency attention from larger, better-funded city programmes

Table 11.6: SWOT analysis: Africa Blue Intelligence Grid. Internal items are drawn from the concept paper [Terhile et al., 2026]; external items from the archived continental and programme sources cited in each cell.

Strengths	A clear integration boundary: observation belongs to the ocean observing community, forecasting to the prediction centres, Earth-observation services to the continental programme, and ABIG sits above them as an application layer; capability-first sequencing in which hardware follows evidence; a governance design with a data and ethics board from the outset; a written risk table with design responses [Terhile et al., 2026]
Weaknesses	No budget, no cost structure and no benefit–cost evidence anywhere in the archive for ABIG or its analogues [WEF and Digital Earth Africa, 2021]*[MarCOSIO, 2025]*; no named operational user yet; a dependence on convening that can substitute for delivery; the concept paper's scale figures are second-hand and self-flagged [Terhile et al., 2026]
Opportunities	A continental political mandate adopted on 24 July 2026 *[African Union, 2026]*; a continental blue economy estimated at USD 296 billion with 49 million jobs in 2018 and projected to reach USD 405 billion and 57 million jobs by 2030 [AU-IBAR, 2019]; an existing open data cube covering more than 30 million km^2 with projected benefits above USD 2 billion a year by 2024 [WEF and Digital Earth Africa, 2021]; demonstrated enforcement value from vessel intelligence *[Global Fishing Watch, 2026]*; a regional seascape network reporting support to 83,000 km^2 and more than USD 80 million mobilised *[IUCN, 2026]*; sovereign blue-finance precedent [World Bank, 2025]
Threats	Divergent national data law; donor fatigue and fragmented finance; sanctions and geopolitical rupture among partners; the reputational consequence of any misuse of the security layer; and the structural threat that the archive itself documents: every comparator programme reports outputs and none reports a cost per decision improved [WEF and Digital Earth Africa, 2021, AU-IBAR, 2019]

Table 11.7: SWOT analysis: the combined portfolio. The two programmes are treated as one portfolio sharing an architecture, a method and a stakeholder group.

Strengths	One architecture (physical system, data, semantic layer, knowledge graph, intelligence, decision twin, action, learning), applied in two domains, so that method, identifiers and evaluation design transfer; a stakeholder group spanning the Gulf, Africa, South Asia and Australia; a shared evidence discipline in which every claim carries a grade
Weaknesses	Both programmes are pre-funding; neither has a signed operational user; the shared team is small, so key-person loss is correlated across both; the portfolio's two schedules compete for the same scarce attention
Opportunities	Diversification: Dubai's evidence strengthens ABIG's case and ABIG's convening strengthens Dubai's policy route; one method document serves both; the arid-region laboratory argument and the continental mandate reinforce each other in the same international forums [UN Secretary-General, 2024]*[African Union, 2026]*
Threats	Correlated failure through the shared team; the appearance of overreach if two continental ambitions are presented by one small group; and portfolio-level budgeting error, since risk-adjusting each programme separately at a high confidence level over-provisions the portfolio as a whole [GAO, 2020]

Table 11.8: TOWS strategy matrix for the combined portfolio. Each cell pairs an internal with an external factor from Tables 11.5–11.7 and states the strategy it generates; every strategy names the register entry it treats.

	Opportunities	**Threats**
Strengths	*Offensive (SO).* Anchor the twin on the municipal platform's existing layers instead of building geometry, and spend the saving on the corridor's instruments *[Dubai Media Office, 2026]*; offer the walking-network programme an evidence partnership on the first-and-last-mile links *[Dezeen, 2024]*; present the metric set as the heat-adaptation indicator the international agenda lacks [UN Secretary-General, 2024]. Treats D8, D4.	*Defensive (ST).* Use the gate's four conditions as the public standard of proof so that a change of agency personnel does not reset the argument; rank shade first and treat reflective pavement as conditional on the archived radiant-load evidence [ASU and City of Phoenix, 2021]; publish the method so a comparator's silence on cost and effect becomes this programme's advantage. Treats D4, D16.
Weaknesses	*Adaptive (WO).* Close the instrument gap by adding black-globe and wet-bulb sensors, or by estimating wet-bulb globe temperature and declaring the error, so that results are comparable with the international labour series [ILO, 2019]; use the continental mandate to obtain the named operational users ABIG lacks [Terhile et al., 2026]*[African Union, 2026]*; record unit costs from month one to create the reference class nobody has [GAO, 2020]. Treats D2, A1.	*Contingent (WT).* Confirm the funding cycle in writing before any partner is asked for a date, and write milestones in relative time *[Dubai RDI Programme, 2025]*; require a written sustainment commitment before any intervention or pilot is selected [Boehm et al., 1998]; budget the portfolio at portfolio level rather than each programme at its own high percentile [GAO, 2020]; hold the costed gate fallback as the floor of the downside. Treats D9, A8, D15.

Understanding: Recovery point objective

TIER 1: IN PLAIN WORDS

How much recent work you are willing to lose if something breaks. If you copy your data every night, a failure at lunchtime costs you the morning.

TIER 2: IN PROFESSIONAL TERMS

The maximum period of data, measured backwards from an incident, that an organisation accepts losing; it sets the required frequency of copying and buffering, and is agreed alongside the recovery time objective, which is the maximum period an activity may remain unavailable [ISO, 2019].

TIER 3: IN TECHNICAL TERMS

A parameter of the continuity design fixed by a business impact analysis rather than by technology: for a sensor network it is satisfied at the edge, by on-node buffering that survives a link outage, since a telemetry gap that cannot be back-filled is an irrecoverable hole in the gate's uptime evidence [Iyyunni, 2026].

AS A MATHEMATICAL MODEL

Data loss $= \min\{t_{\text{incident}} - t_{\text{last recoverable record}},\ \text{RPO}\}$ is the design requirement; for a node with local storage capacity B and sampling interval τ, the achievable objective is $\text{RPO} \leq B\tau$.

11.10.1 Six playbooks

Table 11.9 sets out the six losses that either programme is most likely to suffer, with the first action, the recovery route and the decision that must be pre-delegated so that nobody waits for a meeting.

11.10.2 Recovery objectives for the data platform

The data platform is the asset whose loss is least dramatic and most damaging, because the gate's evidence lives in it. Table 11.10 proposes recovery time objectives and recovery point objectives by component; Figure 11.4 shows what the two objectives mean on a timeline.

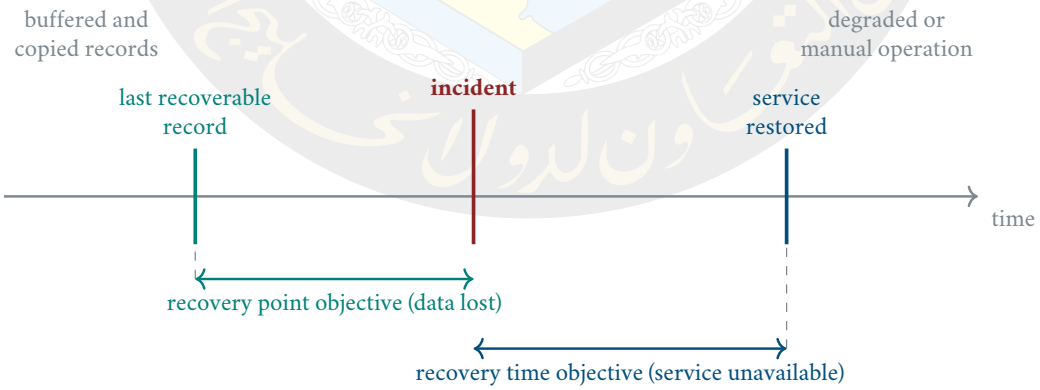


Figure 11.4: The two continuity objectives on one timeline. The recovery point objective is measured backwards from the incident and is satisfied by copying and buffering; the recovery time objective is measured forwards and is satisfied by standby capacity and rehearsed procedure. For a microclimate network the backward interval is the binding one, because a telemetry gap is the one loss that cannot be recreated [ISO, 2019, Iyyunni, 2026].

Table 11.9: Recovery playbooks. Each names the immediate action, the recovery path and the decision delegated in advance. All are this report's proposals, to be adopted, amended and rehearsed by the stakeholders.

Loss	Immediate action (first 48 hours)	Recovery and pre-delegated decision
Data loss	Freeze writes to the affected store; establish the last recoverable record and the size of the gap in node-hours; notify the data and ethics lead	Restore from the most recent copy; back-fill from node buffers; record the gap explicitly in the uptime evidence rather than interpolating it. *Pre-delegated:* the systems Co-I may take the platform offline without approval
Sensor-network loss	Identify whether the loss is power, link, instrument or access; deploy spares to the highest-value positions first (the corridor before the heritage zone)	Replace, recalibrate and document the calibration version; if the loss exceeds two weeks in summer, convene the gate-risk review. *Pre-delegated:* spares may be drawn without a purchase cycle
Partner withdrawal	Establish whether the withdrawal is of data, of permission or of the institution; check which gate condition it touches	Activate the public-baseline fallback for data; seek the alternative confirming agency for condition 6.2.4; re-scope rather than re-promise. *Pre-delegated:* the principal investigator may substitute a public dataset without a budget change
Funding shortfall	Establish the shortfall against the phase, not the total; identify the deliverables at risk by month	Re-profile within the annual cap; reduce scope in the order fixed in advance (city stretch goal first, second summer compute second, never the validation campaign) *[Dubai RDI Programme, 2025]*. *Pre-delegated:* the stretch goal may be dropped by the steering group alone
Key-person loss	Confirm data, credential and calibration-procedure access within 24 hours; notify partners of the acting contact	Named deputy assumes the role; recruitment starts in parallel; no workstream depends on undocumented knowledge. *Pre-delegated:* the deputy is authorised in advance, in writing
Reputational loss	Establish what was claimed, by whom, and on what evidence; stop further circulation of the claim	Correct publicly, in the same channel, with the evidence grade attached; if the loss follows a failed intervention, publish the negative result as a finding. *Pre-delegated:* the steering group may issue a correction without partner sign-off

Table 11.10: Proposed recovery time objectives (RTO, maximum tolerable unavailability) and recovery point objectives (RPO, maximum tolerable data loss) for the Dubai data platform. These are this report's design targets, not measurements or vendor commitments; they should be agreed with the platform partner and written into the contract [ISO, 2019, Iyyunni, 2026].

Component	RTO	RPO	Rationale
Node telemetry capture	24 hours	0 (buffered)	A gap cannot be recreated; on-node storage must cover the longest plausible link outage
Ingestion and quality-control pipeline	48 hours	1 hour	Reprocessing is possible from raw records if the raw store is intact
Spatial–temporal store (raw)	72 hours	1 hour	The gate's primary evidence; geographically separated copy required
Derived products and comfort maps	5 working days	24 hours	Recomputable from raw data and a recorded model-run configuration
Application interface and dashboard	5 working days	24 hours	Presentation layer; its loss does not threaten the evidence
Model-run configurations and code	24 hours	0	Version-controlled; loss would make results irreproducible

11.10.3 Insurance, warranty and other transfers

Transfer is the treatment that moves the financial consequence of a loss to a party better placed to bear it, and it is under-used in research programmes because the instruments are unfamiliar rather than unavailable. Five are relevant here and none is costed in the archive, so each is listed as an option to price rather than as a recommendation. Equipment and property insurance cover for the deployed nodes, the reference station and the field workstation, written for outdoor installation in a desert climate rather than for office equipment. Manufacturer warranty terms specified at procurement (length, environmental envelope, replacement rather than repair, and spares availability for the full 36 months), which is a cheaper transfer than insurance and is bought at the same moment as the instrument. Public-liability cover for any structure built in a public space in Year 3, which the owning agency may require in any case as a condition of the build permission. Professional indemnity for advice given to agencies on the strength of model output, which becomes material the moment a decision brief influences capital spending. And contractual transfer through the platform contract: service levels aligned to the objectives of Table 11.10, an escrow or export provision for the data model and the code, and a termination-for-convenience clause that lets the programme move to the in-house dashboard without losing the data. The guide's distinction between contingency, held above the programme office for unknown unknowns, and management reserve, held at the delivery level for known unknowns, decides where each residual sits once transfers are in place [GAO, 2020].

11.10.4 The gate-failure fallback

The storyboard's provision that, if the gate fails, the project closes with the dataset and platform as deliverables is the single most valuable risk instrument either programme possesses [Iyyunni, 2026]. Its logic is the spiral model's: each cycle carries a termination plan, and stopping on evidence at an anchor point is a success of the process rather than a failure of the project, the stud-poker discipline of committing incrementally instead of betting everything on the most incomplete information a programme will ever have [Boehm and Hansen, 2000]. This report notes, as Chapter 7 did, that reading the fallback as a deliverable is an inference from that terminate-or-fold guidance rather than a phrase used by Boehm.

To function, the fallback must be costed and specified in advance, not improvised at month twelve. Specification means: a documented and licensed dataset covering at least one full summer of twenty-node observations at ten-minute resolution with quality flags and calibration versions; a platform release, either the partner's or the in-house dashboard, with its data model and interface documented; a method document that another city could follow; a provenance record for every derived series [W3C, 2013]; and a published account of why the gate was not passed, which is itself a contribution, because no archived comparator publishes a negative result. Costing means: naming, before award, which budget lines complete those deliverables and which stop, so that the closure is orderly and the residual value is realised. A fallback that is specified in this way changes the shape of the downside: the worst case ceases to be a lost grant and becomes a dataset, a platform and a method in the public record, and, as the megaproject literature observes of large projects generally, the failure mode to avoid is not stopping early but the escalation and lock-in that make stopping impossible [Flyvbjerg, 2014].

11.11 Force majeure

11.11.1 What force majeure is

Force majeure is a contractual allocation of the consequences of events that prevent performance and that no party can reasonably control, foresee or overcome. It is not a synonym for hardship, which concerns performance that has become excessively onerous rather than impossible, and the two are conventionally drafted as separate clauses: the International Chamber of Commerce's March 2020 publication covers both [ICC, 2020]. It is not a risk treatment either: it does not reduce a probability or a consequence, it decides who bears them, and it is therefore the correct instrument for precisely the class of events that Section 11.8 showed the quantitative techniques cannot capture [GAO, 2020].

Understanding: Force majeure

TIER 1: IN PLAIN WORDS

A clause saying that if something nobody could control or expect stops the work (a disaster, a shutdown, a war), neither side is punished for not doing what they promised, provided they tell the other side promptly and try to limit the damage.

TIER 2: IN PROFESSIONAL TERMS

A contractual provision excusing a party from performance, and from liability for non-performance, where an impediment beyond its reasonable control arises that it could not reasonably have foreseen at the time of contracting and whose effects it could not reasonably have avoided or overcome, subject to notice, mitigation and, where the impediment persists, termination [ICC, 2020].

TIER 3: IN TECHNICAL TERMS

An allocation of unpriceable tail risk between parties, operating outside the contingency and simulation machinery of cost estimation [GAO, 2020]; its construction, the availability of a presumed-event list, the treatment of foreseeability at the date of contracting and the interaction with any overriding provision of the applicable law are questions of the governing law and must be settled with counsel in that jurisdiction.

AS A MATHEMATICAL MODEL

Performance is excused where the impediment satisfies the conjunction of three tests (beyond reasonable control, not reasonably foreseeable at contracting, and effects not reasonably avoidable) and the notice and mitigation duties are discharged; relief is suspensive while the

impediment lasts, with termination rights on persistence beyond an agreed period.

11.11.2 The elements a clause must contain

Table 11.11 lists the elements that this report recommends be present in every agreement either programme signs, and shows how each attaches to the three instruments that matter: the grant agreement with the funder, the collaboration agreement with the partner university, and the data-sharing letters with the agencies. The list is this report's drafting checklist, compiled as common practice; the International Chamber of Commerce's model clause should be obtained and used as the drafting base rather than this table [ICC, 2020].

Table 11.11: Force-majeure clause elements and where each attaches. The elements are this report's drafting checklist; the model clause itself was not reviewed for this report and is cited by title [ICC, 2020]. 'Grant' = the funder's grant agreement; 'Collab.' = the collaboration agreement with the partner university; 'Data' = the agency data-sharing and site-access letters.

Element	What it must say	Grant	Collab.	Data
Definition	The three-part test: beyond reasonable control; not reasonably foreseeable at the date of the agreement; effects not reasonably avoidable or surmountable	•	•	•
Presumed events	A non-exhaustive list of events presumed to satisfy the test, with the burden on the other party to rebut	•	•	•
Exclusions	What is expressly not force majeure: shortage of funds as such, the affected party's own default, and any event whose effects could have been overcome at reasonable cost	•	•	
Notice	A period in days from becoming aware, the required contents, and the consequence of late notice (relief from the date of notice rather than the date of the event)	•	•	•
Mitigation duty	An obligation to take reasonable steps to limit the effect and to resume performance	•	•	•
Relief	Suspension of the affected obligations and of liability for delay, with the milestone calendar extended accordingly	•	•	
Milestone and window rules	Explicit treatment of a missed annual measurement window, since a suspension of weeks has a consequence of a year (equation (11.4))	•	•	
Gate interaction	Whether and how the month-12 gate date and its four conditions move when relief is claimed	•	•	
Persistence and termination	The period after which either party may terminate, and what happens to equipment, data, licences and the fallback deliverables on termination	•	•	•
Cost allocation	Who bears costs incurred during suspension, including equipment in the field and staff on continuing contracts	•	•	
Data custody	Continuity of access, custody and licence for shared data during and after a force-majeure period		•	•
Governing law	The applicable law and the forum, and confirmation that the clause is consistent with any mandatory provision of that law	•	•	•

11.11.3 Which events belong in the list

The presumed-event list should be regional, not generic. Six categories are proposed, and a note of evidential discipline applies to all of them: except where a citation appears, *no document in this report's archive addresses these events*, and they are named here as clause categories only. No figure, date or magnitude should be attached to any of them in a signed agreement without a source.

1. **Extreme heat and dust events.** Sustained conditions or dust storms that prevent field work, damage installed instruments or make a site unsafe. The regulatory backdrop is archived and specific: outdoor work in direct sunlight and open spaces is prohibited between 12:30 and 15:00 from 15 June to 15 September under the midday break rule *[The National, 2026]*[4]*[MoHRE, 2023]*[5], which means that a heat event does not merely slow field work but can lawfully stop it in the window the programme most needs. Equipment consequences belong to register entry D11.

2. **Public-health emergency.** A declared epidemic or pandemic, or the public-health measures taken in response, preventing site access, travel or installation. This is named because it is within living institutional memory of every party to these agreements, not because the archive documents it.

3. **Logistics and trade disruption.** Interruption to shipping, air freight, customs clearance or export licensing that prevents delivery of instruments or spares. Disruption to Red Sea shipping routes during 2024 is widely reported and is not covered by any document in this archive; it is named as a clause category and as a reason to require, at procurement, that the supplier state the origin and the routing of the goods and that spares be held locally rather than ordered on demand.

4. **Sanctions, export controls and geopolitical rupture.** Measures that make a partner's contribution unlawful or impossible, particularly relevant to a programme designed around a South–South capability network whose members span several sanctions regimes [Terhile et al., 2026]; this is the contractual counterpart of register entry A12.

5. **Government and funder action.** Withdrawal of a permission, a change of programme rules, a funding freeze or the non-opening of a funding cycle. The last of these is register entry D9, and it illustrates why the exclusions matter: a party's own shortage of funds is conventionally excluded from force majeure, so a funding-cycle failure must be handled by a suspension or re-scoping mechanism in the grant agreement rather than by a force-majeure claim *[Dubai RDI Programme, 2025]*[1].

6. **Civil disturbance, conflict and infrastructure failure.** Including prolonged failure of power or telecommunications affecting a site or a data centre, which is the event the recovery objectives of Table 11.10 are sized against.

11.11.4 Drafting for the three instruments, and the question for counsel

For the **grant agreement**, the provisions that matter most are the milestone and window rules and the gate interaction. A grant whose gate date is fixed in absolute time and whose evidence depends on an annual window needs an express rule that relief moves the gate date rather than

[4] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[5] Ministry of Human Resources and Emiratisation, UAE, *UAE to implement midday work break from 15 June to 15 September for 19th consecutive year*, MoHRE media centre, 2023. https://mohre.gov.ae/en/media-center/news/1/6/2023/uae-to-implement-midday-work-break-from-15-june-to-15-september-for-19th-consecutive-year

excusing the conditions: the conditions are the programme's standard of proof and should survive any suspension intact [Iyyunni, 2026]. The agreement should also state what happens to the fallback deliverables of Section 11.10.4 on early termination, so that closure realises the dataset and platform rather than stranding them.

For the **collaboration agreement** with the partner university, the provisions that matter most are cost allocation during suspension, data custody, and a substitution mechanism. A collaborator with its own budget line and its own institutional obligations may be prevented from performing while the lead institution is not, and the agreement should say whose costs continue, who holds the jointly generated data, and on what terms the workstream can be brought in-house or placed elsewhere [Iyyunni, 2026].

For the **data-sharing and site-access letters**, the provisions that matter most are notice, continuity of access and custody. These letters are short instruments and will not carry a full clause; what they must carry is a statement that access granted survives a temporary interruption, that data already transferred may continue to be held and processed for the agreed purpose, and that either party will give notice of an impediment. Without that, a suspension of weeks can silently become a renegotiation of the permission that took a year to obtain.

One question must go to counsel rather than to this report. The treatment of force majeure and of impossibility or supervening events under the applicable law of the United Arab Emirates (including how a contractual clause interacts with any mandatory provision of the UAE Civil Transactions Law, whether relief operates automatically or on claim, and what a court or tribunal may do to the parties' obligations when an event of this kind occurs) must be verified with qualified counsel in the jurisdiction before any of these agreements is signed. No article number, statutory test or authority is asserted here, because this report has reviewed no legal source on the point. The equivalent question arises separately for each African jurisdiction in which ABIG operates and for the law governing any continental data-sharing instrument [Terhile et al., 2026].

What this chapter establishes

The two programmes are exposed in different ways and require different instruments. Dubai's exposure is concentrated, time-bound and largely treatable: four of its highest exposures (the agency confirmation, the summer window, the undocumented funding cycle and the build permission) can each be reduced substantially by an action taken before award, and its lowest gate probability attaches to the condition that depends least on technology. ABIG's exposure is structural: its register's highest entries are the risks of producing documents instead of decisions, of losing legitimacy through sovereignty or misuse, and of financing that never reaches operations; none of these is treated by better modelling and all are treated by naming operational users and writing pathways to operational budgets. For both, the quantitative machinery ends where the catastrophic begins [GAO, 2020], which is why the gate fallback, the six playbooks, the recovery objectives and a regionally drafted force-majeure clause are not administrative appendices but the part of the risk apparatus that decides what a bad year costs.

11.12 Implications for the programmes

For the Dubai Decision Twin. Nine things should change, each at the moment named. Open a dated risk register for the Al Fahidi corridor testbed in the form of Tables 11.1 and 11.2, with an owner against every line and a review at each anchor point rather than annually [ISO, 2018, Boehm and Hansen, 2000]. Name an individual decision owner inside the Roads and Transport Authority and inside the Dubai Electricity and Water Authority, and agree with each, after the first comfort map and before the gate, what 'useful' will mean, because equation (11.5) shows that condition 6.2.4

governs the gate more than the modelling does [Iyyunni, 2026]. Confirm the funding cycle's calendar in writing and re-express every partner milestone in relative time, since the archived rules document only the 2025 cycle *[Dubai RDI Programme, 2025]*[1]*[Dubai Future Foundation, 2025]*[2]. Buy schedule contingency in the first eight weeks after award, because equation (11.4) makes moderate delay catastrophic. Specify and cost the gate fallback now (dataset, platform, method document, provenance record and a published account of non-passage [W3C, 2013]), so that the worst case has a floor. Write the recovery objectives of Table 11.10 into the platform contract, with an export or escrow provision and a termination-for-convenience clause that preserves the in-house alternative. Adopt the six recovery playbooks and rehearse two of them (data loss and sensor-network loss) before the first summer. Price the five transfers of Section 11.10.3 at procurement, when warranty terms are still negotiable. And put the clause elements of Table 11.11 into the grant agreement, the Curtin collaboration agreement and the site and data letters, with the governing-law question referred to counsel [ICC, 2020].

For the Africa Blue Intelligence Grid. The concept paper's risk table should be promoted from an annexe to an instrument: scored, owned, indicated and reviewed quarterly by the programme office, with the three additions of Table 11.3 incorporated [Terhile et al., 2026]. The indicator against the first risk (named operational users with signed use-case briefs, and decisions demonstrably changed) is the one number that distinguishes this programme from the coordination forum it fears, and it should appear in every quarterly report from month three. Data sovereignty should be treated as a legal workstream running from month zero beside the capability map, because a federated architecture's information boundary is a set of agreements and every later loop inherits them. The security layer needs an audited referral pathway, with uncertainty and provenance carried on every output and human review mandatory before any referral, on the model of the archived enforcement case in which intelligence supported a trained national patrol rather than replacing it [W3C, 2013]*[Global Fishing Watch, 2026]*[3]. Pilots should be risk-adjusted at portfolio level rather than individually [GAO, 2020], and each should carry a written sustainment commitment and a pathway to an operational budget before it starts [Boehm et al., 1998]. Continuity planning should follow the same six playbooks, with partner withdrawal and funding shortfall the two most likely. And force-majeure and data-custody provisions should be drafted once, as a model annexe for the whole federation, with the governing-law question answered separately for each jurisdiction [ICC, 2020, ISO, 2019]. The next chapter turns from protecting the programmes to what they could change: the policy routes by which a metric proven in one corridor and one continental network becomes an instrument others use.

۱۲ {12}

Policy

Abstract

If the work is done by members of the Gulf Co-operation Council, how can it lead planetary policy? The chapter argues that the route runs through measurement rather than advocacy, and states the route as a proposition this report proposes rather than a law it has verified: a metric, an openly published method, and replication by independent teams in at least three cities. It sets out the laboratory argument (that the Gulf runs the largest recurring heat-management operations in the world and therefore produces evidence that transfers to the arid world) and then inventories the instruments the United Arab Emirates and its neighbours already own, chief among them the Global Cooling Pledge, whose three published headline targets concern emissions, access and equipment efficiency and contain no adaptation indicator. It fixes the science–policy calendar: the IPCC Special Report on Climate Change and Cities closed its Second Order Draft review in July 2026 and is scheduled for approval in March 2027, so no output of either programme can enter it, and the Seventh Assessment Report's Working Group II contribution is the window. It then treats South–South transfer, indigenous knowledge, the practice of *nemawashi*, and the planetary-boundaries framing, and closes with a table of policy asks by stakeholder. Readers should use it as the sequencing document for everything that happens after the evidence exists, and not before.

Policy influence is usually sought in the wrong order. The common sequence is to convene, to declare, and then to look for evidence that supports the declaration; the sequence proposed here is to measure, to publish the method, to have the method repeated by people with no stake in the result, and only then to ask an institution to adopt what has survived. The distinction matters for two small programmes. The Climate Intelligence and Decision Twin for Dubai is an [1,400,000.00:388,888.89](AED:USD) study of one corridor over 36 months [Iyyunni, 2026]; the Africa Blue Intelligence Grid (ABIG) is a concept paper with no budget [Terhile et al., 2026]. Neither can buy influence, and neither will be given influence on the strength of its ambitions. What both can do is produce a number that a ministry, a pledge secretariat or an assessment author does not currently have, in a form that can be recomputed by someone else. This chapter sets out how that number travels, which instruments can carry it, when each is open, and what each will want to see first.

12.1 The laboratory argument

The first claim is about where the evidence should come from. This report does not assert that the Gulf is the hottest inhabited region on Earth; no archived source in its evidence base measures

that. What the archive does support is narrower and more useful: the Gulf runs the largest recurring heat-management operations in the world, and it runs them under legal and operational obligation rather than as research.

Three examples stand in the archive. First, the United Arab Emirates has prohibited work in direct sunlight and in open spaces between 12:30 and 15:00 from 15 June to 15 September for more than two decades (a national newspaper dates the rule's introduction to 2004, while the labour ministry's own 2023 announcement described that year as the nineteenth consecutive year of the ban), with fines of [5,000.00:1,388.89](AED:USD) per worker up to [50,000.00:13,888.89] (AED:USD) *[The National, 2026]*[1] *[MoHRE, 2023]*[2]. This is a government that already encodes hours lost to heat in law, by clock rather than by measurement. Second, the Hajj road-cooling at Makkah and Arafat: Saudi Arabia's Roads General Authority reported paving more than 84,000 m^2 of heat-reducing road surface across the Arafat plains for the 2025 pilgrimage, an expansion of 82 per cent on the previous year, with an agency-claimed surface-temperature reduction of about 12 to 15 °C *[Gulf News, 2025]*[3] *[Arab News, 2024]*[4]; misting installations and a tree corridor are reported separately for the 2024 pilgrimage *[Saudi Gazette, 2024]*[5]. These are case-reports: protocol, instruments, sites and controls are unpublished, and no pedestrian-level, cost or durability figure exists in them. Third, Riyadh has a planting programme targeting more than 7.5 million trees by 2030 *[Saudipedia, 2025]*[6], whose claimed temperature benefits are programme claims rather than outcomes.

The pattern is diagnostic. The region operates at a scale no research programme can match and publishes almost nothing that meets an evidence standard. The one archived exception is instructive: a twenty-six-station urban weather network on Abu Dhabi island, mounted on lamp-posts with municipal approval, measured daily-mean urban-minus-rural air temperature differences from −0.23 to +3.21 °C across five districts and four seasonal windows, with a street model at 3 m resolution validated on a held-out week [Manandhar et al., 2019]. That is a measured study, and it exists because a university asked for permission to mount instruments. The laboratory argument is therefore not that the Gulf is special because it is hot; it is that the Gulf is where the largest heat interventions are already being made, where the institutional appetite for them is proven, and where the evidence base is thinnest.

The economic frame is supplied by two projections. The International Labour Organization projects that heat stress will cost 2.2 per cent of global working hours in 2030 (80 million full-time equivalent jobs, some USD 2,400 billion at purchasing-power parity) with Bahrain and the United Arab Emirates among the countries whose projected loss exceeds 2 per cent of gross domestic product [ILO, 2019]. A twelve-city study puts heat-related worker-productivity losses at about USD 44 billion a year in 2020, rising to USD 84 billion by 2050 under a high-emissions pathway,

[1] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[2] Ministry of Human Resources and Emiratisation, UAE, *UAE to implement midday work break from 15 June to 15 September for 19th consecutive year*, MoHRE media centre, 2023. https://mohre.gov.ae/en/media-center/news/1/6/2023/uae-to-implement-midday-work-break-from-15-june-to-15-september-for-19th-consecutive-year

[3] Gulf News, *Hajj 2025: Saudi Arabia expands cooled, rubberised roads*. https://gulfnews.com/world/gulf/saudi/hajj-2025-saudi-arabia-expands-cooled-rubberised-roads-1.500139040

[4] Arab News, *Road-cooling tech will provide Hajj pilgrims with respite from heat*, Saudi Arabia section, 2024. https://www.arabnews.com/node/2530731/saudi-arabia

[5] Saudi Gazette, *Hajj 2024: Saudi Arabia implements advanced cooling techniques in holy sites amid scorching heat*. https://saudigazette.com.sa/article/643620/SAUDI-ARABIA/Hajj-2024-Saudi-Arabia-implements-advanced-cooling-techniques-in-holy-sites-amid-scorching-heat

[6] Saudipedia, *Green Riyadh Project*, encyclopaedia article (sources: Royal Commission for Riyadh City; Saudi Press Agency), 2025. https://saudipedia.com/en/green-riyadh-project

with Abu Dhabi among the cities assessed [Arsht-Rock and Vivid Economics, 2022]. Both are projections, and both model workers only. The visitor, the resident walking to a metro station and the outdoor service economy are absent from every archived economic series: the gap the Outdoor Tourism Capacity construct is built to fill.

12.2 Policy follows measurement

Proposition 12.2.1 Policy follows measurement

A metric becomes an international norm when three conditions hold together: the metric is defined precisely enough to be computed by someone who did not invent it; the method is published openly, including its instruments, its failure modes and its uncertainty; and independent teams have applied it in at least three cities and reported effect sizes with their dispersion. This report advances the three-city rule as a working criterion for its own planning, not as an empirical finding: no archived source in this evidence base establishes a threshold number of replications, and none is claimed.

The rule is proposed because the alternative routes visibly do not work. The archive contains several well-funded, politically endorsed programmes whose measured effect on any indicator cannot be recovered from their own publications. Paris OASIS instrumented ten pilot schoolyards with a controlled before-and-after design and a paired control station 200 m outside one site, published its method, and scaled to further yards on the city's own funds, yet no archived OASIS document reports a measured change in air temperature, mean radiant temperature or thermal comfort for a renovated yard [Urban Innovative Actions, 2023, OASIS research team, 2024a][*EEA Climate-ADAPT, 2023*][7]. Cooling Singapore has run since 2017 with national research funding and produced a modelled whole-island cooling of up to 0.13 °C on one heat-wave day under a green-plan scenario, against an urban heat island quoted as up to 7 °C without a stated method [Cooling Singapore 2.0, 2024][*Centre for Liveable Cities, 2024*][8]. Phoenix's cool-pavement programme is the most rigorously measured of the three and delivers the most awkward result: surface temperatures fell by about 12 °F at noon and 10.5 °F in the afternoon against adjacent untreated asphalt, while mean radiant temperature at the road centre rose by approximately 5.5 °F (range 2.6 to 9.2 °F), and reflectivity fell from 33–38 per cent at installation to 19–30 per cent within ten months [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. A pedestrian standing on the coated surface at midday is hotter, not cooler, a finding that exists only because somebody measured at pedestrian height with controls, and one that directly qualifies the generic endorsement of reflective surfaces in the United Nations Secretary-General's Call to Action [UN Secretary-General, 2024]. Measurement is what corrects policy, and only measurement with controls does it.

Understanding: Policy pathway

TIER 1: IN PLAIN WORDS

A number only changes what governments do if other people can work it out for themselves, in their own cities, and get something similar.

[7] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

[8] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

TIER 2: IN PROFESSIONAL TERMS

An indicator enters international practice through a sequence (definition, open method, single-site result, independent replication, adoption by an instrument that already has an owner and a reporting cycle) and stalls at whichever step is skipped. Advocacy without the earlier steps produces declarations; measurement without the later steps produces papers.

TIER 3: IN TECHNICAL TERMS

Adoption is a transmission problem between agents with different objective functions and no shared information set [Powell, 2019]. The programme controls only the first three steps; the fourth requires teams it does not employ, and the fifth requires an instrument owner for whom the indicator resolves a decision it already faces. The written confirmation demanded at the Dubai month-12 gate is the smallest possible instance of step five [Iyyunni, 2026].

AS A MATHEMATICAL MODEL

Let I be the candidate indicator, M its published method and $R = \{c_1, \dots, c_n\}$ the set of cities in which independent teams have applied M and reported an effect $\hat{\Delta}_c$ with its dispersion. The adoption condition adopted here is $n \geq 3$ together with $V_a(I) > 0$ for at least one instrument owner a, where V_a is the value of information in the classical sense [Howard, 1966]: $V_a(I) = \max_x \mathbb{E}[u_a(x) \mid I] - \max_x \mathbb{E}[u_a(x)]$. Both parts are criteria this report sets itself, not measured thresholds.

Figure 12.1 draws the pathway with the two programmes' own milestones attached; step 1, shaded, is the only one already done.

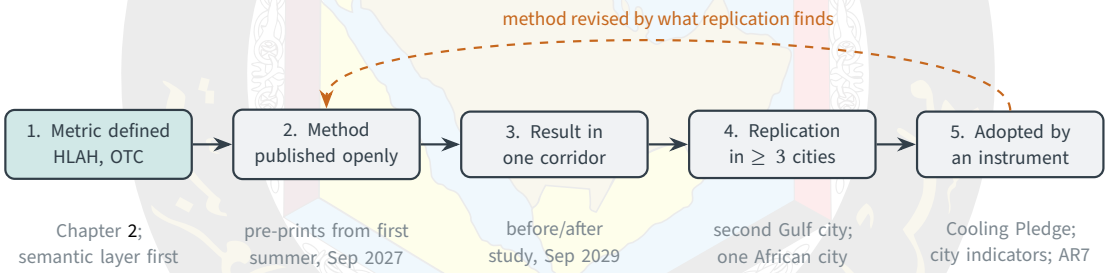


Figure 12.1: The policy pathway proposed in Proposition 12.2.1, with the two programmes' milestones beneath each step. Only step 1 is inside the control ring of Chapter 6; steps 2 and 3 are inside the grant; steps 4 and 5 require institutions the programmes can only influence. The dashed return is the condition that makes the sequence a method rather than a campaign.

12.3 The instruments the Gulf already holds

The second claim is about vehicles. A small programme does not need to create an instrument; it needs one already owned by an actor within reach, already carrying a reporting cycle, and lacking exactly the thing the programme can supply.

The strongest candidate is the Global Cooling Pledge. It is an initiative of the United Arab Emirates as host of the 2023 climate conference, one of nine non-negotiated declarations, pledges and charters constituting the COP28 Presidential Action Agenda, and its three published headline targets are to reduce cooling-related emissions by 68 per cent from 2023 levels by 2050, to increase access to sustainable cooling significantly by 2030, and to raise the global average efficiency of new air conditioners by 50 per cent [UNEP, 2023]. More than sixty countries signed at launch in Dubai on 5 December 2023 *[UN News, 2023]*[9]; by the ministerial roundtable at COP29

[9] UN News, *COP28: Over 60 countries pledge to slash cooling emissions amid rising temperatures*, December

in Baku on 19 November 2024 the Pledge united over 71 countries and 60 non-state actors around fourteen actions, with fifteen cities joining as subnational signatories, seventeen signatories having developed National Cooling Action Plans and four more in preparation, the United Arab Emirates and Brazil as co-chairs for 2025, a 2030 Implementation Strategy adopted, and an Intergovernmental Committee on Cooling announced for COP30 *[UNEP, 2024]*[10]. The secretariat is the UNEP-led Cool Coalition.

Every one of those targets is an emissions, access or equipment-efficiency target; none measures whether a person can be outdoors. The Pledge's evidence base concerns cooling demand (passive cooling alone could cut demand by 24 per cent by 2050, avoiding 1.3 billion tonnes of carbon dioxide annually, the Pledge's own projection relayed in a press release *[UNEP, 2024]*[10]) and its passive-cooling category already includes shading and reflective surfaces, the very interventions the Dubai corridor will rank. The gap is precise and the claim about it should be phrased with equal precision: on the published summary of its headline targets, the Pledge carries no heat-adaptation indicator; the full Pledge report is not in this report's archive and no statement is made about its contents. That is the opening for HLAH and OTC: offered not as a replacement for the emission targets but as the adaptation column the same signatories already need for their National Cooling Action Plans.

Understanding: Adaptation indicator

TIER 1: IN PLAIN WORDS

A way of counting the things people could not do outdoors because it was too hot, and how many of those hours a piece of shade gives back.

TIER 2: IN PROFESSIONAL TERMS

An outcome indicator for heat adaptation, expressed in activity-hours rather than degrees, computed per place and hour for a named population group, so that a shade structure, a tree and a cool surface can be compared on the same scale and against their cost, water and energy.

TIER 3: IN TECHNICAL TERMS

Heat-Lost Activity Hours is an exposure-weighted deficit integral over a place–hour grid, evaluated with a thermal-comfort index at pedestrian height. The Universal Thermal Climate Index is used for visitors and residents; Wet Bulb Globe Temperature is reported alongside it because every archived labour series (the ILO's projections and the Lancet Countdown's hindcast) is built on WBGT, and comparability with them is a condition of policy uptake [ILO, 2019, Romanello et al., 2024]. The Call to Action names both indices, with Heat Index and Humidex, as the recognised family [UN Secretary-General, 2024].

AS A MATHEMATICAL MODEL

$\mathrm{HLAH} = \sum_{p\in\mathcal{P}} \sum_{h\in H} \sum_{c} A_{p,h,c}\big(1-\phi_c(U_{p,h})\big)\Delta h$, with $A_{p,h,c}$ the activity that would otherwise occur and ϕ_c the cohort acceptability function; the policy-facing forms are $\mathrm{RAH}(x) = \mathrm{HLAH}^{0} - \mathrm{HLAH}^{x}$ for an intervention x, and $\mathrm{RAH}(x)$ per AED, per litre and per kilowatt-hour.

The second instrument is the Secretary-General's Call to Action on Extreme Heat, issued in July 2024 (the archived paper carries no printed issue date) with inputs from ten United Nations entities and twenty-three action items across four pillars: care for the vulnerable; protect workers; boost the resilience of economies and societies using data and science; and limit temperature rise to 1.5 °C [UN Secretary-General, 2024]. The third pillar is the one that fits. Its items ask governments

2023. https://news.un.org/en/story/2023/12/1144382

[10] United Nations Environment Programme, *Cooling the Planet: COP29 affirms global climate action*, UNEP press release, November 2024. https://www.unep.org/news-and-stories/press-release/cooling-planet-cop29-affirms-global-climate-action

to develop risk-informed heat action plans and to stress-test the heat resilience of current policies; to ensure standardised reporting of extreme heat risk and impact data; and to appoint champions or create an office with a cross-sectoral heat-resilience mandate. The paper also records that 84 per cent of the Arab States' workforce is affected by excessive heat, that labour productivity begins to decline above 24–26 °C and halves at 33–34 °C on the wet-bulb globe temperature scale, and (decisively for this report), that extreme heat puts pressure on tourism, a statement made once and without a number. The absent number is the report's own case.

Third, the standards route. The Gulf Co-operation Council maintains a regional standardisation body, and the international technical committee on sustainable cities and communities maintains a family of city-indicator standards. Neither body's documents are in this report's archive and neither has been reviewed; they are named as the route the report proposes, not as a source. The argument is that an indicator adopted by a pledge is voluntary, whereas one written into a city-indicator standard is reported annually by every city that certifies against it, and that a regional standardisation body is a plausible sponsor for a hot-arid indicator temperate-climate cities have had no reason to propose. The prerequisite is a stable definition with a measurement protocol: a Year-3 deliverable, not a Year-1 one.

Table 12.1 sets out the instruments with their owners, entry points, evidence requirements and windows.

12.4 The science–policy window

The third claim is about timing, and it is the part of this chapter with the least room for interpretation. The IPCC approved the outline of a Special Report on Climate Change and Cities at its sixty-first session in Sofia on 2 August 2024, with more than 230 delegates from 114 member governments agreeing the outline and the timeline, and approval and publication scheduled for March 2027 *[IPCC news, 2024]*[11]. The proposal placed before that session set out the schedule in detail: a Government and Expert Review of the Second Order Draft from 8 May to 3 July 2026, a fourth lead-author meeting in August 2026, final government distribution from December 2026, and approval on 15–19 March 2027 [IPCC, 2024]. The schedule has been executed. More than a hundred scientists from 56 countries met in Bhopal from 3 to 7 August 2026 for the fourth lead-author meeting, working on a Second Order Draft that had just completed its review with over 25,000 comments from 960 government and expert reviewers in 102 countries, following a First Order Draft that drew over 32,000 comments from more than 1,300 reviewers; the report has five chapters, an author team of 97 experts, and is the only special report of the seventh assessment cycle, scheduled for completion in March 2027 *[IPCC, 2026a]*[12].

The consequence is blunt. The Dubai grant starts on 15 March 2027 [Iyyunni, 2026]; the Special Report's drafting closed before it. Nothing either programme produces can enter the assessment that is, on its subject matter, the most relevant the IPCC will publish this decade. Any proposal claiming otherwise should be corrected.

[11] Intergovernmental Panel on Climate Change, *IPCC approves outlines of the first two reports in the seventh assessment cycle*, IPCC news, 2 August 2024. https://www.ipcc.ch/2024/08/02/ipcc-approves-outlines-of-the-first-two-reports-in-the-seventh-assessment-cycle/

[12] Intergovernmental Panel on Climate Change, *Leading scientists gather in India for final meeting to work on the Special Report on Climate Change and Cities*, IPCC news, 3 August 2026. https://www.ipcc.ch/2026/08/03/srcities-lam4/

Table 12.1: Policy instruments within reach of the two programmes: who owns each, where a small programme can enter it, what evidence it will ask for first, and when it is open. Instruments marked 'not archived' are named from the baseline analysis without a reviewed source and carry no numeric claim in this report. Sources used in this table: [UNEP, 2023], *[UNEP, 2024]*, [UN Secretary-General, 2024], *[IPCC, 2026a]*, *[IPCC news, 2024]*, *[IPCC, 2026b]*, [IPCC, 2024], *[Dubai RDI Programme, 2025]*, *[Dubai Future Foundation, 2025]*, *[African Union, 2026]*, [AU-IBAR, 2019], [Terhile et al., 2026], [UNEP, Cool Coalition and RMI, 2021].

Instrument	Owner	Entry point	Evidence needed	Timing
Global Cooling Pledge	UAE as COP28 host; Cool Coalition secretariat	Contact Group; ministerial roundtable; 2030 Implementation Strategy; Intergovernmental Committee on Cooling	An adaptation indicator with an open method and at least one validated corridor result	COP30 onward
National Cooling Action Plans	Signatory governments (17 with plans)	Add an outdoor heat-adaptation column to an existing plan	HLAH computed on the government's own data	After the month-12 gate, 2028
Call to Action on Extreme Heat, pillar 3	UN system; ten entities	Standardised reporting of heat risk and impact; heat-resilience champion or office	A reporting schema and a worked city example	Continuing
IPCC Special Report on Cities	IPCC; Working Group II leadership	Closed: review of the Second Order Draft ended July 2026	None admissible	Approval March 2027
AR7 Working Group II	IPCC	Peer-reviewed corridor results and method papers	Measured before/after against a twin prediction	Year-3 papers, 2029–30
City-indicator standards	GCC Standardization Organization; ISO committee (not archived)	Propose a hot-arid outdoor-activity indicator	A stable definition plus a measurement protocol	2030 onward
Dubai RDI Grant	Dubai Future Foundation	Letter of Intent then full proposal	Budget, deliverables, partnerships, risk mitigation	Third cycle undocumented
World Governments Summit; Expo City	UAE (not archived)	Demonstration and decision brief	A working twin and one agency confirmation	Annual, from 2028
Luanda Declaration; AU Blue Economy Strategy	African Union; AU-IBAR	Capability map tabled at technical level	Pilots with a named operational user	Years 1–3
New Development Bank; BRICS working groups	NDB; BRICS space and ocean groups	Project-preparation pipeline; virtual constellation	Bankable projects with a cost per decision improved	Years 2–3
Beating the Heat handbook	UNEP; Cool Coalition	Submit a case study in the handbook's format	A completed intervention with before/after data	After Year 3

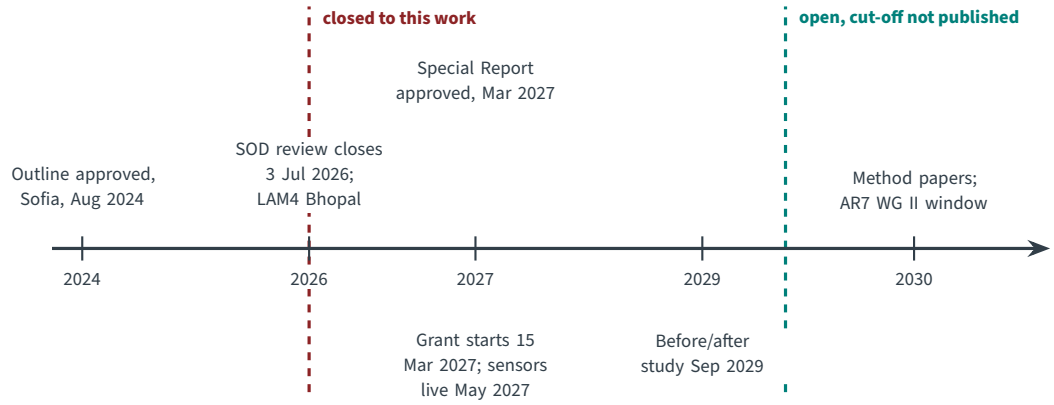


Figure 12.2: The science–policy calendar. The Special Report on Climate Change and Cities closed to new literature before either programme begins; the Seventh Assessment Report's Working Group II contribution is the target, and its literature cut-off is not stated in any archived document, which is itself the reason to publish early rather than late.

Understanding: Science–policy window

TIER 1: IN PLAIN WORDS

Big scientific assessments stop reading new work long before they are published, so a result has to exist years before the report that could quote it.

TIER 2: IN PROFESSIONAL TERMS

Assessment cycles impose a literature cut-off; a research programme that wants to be assessed must schedule its publications against that cut-off rather than against its own funding calendar, and must publish in a citable form (a pre-print or a peer-reviewed paper), rather than a project report.

TIER 3: IN TECHNICAL TERMS

The Special Report's outline includes bullets on urban observation and modelling tools for monitoring and evaluation, on metrics for assessing mitigation and adaptation options, and on the effectiveness and feasibility of city actions: precisely the categories a validated corridor twin populates [IPCC, 2024]. The outline never uses the term 'digital twin'; the nearest wording is artificial intelligence and digitisation, and observation and modelling tools. Any submission must therefore be framed in the assessment's vocabulary, not the programme's.

AS A MATHEMATICAL MODEL

Publication planning reduces to the constraint $t_{\mathrm{pub}} \leq t_{\mathrm{co}}$, where t_{co} is the assessment's literature cut-off. Because t_{co} for the seventh assessment's Working Group II contribution is not published in any archived document, the programme should minimise t_{pub} subject to the evidence being defensible: pre-prints of the method and the first summer's dataset from September 2027, not one paper in 2030.

Three operational consequences follow. Publish the method and the instrument specification before the results, so that the method paper is old enough to be cited when the results exist. Deposit pre-prints, as the Paris OASIS team did for both its microclimate and its paving work [OASIS research team, 2024a, OASIS research team, 2024b]. And write in the assessment's categories (an indicator, a monitoring and evaluation tool, a feasibility assessment, a case study), rather than in the programme's.

12.5 South–South transfer, and whose knowledge counts

The fourth claim is about legitimacy. Evidence produced in a wealthy Gulf city and offered to African and South Asian cities invites the obvious objection, and the answer is architectural rather than rhetorical: the transfer mechanism must be one in which the receiving institutions already hold decision rights.

ABIG is designed that way. It positions itself as an integration and application layer above existing African capability (observation by GOOS-Africa, forecasting by OceanPrediction and OPERA, Earth-observation services by GMES & Africa), rather than as a new observing programme, and it anchors its mandate on the African Union's Luanda Declaration, adopted on 24 July 2026 at the Third Africa Blue Economy Week [Terhile et al., 2026]*[African Union, 2026]*[13]. The continental economic frame is the African Union's Blue Economy Strategy, endorsed in October 2019, which values African blue-economy sectors in 2018 at about USD 296 billion with some 49 million jobs, including USD 21 billion in fisheries and USD 80 billion in coastal tourism [AU-IBAR, 2019]. The concept paper's South–South route runs through the eleven-member BRICS grouping (explicitly a coordination forum rather than a supranational implementing authority), using its space agencies' discussions on reducing technological asymmetries, its Virtual Constellation of Remote Sensing Satellites and its Working Group on Polar and Ocean Science and Technology as technical entry points, with the UAE Space Agency's Space Data Center and Geo-Spatial Analytics platform as the bridge and the New Development Bank, of which the UAE is a member, as the finance rung [Terhile et al., 2026].

Two archived cases show what the receiving end looks like when it works. Satellite intelligence delivered to a national authority led to the interception of three unlicensed trawlers in Gabonese waters on 18 July 2026, after a workshop had trained twenty-five personnel from the fisheries directorate, national parks agency and navy in patrol planning *[Global Fishing Watch, 2026]*[14]. A regional seascape partnership reports more than 83,000 km^2 of marine and coastal areas supported, more than 175,000 hectares under protection, restoration or commitment to protect, over 112,000 livelihoods strengthened and more than USD 80 million mobilised *[IUCN, 2026]*[15]. Both are case-reports without baselines or counterfactuals: evidence of adoption rather than of effect. The continental Earth-observation valuation makes the same point quantitatively, since its projected benefits of more than USD 2 billion a year by 2024 across an open data cube covering more than 30 million km^2 are all scaled by an assumed ten per cent adoption rate [WEF and Digital Earth Africa, 2021]. Adoption, not accuracy, sets realised value, which is why the policy asks in this chapter are written as requests for written commitments rather than for endorsements.

The indigenous-knowledge argument belongs here and should be stated carefully. The discussion notes read the Gulf cities as sites of cultural practice whose Bedouin repertoire for heat, scarcity, mobility, seasonality and community long predates air-conditioning and satellites, ask what contemporary technology can learn from cultures that evolved within extreme environments, and warn explicitly against romanticising traditional practice as a solution to modern climate problems [Iyyunni, 2026]. The caution is the important half. What the framing buys is not a cooling technique but an assessment hook: the Special Report's own outline commits its framing

[13] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

[14] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[15] International Union for Conservation of Nature, *Great Blue Wall report showcases five years of impact on ocean regeneration across the Western Indian Ocean*, IUCN news, June 2026. https://iucn.org/news/202606/great-blue-wall-report-showcases-five-years-impact-ocean-regeneration-across-western

chapter to assessment methodologies drawing on diverse knowledge systems including Indigenous Knowledge, and its chapter on facilitating change to structural inequity, colonialism and justice [IPCC, 2024]. A hot-arid corridor study that treats shade, orientation, water and timing as inherited practice to be measured (rather than as folklore to be illustrated), fits a category the assessment has already opened.

Process completes the argument. The notes name the practice *nemawashi*: preparing the ground with each stakeholder privately before anything is proposed formally [Iyyunni, 2026]. The method literature agrees on the substance. Boehm's WinWin spiral makes win conditions per stakeholder a precondition of each cycle, and its case study found that only the application with budget, people and facilities to sustain it survived into use, after which sustainment commitment became the top selection criterion [Boehm et al., 1998, Boehm and Hansen, 2000]. The Paris programme supplies the tactic: a field visit by fifty stakeholders that helped clear doubts and scepticism before pilots were built [Urban Innovative Actions, 2023]. For this chapter the rule is simple: no instrument in Table 12.1 is to be approached formally before its owner has been briefed informally and has said what it would need to see.

12.6 A note on planetary boundaries

If 'planetary policy' is read in its strict scientific sense (the planetary-boundaries framework, in which six of nine boundaries are assessed as transgressed [Richardson et al., 2023]), then a further observation is available, and it must be labelled as a research agenda rather than a claim. The framework's boundaries are defined on Earth-system processes: climate, biosphere integrity, land-system change, freshwater, biogeochemical flows, novel entities, ocean acidification, aerosol loading, stratospheric ozone. None of them is a habitability boundary. There is no control variable in the framework that says at what point a place stops supporting outdoor human activity, even though that threshold is what heat adaptation is actually about, and even though it is exactly what an activity-hours metric measures.

What the policy analysis shows

Four findings. First, the instrument with the clearest gap and the nearest owner is the Global Cooling Pledge: a UAE-founded instrument with 71 or more country signatories, a secretariat, an implementation strategy and a new intergovernmental committee, whose three published headline targets are all emissions, access or equipment-efficiency targets, none of which measures whether people can be outdoors [UNEP, 2023]*[UNEP, 2024]*[10]. Second, the assessment that would most naturally carry the evidence is closed: the Special Report on Climate Change and Cities completed its Second Order Draft review in July 2026 and is scheduled for approval in March 2027, before the grant produces anything *[IPCC, 2026a]*[12] *[IPCC news, 2024]*[11]; the seventh assessment's Working Group II contribution is the target and its cut-off is unpublished, so publication should be early. Third, the standards route (a regional standardisation body proposing a hot-arid outdoor-activity indicator to an international city-indicator committee) is the only route that converts a voluntary metric into an annually reported one, and it cannot be started before a stable definition and protocol exist in Year 3; neither body's documents were reviewed for this report. Fourth, a human-habitability boundary is absent from the planetary-boundaries framework [Richardson et al., 2023]; a validated activity-hours metric is a candidate contribution to that literature, and is recorded here as a research agenda and not as a deliverable of either programme.

Table 12.2 converts the analysis into the specific request each stakeholder is asked to carry, with the forum, the evidence that must exist first and the earliest realistic date.

Table 12.2: Policy asks by stakeholder: what each is asked to carry, where, on what evidence, and from when. No ask is to be made before the evidence in its row exists; each is preceded by an informal briefing in the sense of Section 12.5.

Stakeholder	Policy ask	Forum	Evidence first	Earliest
Dr Uchi Dominic Terhile	Table the capability map and two use cases with named operational users at African Union technical level, anchored on the Luanda Declaration	AU-IBAR; GOOS-Africa; OPERA	Capability map; one pilot with a signed user	2027
Dr Chakradhar Iyyunni	Convert the month-12 comfort map into a corridor-level heat action plan, and secure the written confirmation that the map served a decision	RTA; DEWA; Dubai Municipality	Sensor uptime; validated model; hour-by-hour map	Mar 2028
Abdullah Al Salmani	Open the UAE Space Agency data route for both programmes and request an introduction to the Cooling Pledge secretariat	UAE Space Agency; Cool Coalition	Method paper and first summer dataset	Sep 2027
Arti Ahluwalia	Carry the activity-hours metric into the civil-society and South Asian replication conversation, and the habitability question into the planetary-boundaries literature	Civil-society networks; one South Asian city	Open method; one replication partner	2028
Mahesh Shastry	Publish the method openly, propose the indicator definition to the standards route, and convene the three-city replication	Standards bodies; Gulf and African partners	Definition, protocol, corridor result	2029

12.7 Implications for the programmes

For the Dubai Decision Twin. The policy work is scheduling, not lobbying, and it has four instructions. Write the indicator definition and its measurement protocol as a standalone, citable document in the first year, not as an appendix to a final report, because every downstream route (the Pledge, the National Cooling Action Plans, the city-indicator standards, the assessment), asks for a definition before it asks for a result. Publish the instrument specification and the first summer's dataset as a pre-print by September 2027 and the method paper before the model validation of January 2028. Report Wet Bulb Globe Temperature alongside the Universal Thermal Climate Index and add the black-globe and natural wet-bulb instruments the current node specification lacks, because comparability with the labour series is the price of entry to the workers' pillar of the Call to Action [UN Secretary-General, 2024, ILO, 2019]. And keep the claims inside the evidence: the twelve-city economic figures are projections [Arsht-Rock and Vivid Economics, 2022], the cool-pavement result is measured with controls and unfavourable to pedestrians [ASU and City of Phoenix, 2021], the Makkah figures are agency claims *[Arab News, 2024]*[4]*[Gulf News, 2025]*[3], and the Holiday Climate Index cannot classify a dry, calm, clear Gulf afternoon as unacceptable because only rain and wind can override its thermal rating [Scott et al., 2016]. That last point is the report's own worked argument for an explicit heat-stress override inside OTC, and the clearest statement of why a new indicator is needed at all. The month-12 gate remains the pivot: a written confirmation from the Roads and Transport Authority or the Dubai Electricity and Water Authority is the first instance of step 5 in Figure 12.1, and without it no later step is available [Iyyunni, 2026]. The municipal digital-twin platform launched on 2 July 2026, holding more than 195,000 buildings and 280,000 infrastructure assets across more than 1,500 geospatial layers, is the natural host for an adopted indicator and should be approached as such rather than rebuilt *[Dubai Media Office, 2026]*[16]; the walkway master plan of December 2024 is the programme a Minimum Cooling Network would inform *[Dezeen, 2024]*[17].

For the Africa Blue Intelligence Grid. ABIG's policy position is stronger in mandate and weaker in evidence, and its sequencing should reflect that asymmetry. The mandate exists: the Luanda Declaration of 24 July 2026 and the African Union's Blue Economy Strategy give the Grid a continental anchor the Dubai study does not have [AU-IBAR, 2019]*[African Union, 2026]*[13]. The evidence does not: no archived source contains a cost, an uptake figure or a measured benefit for the Grid, for the regional marine service it would sit above, or for any candidate application [Terhile et al., 2026]*[MarCOSIO, 2025]*[18]. The first deliverable is therefore the alignment and mapping exercise with the observation, forecasting and Earth-observation programmes, producing a capability map that states what exists rather than what is planned. The second is two or three use cases, each with a named operational user who will confirm in writing that a product changed a decision: the Gabon enforcement case is the shape of the evidence required, and its weakness, a single episode narrated by the provider without a counterfactual, is the shape of what must be improved *[Global Fishing Watch, 2026]*[14]. Financing should follow the concept paper's ladder through project preparation to the New Development Bank and blended instruments, with the missing denominator supplied from the start: a cost per decision improved, in constant United States dollars with the year named [Terhile et al., 2026]. And ABIG should carry the heat indicator too. Africa

[16] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[17] Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

[18] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

and the Arab States are the two regions with the highest share of workforce affected by excessive heat, and the projected growth in urban poor exposed to extreme heat is largest in West Africa [UN Secretary-General, 2024]; a metric proven on a Dubai corridor and replicated in one African coastal city is the concrete form of the South–South claim rather than its rhetoric.

١٣ {13}

Conclusion and Recommendations

Abstract

The findings of the preceding chapters converge here. The evidence of Chapters 1 to 12 is drawn into one argument, and that argument becomes numbered recommendations for the Climate Intelligence & Decision Twin for Dubai, for the Africa Blue Intelligence Grid (ABIG) and for the two programmes jointly. Each of the five stakeholders is then asked to take a named decision. The method is traceability: every recommendation carries the chapter whose evidence supports it, and every decision is tied to a dated milestone of the storyboard: site letters in December 2026; partner letters, the Curtin agreement and the budget in February 2027; grant start on 15 March 2027 [Iyyunni, 2026]. A ninety-day table, counted from 22 September 2026, gives each step an owner and a date and carries the undocumented third funding cycle as a dependency. The chapter ends with the questions the evidence could not settle and with the finding on which the case rests: no archived study ties activity-hours lost to heat, at corridor scale, to a built intervention validated against a twin's prediction. Readers should treat this chapter as the agenda of the first stakeholder meeting.

13.1 Synthesis of findings by chapter

The report answered fourteen questions about two proposals that share one architecture. The answers converge on a small number of conclusions.

The documents (Chapters 1 and 2). Between them the stakeholders hold one finished paper (the ABIG strategic concept paper of 20 September 2026 with its addendum [Terhile et al., 2026]) and one complete but unformalised storyboard for the rescoped Letter of Intent ([1,400,000.00: 388,888.89](AED:USD), 36 months, Cognitive Cities priority) [Iyyunni, 2026], beside a discussion log that still holds a personal telephone number and a private letter. The Dubai proposal moves urban-heat work from mapping exposure to optimising the activity that exposure prevents, through Outdoor Tourism Capacity (OTC), the Minimum Cooling Network (MCN) and a Decision Twin that compares interventions before they are built, on a testbed pairing Al Fahidi as calibration baseline with an approximately 800 m corridor from Al Ghubaiba metro station. ABIG is not a satellite programme but an integration and application layer above the observation, forecasting and Earth-observation services Africa already operates. One pipeline (physical system, data, semantic layer, knowledge graph, intelligence, decision twin, action, learning) serves both; Dubai is the first application, not the goal.

Evidence (Chapters 3 and 4). The landscape is crowded at planetary and city scale and empty at corridor scale. Cooling Singapore, funded since 2017, quotes an urban heat island of up to 7°C

without a method *[Centre for Liveable Cities, 2024]*[1], while its Digital Urban Climate Twin reports a modelled whole-island cooling of up to 0.13°C on one heat-wave day [Cooling Singapore 2.0, 2024]; its 2026 platform paper is not archived and was not reviewed. The Dubai Digital Twin Platform of 2 July 2026 is the host the Decision Twin should sit on rather than rebuild *[Dubai Media Office, 2026]*[2]. Three findings survive the evidence grading. First, no archived study reports activity-hours lost to heat at corridor scale, tied to a built intervention and validated against a prior prediction: that is the defensible gap. Second, cool pavement is conditional rather than default: Phoenix measured surface cooling of about −10.5°F in the afternoon against adjacent aged asphalt [ASU and City of Phoenix, 2021], but also a rise in mean radiant temperature at the road centre of +5.8°F at noon and +4.5°F in the afternoon [ASU and City of Phoenix, 2024], so shade ranks first. Third, most headline figures are projections [ILO, 2019, Arsht-Rock and Vivid Economics, 2022]*[Saudipedia, 2025]*[3]; the designs with controls (Paris OASIS [Urban Innovative Actions, 2023], Phoenix and the Abu Dhabi local climate zone study [Manandhar et al., 2019]) are the templates for the Year-3 campaign.

Method and cost (Chapters 5 to 10). Ordering follows the value of information [Howard, 1966]: semantic layer, metric definitions, re-anchoring on the platform, the paired-control design, then drafting. The circles-of-influence analysis [Covey, 1989] names the mechanism that moves an agency from concern to influence: a metric it did not have, computed on its own data, delivered as a decision brief. The risk-driven spiral [Boehm and Hansen, 2000, Boehm et al., 1998] maps the storyboard onto its anchor points, and the sequential decision formulation [Powell, 2022, Bellman, 1957] of stochastic dynamic programming explains why a citywide product must stay a stretch goal. Identifiers fix H3 place keys *[Uber, 2025]*[4] and PROV-O provenance [W3C, 2013] with a hard `is_synthetic` flag. Life-cycle cost follows ISO 15686-5 vocabulary [ISO, 2017a] with reference-class context [Flyvbjerg, 2014, Cantarelli et al., 2025], and the [1,400,000.00:388,888.89] (AED:USD) over 36 months averages about [467,000.00:129,722.22](AED:USD) a year, just under the programme's annual cap of [500,000.00:138,888.89](AED:USD), of which at least half must be spent in Dubai *[Dubai RDI Programme, 2025]*[5]. That arithmetic is the report's own, and it is the reason money contingency is unavailable.

Risk and policy (Chapters 11 and 12). Dubai's exposure is concentrated and treatable before submission; ABIG's is structural and is treated only by naming operational users. Policy follows measurement: a metric, an open method and replication in at least three cities. The Global Cooling Pledge carries no adaptation indicator on its published summary [UNEP, 2023] and the Secretary-General's Call to Action asks for resilience built on data [UN Secretary-General, 2024]. The Special Report on Cities closed its Second Order Draft review before this grant begins and is approved in

[1] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

[2] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[3] Saudipedia, *Green Riyadh Project*, encyclopaedia article (sources: Royal Commission for Riyadh City; Saudi Press Agency), 2025. https://saudipedia.com/en/green-riyadh-project

[4] Uber Technologies, *H3 Geospatial Indexing System: Table of Cell Areas and Edge Lengths by Resolution*, H3 documentation, 2025. https://h3geo.org/docs/core-library/restable

[5] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

March 2027 *[IPCC, 2026b]*[6] *[IPCC news, 2024]*[7], so AR7 Working Group II is the publication window.

Understanding: The single reporting unit

TIER 1: IN PLAIN WORDS

Every dirham spent should be reported as the hours of outdoor activity it gives back to people.

TIER 2: IN PROFESSIONAL TERMS

Both programmes report one efficiency ratio, so that options of different size and kind can be compared on the same page.

TIER 3: IN TECHNICAL TERMS

Restored Activity Hours per unit of capital and operating resource, computed per corridor-hour and aggregated to the programme, with parallel denominators in litres of water and kilowatt-hours.

AS A MATHEMATICAL MODEL

$$\mathrm{RAH/AED} = \sum_{c,h}(\mathrm{HLAH}^{0}_{c,h} - \mathrm{HLAH}^{1}_{c,h}) \,/\, \mathrm{LCC}$$

13.2 Recommendations

The recommendations are numbered so that the stakeholder meeting can accept, amend or reject each on the record, and each carries its supporting chapter.

13.2.1 Dubai: Climate Intelligence & Decision Twin

D1. **Write the semantic layer before any sensor is bought**: a concept map and data dictionary of exposure, activity, route, place, intervention, resource and decision, on the schema of Chapter 8 and Appendix B. (Chapters 5, 8)

D2. **Fix the metrics and the index now.** Define Heat-Lost Activity Hours (HLAH) per place-hour and Restored Activity Hours (RAH) per AED, litre and kWh; adopt the Universal Thermal Climate Index for pedestrians [Jendritzky et al., 2012] and report Wet Bulb Globe Temperature alongside it [ISO, 2017b, NEA Singapore, 2023]; and give OTC a heat-stress override, since on this report's own worked example from the published equations the Holiday Climate Index scores a dry, calm, clear 37–39°C Gulf afternoon 'Good' (derived) [Scott et al., 2016]. (Chapters 2, 5)

D3. **Re-anchor the twin on the Dubai Digital Twin Platform,** requesting geometry and layers through it *[Dubai Media Office, 2026]*[2]; this converts a platform-build risk into a layer. (Chapters 3, 11)

D4. **Rank shade first and make cool pavement conditional.** The Phoenix radiant-load result [ASU and City of Phoenix, 2024] rules a reflective surface out as the default for a pedestrian corridor; it may enter the MCN optimiser only under a constraint on mean radiant temperature. (Chapter 4)

D5. **Adopt the paired-control design** and add the black-globe and natural-wet-bulb instruments the node list omits [Urban Innovative Actions, 2023]. (Chapters 4, 5)

[6] Intergovernmental Panel on Climate Change, *Special Report on Climate Change and Cities*, IPCC Seventh Assessment Cycle; approval scheduled March 2027. https://www.ipcc.ch/report/special-report-on-climate-change-and-cities/

[7] Intergovernmental Panel on Climate Change, *IPCC approves outlines of the first two reports in the seventh assessment cycle*, IPCC news, 2 August 2024. https://www.ipcc.ch/2024/08/02/ipcc-approves-outlines-of-the-first-two-reports-in-the-seventh-assessment-cycle/

D6. **Treat the month-twelve gate as an option and its four conditions as exit criteria,** adding a written sustainment commitment before the Year-3 intervention is chosen. (Chapters 7, 9)

D7. **Report one unit** (AED per restored activity-hour) across the six phases of the roadmap, holding contingency as scope, phase and partner capital rather than money, with the paired-control design and the black-globe and natural-wet-bulb instruments of D5 excluded from descoping, since without them no health-relevant index can be computed and the gate cannot be tested. (Chapters 9, 10)

D8. **Draft the storyboard as a written proposal** before the December 2026 site letters, and register the undocumented third funding cycle as a dependency, not an assumption *[Dubai RDI Programme, 2025]*[5]. (Chapters 1, 11)

13.2.2 ABIG: Africa Blue Intelligence Grid

A1. **Deliver the capability map first.** The alignment and mapping exercise with GOOS-Africa, the OceanPrediction Decision Collaborative and OPERA is the one first-year deliverable no other programme will produce [Terhile et al., 2026]. (Chapters 5, 7)

A2. **Hold the boundary.** Observation belongs to GOOS-Africa and forecasting to OPERA; drift towards hardware before the evidence shows need turns ABIG into another coordination forum. (Chapter 7)

A3. **Select two or three use cases, each with a named operational user**, for example unlawful-fishing alerts with an authority already working with Global Fishing Watch *[Global Fishing Watch, 2026]*[8]. (Chapters 5, 6)

A4. **Adopt a cost per decision improved and a cost breakdown per regional hub;** these are the two lines the financing ladder lacks, and no cost or benefit measurement exists for the comparable marine consortia *[MarCOSIO, 2025]*[9]. (Chapter 10)

A5. **Anchor legitimacy on the Luanda Declaration** of 24 July 2026 *[African Union, 2026]*[10] and the African Union Blue Economy Strategy [AU-IBAR, 2019], routing financing through the UAE Space Agency and the New Development Bank. (Chapters 6, 12)

13.2.3 Joint recommendations

J1. **One architecture, one identifier schema, one provenance model,** so that method and metric transfer from corridor to coast. (Chapter 8)

J2. **One risk register for two programmes,** following ISO 31000 [ISO, 2018] and ISO 22301 [ISO, 2019], with force-majeure language [ICC, 2020] referred to counsel under UAE law. (Chapter 11)

J3. **Nemawashi before proposal:** consensus with the Dubai agencies and one African partner before any submission beyond the Letter of Intent. (Chapter 12)

J4. **Publish the method openly and plan for three cities,** timing the Year-3 papers for the AR7 window. (Chapter 12)

[8] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

[9] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

[10] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

J5. **Redact the discussion record** before further circulation. (Chapter 1)

Understanding: Gate condition 6.2.4

TIER 1: IN PLAIN WORDS

One of the transport or utility agencies must say, in writing, that the new number is useful to it.

TIER 2: IN PROFESSIONAL TERMS

A written confirmation of usefulness from an operational user is the only gate condition the team cannot satisfy by its own effort, which makes it the governing schedule risk.

TIER 3: IN TECHNICAL TERMS

Adoption evidence from an external principal, obtained through a decision brief computed on that principal's own data, conditions continuation at the Life-Cycle Architecture anchor point.

13.3 Decisions required from each stakeholder

The allocation below follows the authorship of the source documents and the stakeholders' affiliations; it should be confirmed or amended on the record.

Dr Uchi Dominic Terhile (PhD Graduate, Federal University Dutse, Nigeria)
Decide that the capability map (A1) is ABIG's first deliverable and who convenes it; nominate the first two or three operational users and their use cases (A3); confirm that the integration-layer boundary (A2) is written into the charter.

Dr Chakradhar Iyyunni (BITS Pilani Dubai Campus, UAE)
As lead of the Dubai proposal: decide whether to submit the rescoped Letter of Intent to the Dubai Future Foundation RDI Grant *[Dubai Future Foundation, 2025]*[11]; confirm the four gate conditions and the added sustainment condition (D6); approve the metric, index and shade-first decisions (D2, D4); and secure the site letters from Dubai Culture and Dubai Municipality by December 2026 and the letters from the Roads and Transport Authority and the Dubai Electricity and Water Authority, the Curtin University Dubai agreement and the budget by February 2027 [Iyyunni, 2026].

Abdullah Al Salmani (SpacePoint, UAE)
Decide how the UAE hub approaches the UAE Space Agency and the New Development Bank (A5), and which Earth-observation products and municipal layers the twin requests through the Dubai Digital Twin Platform (D3).

Arti Ahluwalia (Country Chair, Ecocivilisation.earth, India)
Decide the design and principles of the data and ethics board for both programmes, including the cohort identifier that holds no personal data (J1), and how the South–South and indigenous-knowledge framing of Chapter 12 enters ABIG governance.

Mahesh Shastry (Kalman Systems Pty Ltd, Sydney, Australia)
As custodian of the working papers: decide the redaction and versioning policy (J5); own the semantic layer, data dictionary and identifier schema (D1, J1) and the conversion of both papers into written proposals (D8); and decide the modelling approach for the cost forecast and the decision problem of Chapters 7 and 9.

[11] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

13.4 The next ninety days

Table 13.1 counts from the date of this report, 22 September 2026, to day 90, which is 21 December 2026. Table 13.1 is the first ninety days of the 180-day calendar of Chapter 5, with one owner and one date for each step; where the two differ, the dates here govern. Dates inside the window are proposed; those beyond it are fixed by the storyboard [Iyyunni, 2026]. The sequence is front-loaded with paper, because no loop may begin without written win conditions [Boehm et al., 1998]. The day-90 checkpoint is a real decision point: if the site letters have not arrived, the December milestone should be recorded as missed and February re-planned.

Understanding: The third-cycle dependency

TIER 1: IN PLAIN WORDS

The money the plan assumes has not yet been announced, and waiting a year would cost a summer of measurement.

TIER 2: IN PROFESSIONAL TERMS

The archived programme documents record only the second funding cycle; a September 2026 Letter of Intent and a March 2027 start presuppose a third cycle that appears in no archived source *[Dubai RDI Programme, 2025]*[5] *[Dubai Future Foundation, 2025]*[11].

TIER 3: IN TECHNICAL TERMS

An exogenous funding-timing dependency with a step-function schedule loss: because the measurement window is annual, slippage past the start of a summer defers validation by a full cycle, not by the length of the delay.

AS A MATHEMATICAL MODEL

$\Delta_{\text{schedule}} = 12\lceil \delta/12 \rceil$ months, for a delay δ in months

13.5 Open questions and further work

The evidence does not settle the following. Each is a research or negotiation task, not a defect of the proposals.

- **Temporal resolution.** Whether ten-minute records justify fifteen-minute operating decisions for misting and route advisories is an empirical question for the first summer (Chapter 8).
- **Transfer of the radiant-load result.** The Phoenix finding was measured on residential streets in a different climate [ASU and City of Phoenix, 2024]; whether it holds on a heritage corridor is what the paired-control design will show, and should not be assumed either way.
- **Behavioural response.** The elasticity of outdoor activity to comfort in a tourism setting is established by no archived source, yet the value of restored hours is most sensitive to it (Chapter 9).
- **City-scale greening.** Green Riyadh's city-level temperature target is a projection and no city-level measurement has been reported *[Saudipedia, 2025]*[3]; a corridor result will not resolve it.
- **Cost per decision improved.** ABIG's financing ladder has no unit cost; defining one, with a breakdown per regional hub, precedes any blended-finance conversation (Chapter 10).
- **Blue-finance outcomes.** The Seychelles blue bond raised USD 15 million in 2018 [World Bank, 2025]*[World Bank, 2018]*[12], but the published critique of its debt treatment was not archived

[12] World Bank, *Seychelles launches World's First Sovereign Blue Bond*, press release, 29 October 2018. https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond

Table 13.1: Proposed ninety-day action plan from 22 September 2026 (day 0), with the fixed storyboard milestones beyond day 90.

Day	Date	Action	Proposed owner
7	29 Sep 2026	Stakeholder meeting: adopt this report; accept or amend D1–D8, A1–A5, J1–J5; confirm owners and the single reporting unit	All five; convened by M. Shastry
8	30 Sep 2026	Submit the rescoped Letter of Intent (storyboard milestone)	C. Iyyunni
14	6 Oct 2026	Redact the discussion log and circulate the redacted record	M. Shastry
21	13 Oct 2026	Nemawashi round one with the four Dubai agencies; request layers through the Dubai Digital Twin Platform; confirm the next funding call's calendar, envelope and topics with the Dubai Future Foundation	C. Iyyunni; A. Al Salmani
28	20 Oct 2026	Semantic-layer concept map and data dictionary v0.1; HLAH and RAH definitions; comfort-index decision	M. Shastry; C. Iyyunni
35	27 Oct 2026	ABIG: scope the GOOS-Africa / Ocean-Prediction / OPERA alignment exercise; draft the capability-map template	U. D. Terhile
42	3 Nov 2026	ABIG: nominate two or three use cases, each with a named operational user	U. D. Terhile; A. Ahluwalia
49	10 Nov 2026	Storyboard drafted as a proposal (v0.1); ABIG concept paper drafted as a proposal	M. Shastry; C. Iyyunni
56	17 Nov 2026	Curtin University Dubai scope and budget line drafted; platform partner decision framed	C. Iyyunni
63	24 Nov 2026	Risk register v1 signed off; force-majeure language sent to counsel; third-cycle dependency logged with its fallback	C. Iyyunni; A. Ahluwalia
70	1 Dec 2026	Data and ethics board design; `is_synthetic` firewall rule adopted for both programmes	A. Ahluwalia; M. Shastry
77	8 Dec 2026	First approach to the UAE Space Agency on data access for both programmes	A. Al Salmani
84	15 Dec 2026	Site letters from Dubai Culture and Dubai Municipality (storyboard milestone: December 2026)	C. Iyyunni
90	21 Dec 2026	Day-90 checkpoint: written win conditions for Loop 1; go / no-go to the full proposal	All five
Fixed milestones beyond day 90 [Iyyunni, 2026]			
n/a	Feb 2027	Agency data letters; Curtin University Dubai agreement; budget of approximately ١٬٤٠٠٬٠٠٠.٠٠ [1,400,000.00: 388,888.89](AED:USD) finalised	C. Iyyunni
n/a	Feb 2027	ABIG: financing and convening conversation with the UAE Space Agency and the New Development Bank	A. Al Salmani; U. D. Terhile

and has not been reviewed; ABIG should propose no bond instrument before its own outcome indicators exist.

- **A human-habitability boundary.** The framework of planetary boundaries sets none for human habitability, although it reports six of nine boundaries already transgressed [Richardson et al., 2023]; a validated heat-lost-activity metric is a candidate contribution, and a research agenda rather than a grant claim.
- **Blocked sources.** Appendix A lists what could not be archived; conclusions resting on those sources should be rechecked when access is obtained.

13.6 Implications for the programmes

Dubai. The proposal is feasible as scoped because its boundary is small (one corridor, one control, one built intervention) and because its gate is honest: the dataset and the platform remain deliverables if the gate fails. Its significance is out of proportion to its budget. The United Arab Emirates already encodes hours lost to heat in law through the midday break *[The National, 2026]*[13] *[MoHRE, 2023]*[14], and the Lancet Countdown reports 512 billion potential work-hours lost to heat in 2023 as an observation-based model estimate [Romanello et al., 2024]. Neither rests on a corridor-scale, intervention-tied, twin-validated measurement. A city that produces one, and reports it in AED per restored activity-hour, owns the adaptation indicator the Global Cooling Pledge lacks.

ABIG. The concept is feasible only as an integration layer. Its value is the capability map, the use cases with named users, and the standards and training that follow; its risk is drift towards hardware and towards being one more forum. The Gabon seizure of 18 July 2026 shows that satellite intelligence already changes decisions in African waters when a named authority receives it *[Global Fishing Watch, 2026]*[8]; ABIG's task is to make that routine across the five domains the concept paper names [Terhile et al., 2026], on the political footing of the Luanda Declaration *[African Union, 2026]*[10].

Closing finding: the claim and its test

No archived feasibility study, in any city, has measured activity-hours lost to heat at corridor scale, tied that measurement to a built intervention, and validated it against a twin's prior prediction. That gap (not a new twin, a new sensor or a new index) is the defensible claim of the Dubai programme, and the method ABIG carries to the coast. The claim has one test, written into the storyboard as gate condition 6.2.4: an agency that did not have the metric confirms in writing that the metric, computed on its own data, is useful [Iyyunni, 2026]. Every recommendation here exists to make that sentence true by March 2028. The stakeholders are asked to adopt this report at the meeting proposed for 29 September 2026 and to treat Table 13.1 as their shared calendar.

[13] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[14] Ministry of Human Resources and Emiratisation, UAE, *UAE to implement midday work break from 15 June to 15 September for 19th consecutive year*, MoHRE media centre, 2023. https://mohre.gov.ae/en/media-center/news/1/6/2023/uae-to-implement-midday-work-break-from-15-june-to-15-september-for-19th-consecutive-year

Next Steps

What the two programmes now need is not more analysis. The survey of comparable efforts is done, the data architecture is specified, the institutional map is drawn, the risks are ranked and the costing machinery is built. What is missing is shorter and harder: two documents that state an amount, name a counterparty and ask for a decision. Neither exists, and until they do the work cannot be acted on by anybody outside the group that produced it. This chapter sets out what each must contain.

A submission to the Dubai grant

The first document is an application, and its audience reads quickly against published criteria. Budget viability is scored twice, at the letter of intent and again at full proposal, which makes the budget the cheapest place to gain ground and the likeliest place to lose it *[Dubai RDI Programme, 2025]*[5]. The envelope runs from [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD), the annual budget for any one project may not exceed [500,000.00:138,888.89](AED:USD), and a project may run up to thirty-six months. In the round verified a day after this study's evidence horizon, the share of grant funding to be spent within Dubai was raised from one half to three-fifths, the split into a first-year phase and a subsequent phase was made a condition of award with progression between them explicitly not automatic, and eligibility was restricted to a Principal Investigator not previously funded as such.

The central problem is not drafting but arithmetic. A staff of eleven is named in the source (a Principal Investigator, four Co-Investigators, a postdoctoral researcher, a doctoral student, the collaborating entity's team, a platform partner and external advisers), and a field technician is required although not named [Iyyunni, 2026]. Against that, [1,400,000.00:388,888.89](AED:USD) over thirty-six months is about [38,900.00:10,805.56](AED:USD) a month (derived) for people, instruments, computing, platform, travel, ethics, evaluation and publication together, which on the salary lines of Table 13.2 is roughly the cost of two research posts and nothing else. The scope as written does not fit the instrument, and no profiling of the cash flow makes it fit. What fits must therefore be decided deliberately and stated in the open. Table 13.2 is a budget that fits.

Four decisions make that budget possible, and each should appear in the text of the application rather than being buried in a spreadsheet.

Ten nodes are instrumented rather than twenty, and one zone rather than two. Dropping the second zone removes both the calibration baseline and the untreated control of the paired design, and it changes gate condition 6.2.3, which as written requires an hour-by-hour comfort map of both zones and which Chapter 11 (risk D3) scores as failed if the map covers one zone. The condition must therefore be restated to one zone in the application itself, and the control re-sited within that zone, rather than discovered by the funder at month twelve. A network of ten sited well, calibrated properly and carrying the radiation and globe instruments that a heat-stress index requires is worth more than twenty that cannot compute the index. No reference station is bought: the network is calibrated against the national meteorological and municipal reference data already being requested under the data letters, which converts a capital line into a partnership and releases the cost of a reference station into the sensing and evaluation lines. No platform is contracted in the first

Table 13.2: A three-year budget within the published envelope and the annual cap, in 2026 prices. Every line is this report's own bottom-up estimate, labelled derived; none is taken from the storyboard's budget section, whose figures are drafting targets. The total exhausts the envelope exactly, which is deliberate: an application that leaves the ceiling unclaimed while descoping the science has chosen the wrong thing to economise on.

Year 1 (March 2027 to March 2028)	AED
Sensing: two research-grade and eight calibrated low-cost nodes, mounts, power, connectivity, spares	180,000
Field workstation	30,000
Postdoctoral researcher, from month three	180,000
Doctoral student, twelve months	96,000
Ethics review, site permissions, field travel, consumables	14,000
Year 1	**500,000**
Year 2 (March 2028 to March 2029)	
Postdoctoral researcher and doctoral student, twelve months	312,000
Collaborating entity share: downscaling and optimisation	90,000
Computing: second summer of street-scale simulation	55,000
Decision interface, built in house	30,000
Field campaigns and partner meetings	13,000
Year 2	**500,000**
Year 3 (March 2029 to March 2030)	
Postdoctoral researcher and doctoral student, twelve months	312,000
Collaborating entity share	60,000
Baseline evaluation with controls: paired instrumented and uninstrumented corridor segments, counts, dwell time	70,000
Computing	30,000
Decision briefs, transferable method document, publication	28,000
Year 3	**500,000**
Total	**1,500,000**

year; the twin's interface is built in house in Year 2, and the platform partner named in the source moves from contractor to an option exercisable in the second phase, which the source itself permits. And the postdoctoral post begins in month three, after procurement, rather than on the first day. Because the grant does not buy the built structure, the Year 3 evaluation line of Table 13.2 buys a baseline rather than a before-and-after: paired instrumented and uninstrumented segments of the Al Ghubaiba corridor, with an untreated segment of comparable geometry serving as the control, measured so that the asset built later under tier T2 of Table 13.3 has something to be measured against.

Two compliance points should be settled before anything is submitted. The Dubai-spend requirement is tested against a budget whose largest capital line is imported instrumentation, AED 180,000 of a first year of AED 500,000, and the first year as costed above sits close to the threshold; procuring through a supplier registered in Dubai, or moving a tranche of nodes into the second year, resolves it, and whether the test applies per year or across the grant as a whole is a question to put to the funder in writing before submission. The eligibility restriction on a previously funded Principal Investigator should be checked against the named individual before any other work is done, because it is the one condition that makes an otherwise strong application inadmissible.

The application itself should run to twelve to eighteen pages and open on the decision being improved rather than on the science: the specific choice that a transport or utility authority faces, and what it currently decides without. It should state in one sentence what the grant does not buy, which is the built structure and everything after March 2030, and name the authority expected to carry each. It should attach a written sustainment commitment to the intervention choice as a condition of selection rather than as a hope, on the evidence that unsustained applications do not survive [Boehm et al., 1998]. And it should present the descopes as decisions taken, not omissions discovered. A reviewer scoring budget viability is looking for exactly that: a proposal that visibly knows what it costs.

A submission for the continental programme

The second programme is at an earlier rung than its own architecture suggests. Its financing ladder ends at development-bank and blended finance, and institutions of that kind commit at a scale the concept paper never names, because it carries no budget at all [Terhile et al., 2026]. A programme with no capability map, no data inventory and no costed hub model cannot be put to them, and doing so would waste the only introduction that matters.

The correct first instrument is the ladder's own second rung, a project-preparation grant, and the scope of that preparation is already written: the paper's list of twelve workstreams for its first quarter is a preparation programme in all but name. It comprises the appointment of a founding team and the definition of its charter, a stakeholder proposition for the continental institutions, the opening of a technical dialogue with GOOS-Africa, OceanPrediction and OPERA, a capability map across the coastal states, a data inventory, five to seven costed use cases, two or three demonstration geographies, responsible-use principles for security analytics, engagement with the relevant space and ocean partners, a first fellowship cohort, a published baseline and a technical workshop that fixes a twelve-month demonstration portfolio.

The document is therefore a concept note of ten to fifteen pages with one ask on its first page and one deliverable on its last. The deliverable is specific and should be named as such: an investment-grade programme document comprising the capability map, the data inventory, two or three pilots each with a named operational user and a measured decision outcome, a governance charter, and a cost breakdown per regional hub standardised at a common third level so that a continental total becomes computable [GAO, 2020]. That document is what makes the upper rungs addressable, and saying so converts a diffuse request into a defined piece of work with an end.

The persuasion is done by a comparator rather than by adjectives. A long-running intercontinental Earth-observation service programme already delivers coastal and marine services across African states through institutional consortia (grade: policy-fact) [Terhile et al., 2026] *[MarCOSIO, 2025]*[9], and its scope is narrower than what is now proposed, which carries seven governance layers, eight regional hubs, joint mission definition and a federated data platform besides. Anchoring the preparation ask against that programme makes a modest figure look measured instead of arbitrary, and makes the later rungs look inevitable instead of hopeful.

An investment case

The third document is different in kind from the other two, and it is the one that changes what the work is.

Understanding: Preparation spend

TIER 1: IN PLAIN WORDS

Before you spend a fortune building something, it is worth spending a little to find out where to put it.

TIER 2: IN PROFESSIONAL TERMS

The study, survey and design work bought before capital is committed, conventionally a small percentage of the eventual construction cost, and justified not by what it produces but by the misallocation it prevents in the much larger sum that follows.

TIER 3: IN TECHNICAL TERMS

Front-end expenditure whose return is the reduction in expected capital misallocation across the programme it precedes; it is worth buying while its cost remains below the expected value of the allocation error it removes, which at large capital scale it almost always does [GAO, 2020].

AS A MATHEMATICAL MODEL

Buy the study while $\delta C > S$, where C is the eventual capital programme, δ the share of it that would be misallocated without the study, and S the cost of the study.

Stated as a research project seeking a grant, the Dubai programme asks a university fund for a sum of the order of a million dirhams. Stated accurately, it is the preparation spend on a capital programme of a different order entirely. A city that intends to make its outdoor spaces usable in summer will commit capital in the hundreds of millions to billions of dirhams, as regional peers already have; preparation and design work on capital programmes conventionally runs at a low single-digit percentage of the eventual capital, which on a programme of that size is tens of millions of dirhams (indicative; the percentage is this report's own assumption rather than a figure from the archive, and [GAO, 2020] is cited for the principle, not for a share). The instrument that tells a city where to put its shade, how much of it, and what it will cost to keep, is not an academic curiosity competing for research funding. It is ordinary and cheap due diligence on a very large commitment, and it is currently being offered at a fraction of one per cent of the sum it protects.

That reframing requires a document that the numbered chapters do not yet contain, running to thirty to fifty pages and addressed to sovereign and institutional capital and to senior government rather than to a grant committee. Six things must be in it.

The liability must be stated in money rather than in degrees: the hours of outdoor activity that heat removes from the city, valued through the labour, tourism, retail and transit channels, using the method the archived city studies already establish [ILO, 2019]*[Arsht-Rock, 2022a]*[15]. The cost of

[15] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Bangkok,*

doing nothing must be projected against it. The cost of cooling at city scale must be built from the corridor upward, from cost per square metre to cost per corridor to a minimum cooling network. The operating liability must be given its own section, because it is the finding that a life-cycle analysis exists to produce: on the one comparator with published operating costs, running such an asset costs a large fraction of its capital cost every year, so that over a service life the cost of ownership exceeds the cost of construction several times over, and no party has yet been asked to carry any of it (grade: case-report with policy-fact content) *[EEA Climate-ADAPT, 2023]*[16]. The single reporting unit, cost per restored activity-hour, must be computed with a real numerator and a measured denominator, so that shade can be compared with district cooling, with mechanical cooling and with inaction on one scale. And an ask must appear against each rung of the ladder, with a named sponsor and a named owner, as Table 13.3 sets out.

Table 13.3: The ask ladder. Amounts are this report's own estimates in 2026 prices, labelled derived; sponsors are proposed, not agreed. Each tier is a distinct instrument with its own owner and its own period of analysis, and the upper tiers are the ones at which the financiers named in the concept paper actually commit.

Tier	What it buys	AED	Sponsor
T1	Research only, thirty-six months, descoped to fit	1.4 to 1.5 million	Dubai research grant
T1+	The same scope costed honestly	2.1 to 4.4 million	Grant, institutional and partner contributions
T2	Grant, built asset and its operation to 2033	6 to 15 million	Municipal and heritage capital budgets
T3	Second corridor, platform transferred, several assets	60 to 180 million	Agency procurement within the city research programme
T4	Minimum cooling network at city scale	700 million to 3 billion	Sovereign and blended finance
A1	Continental programme office, baseline, two to three pilots	14 to 33 million	Project-preparation grant and catalytic capital
A2	Regional hubs, federated data platform, validation	69 to 188 million	Blended finance and member states
A3	Repeatable continental services and infrastructure	900 million to 4 billion	Development bank, sovereign and blended

The evidence discipline carries across unchanged, and it is the reason a reader outside the group can act on any of these figures. Every quantity keeps its grade and its arithmetic, and a derived figure remains labelled derived however confident it looks. A document that establishes a problem in the billions and then asks for a fraction of a million has argued itself out of the room; a document that carries the liability, the ratio and the ask on its first three pages has asked a question the counterparty recognises.

Thailand, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-bangkok/

[16]European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

Sequence and dates

The grant application is governed by a calendar and goes first. The continental concept note is not date-bound and can follow within the same quarter, since most of its content is already drafted in the numbered chapters and needs only costing and compression. The investment case is the longest piece of work, because its opening section requires the existing heat-cost method to be applied to Dubai rather than merely cited, and that is the one genuinely new analysis any of these documents demands.

All three draw on the same body of work and none of them is this report. Two are short, specific and addressed to a decision-maker; the third is the argument this study has been building towards without yet stating. The material for all of them is assembled. What remains is to name the amounts, name the counterparties and ask.

A {A}

Source Archive and Evidence Register

Abstract

Evidence that cannot be re-checked is not evidence. Registered here is every document secured for the feasibility analysis on 22 September 2026 and held for review: sixty-seven Portable Document Format files in three groups, each with its extent, the evidence grade assigned to it and the chapters in which it is cited. It then registers what could *not* be archived (thirteen sources blocked by paywalls, bot walls or connection failures) and the bibliography entries that exist as metadata only, so that a reader can tell at a glance which claims in this report rest on a document the stakeholders hold and which rest on a title. It closes with the evidence horizon, a summary of the known defects in the sources, and the consequences of all of this for the two programmes. The register, not the prose, is the auditable object: if a number in Chapters 1 to 13 cannot be traced through these registers to a source of stated standing, it should not have been written.

Evidence that cannot be re-checked is assertion with a footnote. The stakeholders asked for a feasibility report whose numbers could be challenged, and a challenge is only possible if the challenger can obtain the same document the writer read, confirm that it is bit-for-bit the same document, and see what grade of evidence the writer thought it carried. This appendix provides those three things. It is deliberately mechanical: source, extent, grade, chapters. Nothing in it is an argument. Everything in it is a handle by which an argument made elsewhere in the report can be pulled apart.

A.1 What the register contains and why

The archive was assembled in a single retrieval campaign on 22 September 2026 and has not been added to since. It contains public documents only; copyright remains with the publishers, and the copies exist so that the five stakeholders and any reviewer they nominate can read exactly what the analysis read. The archive is organised in three groups, and the register follows that organisation because the groups differ in what they can support.

The first group, the primary feasibility studies and evidence, holds the seventeen documents closest to a measurement, a model run or a transaction: the two Phoenix cool-pavement reports [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024], the Abu Dhabi local climate zone study [Manandhar et al., 2019], the four Paris OASIS documents [Urban Innovative Actions, 2023, OASIS research team, 2024a, OASIS research team, 2024b], the two Cooling Singapore documents [Cooling Singapore 2.0, 2024, NSCC Singapore, 2024], the labour-economics series [ILO, 2019, Romanello et al., 2024], the Arsht-Rock methodology [Vivid Economics, 2022], the Holiday Climate Index paper [Scott et al., 2016], the African blue-economy strategy [AU-IBAR, 2019], the Digital

Earth Africa valuation [WEF and Digital Earth Africa, 2021], the Seychelles blue bond case study [World Bank, 2025] and the Singapore heat-stress advisory [NEA Singapore, 2023]. These are the documents from which a number may be extracted, each within the limit its grade allows.

The second group, the web captures, holds forty landing pages, press releases and news reports, secured as they stood on the retrieval date. They establish that a programme exists, that a launch happened on a date, that an agency said something. They are the report's evidence for policy facts and for cases; they are almost never evidence for an effect size. The Makkah and Arafat road-cooling reports *[Arab News, 2024]*[1] *[Gulf News, 2025]*[2], the Dubai Digital Twin Platform launch *[Dubai Media Office, 2026]*[3] and the Gabon enforcement narrative *[Global Fishing Watch, 2026]*[4] all sit here, and the register's grade column is what stops them being read as measurements.

The third group, the supporting references, holds the ten method, standards and policy documents that tell the report how to reason rather than what is true of the world: the spiral-development papers [Boehm and Hansen, 2000, Boehm et al., 1998], the sequential-decision framework [Powell, 2019], the cost and forecasting literature [Flyvbjerg, 2014, Flyvbjerg et al., 2016, Cantarelli et al., 2025, GAO, 2020], the life-cycle costing standard preview [ISO, 2017a], the Special Report outline [IPCC, 2024] and the United Nations Call to Action on Extreme Heat [UN Secretary-General, 2024].

A.2 How to read the registers

Each register names the source, its extent in pages, the evidence grade assigned to it and the chapters that use it, with 13 denoting the concluding chapter, Pref. the Preface, and front matter the foreword and executive summary; citations made inside the appendices themselves are not tracked. The grade is the judgement described in the Preface and rests on what the source's own text will support, not on the standing of its author. Sources in the first and third registers, and a few in the second, are bibliography entries: normative documents cited by digital object identifier, by standard designation, or by institution and date. The remainder, chiefly in the second register, are press releases, news reports and programme pages, which have no such identity and are therefore carried as footnotes with a live link at the point of use rather than as bibliography entries.

The third column, the evidence grade, is the only one that carries an opinion, and it is the column that does the work. The scheme has seven labels, used in exactly the sense given in the Preface and repeated here because this appendix is where a reader will look for it.

Understanding: Evidence grade

TIER 1: IN PLAIN WORDS

A grade is a short warning label saying how much weight a number can take. Some numbers come from someone measuring a treated street next to an untreated one on the same afternoon; some come from a computer model; some come from a press release repeating what an agency said. They are not the same kind of thing, and the label says which kind this is.

[1] Arab News, *Road-cooling tech will provide Hajj pilgrims with respite from heat*, Saudi Arabia section, 2024. https://www.arabnews.com/node/2530731/saudi-arabia

[2] Gulf News, *Hajj 2025: Saudi Arabia expands cooled, rubberised roads*. https://gulfnews.com/world/gulf/saudi/hajj-2025-saudi-arabia-expands-cooled-rubberised-roads-1.500139040

[3] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[4] Global Fishing Watch, *In Gabon, Satellite Intelligence Leads to the Seizure of Illegal Trawlers*, success story, July 2026. https://globalfishingwatch.org/success-story/in-gabon-satellite-intelligence-leads-to-the-seizure-of-illegal-trawlers/

TIER 2: IN PROFESSIONAL TERMS

Each archived document is assigned one of seven grades: *measured-with-controls*, *measured*, *modelled*, *projection*, *policy-fact*, *case-report* and *method-reference*. A document may carry more than one grade if its sections differ in kind, in which case the register records the grade of the material the report actually uses and the qualification is stated in the text of Chapter 4.

TIER 3: IN TECHNICAL TERMS

The grade encodes the inferential warrant, not the credibility of the publisher: *measured-with-controls* requires a contemporaneous untreated comparator; *measured* an instrumented observation without an intervention contrast; *modelled* a simulation calibrated on observations; *projection* a forward scenario under assumed parameters; *policy-fact* an institutional or contractual fact; *case-report* a single narrated episode without a counterfactual; *method-reference* a specification of procedure that asserts nothing empirical.

AS A MATHEMATICAL MODEL

Grade $g \in \{\text{mwc, m, mod, proj, pf, cr, mr}\}$ and an empirical claim inherits min over the grades of its supporting sources under the ordering $\text{mwc} \succ \text{m} \succ \text{mod} \succ \text{proj}$; the three categorical grades (policy-fact, case-report, method-reference) are carried as labels, not as ranks.

Table A.1 counts the archive by grade. The counts are the report's own arithmetic over the three registers that follow, and they are worth reading before the registers themselves, because they are the single most useful summary of the evidential position: two documents in sixty-seven support a controlled comparison, and both are about cool pavement in Phoenix.

Table A.1: Distribution of the sixty-seven secured sources by evidence grade. Counts are the report's own tally over Tables A.2, A.3 and A.4; derived. Shares are rounded to the nearest whole number and therefore sum to 99 rather than 100.

Grade	Files	Share	Where they dominate
measured-with-controls	2	3%	Phoenix cool pavement, Years 1 and 2
measured	4	6%	Abu Dhabi stations; OASIS paving laboratory; two Flyvbjerg cost datasets
modelled	6	9%	Arsht-Rock city economics; Cooling Singapore scenario; Lancet Countdown hindcast
projection	5	7%	ILO 2030 hours; Digital Earth Africa 2024 benefits; Green Riyadh; URB concept
policy-fact	24	36%	Agency launches, grant rules, advisories, declarations
case-report	15	22%	Press relays, enforcement episodes, programme self-reports
method-reference	11	16%	Spiral, sequential decisions, cost guidance, indices, standards
Total	**67**	**100%**	

A.3 Register I: primary feasibility studies and evidence

The seventeen documents of Table A.2 are stored at the root of the archive directory. Three entries deserve a note at the point of registration rather than in the chapters. The file `ASU_COPE_CoolPavement_Report_2025` carries a 2025 date in its name and its retrieval path, but its content is the Year-1 report of September 2021 covering the project period 15 July 2020 to 14 July 2021; the bibliography key `asu2021year1` is therefore dated by content, not by file name, and every residential surface-temperature result in this report belongs to that document and not to the Phase II report [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. Second, the two Paris OASIS documents from the Urban Innovative Actions initiative share one bibliography key, so a reader tracing a citation to `uia_oasis` should expect either file [Urban Innovative Actions, 2023]. Third, the Holiday Climate Index appears twice in the archive (as the Atmosphere paper and as a bibliographic record) but carries one key [Scott et al., 2016]; the record is retained because it settles the publication metadata that the paper itself renders ambiguously.

Table A.2: Register I: primary feasibility studies and evidence. Chapter 13 is the concluding chapter.

Source	Pages	Evidence grade	Chapters
ASU and City of Phoenix, 2021	65	measured-with-controls	2–6, 8, 9
AU-IBAR, 2019	48	projection	2–6, 8, 9, 13
Manandhar, 2019	15	measured	1–8, 13
Vivid Economics, 2022	20	method-reference	3–5
Cooling Singapore 2.0, 2024	1	modelled	3, 4
NSCC Singapore, 2024	7	case-report	3, 4
Scott, 2016	17	method-reference	2–5, 7, 13
ILO, 2019	98	projection	2–8, 13
Romanello et al., 2024	136	modelled	2–5, 13
OASIS research team, 2024a	8	method-reference	3–5
OASIS research team, 2024b	18	measured	3, 4
Urban Innovative Actions, 2023	5	case-report	3–6, 13
Urban Innovative Actions, 2023	4	case-report	3–6, 13
ASU and City of Phoenix, 2024	42	measured-with-controls	3–6, 9, 13
World Bank, 2025	2	policy-fact	3, 4, 6, 9
NEA Singapore, 2023	1	policy-fact	2–5, 13
WEF and Digital Earth Africa, 2021	31	projection	3–6, 9, 13

A.4 Register II: announcements and reporting

The forty files of Table A.3 were secured from live pages on the retrieval date; two were fetched as HTML and converted. Web captures age badly in a way that archived publications do not: the page may change, the press release may be re-dated, the page may be revised in place, without notice. Dating is the defence. Where a claim in the chapters rests on an announcement, the wording in the text names the source as the speaker (“per the Roads General Authority”, “per the Media Office”), rather than stating the claim in the report’s own voice *[Gulf News, 2025]*[2] *[Dubai Media Office, 2026]*[3].

Two capture groups carry a standing caution. The Green Riyadh material in the archive is the Saudipedia encyclopaedia article, not the Royal Commission page that the key `rcrc_greenriyadh` addresses; figures are therefore cited to `saudipedia_greenriyadh` and the Commission key is

reserved for the programme's existence *[Saudipedia, 2025]*[5] *[RCRC, 2025]*[6]. The Arsht-Rock city pages were captured after the main analysis because the city-level figures that the baseline analysis carried could not be found in the landing page or the methodology; any city figure used in the report is cited to its own city page *[Arsht-Rock, 2022a]*[7] *[Arsht-Rock, 2022b]*[8] *[Arsht-Rock, 2022c]*[9], and no Abu Dhabi figure is used at all, because the Abu Dhabi city page does not exist.

Table A.3: Register II: landing pages, press releases and news reports.

Source	Pages	Evidence grade	Chapters
African Union, 2026	8	policy-fact	1–6, 13
Arsht-Rock and Vivid Economics, 2022	10	modelled	3–5, 13
Arsht-Rock, 2022a	5	modelled	3, 4
Arsht-Rock, 2022b	7	modelled	3, 4
Arsht-Rock, 2022c	5	modelled	3, 4
Centre for Liveable Cities, 2024	3	case-report	2–4, 6, 13
Dubai Future Foundation, 2025	5	policy-fact	1, 2, 5, 6, 9, 13
ECMWF, 2024	6	method-reference	2, 3, 5
Digital Earth Africa, 2021	6	case-report	3–5
Dubai RDI Programme, 2025	10	policy-fact	1, 2, 5–7, 9, 13
Khaleej Times, 2025	10	policy-fact	3
Dubai Media Office, 2026	5	policy-fact	2, 3, 5–9, 13
Dezeen, 2024	8	policy-fact	2, 3, 6, 13
Copernicus Marine Service, 2024	9	case-report	3, 5
Global Fishing Watch, 2021	7	policy-fact	3
Global Fishing Watch, 2026	6	case-report	2–8, 13
Global Fishing Watch, 2025a	11	policy-fact	3–6, 8, 13
Global Fishing Watch, 2025b	5	policy-fact	3
IUCN, 2026	10	case-report	3, 4, 6
Saudipedia, 2025	4	projection	3, 4
Scott, 2016	3	method-reference	2–5, 7, 13
IPCC news, 2024	11	policy-fact	3
IPCC, 2026a	14	policy-fact	3, 5
IPCC, 2026b	7	policy-fact	3, 13
IUCN, 2022	9	case-report	3
IUCN, 2021	8	case-report	3
Arab News, 2024	10	case-report	3, 4
Gulf News, 2025	11	case-report	3, 4

[5] Saudipedia, *Green Riyadh Project*, encyclopaedia article (sources: Royal Commission for Riyadh City; Saudi Press Agency), 2025. https://saudipedia.com/en/green-riyadh-project

[6] Royal Commission for Riyadh City, *Green Riyadh Program*, project page, 2025. https://www.rcrc.gov.sa/en/projects/green-riyadh-project/

[7] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Bangkok, Thailand*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-bangkok/

[8] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: Los Angeles, United States*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-los-angeles/

[9] Adrienne Arsht-Rockefeller Foundation Resilience Center, *Hot Cities, Chilled Economies: New Delhi, India*, City report page, 2022. https://onebillionresilient.org/hot-cities-chilled-economies-new-delhi/

Source	Pages	Evidence grade	Chapters
Saudi Gazette, 2024	6	case-report	3, 4
MarCOSIO, 2025	8	policy-fact	3–6, 8, 9, 13
EEA Climate-ADAPT, 2023	8	case-report	3–6, 9
Powell, 2022	8	method-reference	7, 9, 13
MSE Singapore, 2023	14	policy-fact	2–6, 9
The National, 2026	2	policy-fact	2–6, 13
UNEP, 2021	1	policy-fact	3, 4
UNEP, 2024	4	policy-fact	3
UNEP, 2023	1	policy-fact	3, 4, 13
UN News, 2023	7	policy-fact	3
URB, 2025	28	projection	3
World Bank, 2018	1	policy-fact	3, 4, 9, 13

A.5 Register III: methods, standards and policy instruments

The ten documents of Table A.4 are the report's methodological backbone, and the register records one fact about them that matters more than their extent: the life-cycle costing standard is present only as a fifteen-page preview ending at clause 4.2.2, so no service life, discount rate or escalation rate anywhere in this report may be attributed to it, and none is [ISO, 2017a]. The Princeton framework appears twice, as the arXiv monograph [Powell, 2019] and as the laboratory web page that describes the published book [Powell, 2022]; the book itself is not archived, and the page is cited only for the framework's description. The cost-estimating guide is the largest file in the archive at 476 pages, and its own statement that its risk techniques do not capture catastrophic events is the reason force majeure is treated contractually in Chapter 11 rather than as a contingency percentage [GAO, 2020].

Table A.4: Register III: methods, standards and policy instruments.

Source	Pages	Evidence grade	Chapters
Boehm, 2000	49	method-reference	1, 2, 4–9, 13
Boehm, 1998	12	case-report	2, 4–9, 13
Cantarelli et al., 2025	20	method-reference	4, 9, 13
Flyvbjerg, 2014	30	measured	4, 5, 9, 13
Flyvbjerg, 2016	28	measured	4, 5, 9
GAO, 2020	476	method-reference	4, 8, 9
IPCC, 2024	7	policy-fact	3, 5, 13
ISO, 2017a	15	method-reference	Pref., 4, 9, 13
Powell, 2019	50	method-reference	2, 5–9
UN Secretary-General, 2024	20	policy-fact	3–6, 13

A.6 Sources that could not be archived

Thirteen sources resisted retrieval on 22 September 2026, for five kinds of reason: bot walls and access challenges at publishers and document servers, paywalls, connection resets, hosting behind file-sharing services that cannot be followed automatically, and documents that were never linked or whose page no longer resolves. Table A.5 records each one with its reason and with the chapters

that nonetheless refer to it. The rule the chapters follow is stated once here: an unarchived source may be named, its title given and its existence asserted, but no number, method or result may be attributed to it, and the text must say that it was not reviewed. The Singapore digital-twin paper [Aydt et al., 2026], the Dubai local climate zone paper [Bande et al., 2020], the Destination Earth climate-services paper [Hoffmann et al., 2023], the Great Blue Wall impact report [Great Blue Wall Initiative, 2026] and the blue-bond critique [Kilic, 2024] are all handled that way.

Table A.5: Sources that could not be archived, with the reason and the chapters that refer to them by title only.

Source	Key	Reason not archived	Chs.
Cooling Singapore, Digital Urban Climate Twin (summary PDF)	n/a	HTTP 403, bot protection	n/a
Aydt et al. 2026, *Tools to manage Singapore's heat*	`duct2026paper`	ScienceDirect bot wall	3, 4, 13
Hoffmann et al. 2023, *Destination Earth, a digital twin in support of climate services*	`destine2023 climate services`	ScienceDirect bot wall	3, 4
Kılıç 2024, *Seychelles blue bond: Indebting ecological restructuring of fisheries*	`seychelles 2024critique`	ScienceDirect bot wall; paywalled	3, 4, 13
Bande et al. 2020, *Local climate zones definition in relation to ENVI-met in the city of Dubai*	`dubai2020lcz`	IOP bot verification	2, 3, 4
Ministry of Human Resources and Emiratisation, midday-break release (2023)	`mohre2023 middaybreak`	Connection reset; The National 2026 report archived instead	3, 4, 13
GMES and Africa programme pages (About; MarCNoWA phase 2)	`gmes_about, marcnowa`	Connection reset	3, 4, 13
Royal Commission for Riyadh City, Green Riyadh page	`rcrc_ greenriyadh`	Connection reset; Saudipedia article archived instead	3–5, 13
United Nations Secretary-General's Call to Action web page	`un_ extremeheat_ page`	Access refused; the Call itself is held	n/a
UNEP *Beating the Heat* handbook and Global Cooling Pledge briefing	`unep2021 beatingheat, unep_global coolingpledge`	Cloudflare challenge on the document server; landing pages archived	3, 4, 13
Great Blue Wall Impact Report 2021–2025 (full PDF)	`gbw2026impact`	Hosted on file-sharing links; IUCN summaries archived	3, 4, 13
Arsht-Rock *Hot Cities, Chilled Economies* main report and Abu Dhabi city page	`arshtrock2022 hotcities`	Main PDF not linked; Abu Dhabi page returns not-found	3–5, 13
ISO 15686-5:2017, full standard	`iso15686`	Paid standard; preview archived	Pref., 4, 9, 13

A.6.1 Bibliography entries held as metadata only

A second, smaller class of entries is cited in the report without any archived capture and without any attempt at one, because the source is a standard, a classic text or a reference work whose content the report uses by definition rather than by quotation. Table A.6 lists them so that no reader mistakes their absence from the registers for an oversight. The thermal indices are the

sharpest case: the Universal Thermal Climate Index and the wet bulb globe temperature standard govern the comfort metric of the Dubai proposal, yet neither document is in the archive, and their threshold values must be taken from the standards themselves before the Letter of Intent fixes an index [Jendritzky et al., 2012, ISO, 2017b].

Table A.6: Sources cited from their publication details alone: standards, classic texts and reference works.

Key	What it is	Status	Chs.
`iso7243`	Assessment of heat stress using the wet bulb globe temperature index	Paid standard; thresholds to be read from the standard	2, 4, 5, 13
`jendritzky 2012utci`	The rationale for the Universal Thermal Climate Index	Journal article; categories to be read from the source	2–5, 13
`stewart2012lcz`	Local climate zones for urban temperature studies	Journal article; used for the standard class scheme	3, 4, 8
`iso31000`	Risk management guidelines	Paid standard; used for vocabulary and process	13
`iso22301`	Business continuity management systems	Paid standard; used for continuity vocabulary	13
`icc2020force majeure`	ICC force majeure and hardship clauses	Model clauses; used for contractual drafting	13
`howard1966voi`	Information value theory	Journal article; the correct source for value of information	2, 5–9
`bellman1957dp`	*Dynamic Programming*	Book; reached through later citations only	7, 9, 13
`powell2022rlso`	*Reinforcement Learning and Stochastic Optimisation*	Book not archived; laboratory page archived instead	7, 9, 13
`covey1989habits`	*The 7 Habits of Highly Effective People*	Book; source of the control/influence/concern rings	6, 13
`richardson2023 planetary`	Earth beyond six of nine planetary boundaries	Journal article; used for the boundaries framing	13
`beck2002words`	*Bringing Words to Life*	Book; source of the tiered vocabulary scheme	Pref.
`h3_restable`	H3 cell areas and edge lengths by resolution	Reference table; used for the place identifier	7, 8, 13
`w3c_provo`	PROV-O, the provenance ontology	Specification; used for the provenance identifier	1, 5, 8, 13
`vision2030_ greenriyadh`	Green Riyadh programme page	Not captured; no figure attributed to it	13

Key	What it is	Status	Chs.
`phoenix_coolpavement_program`	City of Phoenix cool pavement programme page	Not captured; programme facts taken from the two reports	n/a

A.6.2 The stakeholders' own working papers

Four cited documents are working papers of the stakeholders themselves rather than published sources; Table A.7 registers them. They are graded differently from public sources: the storyboard and the concept paper are design documents whose numbers are proposals, not findings [Iyyunni, 2026, Terhile et al., 2026], and the two civil-society documents supply values and policy context rather than technical input [Commons Cluster, 2024, Commons Cluster, 2023]. The remaining working material, the discussion record of September 2026 and the documents that carry no machine-readable text, has no bibliography entry and is not quoted here, because the discussion record in particular contains a personal telephone number and a private letter and must be redacted before it is circulated.

Table A.7: The stakeholders' own documents, cited as working papers.

Key	Document	Grade	Chs.
`iyyunni2026storyboard`	Storyboard for the rescoped Letter of Intent, Climate Intelligence & Decision Twin for Dubai	policy-fact (design document)	1–9, 13, front matter
`terhile2026abig`	Africa Blue Intelligence Grid strategic concept paper with addendum, 20 September 2026	method-reference (design document)	1–9, 13, front matter
`commons2024unity`	*Unity: a pathway to achieving a thriving planet*, post-HLPF 2024 edition	policy-fact	1, 6, 13
`commons2023buildingunity`	*Building unity as a prerequisite to achieving the SDGs*	policy-fact	1, 6, 13

A.7 The evidence horizon

Every source in the three registers was secured on 22 September 2026, and that date is the study's evidence horizon. A claim about the state of the world after it lies outside the evidence assembled here and outside what this study warrants. Where a programme was still running at that date, the registers record its position then, not its position now, and a reader returning to this work later should expect the fast-moving entries (grant calendars, platform launches, enforcement actions) to have moved on.

Two consequences follow for anyone re-checking a figure. The first is that a source may have changed since it was read: announcement pages in particular are revised in place, without notice and without a version history, so a figure quoted here may not appear in the same words at the same address. That is one reason the second register carries a lower evidential weight throughout. The second is that a published document is stable in a way a page is not, which is why every measured result in this study rests on a published document rather than on an announcement.

A.8 Known defects in the sources

Several sources contain defects that make particular numbers unsafe to quote, and they are recorded here because a reader who re-checks a figure and finds something different should know where the difficulty lies. Tables are the usual casualty. The labour-economics projection presents its country rows in a form in which the labels and the numbers can be read apart, so the row for the United Arab Emirates was reconstructed and checked against the printed regional totals; it is used in this study only with that qualification [ILO, 2019]. The local climate zone study's seasonal tables carry displaced season labels, so only season-agnostic ranges are quoted, and its district-area column carries an implausible unit [Manandhar et al., 2019]. The second Phoenix report is internally inconsistent between its narrative and its tables, so table values are preferred [ASU and City of Phoenix, 2024]. The OASIS paving paper's albedo table and the microclimate paper's instrument-uncertainty table cannot be read reliably [OASIS research team, 2024b, OASIS research team, 2024a].

Three further defects are matters of substance rather than presentation. One civil-society document carries no machine-readable text at all, so nothing in it can be verified and nothing in it is cited. The Great Blue Wall figures reach this study through a summary rather than through the underlying report, and are attributed accordingly. And several announcement pages carry figures that appear in no underlying document, which is why they are graded policy-fact and are never used to support a measured claim.

A.9 Implications for the programmes

What the register tells the two programmes

Sixty-seven archived documents yield two with a contemporaneous untreated comparator, and both concern a surface treatment that Chapter 4 recommends against ranking first for a pedestrian corridor. The evidential centre of gravity of the field is policy-fact and case-report, at fifty-eight per cent of the archive. That is the gap both programmes are designed to close, and it is also the standard they must meet.

For the Climate Intelligence & Decision Twin the register has three consequences. First, the Year-3 before/after campaign should be designed to produce the archive's third *measured-with-controls* entry, which means a paired control site instrumented from the start: Al Fahidi serves as the district control, an untreated corridor segment of comparable geometry as the local control, and the Paris OASIS design of a control station outside the intervention site is the template [OASIS research team, 2024a][*EEA Climate-ADAPT, 2023*][10]. Second, the month-12 gate should require that the project's own outputs be registered the way this appendix registers its sources: every sensor series, model run and comfort map carrying a date of origin and a grade, so that the dataset deliverable which the gate makes the fallback is auditable by a third party on the day the project closes [Iyyunni, 2026, Boehm and Hansen, 2000]. Third, the standards gap recorded in Table A.6 is a procurement item, not a footnote: the comfort index cannot be fixed until the two threshold documents are held, and the sensor specification in the storyboard lacks the black-globe and natural wet-bulb instruments that the wet bulb globe temperature requires, so buying the standards and the instruments belongs in the pre-award phase [Jendritzky et al., 2012, ISO, 2017b, NEA Singapore, 2023].

[10] European Environment Agency, *Paris OASIS schoolyard programme, France*, Climate-ADAPT case study, 2023. https://climate-adapt.eea.europa.eu/en/metadata/case-studies/paris-oasis-schoolyard-programme-france

For the Africa Blue Intelligence Grid the consequences differ in kind. The African sources in the archive are almost entirely policy-fact and case-report (a declaration *[African Union, 2026]*[11], a strategy with its projection method in unarchived annexes [AU-IBAR, 2019], a consortium page *[MarCOSIO, 2025]*[12], an enforcement narrative *[Global Fishing Watch, 2026]*[4], a continent-scale valuation built from single-site analogues [WEF and Digital Earth Africa, 2021]) and not one of them reports a cost, an uptake figure or a measured benefit for an ocean-intelligence service. The programme therefore cannot justify itself by citing the field; it must generate the evidence the field lacks. The practical implication is that the capability-mapping exercise of the first twelve months should be published with a register of exactly this form, that each pilot should record the decision it changed alongside its cost so that a cost per decision improved can eventually be computed, and that ABIG's own documents should be dated and secured from the outset, because a continental programme assembling data from many agencies will be asked, sooner than a city project will, to prove that the document it acted on is the document it received [Terhile et al., 2026].

Both programmes share one implication. The register should be kept current as sources are added, so that it and the evidence it describes never drift apart. A provenance record that is maintained manually will eventually be wrong, and a provenance record that is wrong is worse than none, because it transfers the reader's trust to a claim that no longer holds.

[11] African Union, *AU Member States adopt Luanda Declaration to accelerate Africa's blue economy transformation*, press release, 24 July 2026. https://au.int/en/pressrelease/au-member-states-adopt-luanda-declaration-accelerate-africas-blue-economy

[12] MarCOSIO consortium, *MarCOSIO – Marine and Coastal Operations for Southern Africa and the Indian Ocean*, GMES & Africa marine consortium site, 2025. https://marcosio.org/

B {B}

Data Dictionary and Identifier Schema

Abstract

This appendix is the field-level reference for the identifier schema introduced in Chapter 8 and used by the cost and gate arithmetic of Chapters 9 and 10. It gives every canonical key and every synthetic-record flag a name, a type, a format with an example, a cardinality, the architecture layer in which it is first written, a proposed steward, an indicative retention period and a statement of whether the field may ever carry a synthetic value. It then does the same for the four named metrics (Heat-Lost Activity Hours, Restored Activity Hours with its three efficiency ratios per dirham, per litre and per kilowatt-hour, Outdoor Tourism Capacity and the Minimum Cooling Network), giving each a formula, a unit, a reporting cadence and the identifiers it joins on. The schema is the report's own design; only the spatial indexing scheme and the provenance vocabulary are taken from published standards. Readers should treat this appendix as a procurement document: it is the specification that must be fixed before sensors are ordered, because identifiers inherit upward and cannot be added retrospectively.

This appendix exists because the most expensive mistakes in a measurement programme are made in its first month, in decisions that look administrative. Whether a sensor record carries the version of the calibration under which it was taken, whether a pedestrian count is stamped with a fifteen-minute bin or an hour, whether a generated series is marked as generated: none of these questions costs anything to answer correctly at the outset, and each of them, answered wrongly, silently destroys an indicator two years later. Chapter 8 established the architecture of the schema: twelve canonical keys in four families and seven flags that mark a record as synthetic, with one sticky boolean acting as the firewall between data that may train the Decision Twin and data that may stand as evidence at the month-twelve gate. What follows is the same schema written as a dictionary, field by field, so that a firmware engineer, a database administrator and an auditor can each read their own obligations out of one table.

B.1 How to read this dictionary

A data dictionary is a contract rather than a description. Each row states what a field is called, what kind of value it holds, what a valid value looks like, how many of it there are relative to the records it keys, which layer of the architecture first writes it, who is accountable for it, how long it is kept, and whether it may ever hold a value that was generated rather than observed. Nothing in the dictionary is negotiable after procurement without a versioned change to the dictionary itself, recorded with the same provenance machinery that governs the data.

Understanding: Field-level data dictionary

TIER 1: IN PLAIN WORDS

A list of every piece of information the system stores, saying what it is called, what it looks like, who looks after it and how long it is kept.

TIER 2: IN PROFESSIONAL TERMS

A controlled specification of the data model: for every attribute, its name, data type, permitted format, relationship cardinality, originating layer, accountable steward, retention period and admissibility status, maintained under version control and treated as a change-controlled artefact.

TIER 3: IN TECHNICAL TERMS

The schema-level contract between the sensing, data, semantic, model and decision layers of the architecture, in which each attribute's definition, domain and lineage obligations are fixed before instrumentation, so that every downstream indicator is a declarative join over named keys rather than an ad-hoc reconciliation.

AS A MATHEMATICAL MODEL

An indicator I is admissible only if it can be written as $I = f(\pi_K(D))$ for a projection π_K onto keys K all of which appear in this dictionary, evaluated over the view in which `is_synthetic` is false throughout the lineage closure [W3C, 2013].

Three columns need a definition before the table is readable, because each of them carries a decision rather than a fact.

B.1.1 Cardinality

Cardinality in this dictionary means the relationship between the field and the fact records it keys, not a count of rows. It is written in the conventional form: `1:N` where one value of the field keys many records, `N:1` where many values roll up into one parent, and `1:1` where the field is an attribute of a single record. No row of the dictionary states how many places, slots or sensors the corridor will contain, because those counts depend on a site survey that has not been done; the storyboard commits only to twenty microclimate nodes and one reference station on an approximately 800 m corridor from the Al Ghubaiba metro station to the Al Fahidi Historical Neighbourhood, with a further walkable waterfront of about one kilometre [Iyyunni, 2026].

Understanding: Cardinality

TIER 1: IN PLAIN WORDS

How many things of one kind belong to one thing of another kind: one street cell has many hourly readings, but every reading belongs to exactly one cell.

TIER 2: IN PROFESSIONAL TERMS

The multiplicity of a relationship in the data model, stated as one-to-one, one-to-many or many-to-one, which determines whether a join duplicates rows and whether an aggregate can be computed without double counting.

TIER 3: IN TECHNICAL TERMS

The functional dependency structure of the schema: a key with `1:N` cardinality to a fact table is a legitimate grouping variable for sums, while a many-to-many association must be resolved through an explicit bridge relation before any hours-based metric is aggregated over it.

AS A MATHEMATICAL MODEL

For key k and fact set F, aggregation $\sum_{r \in F} v(r)$ grouped by k is sound when the map $F \to \mathrm{dom}(k)$ is total and single-valued; where it is not (a pedestrian traversing several cells within one bin), the report allocates by dwell fraction and records the allocation rule as a versioned transformation.

B.1.2 Stewardship

The steward column names the party accountable for a field's definition, quality and release, not the party that owns the underlying asset. The assignments are the report's own proposal and depend on agreements that do not yet exist: the storyboard's partner list names the Roads and Transport Authority for station exit counts and bus and abra ridership, the Dubai Electricity and Water Authority for the Bur Dubai district-cooling load, and Dubai Culture together with Dubai Municipality for sensor permissions and one shade structure, with the National Center of Meteorology supplying reference station data [Iyyunni, 2026]. Geometry and base layers should be requested through the Dubai Digital Twin Platform launched on 2 July 2026, which holds more than 1,500 geospatial layers and three-dimensional models of more than 195,000 buildings and more than 280,000 infrastructure assets *[Dubai Media Office, 2026]*[1], so that the programme consumes an existing municipal layer rather than building a competing one. Where a steward is listed as the programme office, the accountability sits with the principal investigator's team at BITS Pilani Dubai and cannot be delegated to a platform vendor.

B.1.3 Retention

Retention periods in Table B.1 are the report's own indicative proposal. No archived source in this study states a retention rule, and the storyboard records only that ethics review, data-governance and publication lines were added to the budget [Iyyunni, 2026]; the periods below are therefore design assumptions to be confirmed with the ethics review and with each data provider before any agreement is signed. Two principles shape them. Keys and ledgers are kept permanently because they are what makes an old measurement defensible; person-adjacent detail is kept for the shortest period that supports the analysis, because the cohort key is deliberately coarse (visitor, resident, worker) and carries no personal data by construction.

Understanding: Retention

TIER 1: IN PLAIN WORDS

How long each kind of record is kept before it is deleted, and why.

TIER 2: IN PROFESSIONAL TERMS

A documented schedule setting a retention period for every class of data, balancing the evidentiary needs of the programme against data-protection obligations and storage cost, and approved by the ethics review before collection begins.

TIER 3: IN TECHNICAL TERMS

A per-attribute retention policy enforced in the data layer, in which raw instrument series, derived fields, provenance records and privacy ledgers carry different periods, and in which the deletion of a person-adjacent attribute must not invalidate the lineage closure of any published indicator.

[1] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

AS A MATHEMATICAL MODEL

Retention is a constraint on the admissible evidence set at time t: an indicator published at t must be recomputable from records whose retention window contains t, which is why every field feeding a gate condition carries a period at least as long as the programme plus its defence.

B.2 The field dictionary

Table B.1 is the dictionary proper. Group A covers the spatial and temporal keys, group B the subject keys, group C the instrument and model keys, group D the purpose keys, and group E the seven synthetic-record flags, of which the validity pair `valid_from` and `valid_to` takes two rows. The place key is an H3 cell index at two resolutions, resolution 11 with an average edge length of about 24.9 m and resolution 13 with an average edge length of about 3.6 m *[Uber, 2025]*[2]; these are the hexagonal equivalents of the district model at 25 to 50 m and the street model at 1 to 5 m that the storyboard commits to [Iyyunni, 2026], and of the nesting the Abu Dhabi precedent demonstrated between classification at 30 m, with a thermal band at 100 m, and street simulation at 3 m per pixel [Manandhar et al., 2019]. The provenance key follows the PROV-O vocabulary, in which a derived entity is linked to its inputs by `wasDerivedFrom`, to the activity that produced it by `wasGeneratedBy` and to a responsible agent by `wasAttributedTo` [W3C, 2013]. Every other scheme in the table is the project's own.

[2]Uber Technologies, *H3 Geospatial Indexing System: Table of Cell Areas and Edge Lengths by Resolution*, H3 documentation, 2025. https://h3geo.org/docs/core-library/restable

Table B.1: Field-level data dictionary for the canonical keys and synthetic-record flags. Types are logical types, not storage types. Cardinality is stated relative to the fact records the field keys. Stewards and retention periods are the report's own proposal and are subject to the ethics review and to agreements that do not yet exist; only the H3 scheme *[Uber, 2025]* and the provenance vocabulary [W3C, 2013] are taken from published standards.

Field	Type	Format and example	Card.	Source layer	Steward	Retention	Syn.?
A. Where and when							
`place_id`	string	H3 index, 15 hexadecimal characters; `8b…` at resolution 11, `8d…` at resolution 13 *[Uber, 2025]*	1:N	Data	Programme office	Permanent	Never
`place_res`	integer	11 or 13 only	1:N	Data	Programme office	Permanent	Never
`place_parent_id`	string	H3 index at resolution 11 containing a resolution 13 cell	N:1	Data	Programme office	Permanent	Never
`place_class`	string	Local climate zone label plus scheme version; classes are project composites and are not Stewart and Oke standard classes [Stewart and Oke, 2012, Manandhar et al., 2019]	N:1	Semantic	Programme office	Permanent	Yes
`timeslot_id`	string	ISO 8601 instant, Gulf Standard Time, plus 15-minute bin index; `2027-08-14T13:30+04/B54`	1:N	Data	Programme office	Permanent	Never
`timeslot_rule`	string	Versioned rule mapping 10-minute sensor records into 15-minute bins	N:1	Data	Programme office	Permanent	Never
B. What, and for whom							
`activity_id`	enum	`walk`, `wait`, `visit`, `work`	1:N	Semantic	Programme office	Permanent	Yes
`route_id`	string	Segment code plus direction; `AGB-FAH-07/NE`	1:N	Semantic	Programme office with the Roads and Transport Authority	Permanent	Yes
`cohort_id`	enum	`visitor`, `resident`, `worker`; no personal data by construction	1:N	Semantic	Programme office, under ethics review	Programme + 10 yr	Yes
C. By what means							

continued on the next page

Table B.1, continued from the previous page

Field	Type	Format and example	Card.	Source layer	Steward	Retention	Syn.?
`sensor_id`	string	Node serial; `DXB-NODE-07`; 20 nodes plus 1 reference station [Iyyunni, 2026]	1:N	Sensing	Programme office	Permanent	Never
`calibration_version`	string	Semantic version plus calibration date; `v2.1/2027-06-03`	1:N	Sensing	Programme office	Permanent	Never
`observation_value`	decimal	Air and surface temperature, humidity, wind, radiation at 10-minute cadence [Iyyunni, 2026]	1:1	Sensing	Programme office	Programme + 10 yr	Yes
`occupancy_count`	integer	Persons present in a place during one bin, by activity and cohort; the $n(p,t)$ of the Heat-Lost Activity Hours definition	1:1	Sensing	Programme office with the Roads and Transport Authority	Programme + 10 yr	Yes
`model_run_id`	string	Model name, version and configuration hash; `envimet-4.x/cfg-9ab3d1`	1:N	Model	Curtin Dubai and programme office	Programme + 10 yr	Yes
`provenance_id`	string	PROV-O entity, activity and agent triple [W3C, 2013]	1:1	All layers	Programme office	Permanent	Never
D. For what purpose							
`intervention_id`	string	Type, site and instance; `SHADE/AGB-07/01`	1:N	Decision Twin	Programme office with Dubai Municipality	Permanent	Never
`intervention_version`	string	Monotone version; `v3` distinguishes as-designed from as-built	1:N	Decision Twin	Programme office	Permanent	Yes
`scenario_id`	string	Weather year, demand path and price path; `WY2027/DEM-base/PR-high`	1:N	Decision Twin	Programme office	Programme + 10 yr	Yes
`cost_element_id`	string	Code in a breakdown structure of at least three levels, children summing to their parent [GAO, 2020]; `1.2.3`	1:N	Economics	Programme office	Permanent	Yes

continued on the next page

Table B.1, continued from the previous page

Field	Type	Format and example	Card.	Source layer	Steward	Retention	Syn.?
`decision_id`	string	Brief, recipient, date and outcome; `BR-2028-04/RTA/adopted`	1:N	Decision	Programme office	Permanent	Never
E. Synthetic-record flags							
`is_synthetic`	boolean	True if the record or any ancestor was generated; sticky through the lineage closure	1:1	All layers	Programme office	Permanent	n/a
`generator_version`	string	Program name and semantic version; `tracegen/1.4.0`	1:N	Model	Programme office	Permanent	n/a
`seed`	integer	64-bit random seed making a stochastic draw reproducible	1:1	Model	Programme office	Permanent	n/a
`source_dataset_id`	string	Observed dataset the generator was fitted or conditioned on; the field that makes leakage detectable	N:1	Model	Programme office	Permanent	n/a
`privacy_budget`	decimal	Differential-privacy parameter ε spent, with a cumulative ledger per release	1:1	Model	Ethics review and programme office	Permanent ledger	n/a
`fidelity_score`	decimal	Metric, variable and held-out window named; reported as a range, after the Abu Dhabi validation design [Manandhar et al., 2019]	1:1	Model	Curtin Dubai and programme office	Programme + 10 yr	n/a
`valid_from`	timestamp	ISO 8601; opens the period for which the record is asserted true	1:1	All layers	Programme office	Permanent	n/a
`valid_to`	timestamp	ISO 8601 or open; closes the period, superseding without deleting	1:1	All layers	Programme office	Permanent	n/a

Four rows repay a second reading. The `timeslot_rule` field exists because the sensors record every ten minutes while the schema stores fifteen-minute bins [Iyyunni, 2026]; the mapping between them is an analytical choice that changes values, so it is stored as a versioned rule rather than left to convention. The `intervention_version` field is what allows an as-designed geometry to be superseded by an as-built one without either being deleted, which is the precondition for the Year-3 comparison between predicted and measured change. The `observation_value` row is marked as admitting synthetic values because gap-filled series are unavoidable in a dusty climate; the point of the firewall is not that gaps are never filled but that filled values are never counted as uptime or as validation evidence. And `place_class` is marked synthetic-admissible because a classified zone is a model output, not a measurement: the Abu Dhabi study's classes were its own composites and did not correspond to the standard numbering [Stewart and Oke, 2012], which is exactly why a scheme version belongs inside the key [Manandhar et al., 2019].

B.3 Controlled vocabularies

Four fields take values from closed lists. Table B.2 gives them. Closed lists are what make an hours-based metric summable: if `activity_id` admits free text, then walking, strolling and promenading become three activities and the Heat-Lost Activity Hours of a place stop adding up.

Table B.2: Controlled vocabularies. The intervention types are those the storyboard names as candidates for the Minimum Cooling Network [Iyyunni, 2026]; the comfort bands are reported on two indices because the report recommends the Universal Thermal Climate Index for pedestrians while retaining wet-bulb globe temperature for comparability with the labour literature.

Field	Permitted values
`activity_id`	`walk`, `wait`, `visit`, `work`
`cohort_id`	`visitor`, `resident`, `worker`
`intervention_id` type prefix	`SHADE` (shade canopy), `TREE` (tree planting), `PAVE` (cool pavement), `MIST` (misting rest point), `ROUTE` (revised route)
`comfort_index`	`UTCI` (Universal Thermal Climate Index, the recommended pedestrian index [Jendritzky et al., 2012]) and `WBGT` (wet-bulb globe temperature [ISO, 2017b]), both stored, neither substituted for the other
`evidence_grade`	`measured-with-controls`, `measured`, `modelled`, `projection`, `policy-fact`, `case-report`, `method-reference`

Two cautions attach to the comfort vocabulary. The first is instrumental: the storyboard's node specification lists air temperature, humidity, wind, radiation and surface temperature [Iyyunni, 2026], and that set does not include the black-globe and natural wet-bulb instruments that a directly measured wet-bulb globe temperature requires. Reporting both indices therefore implies either additional instrumentation or an explicitly documented estimation procedure, and the choice belongs in the equipment line of the budget rather than in a footnote. The second is comparative: the labour literature that gives the metric its economic meaning is built on wet-bulb globe temperature, with productivity loss beginning at 24 to 26 °C and reaching 50 per cent at 33 to 34 °C for work at 300 W, and an annual denominator of 4,320 daylight hours (grade: method-reference) [ILO, 2019]. Singapore's public bands (low below 31 °C, moderate 31 to below 33 °C, high at 33 °C and above) are population guidance rather than occupational limits (grade: policy-fact) and must not be presented as standard thresholds [NEA Singapore, 2023][*MSE Singapore, 2023*][3].

There is also an acronym collision to record in the dictionary itself, because it will otherwise propagate into field names: in the Singapore literature OTC denotes outdoor thermal comfort

[3] Ministry of Sustainability and the Environment, Singapore, *New Heat Stress Advisory Launched To Guide Public On Minimising Risk Of Heat-Related Illnesses*, Press release, 24 July 2023. https://www.mse.gov.sg/latest-news/press-release-heat-stress-advisory/

[Centre for Liveable Cities, 2024][4], whereas throughout this report OTC denotes Outdoor Tourism Capacity. Field names in this schema use the expanded form.

B.4 The join path from a sensor record to a decision

Figure B.1 traces one record through the layers and names the keys it must already carry at each step. The figure is the operational argument for the dictionary: at no point downstream can a key be inferred that was not written upstream, so a node shipped without a calibration version produces a summer of data that cannot support the model-error condition of the month-twelve gate, whatever the platform does afterwards.

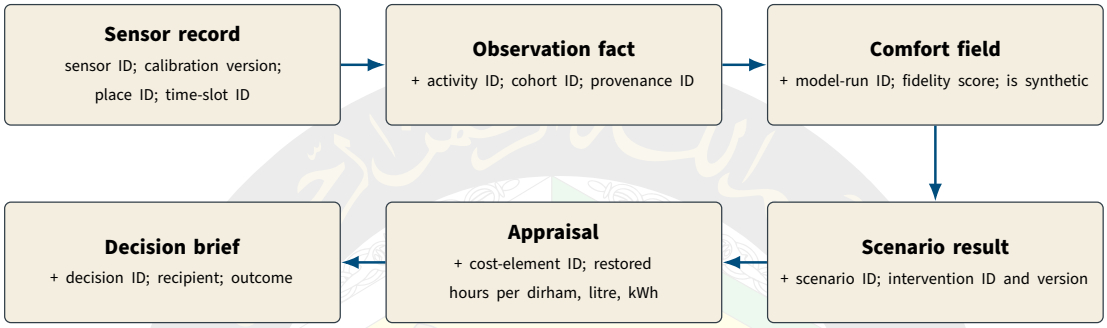


Figure B.1: The join path from an instrument reading to a decision, with the keys each step adds. Keys accumulate along the arrows, left to right across the top row and right to left across the bottom, and are never dropped; the firewall is evaluated at the third step, so that any downstream evidence claim can be restricted to the view in which the synthetic flag is false throughout the lineage [W3C, 2013].

B.5 The metric dictionary

Table B.3 defines the four named metrics of Chapter 7 in the same manner, with the three efficiency ratios of Restored Activity Hours set out as separate rows. Every one of them is a join over keys in Table B.1 and the controlled vocabularies of Table B.2, and the column headed "joins on" is therefore also a dependency statement: the metric named in a row is uncomputable if any key in that row was not written at collection time.

Table B.3: Metric dictionary. Formulas are given in compact form; the full statements, with constraints, follow in the text. Every row is reported on the observed-only view unless the accompanying note says otherwise, and each carries its evidence grade and the model-run identifier of any modelled input.

Metric	Formula (compact)	Unit	Cadence	Joins on
HLAH Heat-Lost Activity Hours	$\sum_t \Delta t \, n \, \mathbf{1}[U > c^*]$	person-hours per place per day	computed per 15-minute bin; reported daily and per summer season	`place_id`, `timeslot_id`, `activity_id`, `cohort_id`, `comfort_index`, `occupancy_count`

continued on the next page

[4]Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

Table B.3, continued from the previous page

Metric	Formula (compact)	Unit	Cadence	Joins on
RAH Restored Activity Hours	$\mathrm{HLAH}_0 - \mathrm{HLAH}_v$	person-hours per intervention version	per intervention version; per summer season; once for the before-and-after campaign	`intervention_id`, `intervention_version`, `place_id`, `timeslot_id`, `scenario_id`, `occupancy_count`
RAH per AED	$\mathrm{RAH}(v)/\mathrm{LCC}(v)$	person-hours per dirham; reciprocal reported as AED per restored hour	annually and at life-cycle close; percentile statement where costs are drawn	`cost_element_id`, `intervention_id`, `intervention_version`, `scenario_id`
RAH per litre	$\mathrm{RAH}(v)/W(v)$	person-hours per litre of water	annually and per summer season	`cost_element_id` (water), `intervention_id`, `intervention_version`, `scenario_id`
RAH per kWh	$\mathrm{RAH}(v)/E(v)$	person-hours per kilowatt-hour	annually and per summer season	`cost_element_id` (energy), `intervention_id`, `intervention_version`, `scenario_id`
OTC Outdoor Tourism Capacity	$\sum_h \kappa_p \, \phi_c(U_{p,h})$	visitor-hours per place	hourly per place; reported by day type and season	`place_id`, `timeslot_id`, `cohort_id`, `comfort_index`
MCN Minimum Cooling Network	$\operatorname{supp} \arg\min_x w^\top C(x)$	a set of intervention-location pairs, with its cost vector	once per capital epoch; re-solved at each gate	`intervention_id`, `place_id`, `route_id`, `cost_element_id`, `scenario_id`

The full statements are these. Heat-Lost Activity Hours counts, for place p over the slots T_d of day d, the hours in which a comfort threshold is exceeded, weighted by the population that would otherwise have been there:

$$\mathrm{HLAH}(p,d) \;=\; \sum_{t \in T_d} \Delta t \; n(p,t) \; \mathbf{1}\!\left[\, U_{p,t} > c^{*} \,\right],$$

with Δt the bin width, $n(p,t)$ the exposed population and $U_{p,t}$ the comfort index of Chapter 7 against threshold c^{*}. The indicator is the acceptability function ϕ_c of Equation (7.1) hardened to a threshold at c^{*}; where ϕ_c is estimated as a graded function, $\big(1 - \phi_c(U_{p,t})\big)$ replaces the indicator, and the two forms are never averaged. Restored Activity Hours is the same quantity differenced across an intervention version, $\mathrm{RAH}(v) = \mathrm{HLAH}|_{x=0} - \mathrm{HLAH}|_{x=v}$, evaluated at the same places and slots with the untreated site as the paired control. Outdoor Tourism Capacity sums, over the hours of a day, the physical capacity κ_p of a place weighted by an acceptability function $\phi_c : U \mapsto [0,1]$ that is zero above a cohort's heat-stress limit,

$$\mathrm{OTC}_{p,c} \;=\; \sum_{h \in H} \kappa_p \, \phi_c\!\big(U_{p,h}\big),$$

and the Minimum Cooling Network is the support of the solution to

$$x^{\star} = \arg\min_{x \in \{0,1\}^{n}} w^{\top} C(x) \quad \text{subject to} \quad \Delta \mathrm{RAH}_{p}(x) \geq R_{p}^{\star} \;\; \forall p \in \mathscr{P}_{\text{priority}},$$

with $C(x)$ the cost vector over capital, energy, water and maintenance and w the weights agreed with the partner agencies.

Three properties of this set of definitions deserve to be recorded in the dictionary rather than argued in the text. First, $n(p,t)$ is the quantity most likely to be synthetic in the early years, because the Roads and Transport Authority series covers station exits and ridership rather than street occupancy [Iyyunni, 2026]; Heat-Lost Activity Hours will therefore be published in two versions, one restricted to observed counts and one using imputed occupancy, distinguished by the synthetic flag and never averaged together. Second, the denominators of the three efficiency ratios are assembled from cost elements, and the requirement that a breakdown structure have at least three levels with children summing to their parent is what makes them auditable; where a denominator is drawn from a distribution rather than observed, the result is a percentile statement, and the 50th, 70th and 80th percentile language of the cost-estimating literature is the right vocabulary for reporting it [GAO, 2020]. Third, the ratios are life-cycle quantities in the sense of service-life planning [ISO, 2017a], which means that the service lives used in the denominator (this report's own indicative assumptions, not values taken from any archived source) must be stated in the same table as the result. The standing warning is the Phoenix reflectivity series: solar reflectivity measured at 33 to 38 per cent at installation had fallen to 19 to 30 per cent within ten months (grade: measured-with-controls) [ASU and City of Phoenix, 2021], so a performance parameter recorded once and never re-measured will misstate every ratio that depends on it.

Understanding: Restored Activity Hours per dirham, litre and kilowatt-hour

TIER 1: IN PLAIN WORDS

How many extra hours of comfortable outdoor use each dirham, each litre of water and each unit of electricity buys.

TIER 2: IN PROFESSIONAL TERMS

Efficiency ratios that put a cooling intervention's benefit over its life-cycle cost and over the two scarce physical resources it consumes, so that shade, trees, surfaces and misting can be compared on one page rather than argued about by preference.

TIER 3: IN TECHNICAL TERMS

Quotients of a differenced hours-metric by discounted life-cycle cost and by metered water and energy consumption, computed per intervention version over a stated analysis period, with the denominator assembled from a three-level cost breakdown structure and reported as a percentile where cost inputs are distributions [GAO, 2020, ISO, 2017a].

AS A MATHEMATICAL MODEL

$\mathrm{RAH}_{\mathrm{AED}}(v) = \mathrm{RAH}(v)/\mathrm{LCC}(v)$, $\mathrm{RAH}_{\mathrm{L}}(v) = \mathrm{RAH}(v)/W(v)$ and $\mathrm{RAH}_{\mathrm{kWh}}(v) = \mathrm{RAH}(v)/E(v)$; the reporting unit adopted by this report is the reciprocal of the first, dirhams per restored activity-hour.

B.6 What is design and what is evidence

The status of this appendix must not be misread. The schema is a design. No archived source in this study tests it, no archived source states a retention rule, and the stewardship column describes agreements that do not yet exist. Of everything in Table B.1, only two elements are taken from published standards: the H3 resolutions and their edge lengths of about 24.9 m and about 3.6 m *[Uber,*

2025][2], and the PROV-O relations [W3C, 2013]. Two further elements are borrowed as practice rather than as fact: the breakdown-structure discipline [GAO, 2020] and the held-out validation design whose reported errors (1.33 to 2.71 °C uncorrected, 0.27 to 1.62 °C after empirical correction, and 0.24 to 0.68 °C when a model calibrated on one district predicted another of the same class (grade: measured)) show what a fidelity score looks like when it is done properly [Manandhar et al., 2019].

The cost of the schema has not been estimated. It falls in firmware, in platform engineering and in analyst discipline, and it belongs in the cost elements of Chapter 9 rather than being treated as free. Against that cost stands the argument of Chapter 7: the first year is the purchase of information about a poorly understood state, in the sense of information value theory [Howard, 1966], and identifiers are what convert that purchase into an asset. A summer of records keyed to place, time, sensor and calibration remains a usable deliverable if the gate fails and the project closes with the dataset and platform as its outputs [Iyyunni, 2026]; the same summer stored without calibration versions is a collection of numbers nobody can defend.

Schema risk

The single failure mode this appendix is written to prevent is retrospective keying. Identifiers inherit upward: a field omitted at the sensing layer cannot be recovered at the decision layer, and no amount of platform engineering will reconstruct the calibration under which a reading was taken. The mitigation is procedural and cheap: the dictionary is frozen before the equipment order, every deviation is a versioned change to the dictionary, and acceptance testing of the nodes includes a check that each delivered record carries the fields of group C in Table B.1.

B.7 Implications for the programmes

For Dubai, three consequences follow directly and all of them fall before the grant starts. The dictionary must be frozen before the equipment order, because the storyboard places roughly [250,000.00:69,444.44](AED:USD) of sensing capital in the first year of an approximately [1,400,000.00:388,888.89](AED:USD), 36-month budget [Iyyunni, 2026], and the Dubai Research, Development and Innovation grant rules cap the annual budget per project at [500,000.00:138,888.89](AED:USD) within an envelope of [500,000.00:138,888.89](AED:USD) to [1,500,000.00:416,666.67](AED:USD), with at least 50 per cent of the funding to be spent within Dubai *[Dubai RDI Programme, 2025]*[5]; on the report's own arithmetic, [1,400,000.00:388,888.89](AED:USD) over 36 months averages about [466,667.00:129,629.72](AED:USD) a year, which sits just under that cap and leaves no headroom for a mid-project re-instrumentation caused by a missing field. The second consequence is that the four month-twelve gate conditions are each a query over this dictionary (uptime over `sensor_id` and `timeslot_id`, model error over `model_run_id` and `calibration_version`, the hour-by-hour comfort map over `place_id` and `timeslot_id`, and written confirmation of usefulness over `decision_id`), so the dictionary is the gate's machine-readable form and should be circulated to the Roads and Transport Authority and the Dubai Electricity and Water Authority with the data-sharing request, not after it. The third is that geometry and base layers should be drawn from the Dubai Digital Twin Platform *[Dubai Media Office, 2026]*[1], which converts a platform-build risk into an integration task and makes `place_id` the one key that must be reconcilable with a municipal layer the programme does not control.

For the Africa Blue Intelligence Grid the same dictionary is read at a different scale but with the same discipline. The concept paper's federated model names discovery, metadata interoperability,

[5] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

common standards, controlled access, provenance, quality assurance and reproducibility as the properties African institutions must retain when they share data [Terhile et al., 2026], and those seven properties are the ocean statement of Table B.1: a continental grid whose members keep custody of their data can only interoperate through agreed keys and agreed provenance. Three adaptations are needed. The place key becomes a marine cell index with a named vertical coordinate, because an ocean state has depth where a corridor does not. The cohort key is replaced by a vessel and fleet key, with the ethical constraint stated in the concept paper (surfacing patterns for trained analysts, preserving uncertainty, documenting provenance, minimising unnecessary personal data and maintaining auditable trails [Terhile et al., 2026]), carried in the same fields that carry the privacy budget here. And the decision key becomes the grid's primary outcome measure rather than a secondary one, because no archived source in this study reports a cost, an uptake figure or a benefit measurement for the existing African marine services, which makes "decisions improved per dollar" a quantity the grid will have to originate rather than inherit. In both programmes the conclusion is the same and it is not a technical one: the cheapest work in the whole life cycle is agreeing, in advance and in writing, what each field is called.

Glossary

The glossary is grouped by subject rather than run together as a single alphabet, so that a reader who knows one domain but not another can find a term by the family it belongs to: heat and comfort, cities, oceans, data, decisions, money, delivery, policy and institutions. Within each of the nine groups the entries are alphabetical, and acronyms are listed as cross-references to the spelt-out term rather than defined twice. Terms are defined as this report uses them; where a term carries a wider meaning elsewhere, the narrower sense intended here is stated, and where a definition rests on an argument made in the body, the chapter is named. Page numbers for these definitions appear in bold in the Index, so the Index and the Glossary may be used as one instrument.

G.1 Climate and thermal comfort

Air temperature The temperature of the air measured in shade at a stated height, usually two metres. It is the quantity most often reported in climate statistics and the least informative about how hot a person feels, because it excludes radiation and wind. Chapter 4 shows interventions that move surface temperature by whole degrees while moving air temperature by a fraction of one.

Black-globe thermometer A thermometer housed inside a matt black sphere, used to capture the radiant heat load falling on a body. It is one of the three instruments required to compute wet-bulb globe temperature, with a natural wet-bulb and a shaded air thermometer. Chapter 5 records its absence from the Dubai sensor specification as an equipment gap that must be closed before any occupational threshold can be reported.

Comfort index A single number combining several environmental variables into an estimate of thermal comfort or heat stress. This report adopts the Universal Thermal Climate Index for pedestrians and reports wet-bulb globe temperature alongside it, so that the same hour can be read against both public-realm and occupational guidance.

Cool pavement A road or footway surface treated to reflect more solar radiation than conventional asphalt. Controlled field evidence summarised in Chapter 4 shows that it lowers surface temperature substantially while raising the radiant load on a person standing on it. That asymmetry is why the report ranks it below shade for pedestrian routes.

Dust storm An episode of airborne dust that reduces visibility, degrades air quality and suppresses outdoor activity independently of temperature. Chapter 7 treats dust events as exogenous information within the decision model rather than as noise, because they change the same activity hours that heat removes.

Heat action plan A published municipal or national protocol setting out who does what at each level of heat warning: advisories, cooling centres, work restrictions and outreach to exposed groups. Chapter 12 treats it as the existing instrument into which a validated loss metric can be inserted, rather than as a new instrument to be created.

Heat-Lost Activity Hours The report's loss metric: the hours of intended outdoor activity (walking, waiting, visiting, working) that heat prevents at a given place and hour, summed over a stated period. It is the quantity the Dubai proposal exists to measure and the unit from which Restored Activity Hours and the cost-effectiveness ratios are derived. It is defined formally in Chapter 7 and its candidacy as a policy indicator is argued in Chapter 12.

Heat-stress override A rule that forces a comfort or suitability score to the unacceptable class when a physiological threshold is crossed, whatever the composite score would otherwise report. Chapter 4 shows that without such an override the established tourism indices cannot classify a Gulf summer afternoon as unacceptable, which is the central argument for Outdoor Tourism Capacity.

Local climate zone A classification of urban land into standard types by built form and land cover, used to structure thermal analysis and to make measurements comparable between cities. The Abu Dhabi study discussed in Chapter 3 defines city-specific composite classes that do not correspond to the standard numbered scheme, a distinction this report preserves rather than silently reconciles.

Mean radiant temperature The uniform surface temperature of an imaginary enclosure that would give a person the same net radiant heat exchange as the real, non-uniform environment. It is the component of thermal comfort that reflective surfaces tend to worsen and that shade improves, and it is therefore the variable that separates the two families of intervention in Chapter 4.

Midday break (UAE) The national rule prohibiting work in direct sunlight between 12:30 and 15:00 from 15 June to 15 September. Chapter 4 treats it as a policy fact and as evidence that a government already encodes hours lost to heat in law. The derived figure of roughly 232.5 hours per worker per summer is this report's own arithmetic and an indicative upper bound, not a measured loss.

Misting The release of fine water droplets to cool air by evaporation in a confined outdoor space. Chapter 4 finds only case-report evidence for it, from the holy sites, where the relayed hospital study is confounded. Because it cools by consuming water, the report insists that Restored Activity Hours be reported per litre of water as well as per dirham.

Natural wet-bulb temperature The temperature read by a thermometer with a wetted wick exposed to ambient air movement and radiation. It is the dominant term in wet-bulb globe temperature and the reason that index responds to humidity in a way that air temperature alone does not.

Reflectivity (albedo) The fraction of incoming solar radiation that a surface returns. Raising it cools the surface and the air above it but redirects radiation towards anyone standing on the surface, which is the trade-off Chapter 4 records from the only controlled pavement trial in the archive.

Shade canopy An engineered overhead structure that removes direct solar radiation from a route or space. It is the intervention the Dubai storyboard expects to build in Year 3, the one the archived evidence supports most directly, and the anchor of the Minimum Cooling Network described in Chapter 2.

Surface temperature The temperature of a material surface, typically measured by infrared sensing from the ground or from satellite. It responds strongly to treatment and is therefore the figure most often quoted in press accounts of cooling projects, which Chapter 4 warns is not a measure of human comfort.

Universal Thermal Climate Index A physiologically based index expressing the combined effect of air temperature, mean radiant temperature, humidity and wind as an equivalent temperature. The report adopts it as the pedestrian comfort index and pairs it with an occupational index so that the same hour can be assessed for a visitor and for a worker.

Urban heat island The tendency of an urban area to be warmer than its rural surroundings, chiefly at night, because of heat stored in built fabric and waste heat released by human activity. Chapter 3 warns that citywide island averages conceal precisely the street-level variation that a corridor intervention is meant to address.

Wet Bulb Globe Temperature An occupational heat-stress index combining natural wet-bulb, black-globe and air temperatures in fixed proportions. It underpins the Singapore heat-stress advisory and the labour-productivity studies used in Chapter 4, and it requires instruments that the current Dubai sensor specification lacks.

HLAH See *Heat-Lost Activity Hours*.

LCZ See *Local climate zone*.

MRT See *Mean radiant temperature*.

UHI See *Urban heat island*.

UTCI See *Universal Thermal Climate Index*.

WBGT See *Wet Bulb Globe Temperature*.

G.2 Urban systems, mobility and tourism

Al Fahidi Historical Neighbourhood The heritage core of the Dubai testbed: narrow lanes, coral-stone walls and wind towers, already passively cooled by its own geometry. The storyboard uses it as a calibration baseline with no interventions, which makes it in effect the control site of the experiment described in Chapter 5.

Al Ghubaiba corridor The route of roughly 800 metres between Al Ghubaiba metro station and the heritage core that the Dubai proposal designates as its intervention site. It is the unit on which the Minimum Cooling Network, the sensor plan and the Year-3 before-and-after study are all defined.

Citywide 100 m product The stretch goal of a hundred-metre-resolution comfort surface for the whole emirate, distinguished in Chapter 7 from the corridor product that the programme commits to deliver. Chapter 3 warns that it must be reported per district rather than as a city mean, or it will reproduce the island-average problem it is meant to solve.

Climate-resilient urban cells The wider concept recorded in the discussion notes: organising a city into interconnected low-exposure areas joined by low-exposure mobility corridors, so that essential activity remains reachable without continuous outdoor exposure. Chapter 2 treats it as the long-horizon ambition that the corridor pilot is a first instance of.

Comfort map An hour-by-hour, spatially explicit surface of thermal comfort over a route or district. Producing one for both testbed zones is the third of the four conditions of the month-12 gate set out in Chapter 7.

District cooling Centralised production of chilled water distributed to many buildings through a pipe network. It is the mechanism by which outdoor heat becomes an energy and water load, and the reason cooling demand appears in Chapter 7 as a constraint shared with the utility rather than as an externality.

Dubai Creek The tidal inlet along which the testbed sites lie, and the historic reason for the concentration of heritage fabric, waterfront walking and visitor activity in the study area described in Chapter 5.

Dubai Walk Master Plan The emirate's pedestrian-network plan, comprising new and rehabilitated walkways together with the bridges and underpasses that connect them. It is the demand-side programme that a Minimum Cooling Network would serve, and the reason Chapter 6 treats the roads authority as the decisive audience for the metric.

Green Riyadh The Saudi programme of large-scale urban planting with stated targets for temperature reduction and green-space provision per inhabitant. Chapter 3 uses it as the regional comparator for planting at scale and notes that its published targets are projections rather than measured outcomes.

Holiday Climate Index A composite index rating the climatic suitability of a destination for tourism, built from stated tourist preferences for temperature, precipitation, cloud and wind. Chapter 4 shows that it cannot classify a Gulf summer afternoon as unacceptable, which is the argument for a heat-stress override and for Outdoor Tourism Capacity.

Labour productivity Output per hour worked, used here as the channel through which heat becomes an economic loss. The international estimates summarised in Chapter 4 are modelled rather than measured, and the report keeps that grade attached to them wherever they are quoted.

Low-exposure mobility network A connected set of shaded, cooled or indoor links along which a person can reach essential destinations without sustained heat exposure. It is the mobility counterpart of climate-resilient urban cells and the organising idea behind the Minimum Cooling Network.

Minimum Cooling Network The smallest set of shade, planting, material and rest-point interventions that makes a priority route or space comfortable enough to use. It is the output of the multi-objective optimisation described in Chapter 2 and the object of the capital-decision problem stated in Chapter 7.

Outdoor activity Intended use of outdoor space (walking, waiting, visiting, trading, working), measured in person-hours and disaggregated by cohort. It is the denominator of the report's metrics: heat is counted not by degrees but by the activity those degrees prevent.

Outdoor Tourism Capacity A measure of how many people can comfortably use a destination, for how long and at which hours, under given conditions. Chapter 4 positions it as an hourly, spatially explicit successor to the Holiday Climate Index. Note the collision of acronyms: in the Singapore sources the same three letters denote outdoor thermal comfort.

Paris OASIS The Paris programme converting schoolyards into cool islands, whose paired control-station evaluation design Chapter 3 recommends copying for the Year-3 before-and-after study. It is also the only comparable programme in the archive with a published annual operating cost.

Restored Activity Hours The activity hours returned by an intervention, that is, the reduction in Heat-Lost Activity Hours attributable to it. Expressed per dirham, per litre of water and per kilowatt-hour, it is the report's single measure of cost-effectiveness and the bridge between Chapter 7 and Chapter 10.

Testbed The pair of sites on which the Dubai proposal is to be proved: the heritage core as baseline and the metro corridor as intervention site, together about one kilometre of walkable

waterfront and lanes. Chapter 5 argues that the pairing, not the instrumentation, is what makes the result defensible.

Tourism Climate Index An earlier composite index of climatic suitability for tourism, superseded for this purpose by the Holiday Climate Index and, in this report's proposal, by Outdoor Tourism Capacity.

HCI See *Holiday Climate Index*.

MCN See *Minimum Cooling Network*.

OTC See *Outdoor Tourism Capacity*.

RAH See *Restored Activity Hours*.

TCI See *Tourism Climate Index*.

G.3 Ocean, Earth observation and the blue economy

Africa Blue Intelligence Grid The proposed continental capability programme that connects existing African ocean-observation, forecasting and Earth-observation capabilities into a decision-oriented architecture across five domains. Its defining constraint, stated in its own addendum and restated throughout Chapter 2, is that it is an integration and application layer rather than a new observing or forecasting system.

Automatic identification system The transponder system by which vessels broadcast their identity, position and course. Gaps in its coverage (vessels that go dark) are a principal signal in maritime-security analytics and the reason radar and optical imagery are needed alongside it, as Chapter 8 explains.

Blue bond A debt instrument whose proceeds are earmarked for marine conservation and the sustainable use of ocean resources. Chapter 3 treats it as a demonstration that ocean outcomes can be financed, and its academic critique as a caution about what such instruments do not by themselves achieve.

Blue economy Economic activity derived from oceans, seas, coasts, rivers and lakes, treated by the African Union as a strategic domain requiring standardised accounting and shared data systems. Chapter 12 identifies that accounting requirement as the opening through which the programme's indicators can enter continental policy.

Capability map An inventory of what observing, forecasting, modelling, data and training capability exists, where it sits, who operates it and where the gaps are. The addendum to the concept paper makes it the programme's first deliverable, and Chapter 5 agrees that nothing should be procured before it exists.

Dark vessel A vessel operating without a working or truthful position transponder. Detecting one requires fusing radar or optical imagery with the absence of an expected transponder signal, which is why Chapter 8 treats dark-vessel detection as the clearest case for synthetic training data under a strict validation firewall.

Earth observation The systematic collection of data about the planet's surface, atmosphere and oceans from satellites, aircraft and in-situ instruments. Chapter 2 stresses that the programme's contribution lies in turning such observations into decisions, not in acquiring new sensors.

Great Blue Wall The Western Indian Ocean initiative creating a connected network of conserved and restored seascapes. Chapter 3 cites it as the regional precedent for programme-level

reporting across many jurisdictions and as evidence that regional co-ordination of this kind is already accepted politically.

Illegal, unreported and unregulated fishing Fishing that evades licensing, reporting or conservation rules. It is the use case in which satellite intelligence has most clearly changed enforcement outcomes, and Chapter 5 therefore proposes it as the strongest candidate for an early demonstration.

Large Marine Ecosystems Coherent ocean regions defined by bathymetry, hydrography, productivity and trophic relationships. Chapter 7 adopts them as the natural units for regionalising the programme, in preference to imposing a single continental structure on very different seas.

Luanda Declaration The African Union declaration of 24 July 2026 recognising oceans, seas, rivers and lakes as strategic assets and committing member states to standardised blue accounting, stronger data systems and co-ordinated maritime surveillance. Chapter 12 treats it as the mandate the programme should attach itself to rather than seek to replace.

Open data cube An indexed, analysis-ready store of satellite observations that allows time-series analysis over large areas without handling individual scenes. It is the standard pattern by which national Earth-observation holdings become usable by non-specialists.

Port State Measures Agreement The international instrument obliging ports to deny entry or service to vessels suspected of illegal fishing. Chapter 12 identifies it as an existing enforcement channel with an owner and a reporting cycle, and therefore as a realistic destination for the programme's vessel intelligence.

Regional hub One of the distributed centres proposed to host the programme's regional operations, aligned to marine systems rather than to political boundaries. Chapter 10 notes that a cost breakdown per hub is the figure the concept paper leaves unstated and the one funders will ask for first.

Seychelles blue bond The first sovereign bond issued to finance marine conservation and sustainable fisheries, used in Chapter 3 as the reference case for blue finance and, in the light of its academic critique, as a caution about attributing outcomes to instruments.

Synthetic aperture radar Satellite or airborne radar that synthesises a large antenna from the platform's motion to produce high-resolution imagery irrespective of cloud or darkness. It is the observation type on which vessel detection depends when optical imagery and transponder data fail, as Chapter 8 describes.

ABIG See *Africa Blue Intelligence Grid*.

AIS See *Automatic identification system*.

EO See *Earth observation*.

IUU See *Illegal, unreported and unregulated fishing*.

SAR See *Synthetic aperture radar*.

G.4 Data, semantics and synthetic data

Bias correction The adjustment of model output so that its statistical distribution matches observations over a reference period. Chapter 8 classifies the corrected values as synthetic and requires them to be flagged, because a corrected series is no longer an observation.

Data dictionary A field-level specification of every identifier and metric: name, type, format, cardinality, source layer, owner, retention rule and whether the field may be synthetic. Appendix B holds this report's dictionary, derived from the schema of Chapter 8.

Data governance The arrangements that decide who may see, change, publish and retire data, and who answers for those decisions. Chapter 12 argues that governance must be written before collection begins, because retrofitting consent to an existing dataset is rarely possible.

Data lineage The recorded path from raw observation through every transformation to a reported figure, so that any number can be traced back to its source. Closure of lineage (no untraceable value anywhere in the chain) is one of the auditable properties Chapter 8 asks the ethics board to test.

Data sovereignty The principle that data about a jurisdiction remains under that jurisdiction's ownership and control. Chapter 2 treats it as a design constraint on the continental architecture rather than a legal formality, and it is the reason a federated hub is proposed instead of a central database.

Differential privacy A formal guarantee limiting how much any single record can influence a published result, quantified by a privacy budget. It is the mechanism by which behavioural data can inform the report's metrics without exposing individuals, and the budget itself is a field in the data dictionary.

Downscaling Deriving fine-resolution fields from coarse model output by statistical or dynamical means. Chapter 8 places its products on the synthetic side of the firewall, however well calibrated they are.

Federated data hub A data architecture in which institutions retain ownership and control of their datasets while exposing discovery, metadata, standards and controlled access. It is proposed in place of a central database for reasons of sovereignty as much as engineering, as Chapter 2 sets out.

Fidelity score A recorded measure of how closely a synthetic record reproduces the statistical properties of the real data it stands for. It is one of the mandatory flags carried by every synthetic record under the scheme of Chapter 8.

Firewall The rule separating synthetic data from validation data: synthetic records may train, test and stress a system but may never enter the evidence on which a gate decision rests. Chapter 8 states the rule and Chapter 7 applies it to the month-12 gate.

Gap filling Estimating missing values in an observation series. The result is synthetic data and must be flagged as such, because gate validation may not rest upon it.

H3 cell A cell of a hierarchical hexagonal global grid, adopted here as the place identifier so that district-scale and street-scale layers share a single key space. Appendix B fixes the resolutions used.

Identifier schema The set of canonical keys (place, time slot, activity, route, intervention, scenario, sensor, model run, provenance, cohort, cost element and decision) that allow observations, models, costs and decisions to be joined. Chapter 8 derives it and Appendix B specifies it field by field.

Knowledge graph A structured representation of entities and the relations between them, supporting inference across domains. Chapter 2 insists that it is distinct from a data model, a semantic model, a statistical model, a digital twin and a decision system, and that conflating them is the commonest failure in programmes of this kind.

Ontology A formal specification of the concepts in a domain and the relationships between them. In this report the ontology of heat exposure, activity, route, place, intervention, resource and decision is the first deliverable of the Dubai programme, ahead of any hardware.

Personal data Any data relating to an identified or identifiable person. The report's position, set out in Chapter 12, is that the metrics must be computable from aggregated and privacy-protected inputs, so that no operational decision depends on identifying anyone.

Provenance The record of where a value came from: which instrument or program produced it, from what inputs, when, and under whose responsibility. Chapter 8 follows the established provenance vocabulary of agents, activities and entities so that the record is machine-readable.

Semantic layer The layer that gives consistent meaning to terms such as heat exposure, pedestrian, shade, intervention and decision, and defines how they relate. It sits between the data and everything that reasons over them, and Chapter 5 argues it must be written before a single sensor is bought.

Source archive The body of evidence secured and dated on 22 September 2026, each source registered in Appendix A with its retrieval address and the evidence grade it carries. Claims resting on sources that could not be obtained are marked as not reviewed.

Synthetic data Data generated by a model or procedure rather than observed. It may train, test and stress a system, but under the firewall rule of Chapter 8 it may never enter the validation on which the month-12 gate depends.

G.5 Decision science and modelling

Anchor point A milestone in the spiral model at which all stakeholders commit: Life-Cycle Objectives, Life-Cycle Architecture and Initial Operational Capability. Chapter 7 maps the three Dubai stage gates onto these anchor points so that the programme's calendar and its development model use one vocabulary.

Approximate dynamic programming A family of methods that replace the exact solution of a sequential decision problem with an approximation of the value function or of the policy. It is the only tractable route to the citywide problem, and Chapter 7 notes the caution attached to it: approximate value functions can fail in ways that are hard to detect.

Belief state The probability distribution over the true state of the system, given everything observed so far. It is the object on which decisions are actually made when the state itself cannot be seen, and it is what the sensor network in Chapter 7 exists to sharpen.

Bellman equation The recursion expressing the value of being in a state as the best immediate reward plus the discounted expected value of the state that follows. It is the formal backbone of the sequential decision problem stated in Chapter 7.

Curse of dimensionality The explosive growth of a state space as variables are added, which makes exact dynamic programming infeasible at scale. It is the technical reason the corridor problem is solvable while the citywide product remains a stretch goal.

Decision Twin A simulation environment for comparing interventions before they are built, as distinct from a three-dimensional rendering of a city. It is the central instrument of the Dubai proposal, and Chapter 3 recommends building it as a layer on the existing municipal digital twin rather than duplicating that platform.

Digital twin A computational model of a physical system kept in correspondence with it by a flow of data. The term spans everything from geometry viewers to coupled physical simulations, and Chapter 2 keeps the distinction explicit because procurement decisions turn on it.

Exogenous information Information arriving from outside the decision-maker's control: weather, dust events, visitor demand, prices and the decisions of other agencies. It is one of the five elements of the sequential decision model set out in Chapter 7.

Expected value of information The gain in expected outcome from making an observation before deciding. It is the rigorous justification for spending Year 1 on measurement, and it is zero for any observation that would change no decision: a test Chapter 5 applies to every proposed instrument.

Initial Operational Capability The anchor point at which a system is first usable in service by its intended operators, however limited its scope. In this programme it is the point at which an operating agency takes the output into a live decision, not the point at which the software runs.

Life-Cycle Architecture The anchor point at which the architecture is committed, the major risks are resolved and the build may proceed. Chapter 7 places it at the end of the modelling phase, after the semantic layer and the identifier schema are fixed.

Life-Cycle Objectives The anchor point at which the scope, the stakeholders' win conditions and the business case are agreed and a feasible concept is shown to exist. It is the gate that this report is written to inform.

Month-12 gate The stop-or-go decision at which second-phase funds are released only if four conditions are met: sensor uptime, model error within threshold, an hour-by-hour comfort map of both testbed zones, and written confirmation of usefulness from an operating agency. Its failure branch is a deliverable in its own right, as Chapter 11 sets out.

Partially observable Markov decision process A sequential decision problem in which the state is not directly observed but inferred from measurements. The Dubai problem is of this kind: the sensor network is the information boundary, and Chapter 7 treats the placement of sensors as part of the decision rather than as a precondition for it.

Spiral model A risk-driven development model in which each cycle sets objectives, identifies and resolves the dominant risks, develops and verifies, then plans the next cycle. Its win-win variant begins each cycle by eliciting and reconciling the success conditions of every stakeholder, which is the discipline behind this report's insistence on written confirmations.

State variable Everything that must be known at a decision epoch to make the decision and compute what follows: here the microclimate field, the installed assets, observed behaviour, remaining resources and the model's own uncertainty.

Stochastic dynamic programming The framework for making sequential decisions under uncertainty, in which a policy rather than a plan is sought. Chapter 7 uses it to state the system boundaries of both programmes precisely, and to show which questions lie inside them.

Transition function The rule describing how the state evolves from one epoch to the next given the decision taken and the exogenous information received. Building a credible transition function for the corridor is the substance of the modelling year.

Value function The expected discounted value of following the best policy from a given state onwards. Under this report's objective it is also the expected life-cycle cost net of restored activity, which is what links Chapter 7 to Chapter 9.

ADP See *Approximate dynamic programming*.

EVI See *Expected value of information*.

IOC See *Initial Operational Capability*.

LCA See *Life-Cycle Architecture*.

LCO See *Life-Cycle Objectives*.

POMDP See *Partially observable Markov decision process*.

SDP See *Stochastic dynamic programming*.

G.6 Economics, costs and finance

Blended finance The use of public or concessional capital to make a project bankable for private investors. Chapter 10 places it on the fifth rung of the financing ladder, reachable only after independent validation has been obtained.

Contingency Budget held against unknown unknowns and, in the arrangement recommended in Chapter 10, held above the project by the funder rather than by the delivery team. Setting it from the outturn of comparable completed projects is preferred to a flat percentage uplift.

Cost breakdown structure The hierarchical decomposition of a programme's costs into elements that sum to their parent, with programme management carried as an element in its own right rather than buried in overhead. Chapter 10 builds one for each of the two programmes.

Cost estimating relationship A fitted relationship between a cost and one or more physical or performance drivers, used to estimate cost parametrically. Its credibility depends entirely on the comparability of the data from which it was fitted, a point Chapter 9 makes repeatedly.

Cost per decision improved The ratio of programme cost to the number of decisions demonstrably changed by its outputs. Chapter 10 identifies it as the missing denominator in the continental proposal and recommends recording the decision alongside its cost from the first month, so that the ratio becomes computable later.

Discount rate The rate at which future costs and benefits are converted to present value. Every present value in this report states its rate, its base year and its period of analysis, so that comparisons remain valid when any of the three changes.

Escalation rate The rate at which the price of an input rises over time, applied so that a multi-year estimate is expressed consistently in either constant or current terms. Confusing escalation with contingency is one of the recurring estimating errors catalogued in Chapter 9.

Financing ladder The sequence from policy mandate through project preparation, pilot and independent validation to blended finance and commercial service, by which a demonstrated capability becomes a funded operation. Chapter 10 treats each rung as a gate with its own evidence requirement.

Life-cycle cost The whole cost of an asset or programme across acquisition, operation, maintenance, renewal and disposal, assessed over an agreed period of analysis. Chapter 9 sets out the methods for forecasting it and Chapter 10 applies them.

Management reserve Budget held against known unknowns at delivery level, distinct from contingency and held by the delivery organisation. Keeping the two separate, each at a named level, is one of the estimating disciplines recommended in Chapter 9.

Monte Carlo simulation Repeated random sampling of uncertain inputs to produce a distribution of outcomes rather than a single figure. Chapter 9 uses it to convert the cost model into a percentile range, from which a declared confidence level can be chosen as a management decision.

Net present value The sum of discounted costs and benefits over the period of analysis, and the standard basis for comparing options whose cash flows differ in timing as well as in size.

Optimism bias The systematic tendency to underestimate cost and duration and to overestimate benefit. It is the reason Chapter 9 prefers an outside view to a bottom-up estimate taken alone, and the reason uplifts are applied to the latter.

Parametric estimating Estimating cost from measured drivers using a fitted relationship, rather than by building the estimate from its components. It is fast and auditable but only as good as its underlying dataset.

Real option The value of retaining the right, without the obligation, to continue, expand or abandon at a future decision point. Chapter 9 treats the month-12 gate as exactly such an option, which is what makes a modest first phase economically rational.

Reference class forecasting Forecasting by comparison with the distribution of outcomes of similar completed projects, rather than by building an estimate for the project in isolation. It is the only method with documented evidence of reducing optimism bias, and Chapter 9 records the difficulty of assembling a reference class for a programme of this novelty.

Residual value The value remaining in an asset at the end of the period of analysis, subtracted from life-cycle cost where it can actually be realised.

Total cost of ownership The full cost of running a system, including the recurring cost of cloud computation, data licences, retraining and accumulated technical debt. Chapter 10 argues that these recurring items, not the capital works, determine whether the capability survives the funded period.

Whole-life cost Life-cycle cost extended to include income and externalities where the agreed scope admits them. The report uses the narrower life-cycle cost for its headline figures and states explicitly where the wider boundary is used instead.

Work breakdown structure The hierarchical decomposition of the work to be performed, against which cost and schedule are estimated and controlled. It is the companion of the cost breakdown structure and must map onto it element by element.

CBS See *Cost breakdown structure*.

LCC See *Life-cycle cost*.

NPV See *Net present value*.

RCF See *Reference class forecasting*.

TCO See *Total cost of ownership*.

WBS See *Work breakdown structure*.

G.7 Programme management and risk

Business continuity The capability to continue delivering at an acceptable level following a disruptive incident, together with the planning and rehearsal that make it possible.

Chapter 11 argues that continuity is a design property of the programme, not an annex to its contract.

Capability-first sequencing The rule that capability, standards and use cases are established before hardware is procured. Chapter 5 expresses it as a value-of-information test: no instrument is bought unless the observation it yields would change a decision.

Continuity exercise A rehearsed simulation of a disruptive event, run to test whether the recovery playbooks work in practice. Chapter 11 recommends rehearsing at least the data-loss and sensor-loss scenarios before the first summer of measurement.

Escrow A contractual arrangement placing source code, models or data with a third party so that the client can recover them if the supplier fails or the relationship ends. Chapter 11 recommends it, with an export provision and a termination-for-convenience clause, to preserve the in-house alternative.

Force majeure A contractual provision excusing performance prevented by an impediment beyond a party's control that was unforeseeable and unavoidable. Chapter 11 sets out the clause elements, distinguishes force majeure from hardship, and notes that its treatment under the applicable law must be confirmed with counsel.

Funding-timing risk The risk, specific to this portfolio, that the grant cycle assumed by the storyboard's calendar is not the cycle actually offered. Only one round is documented in the archived rules, which Chapter 11 records as an assumption rather than a fact.

Gate-failure fallback The deliverables that survive if the month-12 gate is not passed: the dataset and the platform. Naming them in advance converts an open-ended programme risk into a bounded one and makes the gate genuinely decidable.

Hardship clause A contractual provision addressing performance that has become excessively onerous rather than impossible. Chapter 11 distinguishes it from force majeure and notes that the two are conventionally drafted as separate clauses with different remedies.

Key performance indicator A measurable signal of whether the programme is achieving what it set out to achieve, chosen so that it can be read by the funder without interpretation. Chapter 10 pairs each indicator with the decision it is meant to inform.

Key risk indicator A measurable signal that a risk is materialising, chosen so that it moves before the loss does. The distinction from a performance indicator is one of timing: the one leads, the other reports.

Pilot fatigue The condition of a region or agency that has hosted many demonstrations and retained none of them. Chapter 11 treats it as a reputational and political risk, to which the stated answer is a written sustainment commitment and an operational budget rather than a further pilot.

Recovery playbook A written, rehearsed procedure for restoring a specific capability after a specific class of failure. Chapter 11 proposes six, covering data loss, sensor-network loss, partner withdrawal, funding shortfall, key-person loss and reputational loss.

Recovery point objective The maximum period of data loss a system may sustain, expressed as the age of the most recent recoverable state. It is set per dataset and written into the platform contract.

Recovery time objective The maximum acceptable time to restore a service after an interruption. Chapter 11 sets one for each component and notes that an objective that is not rehearsed is an aspiration.

Risk exposure The probability of a loss multiplied by its size. It is the quantity the spiral model seeks to minimise when deciding how much effort to spend on any activity, and it is the reason the report front-loads measurement rather than construction.

Risk register The maintained list of identified risks with cause, consequence, likelihood, impact, owner, treatment and residual position. Chapter 11 presents the portfolio register and insists that every entry carries a named owner and a review date.

Stage gate A point in the programme at which funds are released only against written pass conditions. The three stage gates of this programme correspond to the spiral anchor points, as Chapter 10 shows on a single calendar.

Sustainment commitment A written undertaking by an operating agency to carry a capability beyond the funded period. Chapter 11 treats it as the only reliable test of whether a pilot has a future, and the answer to pilot fatigue.

SWOT analysis A structured appraisal of strengths, weaknesses, opportunities and threats. Chapter 11 extends it into a matrix that pairs internal with external factors to generate strategies, so that the appraisal produces actions rather than lists.

Win conditions The success criteria of each stakeholder, elicited and reconciled before a development cycle begins. Unreconciled win conditions are, in the spiral literature, the commonest single cause of programme failure.

KPI See *Key performance indicator*.

KRI See *Key risk indicator*.

RPO See *Recovery point objective*.

RTO See *Recovery time objective*.

SWOT See *SWOT analysis*.

TOWS matrix See *SWOT analysis*.

G.8 Policy and governance

Adaptation indicator A measure of how well a place is adapting to a hazard, as distinct from a measure of emissions or of energy. Chapter 12 argues that the principal international cooling instrument lacks such an indicator and that Heat-Lost Activity Hours is a candidate to fill the gap.

Circle of concern Those matters a party cares about but can neither control nor affect. Chapter 6 uses the distinction to keep the programme's effort inside the ground it can actually move.

Circle of control Those matters a party decides directly: its own methods, publications, data practices and commitments. Chapter 6 maps the programme's deliverables onto this circle first, because they are the only things it can promise.

Circle of influence Those matters a party does not control but can affect, typically by supplying evidence that changes another party's decision. Chapter 6 treats the decision brief as the instrument of influence and names, for each actor, the lever and the evidence expected to move it.

Co-design Developing a service with the people who will use it operationally, beginning from the decision they must make rather than from the data that happen to exist. Chapter 5 makes it the first activity of the programme, ahead of procurement.

Data and ethics board The body proposed to govern data access, privacy budgets, synthetic flags and lineage closure across the continental programme. Chapter 8 gives it concrete objects to audit rather than principles to endorse.

Decision brief A short document putting a comparison and a recommendation in front of a decision-maker. In this report it is the unit of influence: the artefact that converts a metric into an institutional commitment, and the thing that is counted when uptake is measured.

Decision uptake The recorded joining of a decision brief to the decision it produced. It is computed rather than observed, and Chapter 8 specifies the identifiers that make the join possible.

Evidence grade This report's classification of how far a source's claims are supported: measured-with-controls, measured, modelled, projection, policy-fact, case-report or method-reference. Every substantive claim in the report carries one, and Chapter 1 explains why the discipline matters more than the label.

Governance The arrangements by which authority, accountability and review are distributed across a programme. Chapter 12 argues that governance designed after delivery has begun tends to ratify what already exists rather than direct it.

Indigenous knowledge Accumulated local understanding of living in an environment, recorded in practice rather than in literature. Chapter 12 treats the Gulf's pre-modern repertoire of heat adaptation as a legitimate design input rather than as decoration, and Chapter 5 points to the heritage core as its surviving demonstration.

Influence plan A table naming, for each actor to be reached, the lever, the evidence expected to move it, the proposed owner and the date. Chapter 6 closes with one and intends it to be used as a checklist rather than read as an argument.

Nemawashi The practice of building consent privately and incrementally with each party before a proposal is formally tabled. Chapter 12 recommends it as the route by which a metric becomes an accepted standard, on the grounds that standards bodies ratify agreements rather than create them.

Open method A method published in full, with its data and code, so that others can apply it without permission. Chapter 12 makes openness the precondition of replication and therefore of the metric ever becoming a norm.

Planetary boundaries A framework of biophysical limits within which humanity can operate safely. Chapter 12 notes that it contains no boundary for human habitability, which is the conceptual space a validated heat-loss metric might eventually occupy.

Policy pathway The route by which a technical result becomes an instrument of governance: definition, open method, a single-site result, independent replication, then adoption by a body that already has an owner and a reporting cycle. Chapter 12 sets out the sequence and its realistic timescale.

Replication Independent reproduction of a result by another team in another city. It is the step between a local demonstration and a general claim, and Chapter 12 treats three replications as the practical threshold for a norm.

Science-policy window The period during which a published result can still enter an assessment cycle. Chapter 12 shows that the current special report on cities closes too early for this programme and that the next full assessment is the realistic target.

South–South co-operation Exchange of capability, finance and technology between developing countries. It is the model on which the continental programme's partnership architecture rests, and Chapter 2 notes that it is a political commitment as much as a technical one.

Written confirmation A signed statement by an operating agency that an output is useful for a decision it actually faces. It is the fourth condition of the month-12 gate and, in this report's judgement, the single most consequential line in the Dubai proposal.

G.9 Institutions and programmes

African Union The continental body whose declarations and strategies provide the mandate the blue-intelligence programme attaches itself to. Chapter 12 treats alignment with its instruments as the lever that reaches national institutions indirectly.

BRICS The grouping of major emerging economies whose space and ocean working groups, and whose development finance, the continental concept paper identifies as co-financing and co-operation channels. Chapter 10 treats these as a rung of the financing ladder rather than as a committed source.

Cooling Singapore The multi-institution research programme that built the Digital Urban Climate Twin. Chapter 3 treats it as the closest existing analogue to the proposed Decision Twin and as the best available guide to what such a system costs to sustain.

Digital Urban Climate Twin The coupled climate, anthropogenic-heat and building-energy modelling environment built for evaluating heat-mitigation scenarios in a tropical city. It is the reference implementation against which Chapter 3 measures the Dubai proposal's ambition.

Dubai Digital Twin Platform The municipal platform carrying the emirate's geospatial layers, three-dimensional assets and simulation applications. Chapter 3 recommends that the Decision Twin be built as a layer upon it rather than as a parallel system, both for cost and for institutional acceptance.

Dubai Electricity and Water Authority The emirate's utility, holder of cooling-demand, energy and water data and a natural counterpart for the resource side of the metrics. Chapter 6 places it in the circle of influence and names the evidence expected to reach it.

Dubai Future Foundation The emirate body whose research, development and innovation grant programme frames the Dubai proposal. Chapter 1 describes the documents it publishes and Chapter 11 the timing risk that its grant cycle creates.

Dubai Future Foundation RDI Grant The competitive research grant, under its Cognitive Cities priority, to which the Dubai proposal is to be submitted. Its published rules cap the annual budget per project, a constraint Chapter 10 reconciles with the storyboard's stated budget.

Global Fishing Watch The open platform publishing vessel activity derived from transponder and satellite data. Chapter 3 identifies the fisheries authorities already using it as the readiest partners for an early demonstration.

Global Ocean Observing System The international ocean-observing framework whose African regional alliance provides the observation foundation that the continental programme is designed to connect rather than replace, as Chapter 2 emphasises.

GMES & Africa The African Union and European Union programme building operational Earth-observation services and institutional capacity across the continent, including the marine

and coastal consortia on which the blue-intelligence programme would build rather than compete.

Intergovernmental Panel on Climate Change The body whose assessment and special reports set the vocabulary and the evidence standard for climate policy. Chapter 12 analyses its assessment calendar to establish which window this programme's results could realistically enter.

National Center of Meteorology The UAE national meteorological authority, source of the reference station data against which the corridor sensors are to be calibrated, as Chapter 5 sets out.

New Development Bank The multilateral development bank identified in the continental concept paper as a project-preparation and financing counterpart. Chapter 10 places contact with it on the early rungs of the financing ladder, before any capital request.

Ocean Prediction Enhancement in Regions of Africa The regional ocean-forecasting programme that the continental proposal is designed to serve rather than duplicate. It is one of the three existing capabilities named in Chapter 2 as the programme's foundation.

Phoenix cool pavement The municipal and university evaluation of reflective road coatings in a desert city. It is the only controlled field evidence in the archive and the source of the radiant-load warning that shapes the intervention ranking in Chapter 4.

Regional economic community One of the sub-continental groupings of African states through which continental policy is implemented. Chapter 6 identifies them, with the continental commission and the space agency, as the holders of the political mandate.

Roads and Transport Authority The Dubai agency responsible for roads, public transport and the pedestrian network. Chapter 6 treats it as the decisive audience for the metrics, because it is the body whose capital decisions the Minimum Cooling Network would inform.

UAE Space Agency The national agency whose space-data infrastructure and analytics platform the continental proposal identifies as the intended gateway between African demand and partner capability.

United Nations Environment Programme The body convening the principal international cooling instruments and handbooks. Chapter 12 examines those instruments and finds them rich in commitments and poor in adaptation indicators, which is the opening this report's metric is designed to occupy.

AU See *African Union*.

DEWA See *Dubai Electricity and Water Authority*.

DFF See *Dubai Future Foundation*.

DUCT See *Digital Urban Climate Twin*.

GMES See *GMES & Africa*.

GOOS-Africa See *Global Ocean Observing System*.

IPCC See *Intergovernmental Panel on Climate Change*.

NCM See *National Center of Meteorology*.

NDB See *New Development Bank*.

OPERA See *Ocean Prediction Enhancement in Regions of Africa*.

RDI See *Dubai Future Foundation RDI Grant*.

REC See *Regional economic community*.

RTA See *Roads and Transport Authority*.

UNEP See *United Nations Environment Programme*.

Bibliography

[Arsht-Rock and Vivid Economics, 2022] Adrienne Arsht-Rockefeller Foundation Resilience Center and Vivid Economics (2022). Hot Cities, Chilled Economies: Impacts of Extreme Heat on Global Cities. Report, September 2022, Atlantic Council.

[AU-IBAR, 2019] African Union – Inter-African Bureau for Animal Resources (2019). Africa Blue Economy Strategy. Technical report, African Union, Nairobi.

[ASU and City of Phoenix, 2021] ASU Cool Pavement Team and City of Phoenix Street Transportation Department (2021). City of Phoenix Cool Pavement Pilot Program: Year 1 Report (15 July 2020 – 14 July 2021). Report, September 2021 (file published as COPE-Report_FULLFINAL.pdf), Arizona State University and City of Phoenix.

[ASU and City of Phoenix, 2024] ASU Cool Pavement Team and City of Phoenix Street Transportation Department (2024). City of Phoenix Cool Pavement Pilot Program, Phase II – 2022. Final report, October 2024, Arizona State University and City of Phoenix.

[Aydt et al., 2026] Aydt, H., Acero, J. A., Ivanchev, J., Nevat, I., Adelia, A. S., Benny, J., others, and Orehounig, K. (2026). Tools to manage Singapore's heat: Coupled climate and anthropogenic heat emission models for urban comfort in a digital twin framework. *City and Environment Interactions*. DOI: 10.1016/j.cacint.2026.100301.

[Bande et al., 2020] Bande, L. et al. (2020). Local Climate Zones Definition in Relation to ENVI-met in the City of Dubai, UAE. *IOP Conference Series: Materials Science and Engineering*, 829:012013. DOI: 10.1088/1757-899X/829/1/012013.

[Beck et al., 2002] Beck, I. L., McKeown, M. G., and Kucan, L. (2002). *Bringing Words to Life: Robust Vocabulary Instruction*. Guilford Press, New York.

[Bellman, 1957] Bellman, R. (1957). *Dynamic Programming*. Princeton University Press, Princeton, NJ.

[Boehm et al., 1998] Boehm, B., Egyed, A., Kwan, J., Port, D., Shah, A., and Madachy, R. (1998). Using the WinWin Spiral Model: A Case Study. *IEEE Computer*, 31(7):33–44. DOI: 10.1109/2.689675.

[Boehm and Hansen, 2000] Boehm, B. and Hansen, W. J. (2000). Spiral Development: Experience, Principles, and Refinements. Technical Report CMU/SEI-2000-SR-008, Software Engineering Institute, Carnegie Mellon University.

[Cantarelli et al., 2025] Cantarelli, C. C., Davis, K., Pinto, J. K., and Turner, N. (2025). Reference class forecasting: promises, problems, and a research agenda moving forward. *Production Planning & Control*. DOI: 10.1080/09537287.2025.2578708.

[Commons Cluster, 2023] Commons Cluster of the NGO Major Group (2023). Building Unity as a Prerequisite to Achieving the SDGs: Proposal for UN Member States and Civil Society in Preparation for the SDG Summit 2023. Technical report, United Nations NGO Major Group.

[Commons Cluster, 2024] Commons Cluster of the NGO Major Group (2024). Unity: A Pathway to Achieving a Thriving Planet. Eradicating the Roots of Poverty in Times of Multiple Crises to Achieve the SDGs (Post-HLPF 2024 edition). Technical report, United Nations High Level Political Forum side event.

[Cooling Singapore 2.0, 2024] Cooling Singapore 2.0 (2024). Cooling Singapore 2.0: Digital Urban Climate Twin (poster). World Cities Summit 2024, Science of Cities poster P9.

[Covey, 1989] Covey, S. R. (1989). *The 7 Habits of Highly Effective People*. Free Press, New York.

[Flyvbjerg, 2014] Flyvbjerg, B. (2014). What You Should Know About Megaprojects and Why: An Overview. *Project Management Journal*, 45(2):6–19. DOI: 10.1002/pmj.21409.

[Flyvbjerg et al., 2016] Flyvbjerg, B., Stewart, A., and Budzier, A. (2016). The Oxford Olympics Study 2016: Cost and Cost Overrun at the Games. Saïd Business School Working Papers, arXiv 1607.04484.

[Great Blue Wall Initiative, 2026] Great Blue Wall Initiative (2026). Great Blue Wall Impact Report 2021–2025. Technical report, IUCN and Western Indian Ocean partners.

[Hoffmann et al., 2023] Hoffmann, J. et al. (2023). Destination Earth – A digital twin in support of climate services. *Climate Services*, 30:100394. DOI: 10.1016/j.cliser.2023.100394.

[Howard, 1966] Howard, R. A. (1966). Information Value Theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1):22–26. DOI: 10.1109/TSSC.1966.300074.

[IPCC, 2024] Intergovernmental Panel on Climate Change (2024). Seventh Assessment Report: Outline of the Special Report on Climate Change and Cities. IPCC-61 decision document, 2 August 2024.

[ICC, 2020] International Chamber of Commerce (2020). ICC Force Majeure and Hardship Clauses, March 2020. International Chamber of Commerce.

[ILO, 2019] International Labour Organization (2019). Working on a warmer planet: The impact of heat stress on labour productivity and decent work. Technical report, International Labour Office, Geneva.

[ISO, 2017a] International Organization for Standardization (2017a). ISO 15686-5:2017 Buildings and constructed assets – Service life planning – Part 5: Life-cycle costing. ISO.

[ISO, 2017b] International Organization for Standardization (2017b). ISO 7243:2017 Ergonomics of the thermal environment – Assessment of heat stress using the WBGT (wet bulb globe temperature) index. ISO.

[ISO, 2018] International Organization for Standardization (2018). ISO 31000:2018 Risk management – Guidelines. ISO.

[ISO, 2019] International Organization for Standardization (2019). ISO 22301:2019 Security and resilience – Business continuity management systems – Requirements. ISO.

[Iyyunni, 2026] Iyyunni, C. (2026). Storyboard: Climate Intelligence & Decision Twin for Dubai. Rescoped Letter of Intent, Dubai Future Foundation RDI Grant, Cognitive Cities priority. Internal working document, BITS Pilani Dubai Campus.

[Jendritzky et al., 2012] Jendritzky, G., de Dear, R., and Havenith, G. (2012). UTCI – Why another thermal index? *International Journal of Biometeorology*, 56(3):421–428. DOI: 10.1007/s00484-011-0513-7.

[Kilic, 2024] Kılıç, A. O. (2024). Seychelles blue bond: Indebting ecological restructuring of fisheries. *Marine Policy*, 165:106144. DOI: 10.1016/j.marpol.2024.106144.

[Manandhar et al., 2019] Manandhar, P., Bande, L., Tsoupos, A., Marpu, P. R., and Armstrong, P. (2019). A Study of Local Climate Zones in Abu Dhabi with Urban Weather Stations and Numerical Simulations. *Sustainability*, 12(1):156. DOI: 10.3390/su12010156.

[NEA Singapore, 2023] National Environment Agency, Singapore (2023). Heat Stress Advisory: WBGT levels and recommended actions. Meteorological Service Singapore.

[NSCC Singapore, 2024] National Supercomputing Centre Singapore (2024). Cooling Singapore 2.0: NSCC Singapore's supercomputing supports the Digital Urban Climate Twin. NSCC Newsbytes, June 2024.

[OASIS research team, 2024a] OASIS research team (2024a). Spatial Microclimatic Characterization of a Parisian "Oasis" Schoolyard. arXiv preprint 2408.07284.

[OASIS research team, 2024b] OASIS research team (2024b). Thermal and microclimatic behavior of OASIS schoolyard paving materials. arXiv preprint 2408.08317.

[Powell, 2019] Powell, W. B. (2019). From Reinforcement Learning to Optimal Control: A unified framework for sequential decisions. arXiv preprint 1912.03513.

[Powell, 2022] Powell, W. B. (2022). *Reinforcement Learning and Stochastic Optimization: A Unified Framework for Sequential Decisions*. Wiley, Hoboken, NJ.

[Richardson et al., 2023] Richardson, K., Steffen, W., Lucht, W., et al. (2023). Earth beyond six of nine planetary boundaries. *Science Advances*, 9(37):eadh2458. DOI: 10.1126/sciadv.adh2458.

[Romanello et al., 2024] Romanello, M. et al. (2024). The 2024 report of the Lancet Countdown on health and climate change: facing record-breaking threats from delayed action. *The Lancet*, 404. DOI: 10.1016/S0140-6736(24)01822-1.

[Scott et al., 2016] Scott, D., Rutty, M., Amelung, B., and Tang, M. (2016). An Inter-Comparison of the Holiday Climate Index (HCI) and the Tourism Climate Index (TCI) in Europe. *Atmosphere*, 7(6):80. DOI: 10.3390/atmos7060080.

[Stewart and Oke, 2012] Stewart, I. D. and Oke, T. R. (2012). Local Climate Zones for Urban Temperature Studies. *Bulletin of the American Meteorological Society*, 93(12):1879–1900. DOI: 10.1175/BAMS-D-11-00019.1.

[Terhile et al., 2026] Terhile, U. D., Iyyunni, C., Al Salmani, A., and Ahluwalia, A. (2026). Africa Blue Intelligence Grid: A Pan-African Ocean, Environment, Security and Space-Data Capability Programme. Strategic concept paper with addendum on alignment with GOOS-Africa, OceanPrediction DCC and OPERA. Strategic concept paper, 20 September 2026, Federal University Dutse; BITS Pilani Dubai; SpacePoint; Ecocivilisation.earth.

[UNEP, 2023] United Nations Environment Programme (2023). Global Cooling Pledge. UNEP / COP28 UAE Presidency / Cool Coalition.

[UNEP, Cool Coalition and RMI, 2021] United Nations Environment Programme, Cool Coalition, and RMI (2021). Beating the Heat: A Sustainable Cooling Handbook for Cities. Technical report, UNEP, Nairobi.

[UN Secretary-General, 2024] United Nations Secretary-General (2024). United Nations Secretary-General's Call to Action on Extreme Heat. Joint product of FAO, ILO, OCHA, UNDRR, UNEP, UNESCO, UN-Habitat, UNICEF, WHO and WMO, 25 July 2024, United Nations.

[Urban Innovative Actions, 2023] Urban Innovative Actions (2023). Paris OASIS: School yards – Openness, Adaptation, Sensitisation, Innovation and Social ties. Urban Innovative Actions project page and project journals, European Urban Initiative.

[GAO, 2020] U.S. Government Accountability Office (2020). Cost Estimating and Assessment Guide: Best Practices for Developing and Managing Program Costs. Technical Report GAO-20-195G, GAO.

[Vivid Economics, 2022] Vivid Economics (2022). Hot Cities, Chilled Economies: Public Methodology. Technical report, Adrienne Arsht-Rockefeller Foundation Resilience Center.

[World Bank, 2025] World Bank (2025). Introducing the World's First Sovereign Blue Bond: Seychelles. Case study dated 27 June 2025 describing the October 2018 issuance, World Bank Treasury.

[WEF and Digital Earth Africa, 2021] World Economic Forum and Digital Earth Africa (2021). Unlocking the potential of Earth Observation to address Africa's critical challenges. Insight report, January 2021, World Economic Forum.

[W3C, 2013] World Wide Web Consortium (2013). PROV-O: The PROV Ontology. W3C Recommendation, 30 April 2013.

Index

180-day action table, 79
90-day action list, 78

ABIG, *see* Africa Blue Intelligence Grid
active learning, 16, 113
Activity identifier, 133, 237
Adaptation indicator, 191, **250**
ADP, *see* Approximate dynamic programming
Africa Blue Economy Strategy, 38
Africa Blue Intelligence Grid, v, vi, viii, 5, 15, 52, 76, 101, 119, 195, 237, **242**
  cost structure, 148
  evidence base, 224
  Financing ladder, 161
  hardware sequencing, 158
  integration layer, 200
  Risk register, 186
Africa Ocean Observation, Prediction and Intelligence Capability Map, 77
African Union, v, **252**
African Union Blue Economy Strategy, 195
African Union Commission, 91
Agent-based model, 133
Air temperature, 51, **238**
AIS, *see* Automatic identification system
Al Fahidi Historical Neighbourhood, 3, 13, 66, 73, 89, 103, 133, 185, 200, 223, 226, **240**
Al Ghubaiba, 103
Al Ghubaiba corridor, 3, 13, 74, 200, 210, **240**
Amāna, xv
analogy estimating, 137
Anchor point, 102, 181, **245**
Approximate dynamic programming, **245**
Arafat, 55
Architecture
  eight-layer chain, vi
Arsht-Rock Resilience Center, 218
AU, *see* African Union
auditability, 224
Automatic identification system, 133, 237, **242**

Bayesian updating, 170
BE WELL, 7
Beating the Heat (UNEP handbook), 32
Before-After-Control-Impact design, 30, 74
Belief state, 108, **245**
Bellman equation, 112, 142, **245**
Bias correction, 12, 74, 133, **243**
BITS Pilani Dubai, 14, 87
Black-globe thermometer, 73, 144, 223, **238**
Blended finance, 148, 161, **247**
Blue bond, 37, 160, **242**
Blue economy, viii, **242**
bot wall, 220
boundary taxonomy, 115
BRICS, 16, 91, 195, **252**
Building Climate and Infection Intelligence, 20
Business continuity, 175, **248**

Calibration, 133
calibration drift, 168
Calibration version, 237
Call to Action on Extreme Heat, 39
Capability map, 6, 18, 77, 90, 115, 133, 161, 224, **242**
Capability-first sequencing, vi, 5, 15, 46, 76, **249**
Cardinality, 226
CBS, *see* Cost breakdown structure
Circle of concern, 86, **250**
Circle of control, 86, **250**
Circle of influence, 86, 87, **250**
Citywide 100 m product, 13, **240**
Climate-resilient urban cells, 20, **240**
Co-design, 6, 18, **250**
Cognitive Cities priority, 3, 10

Cohort identifier, 133, 237
Comfort index, 221, **238**
Comfort map, 13, 93, 103, **240**
Commons Cluster, 6
compute, 154
Concept paper, 5
cone of uncertainty, 146
Contingency, 139, 161, 181, 219, **247**
Continuity exercise, 175, **249**
Controlled vocabulary, 232
Conventions
  British English, x
  currency and year, x
  Eastern Arabic numerals, x
Cool Coalition, 191
Cool pavement, 12, 28, 51, 161, 189, 201, 235, **238**
  Phoenix pilot programme, v
Cooling Singapore, v, 25, 92, 161, 189, 215, **252**
Copernicus, 13
correlation, 171
Cost breakdown structure, 147, 149, 235, **247**
Cost estimating relationship, 137, **247**
Cost per decision improved, 156, 161, 224, **247**
Cost-element identifier, 133, 237
Countries Geospatial Readiness Index, 77
Curse of dimensionality, 112, **245**
Curtin University Dubai, 13, 75, 89, 185, 204

Dark vessel, 133, **242**
Data and ethics board, 161, 204, **251**
Data dictionary, 71, 133, 225, 226, **244**
Data governance, 163, **244**
Data licence, 237
Data lineage, 133, **244**
data provenance, 223
data quality, 223
Data retention, 227, 237
Data sovereignty, 91, 186, **244**
Data stewardship, 225, 237
Decision brief, 92, **251**
Decision identifier, 133, 237
Decision Twin, v, vi, xi, 3, 12, 23, 64, 71, 103, 119, 198, 200, 225, **245**
  cost model, 141
  evidence standard, 223
  gate conditions, vi, 3, 13, 93, 133, 186, 204, 233
Decision uptake, 133, 237, **251**
degradation curve, 140
Department of Economy and Tourism, 89
dependency diagram, 82
Derived figures, x
design-to-cost, 144
Destination Earth, 26
DEWA, *see* Dubai Electricity and Water Authority
DFF, *see* Dubai Future Foundation
Differential privacy, 133, 227, **244**
Digital Earth Africa, 35, 59, 77, 195, 215
Digital twin, iv, 24, 120, **246**
Digital Urban Climate Twin, 25, 56, 201, **252**
Discount rate, **247**
  nominal, 145
  real, 145
Discounting, x
District cooling, 13, 72, 111, 161, **240**
Document quality, 7
Downscaling, 12, 133, **244**
Dubai Creek, 13, **241**
Dubai Culture, v, 13, 75, 89, 103, 204
Dubai Decision Twin
  cost breakdown, 160
Dubai Digital Twin Platform, 12, 26, 73, 89, 133, 201, 215, 227, **252**
Dubai Electricity and Water Authority, v, 10, 75, 89, 103, 166, 204, **252**
Dubai Future Foundation, v, 161, **252**
  RDI Grant, 204
Dubai Future Foundation RDI Grant, 3, 75, 89, **252**
Dubai Municipality, v, 10, 73, 89, 103, 204
Dubai RDI Grant, v, 147, 186
Dubai Walk Master Plan, 27, 89, **241**
DUCT, *see* Digital Urban Climate Twin
Dust storm, 184, **238**

Earth observation, **242**
Ecocivilisation, 6
end of life, 161
engineering build-up, 137
ENVI-met, 19, 29, 55
EO, *see* Earth observation
Escalation rate, 145, **247**
escalator, 92
Escrow, 181, **249**

Ethics, 133, 227
European Digital Twin Ocean, 37
EVI, *see* Expected value of information
Evidence grade, v, ix, 23, 146, 199, 201, 214, **251**
  case-report, x, 217
  distribution, 216
  measured, ix
  measured-with-controls, ix
  method-reference, x
  modelled, ix
  policy-fact, ix
  projection, ix
  scheme, 215
Evidence grading, vi, 6
Exogenous information, 109, **246**
Expected value of information, 16, 140, **246**
externality, 135

Feasibility study, 48
Federated data hub, 6, 18, 91, 115, 133, 237, **244**
Fidelity score, 124, 236, **244**
Financing ladder, 5, 16, 100, 148, **247**
Firewall, 129, **244**
Force majeure, 182, 203, 219, **249**
Funding-timing risk, 4, 15, 76, 89, 166, **249**
Future Loop, 27

Gap filling, 133, 232, **244**
Gate-failure fallback, 181, **249**
Gharar, xiv
Global Cooling Pledge, 39, 190, 201
Global Fishing Watch, v, 16, 36, 59, 78, 91, 161, 195, 203, 215, **252**
Global Ocean Observing System, **252**
Glossary, xi
GMES, *see* GMES & Africa
GMES & Africa, 15, 34, 76, 106, **252**
GMES & Africa, v, 5, 91
GOOS-Africa, v, 5, 18, 76, 91, 106, 203, 210, *see* Global Ocean Observing System
Governance, 161, **251**
Great Blue Wall, 36, 60, 161, 195, **242**
Green Riyadh, 27, 57, 72, 188, 205, 218, **241**

H3 cell, 110, 120, **244**
H3 geospatial indexing system, 228
H3 spatial index, 201
Hajj, 188
Hardship clause, 182, **249**
HCI, *see* Holiday Climate Index
Heat action plan, 192, **238**
Heat stress, 232
Heat Stress Advisory (Singapore), 32
Heat-Lost Activity Hours, vi, x, 11, 31, 58, 72, 110, 119, 147, 160, 176, 188, 191, 202, 233, **239**
Heat-stress override, 72, **239**
High-Level Political Forum, 6
HLAH, *see* Heat-Lost Activity Hours
Holiday Climate Index, 11, 33, 59, 72, 198, 202, 217, **241**
Hot Cities, Chilled Economies, 32

Identifier, 119
Identifier schema, 225, 237, **244**
Ijāra, xv
Ikrāh, xvi
Illegal, unreported and unregulated fishing, 16, 36, 60, 78, 91, 131, **243**
impact, 166
Imputation, 133, 235
incremental commitment, 102, 140, 161, 172
Index, xi
Indigenous knowledge, 5, 21, 195, **251**
Influence plan, 94, **251**
Initial Operational Capability, 102, 152, **246**
insurance, 181
Intergovernmental Panel on Climate Change, v, **253**
International Labour Organization, v, 223
International Union for Conservation of Nature, v
Intervention identifier, 133, 237
IOC, *see* Initial Operational Capability
IPCC, *see* Intergovernmental Panel on Climate Change
IPCC Special Report on Climate Change and Cities, 40, 192, 201
ISO 15686-5, 219
Istiṣnāʿ, xv
IUU, *see* Illegal, unreported and unregulated fishing

Key performance indicator, **249**
Key risk indicator, 168, **249**
Knowledge architecture, 4
Knowledge graph, vi, 4, 18, 71, 120, 200, **244**
KPI, *see* Key performance indicator

KRI, *see* Key risk indicator

Labour productivity, 188, **241**
Lancet Countdown, v
Landsat-8, 19
Large Marine Ecosystems, 15, **243**
layered architecture, 19
LCA, *see* Life-Cycle Architecture
LCC, *see* Life-cycle cost
LCO, *see* Life-Cycle Objectives
LCZ, *see* Local climate zone
learning curve, 142
Letter of Intent, 3, 71
Life-Cycle Architecture, 102, 152, **246**
Life-cycle cost, viii, 134, 135, 161, 201, 215, 235, **247**
Life-Cycle Objectives, 102, 152, **246**
likelihood, 166
Local climate zone, 4, 19, 29, 54, 124, 188, 201, 228, **239**
Low-exposure mobility network, 20, **241**
Luanda Declaration, 5, 15, 38, 79, 91, 195, 203, 224, **243**

Macroeconomic feasibility report, viii
Makkah, 215
Management reserve, 139, 161, 181, **247**
Maqāṣid al-sharīʿa, xvi
MarCOSIO, 34, 91
Maysir, xiv
MCN, *see* Minimum Cooling Network
Mean radiant temperature, 12, 28, 51, 73, 201, **239**
Metadata, 237
method-selection matrix, 142
midday break, 161, 184
Midday break (UAE), 11, 33, 59, 72, 89, 188, 207, **239**
Minimum Cooling Network, 3, 12, 27, 52, 87, 111, 133, 142, 156, 160, 176, 198, 200, 233, **241**
Misting, 55, 161, **239**
mobile traverse, 74
Model-run identifier, 133, 237
Monte Carlo simulation, 171, **248**
Month-12 gate, 103, 223, **246**
MRT, *see* Mean radiant temperature
Muḍāraba, xiv
Murābaḥa, xiv
Mushāraka, xiv

National Center of Meteorology, v, 10, **253**
National Cooling Action Plan, 191
Natural wet-bulb temperature, 73, **239**
NCM, *see* National Center of Meteorology
NDB, *see* New Development Bank
Nemawashi, 5, 21, 75, 93, 196, 203, **251**
Net present value, 135, **248**
New Development Bank, 16, 91, 161, 195, 203, **253**
NPV, *see* Net present value
NSPE Code of Ethics for Engineers, iv

OASIS schoolyards (Paris), v
Ocean Prediction Enhancement in Regions of Africa, **253**
OceanPrediction DCC African Seas Regional Team, 76
Ontology, 71, 120, **245**
Open data cube, 35, **243**
Open method, 199, **251**
OPERA, v, 5, 18, 76, 91, 106, 203, 210, *see* Ocean Prediction Enhancement in Regions of Africa
operational user, 78
Optimism bias, 138, 175, **248**
OTC, *see* Outdoor Tourism Capacity
Outdoor activity, 10, **241**
Outdoor Tourism Capacity, x, 3, 11, 26, 56, 72, 87, 110, 133, 148, 161, 176, 189, 191, 200, 233, **241**
outside view, 137

paired-control design, 201
Parametric estimating, 137, **248**
Paris OASIS, 30, 53, 93, 161, 189, 201, 215, **241**
Partially observable Markov decision process, 109, **246**
period of analysis, 135, 160
Personal data, 5, 84, **245**
Phoenix cool pavement, **253**
Phoenix cool pavement programme, 161, 215
  Year-1 report, 217
Pilot fatigue, 100, **249**
Place identifier, 133, 237
Planetary boundaries, 196, **251**
Policy pathway, 189, **251**
POMDP, *see* Partially observable Markov decision process

Port State Measures Agreement, 36, **243**
Privacy budget, 237
professional indemnity, 181
PROV-O, 201
Provenance, 3, 71, 127, 182, 214, 225, 237, **245**

RAH, *see* Restored Activity Hours
RCF, *see* Reference class forecasting
RDI, *see* Dubai Future Foundation RDI Grant
Real option, 140, **248**
REC, *see* Regional economic community
Recovery playbook, 186, **249**
Recovery point objective, 179, **249**
Recovery time objective, 179, **249**
redaction, 222
Reference class forecasting, 137, 161, 168, 215, **248**
Reflectivity (albedo), 30, **239**
Regional economic community, **253**
Regional hub, 158, **243**
Replication, 190, **251**
Residual value, 135, 161, **248**
resolution architecture, 19
Restored Activity Hours, vi, 12, 72, 110, 133, 135, 156, 191, 202, 233, **241**
retrieval date, 214, 222
Ribā, xiv
Riḍā, xvi
risk criteria, 163
Risk exposure, 70, 103, 163, **250**
risk management, 163
Risk register, 162, 203, **250**
Roads and Transport Authority, v, 13, 75, 89, 103, 166, 204, **253**
Route identifier, 133, 237
RPO, *see* Recovery point objective
RTA, *see* Roads and Transport Authority
RTO, *see* Recovery time objective

S-curve, 139
sanctions, 184
SAR, *see* Synthetic aperture radar
Scenario identifier, 133, 237
schedule risk, 168
Science-policy window, 194, **251**
SDP, *see* Stochastic dynamic programming
Seed, 237
Semantic layer, vi, 4, 18, 71, 87, 120, 200, **245**
sensitivity analysis, 146, 170
Sensor identifier, 133, 237
Sensor network, 133, 152, 226
service life, 139, 161, 219
Seychelles blue bond, 60, 146, 161, 205, **243**
Shade canopy, 12, 65, 103, 154, 201, **239**
SIFA Plan, 6
Solar reflectance, 51
Source archive, vii, x, 48, 214, **245**
South–South co-operation, 199, **252**
spiral development, 215
Spiral model, 101, 161, 172, **246**
Stage gate, 160, **250**
stakeholder map, 94
State variable, 108, **246**
Stochastic dynamic programming, 101, 142, 201, **246**
Storyboard, 3
Stress testing, 133
Ṣukūk, xiv
summer measurement window, 163
sunk cost, 135
Surface temperature, 51, **239**
Sustainment commitment, 75, 93, 104, 161, 186, **250**
SWOT, *see* SWOT analysis
SWOT analysis, 175, **250**
Synthetic aperture radar, **243**
Synthetic data, 119, 237, **245**
    Firewall, 201

Taṭhīr, xv
TCI, *see* Tourism Climate Index
TCO, *see* Total cost of ownership
Testbed, **241**
    Dubai Creek corridor, vi
The Loop (URB concept), 27
Tiered explanation, x
Tiered explanations, vii
Time-slot identifier, 133, 237
tornado diagram, 172
Total cost of ownership, 141, **248**
Tourism Climate Index, 33, **242**
TOWS matrix, 175, *see* SWOT analysis
Transition function, 109, **246**
tree planting, 161

UAE Civil Transactions Law, 185
UAE Space Agency, 18, 91, 195, 203, **253**
UHI, *see* Urban heat island

UN Secretary-General's Call to Action on Extreme Heat, 191
unarchived source, 220
uncertainty reporting, 146
UNEP, *see* United Nations Environment Programme
United Nations Environment Programme, v, **253**
Universal Thermal Climate Index, 11, 34, 59, 73, 109, 161, 191, 202, 221, 232, **240**
uplift, 138
Uptime, 133, 232
Urban heat island, 25, 55, **240**
Urban Innovative Actions, v
UTCI, *see* Universal Thermal Climate Index

Value function, 112, 142, **246**
value of information, 70, 87
versioning, 224

Wakāla, xiv
Waqf, xv
warranty, 181
WBGT, *see* Wet Bulb Globe Temperature
WBS, *see* Work breakdown structure
web capture, 217
Wet Bulb Globe Temperature, 11, 32, 56, 73, 110, 150, 168, 191, 202, 221, 232, **240**
Whole-life cost, 135, 149, **248**
Win conditions, 75, 92, **250**
WinWin spiral model, 92, 103
Work breakdown structure, 137, 161, **248**
Written commitments, vi
Written confirmation, 93, **252**

Zakāt, xv

The Kalman Systems company portfolio linked above is copyright © 2026 Kalman Systems Pty Ltd, all rights reserved. No licence is granted to use, copy, modify or distribute that material without explicit written permission.